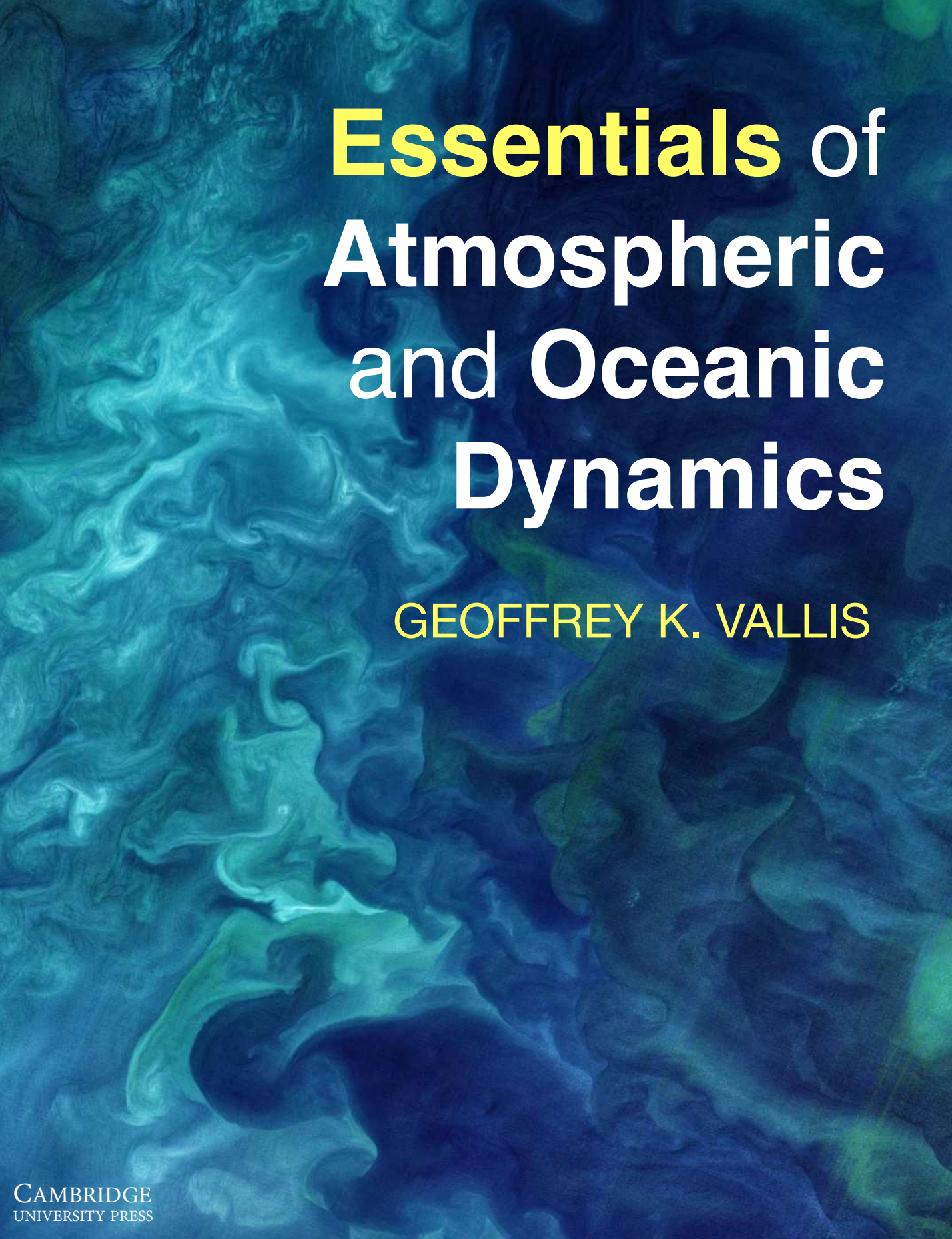




# **Essentials** of **Atmospheric and Oceanic Dynamics**

**GEOFFREY K. VALLIS**

The "big book" [AOFD] by Vallis is a treasure, but I suspect that this new *Essentials* is destined to be used much more widely in classrooms. Vallis does a superb job of communicating the peculiar tensions between deductive reasoning and physical intuition that underlie this science. The new book is more approachable but no less rigorous. I especially appreciate how the various equation sets are derived in succinct but meaningful ways in the first few chapters, and then used as tools to explore the dynamics in the chapters that follow. It's almost the perfect introductory textbook on this subject, and I plan to use it in my own courses.

Brian E. J. Rose,  
University at Albany

He's done it again. In *Essentials*, Geoff Vallis has produced a text that is useful to the student and the experienced scientist alike. While the content is simplified and shortened compared to its parent text, Vallis now provides even more descriptive explanations to support readers in their quest to navigate the physics of fluid flows. These explanations pair well with the theory, serving as an accessible introduction to students while also supporting the more experienced scientist as they put all of the pieces together. This will certainly be a future favourite for reading groups. Even readers with dog-eared versions of the parent book will want a copy of *Essentials*, for in it Vallis has added an entirely new chapter on planetary atmospheres, allowing the interested reader to venture into outer space to apply their newly honed GFD expertise.

Elizabeth A. Barnes,  
Colorado State University

For the past decade, Geoff Vallis' book *Atmospheric and Oceanic Fluid Dynamics* has been the "go to" encyclopaedic resource, but it is too lengthy and comprehensive to use as a course textbook. With this superb new shorter volume, Geoff Vallis provides us with the definitive graduate-level textbook, with just the right balance of essential topics alongside glimpses of more advanced topics at the cutting edge of research. The extensive use of margin notes, diamonds to indicate advanced topics, and a comprehensive set of problems will ensure that *Essentials of Atmospheric and Oceanic Dynamics* has much to offer students and researchers at all levels. The book opens with the quote: "Seek simplicity, accept complexity. Exploit simplification, avoid complication." On all counts, this book succeeds magnificently!

David Marshall,  
University of Oxford

Vallis' insights into the fundamentals and applications go a long way towards making otherwise complex topics readily grasped by those willing to study. He does not shy away from mathematics where needed, nor does he smother the reader with mathematics where pedagogically unnecessary. Those making it through this book will be ready to tackle a huge suite of research questions related to atmosphere and ocean fluid mechanics. Hence, this book serves an incredibly important role to the academic community. In a nutshell, we need more smart researchers who are adept at atmosphere and ocean dynamics to help understand how those dynamics are increasingly being affected by humanity's choices.

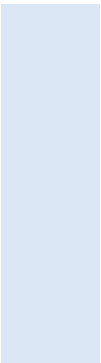
*Essentials of Atmospheric and Oceanic Dynamics* (EAOD) fills an important niche by offering an articulate and authoritative textbook to be worked through by advanced undergraduates and/or entering graduate students taking courses. The inclusion of exercises in EAOD is incredibly valuable for both students and teachers clamouring for more problem sets to test understanding. Whereas Vallis' previous book, *Atmospheric and Oceanic Fluid Dynamics* (AOFD) is the mother reference, EAOD offers a pedagogical entrée for those wishing to test the waters, including some deep waters. I will happily keep both books on my shelf and make use of them for personal study and to support the teaching of geophysical fluid dynamics.

Vallis has a clear writing style that brings the reader into the subject in an authoritative and friendly manner. He is a wise guru and gentle tutor. The subject of ocean and atmosphere fluid mechanics has matured greatly through his efforts at writing AOFD. EAOD furthers that maturation by allowing for a broader readership to tap into his brain. Well done Geoff!

Stephen M. Griffies,  
GFDL, Princeton University.

As its parent book became the bible of the field, but also grew in size and the number of topics it covered in its latest edition, this new book provides a perfect balance and introduction to the essential topics, giving a quick reference without going into all the details. In the Vallis tradition, it is presented clearly, perfectly packaged, and is well organized for both atmospheric and oceanic fluid dynamics. Its simplicity will make it majestically appealing both for people outside the discipline looking for an accessible, yet complete, introduction, and for students within the field at all levels. The inclusion of planetary atmospheres broadens the scope and makes it appealing to a wider and growing audience. Anyone with a background in physics can get the essentials using this book.

Yohai Kaspi,  
Weizmann Institute of Science



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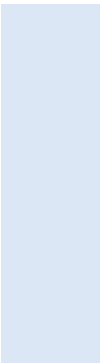
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Note: In the text itself more advanced sections are marked with a diamond, ♦, and may be omitted on a first reading. If a section is so marked then the marking applies to all the subsections within it.



---

## Preface

*Seek simplicity, accept complexity.  
Exploit simplification, avoid complication.*

This is an introductory book on the dynamics of atmospheres and oceans, with a healthy dose of geophysical fluid dynamics. It is written roughly at the level of advanced or upper-division undergraduates and beginning graduate students, but parts of it will be accessible to first- or second-year undergraduates and I hope that practising scientists will also find it useful. The book is designed for students and scientists who want an introduction to the subject but who may not want all the detail, at least not yet, and its prerequisites are just familiarity with some vector calculus and basic classical physics. Thus, it is meant to be accessible to non-specialists and students who will not necessarily go on to become professional dynamists. However, as well as very basic material the book does include some elementary introductions to a few ‘advanced’ topics, such as the residual circulation and turbulence theory, as well as material on the general circulation of the atmosphere and ocean. The more advanced parts could easily be omitted for a first course and, like difficult ski slopes, are marked with a diamond, ♦. Readers may explore these topics more in the references provided, or in this book’s parent, *Atmospheric and Oceanic Fluid Dynamics*. Nearly all the topics in this book, except those in the chapter on planetary atmospheres, are dealt with in greater detail there.

### *What is in the book*

The book is divided into three Parts. The first, and longest, provides the foundation for the study of the dynamics of the atmosphere and ocean. It does not assume any prior knowledge of fluid dynamics or thermodynamics, although readers who have such knowledge may be able to skim Chapter 1. The rest of Part I provides an introduction to ‘geophysical fluid dynamics’, the subject that remains at the heart of atmospheric and oceanic dynamics and without which the subject would be largely qualitative and/or computational. Here we discuss the effects of rota-

tion and stratification, leading into shallow water theory and the quasi-geostrophic and planetary-geostrophic equations. Rossby waves, gravity waves, baroclinic instability and elementary treatments of wave–mean-flow interaction and turbulence round out Part I.

Parts II and III focus on the large scale dynamics and circulation of the atmosphere and ocean, respectively. Our main focus in both Parts is what is sometimes called ‘the general circulation’, meaning the large-scale quasi-steady and/or time-averaged circulation, but this circulation depends on the effects of time-dependent eddies — the atmosphere’s Ferrel Cell may be considered to be ‘driven’ by the effects of baroclinic instability and Rossby waves. And the El Niño phenomenon, described in the final chapter, is explicitly time dependent. One feature of this book that is not in the parent book is a chapter discussing some of the general principles of planetary atmospheres, a topic of increasing interest because of the new, sometimes quite spectacular, observations of the planets in our Solar System and beyond.

### *How to use the book*

The contents of the book are about enough for a two-term course in atmosphere–ocean dynamics. A term-long, first course in geophysical fluid dynamics could, for example, be based on Part I, omitting some of the earlier or later chapters depending on the students’ backgrounds and interests. A term-long course in atmospheric and/or oceanic circulation could be based on Part II and/or Part III, supplementing the material with review articles or research papers as needed, perhaps using data sets to look at the real world (and other planets, if Chapter 13 is to be studied). Alternatively, one could combine aspects of Parts I and II, or Parts I and III, to construct an ‘Atmospheric Dynamics’ or ‘Oceanic Dynamics’ course.

If the book is to be used for self-study it could simply be read from beginning to end, although many other pathways are possible and may be preferable. Parts II and III depend on the material in Part I, but the material is reasonably self-contained, and readers who already have some knowledge of geophysical fluid dynamics should feel free to start at a later chapter, or with Part II or Part III. A few problems are collected at the end of some chapters; these are designed to test understanding as well as to fill in gaps and extend the material in the book itself. Many other problems at varying levels of difficulty can be found on the web site of this book, which can easily be found with a search engine. The reader will also see a number of margin notes throughout the book, rather like the ones to the left. The book itself was typeset using  $\text{\LaTeX}$  with Crimson fonts for text, Cronos Pro for sans serif and Minion Math for equations.

I would like to thank Matt Lloyd, Zoë Pruce and Richard Smith at Cambridge University Press for their expert guidance through the writing and production process, as well as many colleagues and students — too many to list, but they know who they are — for their many comments, corrections and criticisms. If you, the reader, have other comments, major or minor, do please contact me.

Margin notes that are set in a roman (i.e., upright) font emphasize or expand on something that is in the main text.

*Margin notes set in italics are asides or historical anecdotes.*



Part I

# **GEOPHYSICAL FLUIDS**



## Shallow Water Equations

**T**HE *SHALLOW WATER EQUATIONS* are a set of equations that describe, not surprisingly, a shallow layer of fluid, and in particular one that is in hydrostatic balance and has constant density. The equations are useful for two reasons:

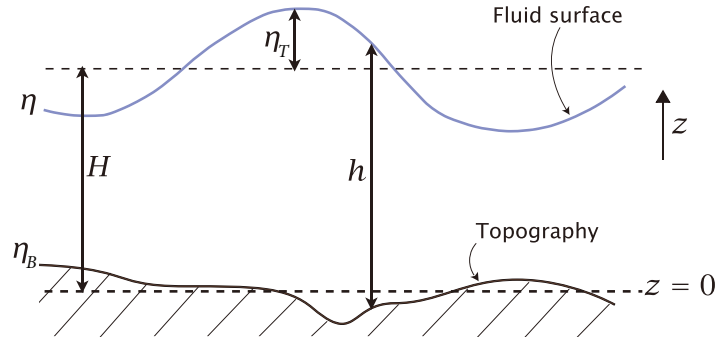
- (i) They are a simpler set of equations than the full three-dimensional ones, and so allow for a much more straightforward analysis of sometimes complex problems.
- (ii) In spite of their simplicity, the equations provide a reasonably realistic representation of a variety of phenomena in atmospheric and oceanic dynamics.

Put simply, the shallow water equations are a very useful *model* for geophysical fluid dynamics. Let's dive head first into the equations and see what they can do for us.

### 4.1 SHALLOW WATER EQUATIONS OF MOTION

The shallow water equations apply, by definition, to a fluid layer of constant density in which the horizontal scale of the flow is much greater than the layer depth, and which have a free surface at the top (or sometimes at the bottom). Because the fluid is of constant density the fluid motion is fully determined by the momentum and mass continuity equations, and because of the assumed small aspect ratio the hydrostatic approximation is well satisfied, as we discussed in Section 3.2.2. Thus, consider a fluid above which is another fluid of negligible density, as illustrated in Fig. 4.1. Our notation is that  $\mathbf{v} = u\hat{\mathbf{i}} + v\hat{\mathbf{j}} + w\hat{\mathbf{k}}$  is the three-dimensional velocity and  $\mathbf{u} = u\hat{\mathbf{i}} + v\hat{\mathbf{j}}$  is the horizontal velocity,  $h(x, y)$  is the thickness of the liquid column,  $H$  is its mean height, and  $\eta$  is the height of the free surface. In a flat-bottomed container  $\eta = h$ , whereas in general  $h = \eta - \eta_B$ , where  $\eta_B$  is the height of the floor of the container.

**Fig. 4.1:** A shallow water system where  $h$  is the thickness of a water column,  $H$  its mean thickness,  $\eta$  the height of the free surface and  $\eta_B$  is the height of the lower, rigid, surface above some arbitrary origin, typically chosen such that the average of  $\eta_B$  is zero. The quantity  $\eta_B$  is the deviation free surface height so we have  $\eta = \eta_B + h = H + \eta_T$ .



#### 4.1.1 Momentum Equations

The vertical momentum equation is just the hydrostatic equation,

$$\frac{\partial p}{\partial z} = -\rho_0 g, \quad (4.1)$$

and, because density is assumed constant, we may integrate this to

$$p(x, y, z, t) = -\rho_0 g z + p_o. \quad (4.2)$$

At the top of the fluid,  $z = \eta$ , the pressure is determined by the weight of the overlying fluid and this is negligible. Thus,  $p = 0$  at  $z = \eta$ , giving

$$p(x, y, z, t) = \rho_0 g(\eta(x, y, t) - z). \quad (4.3)$$

The consequence of this is that *the horizontal gradient of pressure is independent of height*. That is

$$\nabla_z p = \rho_0 g \nabla_z \eta, \quad \text{where} \quad \nabla_z = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y}. \quad (4.4)$$

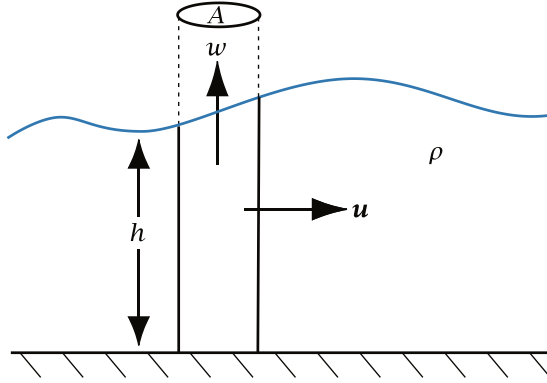
(In the rest of this chapter we drop the subscript  $z$  unless that causes ambiguity; the three-dimensional gradient operator is denoted by  $\nabla_3$ . We also mostly use Cartesian coordinates, but the shallow water equations may certainly be applied over a spherical planet.) The horizontal momentum equations therefore become

$$\frac{D\mathbf{u}}{Dt} = -\frac{1}{\rho_0} \nabla p = -g \nabla \eta. \quad (4.5)$$

The right-hand side of this equation is independent of the vertical coordinate  $z$ . Thus, if the flow is initially independent of  $z$ , it must stay so. (This  $z$ -independence is unrelated to that arising from the rapid rotation necessary for the Taylor–Proudman effect.) The velocities  $u$  and  $v$  are functions of  $x$ ,  $y$  and  $t$  only, and the horizontal momentum equation is therefore

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + u \frac{\partial \mathbf{u}}{\partial x} + v \frac{\partial \mathbf{u}}{\partial y} = -g \nabla \eta. \quad (4.6)$$

The key assumption underlying the shallow water equations is that of a small aspect ratio, so that  $H/L \ll 1$ , where  $H$  is the fluid depth and  $L$  the horizontal scale of motion. This gives rise to the hydrostatic approximation, and this in turn leads to the  $z$ -independence of the velocity field and the ‘sloshing’ nature of the flow.



**Fig. 4.2:** The mass budget for a column of area  $A$  in a flat-bottomed shallow water system. The fluid leaving the column is  $\oint \rho \mathbf{h} \mathbf{u} \cdot \mathbf{n} \, dl$  where  $\mathbf{n}$  is the unit vector normal to the boundary of the fluid column. There is a non-zero vertical velocity at the top of the column if the mass convergence into the column is non-zero.

In the presence of rotation, (4.6) easily generalizes to

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (4.7)$$

where  $\mathbf{f} = f\hat{\mathbf{k}}$ . Just as with the fully three-dimensional equations,  $f$  may be constant or may vary with latitude, so that on a spherical planet  $f = 2\Omega \sin \vartheta$  and on the  $\beta$ -plane  $f = f_0 + \beta y$ .

#### 4.1.2 Mass Continuity Equation

The mass contained in a fluid column of height  $h$  and cross-sectional area  $A$  is given by  $\int_A \rho_0 h \, dA$  (see Fig. 4.2). If there is a net flux of fluid across the column boundary (by advection) then this must be balanced by a net increase in the mass in  $A$ , and therefore a net increase in the height of the water column. The mass convergence into the column is given by

$$F_m = \text{mass flux in} = - \int_S \rho_0 \mathbf{u} \cdot d\mathbf{S}, \quad (4.8)$$

where  $S$  is the area of the vertical boundary of the column. The surface area of the column is composed of elements of area  $h \mathbf{n} \, \delta l$ , where  $\delta l$  is a line element circumscribing the column and  $\mathbf{n}$  is a unit vector perpendicular to the boundary, pointing outwards. Thus (4.8) becomes

$$F_m = - \oint \rho_0 h \mathbf{u} \cdot \mathbf{n} \, dl. \quad (4.9)$$

Using the divergence theorem in two dimensions, (4.9) simplifies to

$$F_m = - \int_A \nabla \cdot (\rho_0 h \mathbf{u}) \, dA, \quad (4.10)$$

where the integral is over the cross-sectional area of the fluid column (looking down from above). This is balanced by the local increase in height of the water column, given by

$$F_m = \frac{d}{dt} \int \rho_0 \, dV = \frac{d}{dt} \int_A \rho_0 h \, dA = \int_A \rho_0 \frac{\partial h}{\partial t} \, dA. \quad (4.11)$$

### The Shallow Water Equations

For a single-layer fluid, and including the Coriolis term, the inviscid shallow water equations are:

$$\text{momentum:} \quad \frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad (\text{SW.1})$$

$$\text{mass continuity:} \quad \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0, \quad (\text{SW.2})$$

$$\text{or} \quad \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \quad (\text{SW.3})$$

where  $\mathbf{u}$  is the horizontal velocity,  $h$  is the total fluid thickness,  $\eta$  is the height of the upper free surface, and  $h$  and  $\eta$  are related by

$$h(x, y, t) = \eta(x, y, t) - \eta_B(x, y), \quad (\text{SW.4})$$

where  $\eta_B$  is the height of the lower surface (the bottom topography). The material derivative is

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y}, \quad (\text{SW.5})$$

with the rightmost expression holding in Cartesian coordinates.

Because  $\rho_0$  is constant, the balance between (4.10) and (4.11) leads to

$$\int_A \left[ \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) \right] dA = 0, \quad (4.12)$$

and because the area is arbitrary the integrand itself must vanish, whence,

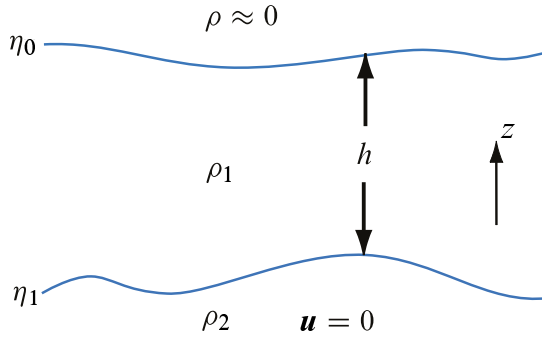
$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0 \quad \text{or} \quad \frac{Dh}{Dt} + h\nabla \cdot \mathbf{u} = 0. \quad (4.13a,b)$$

This derivation holds whether or not the lower surface is flat. If it is, then  $h = \eta$ , and if not  $h = \eta - \eta_B$ . Equations (4.7) and (4.13) form a complete set, summarized in the shaded box above.

#### 4.1.3 Reduced Gravity Equations

Consider now a single shallow moving layer of fluid *on top* of a deep, quiescent fluid layer (Fig. 4.3), and beneath a fluid of negligible inertia. This configuration is often used as a model of the upper ocean: the upper layer represents flow in perhaps the upper few hundred metres of the ocean, the lower layer being the near-stagnant abyss. If we turn the model upside-down we have a perhaps slightly less realistic model of the atmosphere: the lower layer represents motion in the troposphere above which lies an





**Fig. 4.3:** The reduced gravity shallow water system. An active layer lies over a deep, denser, quiescent layer. In a common variation the upper surface is held flat by a rigid lid, and  $\eta_0 = 0$ .

inactive stratosphere. The equations of motion are virtually the same in both cases, but for definiteness we'll think about the oceanic case.

The pressure in the upper layer is given by integrating the hydrostatic equation down from the upper surface. Thus, at a height  $z$  in the upper layer

$$p_1(z) = g\rho_1(\eta_0 - z), \quad (4.14)$$

where  $\eta_0$  is the height of the upper surface. Hence, everywhere in the upper layer,

$$\frac{1}{\rho_1} \nabla p_1 = g \nabla \eta_0, \quad (4.15)$$

and the momentum equation is

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta_0. \quad (4.16)$$

In the lower layer the pressure is also given by the weight of the fluid above it. Thus, at some level  $z$  in the lower layer,

$$p_2(z) = \rho_1 g(\eta_0 - \eta_1) + \rho_2 g(\eta_1 - z). \quad (4.17)$$

But if this layer is motionless the horizontal pressure gradient in it is zero and therefore, at some value of  $z$ ,

$$\rho_1 g \eta_0 = -\rho_1 g' \eta_1 + \text{constant}, \quad (4.18)$$

where  $g' = g(\rho_2 - \rho_1)/\rho_1$  is the *reduced gravity*, and in the world's oceans  $(\rho_2 - \rho_1)/\rho \ll 1$  and  $g' \ll g$ . The momentum equation becomes

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = g' \nabla \eta_1. \quad (4.19)$$

The equations are completed by the usual mass conservation equation,

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0, \quad (4.20)$$

where  $h = \eta_0 - \eta_1$ . Because  $g \gg g'$ , (4.18) shows that surface displacements are *much smaller* than the displacements at the interior interface. We see this in the real ocean where the mean interior isopycnal displacements may be several tens of metres but variations in the mean height of the ocean surface are of the order of centimetres.

## 4.2 CONSERVATION PROPERTIES

There are two common types of conservation property in fluids: (i) material invariants; and (ii) integral invariants. Material invariance occurs when a property ( $\varphi$  say) is conserved on each fluid element, and so obeys the equation  $D\varphi/Dt = 0$ . An integral invariant is one that is conserved after an integration over some, usually closed, volume; energy is an example. The simplicity of the shallow water equations allows us to transparently see how these arise.

### 4.2.1 Energy Conservation: an Integral Invariant

Since we have made various simplifications in deriving the shallow water system, it is not self-evident that energy should be conserved, or indeed what form the energy takes. The kinetic energy density ( $KE$ ), meaning the kinetic energy per unit area, is  $\rho_0 h \mathbf{u}^2/2$ . The potential energy density ( $PE$ ) of the fluid is

$$PE = \int_0^h \rho_0 g z \, dz = \frac{1}{2} \rho_0 g h^2. \quad (4.21)$$

The factor  $\rho_0$  appears in both kinetic and potential energies and, because it is a constant, we will omit it. For algebraic simplicity we also assume the bottom is flat, at  $z = 0$ .

Using the mass conservation equation (4.13b) we obtain an equation for the evolution of potential energy density, namely

$$\frac{D}{Dt} \frac{gh^2}{2} + gh^2 \nabla \cdot \mathbf{u} = 0 \quad (4.22a)$$

or

$$\frac{\partial}{\partial t} \frac{gh^2}{2} + \nabla \cdot \left( \mathbf{u} \frac{gh^2}{2} \right) + \frac{gh^2}{2} \nabla \cdot \mathbf{u} = 0. \quad (4.22b)$$

From the momentum and mass continuity equations we obtain an equation for the evolution of kinetic energy density, namely

$$\frac{D}{Dt} \frac{h\mathbf{u}^2}{2} + \frac{\mathbf{u}^2 h}{2} \nabla \cdot \mathbf{u} = -g\mathbf{u} \cdot \nabla \frac{h^2}{2} \quad (4.23a)$$

or

$$\frac{\partial}{\partial t} \frac{h\mathbf{u}^2}{2} + \nabla \cdot \left( \mathbf{u} \frac{h\mathbf{u}^2}{2} \right) + g\mathbf{u} \cdot \nabla \frac{h^2}{2} = 0. \quad (4.23b)$$

Adding (4.22b) and (4.23b) we obtain

$$\frac{\partial}{\partial t} \frac{1}{2} (h\mathbf{u}^2 + gh^2) + \nabla \cdot \left[ \frac{1}{2} \mathbf{u} (gh^2 + h\mathbf{u}^2 + gh^2) \right] = 0 \quad (4.24)$$

or

$$\frac{\partial E}{\partial t} + \nabla \cdot \mathbf{F} = 0, \quad (4.25)$$

where  $E = KE + PE = (h\mathbf{u}^2 + gh^2)/2$  is the density of the total energy and  $\mathbf{F} = \mathbf{u}(h\mathbf{u}^2/2 + gh^2)$  is the energy flux. If the fluid is confined to a domain

Energy is moved from place to place by pressure forces as well as advection, and is not a material invariant. However, in total energy is conserved and it is an integral invariant.

bounded by rigid walls, on which the normal component of velocity vanishes, then on integrating (4.24) over that area and using Gauss's theorem, the total energy is seen to be conserved; that is

$$\frac{d\hat{E}}{dt} = \frac{1}{2} \frac{d}{dt} \int_A (h\mathbf{u}^2 + gh^2) dA = 0. \quad (4.26)$$

Such an energy principle also holds in the case with bottom topography. Note that, as we found in the case for a compressible fluid in Chapter 2, the energy flux in (4.25) is not just the energy density multiplied by the velocity; it contains an additional term  $guh^2/2$ , and this represents the energy transfer occurring when the fluid does work against the pressure force.

#### 4.2.2 Potential Vorticity: a Material Invariant

The vorticity of a fluid, denoted  $\boldsymbol{\omega}$ , is defined to be the curl of the velocity field. Let us also define the shallow water vorticity,  $\boldsymbol{\omega}^*$ , as the curl of the horizontal velocity. We therefore have:

$$\boldsymbol{\omega} \equiv \nabla \times \mathbf{v}, \quad \boldsymbol{\omega}^* \equiv \nabla \times \mathbf{u}. \quad (4.27)$$

Because  $\partial u/\partial z = \partial v/\partial z = 0$ , only the vertical component of  $\boldsymbol{\omega}^*$  is non-zero and

$$\boldsymbol{\omega}^* = \hat{\mathbf{k}} \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) = \hat{\mathbf{k}} \zeta. \quad (4.28)$$

Considering first the non-rotating case, we use the vector identity

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = \frac{1}{2} \nabla(\mathbf{u} \cdot \mathbf{u}) - \mathbf{u} \times (\nabla \times \mathbf{u}), \quad (4.29)$$

to write the momentum equation, (4.7) as

$$\frac{\partial \mathbf{u}}{\partial t} + \boldsymbol{\omega}^* \times \mathbf{u} + \mathbf{f} \times \mathbf{u} = -\nabla \left( g\eta + \frac{1}{2} \mathbf{u}^2 \right). \quad (4.30)$$

To obtain an evolution equation for the vorticity we take the curl of (4.30), and make use of the vector identity

$$\begin{aligned} \nabla \times (\boldsymbol{\omega}^* \times \mathbf{u}) &= (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}^* - (\boldsymbol{\omega}^* \cdot \nabla) \mathbf{u} + \boldsymbol{\omega}^* \nabla \cdot \mathbf{u} - \mathbf{u} \nabla \cdot \boldsymbol{\omega}^* \\ &= (\mathbf{u} \cdot \nabla) \boldsymbol{\omega}^* + \boldsymbol{\omega}^* \nabla \cdot \mathbf{u}, \end{aligned} \quad (4.31)$$

using the fact that  $\nabla \cdot \boldsymbol{\omega}^*$  is the divergence of a curl and therefore zero, and  $(\boldsymbol{\omega}^* \cdot \nabla) \mathbf{u} = 0$  because  $\boldsymbol{\omega}^*$  is perpendicular to the surface in which  $\mathbf{u}$  varies. A similar expression holds for  $\mathbf{f}$  so that taking the curl of (4.30) gives

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)(\zeta + f) = -(\zeta + f) \nabla \cdot \mathbf{u}, \quad (4.32)$$

where  $\zeta = \hat{\mathbf{k}} \cdot \boldsymbol{\omega}^*$  and  $f$  varies with latitude (and on the beta-plane  $f = f_0 + \beta y$ ). Since  $f$  does not vary with time we can write (4.32) as

$$\frac{D}{Dt} (\zeta + f) = -(\zeta + f) \nabla \cdot \mathbf{u}. \quad (4.33)$$

The mass conservation equation, (4.13b) may be written as

$$-(\zeta + f)\nabla \cdot \mathbf{u} = \frac{\zeta + f}{h} \frac{Dh}{Dt}, \quad (4.34)$$

and using this equation and (4.32) we obtain

$$\frac{D}{Dt}(\zeta + f) = \frac{\zeta + f}{h} \frac{Dh}{Dt}, \quad (4.35)$$

which is equivalent to

$$\frac{DQ}{Dt} = 0 \quad \text{where} \quad Q = \left( \frac{\zeta + f}{h} \right). \quad (4.36)$$

The important quantity  $Q$  is known as the *potential vorticity*, and (4.36) is the potential vorticity equation.

### 4.3 SHALLOW WATER WAVES

Let us now look at the gravity waves that occur in shallow water. To isolate the essence we consider waves in a single fluid layer, with a flat bottom and a free upper surface, in which gravity provides the sole restoring force.

#### 4.3.1 Non-Rotating Shallow Water Waves

Given a flat bottom the fluid thickness is equal to the free surface displacement (Fig. 4.1), and taking the basic state of the fluid to be at rest we let

$$h(x, y, t) = H + h'(x, y, t) = H + \eta'(x, y, t), \quad (4.37a)$$

$$\mathbf{u}(x, y, t) = \mathbf{u}'(x, y, t). \quad (4.37b)$$

The mass conservation equation, (4.13b), then becomes

$$\frac{\partial \eta'}{\partial t} + (H + \eta')\nabla \cdot \mathbf{u}' + \mathbf{u}' \cdot \nabla \eta' = 0, \quad (4.38)$$

and neglecting squares of small quantities this yields the linear equation

$$\frac{\partial \eta'}{\partial t} + H\nabla \cdot \mathbf{u}' = 0. \quad (4.39)$$

Similarly, linearizing the momentum equation, (4.7) with  $f = 0$ , yields

$$\frac{\partial \mathbf{u}'}{\partial t} = -g\nabla \eta'. \quad (4.40)$$

Eliminating velocity by differentiating (4.39) with respect to time and taking the divergence of (4.40) leads to

$$\frac{\partial^2 \eta'}{\partial t^2} - gH\nabla^2 \eta' = 0, \quad (4.41)$$

which may be recognized as a wave equation. We can find the dispersion relationship for this by substituting the trial solution

$$\eta' = \text{Re } \tilde{\eta} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (4.42)$$

where  $\tilde{\eta}$  is a complex constant,  $\mathbf{k} = k\hat{\mathbf{i}} + l\hat{\mathbf{j}}$  is the horizontal wavenumber and  $\text{Re}$  indicates that the real part of the solution should be taken. If, for simplicity, we restrict attention to the one-dimensional problem, with no variation in the  $y$ -direction, then substituting into (4.41) leads to the dispersion relationship

$$\omega = \pm ck, \quad (4.43)$$

where  $c = \sqrt{gH}$ ; that is, the wave speed is proportional to the square root of the mean fluid depth and is independent of the wavenumber — the waves are dispersionless. The general solution is a superposition of all such waves, with the amplitudes of each wave (or Fourier component) being determined by the Fourier decomposition of the initial conditions.

Because the waves are dispersionless, the general solution can be written as

$$\eta'(x, t) = \frac{1}{2} [F(x - ct) + F(x + ct)], \quad (4.44)$$

where  $F(x)$  is the height field at  $t = 0$ . From this, it is easy to see that the shape of an initial disturbance is preserved as it propagates both to the right and to the left at speed  $c$ .

### 4.3.2 Rotating Shallow Water (Poincaré) Waves

We now consider the effects of rotation on shallow water waves. Linearizing the rotating, flat-bottomed  $f$ -plane shallow water equations, (SW.1) and (SW.2) on page 66, about a state of rest we obtain

$$\frac{\partial u'}{\partial t} - f_0 v' = -g \frac{\partial \eta'}{\partial x}, \quad \frac{\partial v'}{\partial t} + f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad (4.45a,b)$$

$$\frac{\partial \eta'}{\partial t} + H \left( \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) = 0. \quad (4.45c)$$

To obtain a dispersion relationship we let

$$(u, v, \eta) = (\tilde{u}, \tilde{v}, \tilde{\eta}) e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (4.46)$$

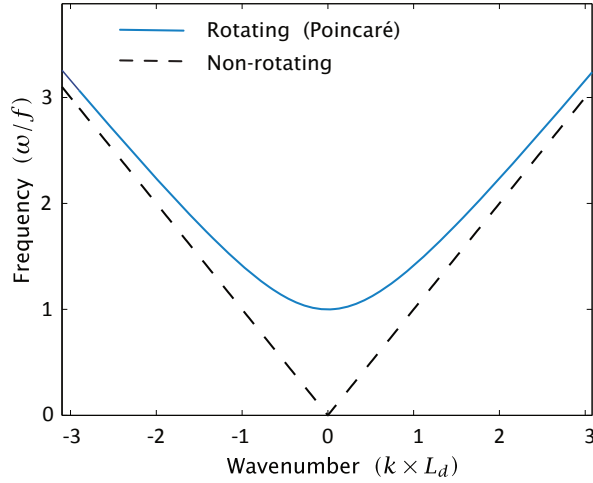
and substitute into (4.45), giving

$$\begin{pmatrix} -i\omega & -f_0 & igk \\ f_0 & -i\omega & igl \\ iHk & iHl & -i\omega \end{pmatrix} \begin{pmatrix} \tilde{u} \\ \tilde{v} \\ \tilde{\eta} \end{pmatrix} = 0. \quad (4.47)$$

This homogeneous equation has non-trivial solutions only if the determinant of the matrix vanishes, and that condition gives

$$\omega(\omega^2 - f_0^2 - c^2 K^2) = 0, \quad (4.48)$$

**Fig. 4.4:** Dispersion relation for Poincaré waves and non-rotating shallow water waves. Frequency is scaled by the Coriolis frequency  $f$ , and wavenumber by the inverse deformation radius  $\sqrt{gH}/f$ . For small wavenumbers the frequency of the Poincaré waves is approximately  $f$ , and for high wavenumbers it asymptotes to that of non-rotating waves.



where  $K^2 = k^2 + l^2$  and  $c^2 = gH$ . There are two classes of solution to (4.48). The first is simply  $\omega = 0$ , i.e., time-independent flow corresponding to geostrophic balance in (4.45). Because geostrophic balance gives a divergence-free velocity field for a constant Coriolis parameter the equations are satisfied by a time-independent solution. The second set of solutions gives the dispersion relation

$$\omega^2 = f_0^2 + c^2(k^2 + l^2), \quad (4.49)$$

or

$$\omega^2 = f_0^2 + gH(k^2 + l^2). \quad (4.50)$$

The corresponding waves are known as *Poincaré waves*, and the dispersion relationship is illustrated in Fig. 4.4. Note that the frequency is always greater than the Coriolis frequency  $f_0$ . There are two interesting limits:

(i) *The short wave limit.* If

$$K^2 \gg \frac{f_0^2}{gH}, \quad (4.51)$$

where  $K^2 = k^2 + l^2$ , then the dispersion relationship reduces to that of the non-rotating case (4.43). This condition is equivalent to requiring that the wavelength be much shorter than the *deformation radius*, defined by

$$L_d = \frac{\sqrt{gH}}{f}. \quad (4.52)$$

Specifically, if  $l = 0$  and  $\lambda = 2\pi/k$  is the wavelength, the condition is  $\lambda^2 \ll L_d^2(2\pi)^2$ . The numerical factor of  $(2\pi)^2$  is more than an order of magnitude, so care must be taken when deciding if the condition is satisfied in particular cases. Furthermore, the wavelength must still be longer than the depth of the fluid, otherwise the shallow water condition is not met.

(Jules) Henri Poincaré (1854–1912) was a remarkable French mathematician, physicist and philosopher. As well as the Poincaré waves of this chapter he is also known as one of the founders of what is now known as chaos theory — he knew that the weather was inherently, and not just in practice, unpredictable.

(ii) *The long wave limit.* If

$$K^2 \ll \frac{f_0^2}{gH}, \quad (4.53)$$

that is if the wavelength is much longer than the deformation radius  $L_d$ , then the dispersion relationship is

$$\omega = f_0. \quad (4.54)$$

These waves are known as *inertial oscillations*. The equations of motion giving rise to them are

$$\frac{\partial u'}{\partial t} - f_0 v' = 0, \quad \frac{\partial v'}{\partial t} + f_0 u' = 0, \quad (4.55)$$

which are equivalent to material equations for free particles in a rotating frame, unconstrained by pressure forces.

### 4.3.3 Kelvin Waves

The Kelvin wave is a particular type of gravity wave that exists in the presence of both rotation and a lateral boundary. Suppose there is a solid boundary at  $y = 0$ ; general harmonic solutions in the  $y$ -direction are not allowable, as these would not satisfy the condition of no normal flow at the boundary. Do any wave-like solutions exist? The answer is yes, and to show that we begin with the linearized shallow water equations, namely

$$\frac{\partial u'}{\partial t} - f_0 v' = -g \frac{\partial \eta'}{\partial x}, \quad \frac{\partial v'}{\partial t} + f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad (4.56a,b)$$

$$\frac{\partial \eta'}{\partial t} + H \left( \frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} \right) = 0. \quad (4.56c)$$

The fact that  $v' = 0$  at  $y = 0$  suggests that we look for a solution with  $v' = 0$  everywhere, whence these equations become

$$\frac{\partial u'}{\partial t} = -g \frac{\partial \eta'}{\partial x}, \quad f_0 u' = -g \frac{\partial \eta'}{\partial y}, \quad \frac{\partial \eta'}{\partial t} + H \frac{\partial u'}{\partial x} = 0. \quad (4.57a,b,c)$$

Equations (4.57a) and (4.57c) lead to the standard wave equation

$$\frac{\partial^2 u'}{\partial t^2} = c^2 \frac{\partial^2 u'}{\partial x^2}, \quad (4.58)$$

where  $c = \sqrt{gH}$ , the usual wave speed of shallow water waves. The solution of (4.58) is

$$u' = F_1(x + ct, y) + F_2(x - ct, y), \quad (4.59)$$

with corresponding surface displacement

$$\eta' = \sqrt{H/g} [-F_1(x + ct, y) + F_2(x - ct, y)]. \quad (4.60)$$

The solution represents the superposition of two waves, one ( $F_1$ ) travel-

*The affirmative answer to the question in the text was provided by William Thomson in the nineteenth century. Thomson later became Lord Kelvin, taking the name of the river flowing near his university in Glasgow, and the waves are now eponymously known as Kelvin waves.*

ling in the negative  $x$ -direction, and the other in the positive  $x$ -direction. To obtain the  $y$  dependence of these functions we use (4.57b) which gives

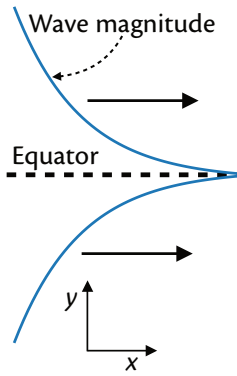
$$\frac{\partial F_1}{\partial y} = \frac{f_0}{\sqrt{gH}} F_1, \quad \frac{\partial F_2}{\partial y} = -\frac{f_0}{\sqrt{gH}} F_2, \quad (4.61)$$

with solutions

$$F_1 = F(x + ct)e^{y/L_d}, \quad F_2 = G(x - ct)e^{-y/L_d}, \quad (4.62)$$

where  $L_d = \sqrt{gH}/f_0$  is the radius of deformation. If we consider flow in the half-plane in which  $y > 0$ , then for positive  $f_0$  the solution  $F_1$  grows exponentially away from the wall, and so fails to satisfy the condition of boundedness at infinity. It thus must be eliminated, leaving the general solution

$$\begin{aligned} u' &= e^{-y/L_d} G(x - ct), & v' &= 0, \\ \eta' &= \sqrt{H/g} e^{-y/L_d} G(x - ct). \end{aligned} \quad (4.63a,b,c)$$



**Fig. 4.5:** Kelvin waves propagating eastward at the equator and decaying rapidly away to either side.

These are Kelvin waves and they decay exponentially away from the boundary. In general, for  $f_0$  positive (negative) the boundary is to the right (left) of the wave direction. Given a constant Coriolis parameter, we could also have obtained a solution on a meridional wall, again with the wave moving with the wall on its the right if  $f_0$  is positive. The equator also acts like a wall, with Kelvin waves propagating eastward along it, decaying to either side, as in Fig. 4.5. (The details of the solution for equatorial Kelvin waves differ from the case with  $f = f_0$ , because now  $f = 0$  at the equator itself, but the structure is similar to the solution above.)

#### 4.4 GEOSTROPHIC ADJUSTMENT

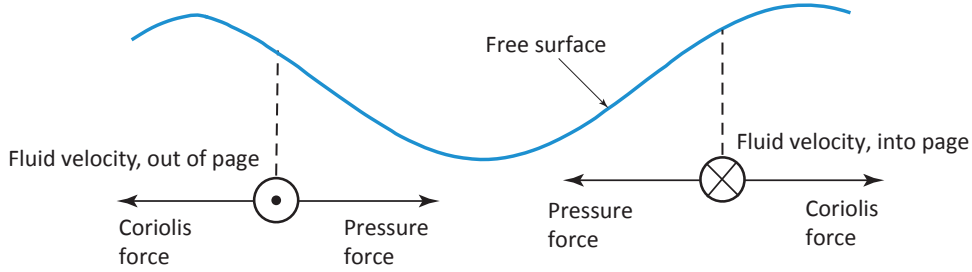
Geostrophic balance occurs in the shallow water equations when the Rossby number  $U/fL$  is small and the Coriolis term dominates the advective terms in the momentum equation. In the single-layer shallow water equations the geostrophic flow is:

$$\mathbf{f} \times \mathbf{u}_g = -g\nabla\eta. \quad (4.64)$$

Thus, the geostrophic velocity is proportional to the slope of the surface, as sketched in Fig. 4.6. The result arises because by hydrostatic balance the slope of an interfacial surface is directly related to the difference in pressure gradient on either side, so that in geostrophic balance the velocity is related to the slope.

But consider the more general question, *why* are the atmosphere and ocean close to geostrophic balance? Suppose that the initial state were not balanced, that it had small scale motion and sharp pressure gradients. How would the system evolve? It turns out that all of these sharp gradients would radiate away, leaving behind a geostrophically balanced state. The process is called *geostrophic adjustment*, and it occurs quite generally in rotating fluids, whether stratified or not. The shallow water equations are an ideal set of equations to explore this process, as we now see.





**Fig. 4.6:** Geostrophic flow in a shallow water system, with a positive value of the Coriolis parameter  $f$ , as in the Northern Hemisphere. The pressure force is directed down the gradient of the height field, and this can be balanced by the Coriolis force if the fluid velocity is at right angles to it. If  $f$  were negative, the geostrophic flow would be reversed.

#### 4.4.1 Posing the Problem

Let us consider the free evolution of a single shallow layer of fluid whose initial state is manifestly unbalanced, and we suppose that surface displacements are small so that the evolution of the system is described by the linearized shallow equations of motion. These are

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} = -g\nabla\eta, \quad \frac{\partial \eta}{\partial t} + H\nabla \cdot \mathbf{u} = 0, \quad (4.65a,b)$$

where  $\eta$  is the free surface displacement and  $H$  is the mean fluid depth, and we omit the primes on the linearized variables.

#### 4.4.2 Non-Rotating Flow

We consider first the non-rotating problem set, with little loss of generality, in one dimension. We suppose that initially the fluid is at rest but with a simple discontinuity in the height field so that

$$\eta(x, t = 0) = \begin{cases} +\eta_0 & x < 0 \\ -\eta_0 & x > 0, \end{cases} \quad (4.66)$$

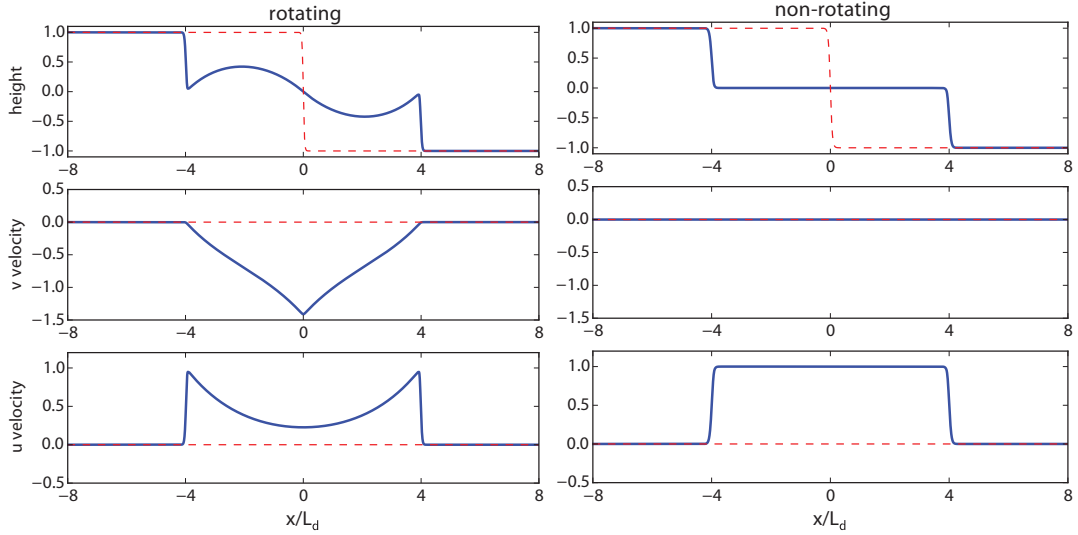
and  $u(x, t = 0) = 0$  everywhere. We can realize these initial conditions physically by separating two fluid masses of different depths by a thin dividing wall, and then quickly removing the wall. What is the subsequent evolution of the fluid? The general solution to the linear problem is given by (4.44) where the functional form is determined by the initial conditions so that here

$$F(x) = \eta(x, t = 0) = -\eta_0 \operatorname{sgn}(x). \quad (4.67)$$

Equation (4.44) states that this initial pattern is propagated to the right and to the left. That is, two discontinuities in fluid height move to the right and left at a speed  $c = \sqrt{gH}$ . Specifically, the solution is

$$\eta(x, t) = -\frac{1}{2}\eta_0 [\operatorname{sgn}(x + ct) + \operatorname{sgn}(x - ct)]. \quad (4.68)$$

The initial conditions may be much more complex than a simple front, but, because the waves are dispersionless, the solution is still simply a sum



**Fig. 4.7:** The solutions of the shallow water equations obtained by numerically integrating the equations of motion with and without rotation. The panels show snapshots of the state of the fluid (solid lines) soon after being released from a stationary initial state (red dashed lines) with a height discontinuity. The rotating flow is evolving toward an end state similar to Fig. 4.8 whereas the non-rotating flow will eventually become stationary. In the non-rotating case  $L_d$  is defined using the rotating parameters.

of the translation of those initial conditions to the right and to the left at speed  $c$ . The velocity field in this class of problem is obtained from

$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x}, \quad (4.69)$$

which gives, using (4.44),

$$u = -\frac{g}{2c} [F(x + ct) - F(x - ct)]. \quad (4.70)$$

Consider the case with initial conditions given by (4.66). At a given location, away from the initial disturbance, the fluid remains at rest and undisturbed until the front arrives. After the front has passed, the fluid surface is again undisturbed and the velocity is uniform and non-zero. Specifically:

$$\eta = \begin{cases} -\eta_0 \operatorname{sgn}(x) \\ 0 \end{cases} \quad u = \begin{cases} 0 & |x| > ct \\ (\eta_0 g/c) & |x| < ct. \end{cases} \quad (4.71)$$

The solution with a discontinuity in the height field, and zero initial velocity, is illustrated in the right-hand panels in Fig. 4.7. The front propagates in either direction from the discontinuity and, in this case, the final velocity, as well as the fluid displacement, is zero. That is, the disturbance is radiated completely away.

#### 4.4.3 Rotating Flow

Rotation makes a profound difference to the adjustment problem of the shallow water system, because a steady, adjusted, solution can exist with non-zero gradients in the height field — the associated pressure gradients being balanced by the Coriolis force — and potential vorticity conservation provides a powerful constraint on the fluid evolution. In a rotating

shallow fluid that conservation is represented by

$$\frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q = 0, \quad (4.72)$$

where  $Q = (\zeta + f)/h$ . In the linear case with constant Coriolis parameter, (4.72) becomes

$$\frac{\partial q}{\partial t} = 0, \quad q = \left( \zeta - f_0 \frac{\eta}{H} \right). \quad (4.73)$$

This equation may be obtained either from the linearized velocity and mass conservation equations, (4.65), or from (4.72) directly. In the latter case we can write

$$Q = \frac{\zeta + f_0}{H + \eta} \approx \frac{1}{H}(\zeta + f_0) \left( 1 - \frac{\eta}{H} \right) \approx \frac{1}{H} \left( f_0 + \zeta - f_0 \frac{\eta}{H} \right) = \frac{f_0}{H} + \frac{q}{H}, \quad (4.74)$$

having used  $f_0 \gg |\zeta|$  and  $H \gg |\eta|$ . The term  $f_0/H$  is a constant and so dynamically unimportant, as is the  $H^{-1}$  factor multiplying  $q$ . Further, the advective term  $\mathbf{u} \cdot \nabla Q$  becomes  $\mathbf{u} \cdot \nabla q$ , and this is second order in perturbed quantities and so is neglected. Thus, making these approximations, (4.72) reduces to (4.73). The potential vorticity field is therefore fixed in space! Of course, this was also true in the non-rotating case where the fluid is initially at rest. Then  $q = \zeta = 0$  and the fluid remains irrotational throughout the subsequent evolution of the flow. However, this is rather a weak constraint on the subsequent evolution of the fluid; it does nothing, for example, to prevent the conversion of all the potential energy to kinetic energy. In the rotating case the potential vorticity is non-zero, and potential vorticity conservation and geostrophic balance are all we need to infer the final steady state, assuming it exists, without solving for the details of the flow evolution, as we now see.

With an initial condition for the height field given by (4.66), the initial potential vorticity is given by

$$q(x, y) = \begin{cases} -f_0 \eta_0 / H & x < 0 \\ f_0 \eta_0 / H & x > 0, \end{cases} \quad (4.75)$$

and this remains unchanged throughout the adjustment process. The final steady state is then the solution of the equations

$$\zeta - f_0 \frac{\eta}{H} = q(x, y), \quad f_0 u = -g \frac{\partial \eta}{\partial y}, \quad f_0 v = g \frac{\partial \eta}{\partial x}, \quad (4.76a,b,c)$$

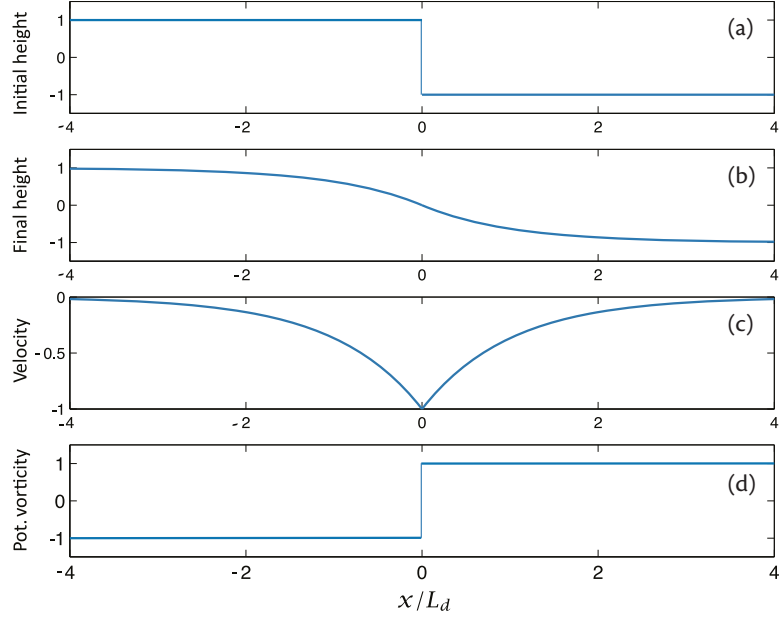
where  $\zeta = \partial v / \partial x - \partial u / \partial y$ . Because the Coriolis parameter is constant, the velocity field is horizontally non-divergent and we may define a streamfunction  $\psi = g\eta / f_0$ . Equations (4.76) then reduce to

$$\left( \nabla^2 - \frac{1}{L_d^2} \right) \psi = q(x, y), \quad (4.77)$$

where  $L_d = \sqrt{gH} / f_0$  is the radius of deformation, as in (4.52), sometimes called the 'Rossby radius'. It is a naturally occurring length scale in problems involving both rotation and gravity, and arises in a slightly different form in stratified fluids.

If an unbalanced flow is left to freely evolve, gravity waves will propagate quickly away from the disturbance. However, the balanced component of the flow is constrained by potential vorticity conservation and it can only evolve advectively, and so much more slowly. Thus, after some time, only the balanced part of the flow remains, with the unbalanced flow having been radiated away and dissipated. This process is *geostrophic adjustment*.

**Fig. 4.8:** Solutions of a linear geostrophic adjustment problem. (a) Initial height field, given by (4.66) with  $\eta_0 = 1$ . (b) Equilibrium (final) height field,  $\eta$  given by (4.79) and  $\eta = f_0\psi/g$ . (c) Equilibrium geostrophic velocity, normal to the gradient of the height field, given by (4.80). (d) Potential vorticity, given by (4.75), and this does not evolve. The distance,  $x$  is nondimensionalized by the deformation radius  $L_d$  and the velocity by  $\eta_0(g/f_0L_d)$ . Changes to the initial state occur within  $\mathcal{O}(L_d)$  of the initial discontinuity.



The initial conditions (4.75) admit of a nice analytic solution, for the flow will remain uniform in  $y$ , and (4.77) reduces to

$$\frac{\partial^2 \psi}{\partial x^2} - \frac{1}{L_d^2} \psi = \frac{f_0 \eta_0}{H} \text{sgn}(x). \quad (4.78)$$

We solve this separately for  $x > 0$  and  $x < 0$  and then match the solutions and their first derivatives at  $x = 0$ , also imposing the condition that the velocity decays to zero as  $x \rightarrow \pm\infty$ . The solution is

$$\psi = \begin{cases} -(g\eta_0/f_0)(1 - e^{-x/L_d}) & x > 0 \\ +(g\eta_0/f_0)(1 - e^{x/L_d}) & x < 0. \end{cases} \quad (4.79)$$

The velocity field associated with this is obtained from (4.76b,c), and is

$$u = 0, \quad v = -\frac{g\eta_0}{f_0 L_d} e^{-|x|/L_d}. \quad (4.80)$$

The velocity is perpendicular to the slope of the free surface, and a jet forms along the initial discontinuity, as illustrated in Fig. 4.8.

The important point of this problem is that the variations in the height and field are not radiated away to infinity, as in the non-rotating problem. Rather, potential vorticity conservation constrains the influence of the adjustment to within a deformation radius (we see now why this name is appropriate) of the initial disturbance. This property is a general one in geostrophic adjustment — it also arises if the initial condition consists of a velocity jump. The time evolution of the rotating flow, obtained by a numerical integration, is illustrated in the left-hand panels of Fig. 4.7.

Fronts propagate away at a speed  $\sqrt{gH} = 1$ , just as in the non-rotating case, but in the rotating flow they leave behind a geostrophically balanced state with a non-zero meridional velocity.

#### 4.5 ♦ A VARIATIONAL PERSPECTIVE ON ADJUSTMENT

In the non-rotating problem, all of the initial potential energy is eventually radiated away to infinity. In the rotating problem, the final state contains both potential and kinetic energy, mostly trapped within a deformation radius of the initial disturbance, because potential vorticity conservation on parcels prevents all of the energy being dispersed. This suggests that it may be informative to think of the geostrophic adjustment problem as a *variational problem*: we seek to minimize the energy consistent with the conservation of potential vorticity. We stay in the linear approximation in which, because the advection of potential vorticity is neglected, potential vorticity remains constant at each point.

The energy of the flow,  $\hat{E}$ , is given by the sum of potential and kinetic energies, namely

$$\hat{E} = \int (Hu^2 + g\eta^2) dA, \quad (4.81)$$

(where  $dA = dx dy$ ) and the potential vorticity field is

$$q = \zeta - f_0 \frac{\eta}{H} = (v_x - u_y) - f_0 \frac{\eta}{H}, \quad (4.82)$$

where the subscripts  $x, y$  denote derivatives. The problem is then to extremize the energy subject to potential vorticity conservation. This is a constrained problem in the calculus of variations, sometimes called an *isoperimetric* problem because of its origins in maximizing the area of a surface for a given perimeter.

The mathematical problem is to extremize the integral

$$I = \int \{H(u^2 + v^2) + g\eta^2 + \lambda(x, y)[(v_x - u_y) - f_0\eta/H]\} dA, \quad (4.83)$$

where  $\lambda(x, y)$  is a Lagrange multiplier, undetermined at this stage. It is a function of space: if it were a constant, the integral would merely extremize energy subject to a given integral of potential vorticity, and rearrangements of potential vorticity (which here we wish to disallow) would leave the integral unaltered.

As there are three independent variables there are three Euler–Lagrange equations that must be solved in order to minimize  $I$ . These are

$$\begin{aligned} \frac{\partial L}{\partial \eta} - \frac{\partial}{\partial x} \frac{\partial L}{\partial \eta_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial \eta_y} &= 0, \\ \frac{\partial L}{\partial u} - \frac{\partial}{\partial x} \frac{\partial L}{\partial u_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial u_y} &= 0, \quad \frac{\partial L}{\partial v} - \frac{\partial}{\partial x} \frac{\partial L}{\partial v_x} - \frac{\partial}{\partial y} \frac{\partial L}{\partial v_y} = 0, \end{aligned} \quad (4.84)$$

where  $L$  is the integrand on the right-hand side of (4.83). Substituting the expression for  $L$  into (4.84) gives, after a little algebra,

$$2g\eta - \frac{\lambda f_0}{H} = 0, \quad 2Hu + \frac{\partial \lambda}{\partial y} = 0, \quad 2Hv - \frac{\partial \lambda}{\partial x} = 0, \quad (4.85)$$

and then eliminating  $\lambda$  gives the simple relationships

$$u = -\frac{g}{f_0} \frac{\partial \eta}{\partial y}, \quad v = \frac{g}{f_0} \frac{\partial \eta}{\partial x}. \quad (4.86)$$

These are the equations of geostrophic balance! Thus, in the linear approximation, *geostrophic balance is the minimum energy state for a given field of potential vorticity.*

### Notes and References

The shallow water equations are sometimes called the Saint-Venant equations after Adhémar Jean Claude Barré de Saint-Venant (1797–1886) who wrote down a form of the equations in 1871. Pierre-Simon Laplace (1749–1825) previously wrote down a linear version of the equations on the sphere, now known as Laplace's tidal equations, in 1776, and one would not be surprised if Euler knew about the equations before that.

### Problems

4.1 Using the shallow water equations:

- (a) A cylindrical column of air at  $30^\circ$  latitude with radius 100 km expands horizontally (shrinking in depth accordingly) to twice its original radius. If the air is initially at rest, what is the mean tangential velocity at the perimeter after the expansion?
- (b) An air column at  $60^\circ$  N with zero relative vorticity ( $\zeta = 0$ ) stretches from the surface to the tropopause, which we assume is a rigid lid, at 10 km. The air column moves zonally on to a plateau 2.5 km high. What is its relative vorticity? Suppose it then moves southwards to  $30^\circ$  N, staying on the plateau. What is its vorticity?

4.2 Show that the vertical velocity within a shallow water system is given by

$$w = \frac{z - \eta_b}{h} \frac{Dh}{Dt} + \frac{D\eta_b}{Dt}. \quad (\text{P4.1})$$

Interpret this result, showing that it gives sensible answers at the top and bottom of the fluid layer.

- 4.3 In the long-wave limit of Poincaré waves, fluid parcels behave as free-agents; that is, like free solid particles moving in a rotating frame unencumbered by pressure forces. Why then, is their frequency given by  $\omega = f = 2\Omega$  where  $\Omega$  is the rotation rate of the coordinate system, and not by  $\Omega$  itself? Do particles that are stationary or move in a straight line in the inertial frame of reference satisfy the dispersion relationship for Poincaré waves in this limit? Explain. [See also Durran (1993) and Egger (1999).]

- 4.4 Linearize the  $f$ -plane shallow water system about a state of rest. Suppose that there is an initial disturbance given in the general form

$$\eta = \iint \tilde{\eta}_{k,l} e^{i(kx+ly)} dk dl, \quad (\text{P4.2})$$

where  $\eta$  is the deviation surface height and the Fourier coefficients  $\tilde{\eta}_{k,l}$  are given, and that the initial velocity is zero.

- (a) Obtain the geopotential field at the completion of geostrophic adjustment, and show that the deformation scale is a natural length scale in the problem.
  - (b) Show that the change in total energy during the adjustment is always less than or equal to zero. Neglect any initial divergence.
- N.B. Because the problem is linear, the Fourier modes do not interact.
- 4.5 If energy conservation is one of the most basic physical laws, how can energy be lost in geostrophic adjustment?
- 4.6 *Geostrophic adjustment of a velocity jump.* In which we consider the evolution of the linearized  $f$ -plane shallow water equations in an infinite domain.

- (a) Show that the *linearized* potential vorticity,  $q'$ , for the shallow water system is given by

$$q' = \zeta' - f_0 \frac{h'}{H}, \quad (\text{P4.3})$$

using standard notation.

- (b) If the flow is in geostrophic balance show that the relative vorticity is given by

$$\zeta' = \nabla^2 \psi,$$

where  $\psi = gh'/f_0$ . Hence show that the potential vorticity is then given by

$$q' = \nabla^2 \psi - \frac{1}{L_d^2} \psi,$$

and write down an expression for  $L_d$ .

- (c) Suppose that initially the fluid surface is flat, the zonal velocity is zero and the meridional velocity is given by

$$v(x) = v_0 \operatorname{sgn}(x). \quad (\text{P4.4})$$

Find the equilibrium height and velocity fields at  $t = \infty$ .

- (d) What are the initial and final kinetic and potential energies?

*Partial solution:*

The potential vorticity is  $q = \zeta - f_0 \eta/H$ , so that the initial state and final state are both given by

$$q = 2v_0 \delta(x). \quad (\text{P4.5})$$

(Why?) The final state streamfunction is thus given by  $(\partial^2/\partial x^2 - L_d^{-2})\psi = q$ , with solution  $\psi = \psi_0 \exp(x/L_d)$  and  $\psi = \psi_0 \exp(-x/L_d)$  for  $x < 0$  and  $x > 0$ , where  $\psi_0 = -L_d v_0$  (why?), and  $\eta = f_0 \psi/g$ . The energy is  $E = \int (Hv^2 + g\eta^2)/2 dx$ . The initial KE is infinite, the initial PE is zero, and the final state has  $PE = KE = gL_d v_0^2/4$ ; that is, the energy is equipartitioned between kinetic and potential energies.

- 4.7 In the shallow water equations show that, if the flow is approximately geostrophically balanced, the energy at large scales is predominantly potential energy and the energy at small scales is predominantly kinetic energy. Define precisely what 'large scale' and 'small scale' mean in this context.

- 4.8 In the shallow water geostrophic adjustment problem, show that at large scales the velocity essentially adjusts to the height field, and that at small scales the height field essentially adjusts to the velocity field. Your derivation may be detailed and mathematical, but explain the result at the end in words and in physical terms.



## Geostrophic Theory

**G** EOSTROPHIC AND HYDROSTATIC BALANCE are the two dominant balances in meteorology and oceanography and in this chapter we exploit these balances to derive various simplified sets of equations. The ‘problem’ with the full equations is that they are *too* complete, and they contain motions that we don’t always care about — sound waves and gravity waves for example. If we can eliminate these modes from the outset then our path toward understanding is not littered with obstacles.

Our specific goal is to derive various sets of ‘geostrophic equations’, in particular the planetary-geostrophic and quasi-geostrophic equations, by making use of the fact that geostrophic and hydrostatic balance are closely satisfied. We do this first for the shallow water equations and then for the stratified, three-dimensional equations. We will use the Boussinesq equations, but a treatment in pressure coordinates would be very similar. The bottom topography,  $\eta_B$ , can be an unneeded complication in the derivations below and readers may wish to simplify by setting  $\eta_B = 0$ .

### 5.1 SCALING THE SHALLOW WATER EQUATIONS

In order to simplify the equations of motion we first *scale* them — we choose the scales we wish to describe, and then determine the approximate sizes of the terms in the equations. We then eliminate the small terms and derive a set of equations that is simpler than the original set but that consistently describes motion of the chosen scale. With the odd exception, we will denote the scales of variables by capital letters; thus, if  $L$  is a typical length scale of the motion we wish to describe, and  $U$  is a typical velocity scale, then

$$\begin{aligned} (x, y) &\sim L & \text{or} & & (x, y) &= \mathcal{O}(L), \\ (u, v) &\sim U & \text{or} & & (u, v) &= \mathcal{O}(U), \end{aligned} \tag{5.1}$$

and similarly for the other variables in the equations.

We then write the equations of motion in a nondimensional form by writing the variables as

$$(x, y) = L(\hat{x}, \hat{y}), \quad (u, v) = U(\hat{u}, \hat{v}), \quad (5.2)$$

where the hatted variables are nondimensional and, by supposition, are  $\mathcal{O}(1)$ . The various terms in the momentum equation then scale as:

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta, \quad (5.3a)$$

$$\frac{U}{T} \quad \frac{U^2}{L} \quad fU \sim g \frac{\mathcal{H}}{L}, \quad (5.3b)$$

where the  $\nabla$  operator acts in the  $x$ - $y$  plane and  $\mathcal{H}$  is the amplitude of the variations in the surface displacement. We choose an ‘advective scale’ for time, meaning that  $T = L/U$  and  $t = \hat{t}L/U$ , and the time derivative then scales the same way as the advection. The ratio of the advective term to the rotational term in the momentum equation (5.3) is  $(U^2/L)/(fU) = U/fL$ ; this is the Rossby number that we previously encountered.

We are interested in flows for which the Rossby number is small, in which case the Coriolis term is largely balanced by the pressure gradient. From (5.3b), variations in  $\eta$  scale according to

$$\mathcal{H} = \frac{fUL}{g} = Ro \frac{f^2 L^2}{g} = Ro H \frac{L^2}{L_d^2}, \quad (5.4)$$

where  $L_d = \sqrt{gH}/f$  is the deformation radius and  $H$  is the mean depth of the fluid. The ratio of variations in fluid height to the total fluid height thus scales as

$$\frac{\mathcal{H}}{H} \sim Ro \frac{L^2}{L_d^2}. \quad (5.5)$$

Now, the thickness of the fluid,  $h$ , may be written as the sum of its mean and a deviation,  $h_D$

$$h = H + h_D = H + (\eta_T - \eta_B), \quad (5.6)$$

where, referring to Fig. 4.1,  $\eta_B$  is the height of the bottom topography and  $\eta_T$  is the height of the fluid above its mean value. Given the scalings above, the deviation height of the fluid may be written as

$$\eta_T = Ro \frac{L^2}{L_d^2} H \hat{\eta}_T \quad \text{and} \quad \eta = H + \eta_T = H \left( 1 + Ro \frac{L^2}{L_d^2} \hat{\eta}_T \right), \quad (5.7)$$

where  $\hat{\eta}_T$  is the  $\mathcal{O}(1)$  nondimensional value of the surface height deviation. We apply the same scalings to  $h$  itself and, if  $h_D = h - H = \eta_T - \eta_B$  is the deviation of the thickness from its mean value, then

$$h = H + h_D = H \left( 1 + Ro \frac{L^2}{L_d^2} \hat{h}_D \right), \quad (5.8)$$

where  $\hat{h}_D$  is the nondimensional deviation thickness of the fluid layer.

The geostrophic theory of this chapter applies when the Rossby number is small. On Earth the theory is generally appropriate for large-scale flow in the mid- and high latitude atmosphere and ocean. On other planets the applicability of geostrophic theory depends on how rapidly the planet rotates and how big it is. Venus has a rotation rate some 200 times slower than Earth and the Rossby number of the large-scale circulation is quite large. Jupiter rotates much faster than Earth (a Jupiter day is about 10 hours), and the planet is also much bigger, and the Rossby number remains small even close to the equator.

### Nondimensional momentum equation

If we use (5.7) and (5.8) to scale height variations, (5.2) to scale lengths and velocities, and an advective scaling for time, then, and since  $\nabla \hat{\eta} = \nabla \hat{\eta}_T$ , the momentum equation (5.3) becomes

$$Ro \left[ \frac{\partial \hat{\mathbf{u}}}{\partial \hat{t}} + (\hat{\mathbf{u}} \cdot \nabla) \hat{\mathbf{u}} \right] + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\eta}, \quad (5.9)$$

where  $\hat{\mathbf{f}} = \hat{\mathbf{k}} \hat{f} = \hat{\mathbf{k}} f / f_0$ , where  $f_0$  is a representative value of the Coriolis parameter. (If  $f$  is a constant, then  $\hat{f} = 1$ , but it is informative to explicitly write  $\hat{f}$  in the equations. Also, where the operator  $\nabla$  operates on a nondimensional variable, the differentials are taken with respect to the nondimensional variables  $\hat{x}, \hat{y}$ .) All the variables in (5.9) will now be assumed to be of order unity, and the Rossby number multiplying the local time derivative and the advective terms indicates the smallness of those terms. By construction, the dominant balance in (5.9) is the geostrophic balance between the last two terms.

### Nondimensional mass continuity (height) equation

The (dimensional) mass continuity equation can be written as

$$\frac{Dh}{Dt} + h \nabla \cdot \mathbf{u} = 0 \quad \text{or} \quad \frac{1}{H} \frac{Dh_D}{Dt} + \left( 1 + \frac{h_D}{H} \right) \nabla \cdot \mathbf{u} = 0, \quad (5.10)$$

since  $Dh/Dt = Dh_D/Dt$ . Using (5.2) and (5.8) the above equation may be written

$$Ro \left( \frac{L}{L_d} \right)^2 \frac{D\hat{h}_D}{D\hat{t}} + \left[ 1 + Ro \left( \frac{L}{L_d} \right)^2 \hat{h}_D \right] \nabla \cdot \hat{\mathbf{u}} = 0. \quad (5.11)$$

Equations (5.9) and (5.11) are the nondimensional versions of the full shallow water equations of motion. Since the Rossby number is small we might expect that some terms in this equation can be eliminated with little loss of accuracy, depending on the size of the second nondimensional parameter,  $(L/L_d)^2$ , as we now explore.

## 5.2 GEOSTROPHIC SHALLOW WATER EQUATIONS

### 5.2.1 Planetary-Geostrophic Equations

We now derive simplified equation sets that are appropriate in particular parameter regimes, beginning with an equation set appropriate for the very largest scales. Specifically, we take

$$Ro \ll 1, \quad \frac{L}{L_d} \gg 1 \quad \text{such that} \quad Ro \left( \frac{L}{L_d} \right)^2 = \mathcal{O}(1). \quad (5.12)$$

The first inequality implies we are considering flows in geostrophic balance. The second inequality means we are considering flows much larger

than the deformation radius. The ratio of the deformation radius to scale of motion of the fluid is called the Burger number; that is,  $Bu \equiv L_d/L$ , so here we are considering small Burger-number flows.

The smallness of the Rossby number means that we can neglect the material derivative in the momentum equation, (5.9), leaving geostrophic balance. Thus, in dimensional form, the momentum equation may be written, in vectorial or component forms, as

$$\begin{aligned} \mathbf{f} \times \mathbf{u} &= -\nabla\eta, \\ \text{or} \\ f v &= g \frac{\partial \eta}{\partial x}, \quad f u = -g \frac{\partial \eta}{\partial y}. \end{aligned} \tag{5.13}$$

The *planetary-geostrophic* equations are appropriate for geostrophically balanced flow at very large scales. In the shallow water version, they consist of the full mass conservation equation along with geostrophic balance.

Looking now at the mass continuity equation, (5.11), we see that there are no small terms that can be eliminated. Thus, we have simply the full mass conservation equation,

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0, \tag{5.14}$$

where  $h$  and  $\eta$  are related by  $\eta = \eta_B + h$ , where  $\eta_B$  is the height of the bottom topography. Equations (5.13) and (5.14) form the *planetary geostrophic shallow water equations*. There is *only one time derivative* in the equations, so there can be no gravity waves. The system is evolved purely through the mass continuity equation, and the flow field is *diagnosed* from the height field.

### Planetary-geostrophic potential vorticity

In the (full) shallow water equations potential vorticity is conserved, meaning that

$$\frac{D}{Dt} \left( \frac{\zeta + f}{h} \right) = 0. \tag{5.15}$$

In the planetary-geostrophic equations we can use (5.13) and (5.14) to show that this conservation law becomes

$$\frac{D}{Dt} \left( \frac{f}{h} \right) = 0, \tag{5.16}$$

as might be expected since  $\zeta$  is smaller than  $f$  by a factor of the Rossby number. An alternate derivation of the planetary-geostrophic equations is to go directly from (5.15) to (5.16), by virtue of the smallness of the Rossby number, and then simply use (5.16) instead of (5.14) as the evolution equation.

### 5.2.2 Quasi-Geostrophic Equations

The *quasi-geostrophic equations* are appropriate for scales of the same order as the deformation radius, and so for

$$Ro \ll 1, \quad \frac{L}{L_d} = \mathcal{O}(1) \quad \text{so that} \quad Ro \left( \frac{L}{L_d} \right)^2 \ll 1. \quad (5.17)$$

Since the Rossby number is small the momentum equations again reduce to geostrophic balance, namely (5.13). In the mass continuity equation, we now eliminate all terms involving Rossby number to give

$$\nabla \cdot \mathbf{u} = 0. \quad (5.18)$$

Neither geostrophic balance nor (5.18) are prognostic equations, and it appears we have derived an uninteresting, static, set of equations. In fact we haven't gone far enough, since nothing in our derivation says that these quantities do not evolve. To see this, let us suppose that the Coriolis parameter is nearly constant, which is physically consistent with the idea that scales of motion are comparable to the deformation scale. Geostrophic balance with a constant Coriolis parameter gives

$$f_0 u = -g \frac{\partial \eta}{\partial y}, \quad -f_0 v = -g \frac{\partial \eta}{\partial z}, \quad \text{giving} \quad \nabla \cdot \mathbf{u} = 0. \quad (5.19)$$

That is to say, the geostrophic flow is divergence-free and we therefore should not suppose that  $\nabla \cdot \mathbf{u} = 0$  is the dominant term in the height equation.

However, with a little more care we can in fact derive a set of equations that evolves under these conditions, and that furthermore is extraordinarily useful, for it describes the flow on the scales of motion corresponding to weather. We make three explicit assumptions:

- (i) The Rossby number is small and the flow is in near geostrophic balance.
- (ii) The scales of motion are similar to the deformation scales, so that  $L \sim L_d$  and  $Ro(L/L_d)^2 \ll 1$ .
- (iii) Variations of the Coriolis parameter are small, so that  $f = f_0 + \beta y$  where  $\beta y \ll f_0$ .

The velocity is then equal to a geostrophic component,  $\mathbf{u}_g$  plus an ageostrophic component,  $\mathbf{u}_a$  where  $|\mathbf{u}_g| \gg |\mathbf{u}_a|$  and the geostrophic velocity satisfies

$$f_0 \times \mathbf{u}_g = -g \nabla \eta, \quad (5.20)$$

which, because of the use of a constant Coriolis parameter (assumption (iii)), implies  $\nabla \cdot \mathbf{u}_g = 0$ .

We proceed from the shallow water vorticity equation which, as in (4.32), is

$$\frac{\partial \zeta}{\partial t} + (\mathbf{u} \cdot \nabla)(\zeta + f) = -(\zeta + f) \nabla \cdot \mathbf{u}. \quad (5.21)$$

The quasi-geostrophic equations are appropriate for geostrophically balanced flow at so-called synoptic scales, or weather scales. This scale is mainly determined by the Rossby radius of deformation which is about 1000 km in the atmosphere and 100 km (and less in high latitudes) in the ocean.

A consequence of (5.20) is that the right-hand side contains only the ageostrophic velocity, which is small, and since  $\zeta$  is smaller than  $f$  by a factor of the Rossby number we can ignore  $\zeta \nabla \cdot \mathbf{u}$  and take  $f$  to be equal to  $f_0$ . On the left-hand side the velocities are well-approximated by using the geostrophic flow, so that we have

$$\frac{\partial \zeta_g}{\partial t} + (\mathbf{u}_g \cdot \nabla)(\zeta_g + f) = -f_0 \nabla \cdot \mathbf{u}_a, \quad (5.22)$$

where on the left-hand side  $f$  can be replaced by  $\beta y$ .

We now use the mass continuity equation to obtain an expression for the divergence. From (5.10) the mass continuity equation is

$$\frac{Dh_D}{Dt} + (H + h_D) \nabla \cdot \mathbf{u} = 0, \quad (5.23)$$

and since  $H \gg h_D$  (using (5.11),  $H$  is bigger by a factor  $(L_d/L)^2 Ro^{-1}$ ), the equation becomes

$$\frac{D}{Dt}(\eta - \eta_B) + H \nabla \cdot \mathbf{u}_a = 0. \quad (5.24)$$

Combining (5.22) and (5.23) gives

$$\frac{D}{Dt} \left( \zeta_g + f - \frac{f_0(\eta - \eta_B)}{H} \right) = 0. \quad (5.25)$$

It appears that we have two dynamical variables here,  $\zeta_g$  and  $\eta$ . However, they are related through geostrophic balance, and the fact that the geostrophic flow is non-divergent. Thus, we may define a streamfunction  $\psi$  such that  $u_g = -\partial\psi/\partial y$ ,  $v_g = \partial\psi/\partial x$ , whence  $\partial u/\partial x + \partial v/\partial y = 0$ . The vorticity and height fields are related to the streamfunction by

$$\zeta_g = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \nabla^2 \psi, \quad \text{and} \quad \eta = \frac{f_0 \psi}{g}, \quad (5.26a,b)$$

where the second relation comes from geostrophic balance. Equation (5.25) may then be written as

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + \beta y - \frac{\psi}{L_d^2} + \frac{f_0 \eta_B}{H}, \quad (5.27)$$

where  $L_d = \sqrt{gH}/f_0$  and the variable  $q$  is the *quasi-geostrophic potential vorticity*.

### 5.2.3 Quasi-Geostrophic Potential Vorticity

#### Connection to shallow water potential vorticity

The quantity  $q$  given by (5.27) is an approximation (except for dynamically unimportant constant additive and multiplicative factors) to the shallow

Both the planetary-geostrophic and the quasi-geostrophic equations can be written in the form of an evolution equation for potential vorticity, along with an 'inversion' to determine the velocity fields and the height field. The difference in the two equation sets lies in the approximations made in the inversion. In the planetary-geostrophic system relative vorticity is ignored and  $Q = f/h$ . In quasi-geostrophy the variations in the height field are small and  $q = \beta y + \zeta - f_0 \eta_T/H$ .

Both equation sets assume a low Rossby number.

water potential vorticity. To see the truth of this statement let us begin with the expression for the shallow water potential vorticity,

$$Q = \frac{f + \zeta}{h}. \quad (5.28)$$

For simplicity we set bottom topography to zero and then  $h = H(1 + \eta_T/H)$  and assume that  $\eta_T/H$  is small to obtain

$$Q = \frac{f + \zeta}{H(1 + \eta_T/H)} \approx \frac{1}{H}(f + \zeta) \left(1 - \frac{\eta_T}{H}\right) \approx \frac{1}{H} \left(f_0 + \beta y + \zeta - f_0 \frac{\eta_T}{H}\right). \quad (5.29)$$

Because  $f_0/H$  is a constant it has no effect in the evolution equation, and the quantity given by

$$q = \beta y + \zeta - f_0 \frac{\eta_T}{H} \quad (5.30)$$

is materially conserved. Using geostrophic balance we have  $\zeta = \nabla^2 \psi$  and  $\eta_T = f_0 \psi/g$  so that (5.30) is identical (except for  $\eta_B$ ) to (5.27).

The approximations needed to go from (5.28) to (5.30) are the same as those used in our earlier, more long-winded, derivation of the quasi-geostrophic equations. That is, we assumed that  $f$  itself is nearly constant, and that  $f_0$  is much larger than  $\zeta$ , equivalent to a low Rossby number assumption. It was also necessary to assume that  $H \gg \eta_T$  to enable the expansion of the height field and this approximation is equivalent to requiring that the scale of motion not be significantly larger than the deformation scale. The derivation is completed by noting that the advection of the potential vorticity should be by the geostrophic velocity alone, and we recover (5.27).

### 5.3 SCALING IN THE CONTINUOUSLY-STRATIFIED SYSTEM

We now apply the same scaling ideas, *mutatis mutandis*, to the stratified primitive equations. We use the hydrostatic Boussinesq equations, which we write as

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad (5.31a)$$

$$\frac{\partial \phi}{\partial z} = b, \quad (5.31b)$$

$$\frac{Db}{Dt} = 0, \quad (5.31c)$$

$$\nabla \cdot \mathbf{v} = 0. \quad (5.31d)$$

We will consider only the case of a flat bottom, but topography is a relatively straightforward extension. Anticipating that the average stratification may not scale in the same way as the deviation from it, let us separate out the contribution of the advection of a reference stratification in (5.31c) by writing

$$b = \tilde{b}(z) + b'(x, y, z, t). \quad (5.32)$$

The thermodynamic equation then becomes

$$\frac{Db'}{Dt} + N^2 w = 0, \quad (5.33)$$

where  $N^2 = \partial \tilde{b} / \partial z$  and the advective derivative is still three-dimensional. We then let  $\phi = \tilde{\phi}(z) + \phi'$ , where  $\tilde{\phi}$  is hydrostatically balanced by  $\tilde{b}$ , and the hydrostatic equation becomes

$$\frac{\partial \phi'}{\partial z} = b'. \quad (5.34)$$

Equations (5.33) and (5.34) replace (5.31c) and (5.31b), and  $\phi'$  is used in (5.31a).

### 5.3.1 Scaling the Equations

We scale the basic variables by supposing that

$$(x, y) \sim L, \quad (u, v) \sim U, \quad t \sim \frac{L}{U}, \quad z \sim H, \quad f \sim f_0, \quad N \sim N_0, \quad (5.35)$$

where the scaling variables (capitalized, except for  $f_0$ ) are chosen to be such that the nondimensional variables have magnitudes of the order of unity, and the constant  $N_0$  is a representative value of the stratification. We presume that the scales chosen are such that the Rossby number is small; that is  $Ro = U/(f_0 L) \ll 1$ . In the momentum equation the pressure term then balances the Coriolis force,

$$|\mathbf{f} \times \mathbf{u}| \sim |\nabla \phi'|, \quad (5.36)$$

and so the pressure scales as

$$\phi' \sim \Phi = f_0 UL. \quad (5.37)$$

Using the hydrostatic relation, (5.37) implies that the buoyancy scales as

$$b' \sim B = \frac{f_0 UL}{H}, \quad (5.38)$$

and from this we obtain

$$\frac{(\partial b' / \partial z)}{N^2} \sim Ro \frac{L^2}{L_d^2}, \quad (5.39)$$

where

$$L_d = \frac{N_0 H}{f_0} \quad (5.40)$$

is the deformation radius in the continuously-stratified fluid, analogous to the quantity  $\sqrt{gH}/f_0$  in the shallow water system, and we use the same symbol for both. In the continuously-stratified system, *if the scale of motion is the same as or smaller than the deformation radius, and the Rossby number is small, then the variations in stratification are small.* The choice of



scale is the key difference between the planetary-geostrophic and quasi-geostrophic equations.

Finally, at least for now, we nondimensionalize the vertical velocity by using the mass conservation equation,

$$\frac{\partial w}{\partial z} = - \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right), \quad (5.41)$$

with the scaling

$$w \sim W = \frac{UH}{L}. \quad (5.42)$$

This scaling will not necessarily be correct if the flow is geostrophically balanced. In this case we can then estimate  $w$  by cross-differentiating geostrophic balance (with  $\bar{\rho}$  constant for simplicity) to obtain the linear geostrophic vorticity equation and corresponding scaling:

$$\beta v \approx f \frac{\partial w}{\partial z}, \quad w \sim W = \frac{\beta UH}{f_0}. \quad (5.43a,b)$$

If variations in the Coriolis parameter are large and  $\beta \sim f_0/L$ , then (5.43b) is the same as (5.42), but if  $f$  is nearly constant then  $W \ll UH/L$ .

Given the scalings above (using (5.42) for  $w$ ) we nondimensionalize by setting

$$\begin{aligned} (\hat{x}, \hat{y}) &= L^{-1}(x, y), & \hat{z} &= H^{-1}z, & (\hat{u}, \hat{v}) &= U^{-1}(u, v), & \hat{t} &= \frac{U}{L}t, \\ \hat{w} &= \frac{L}{UH}w, & \hat{f} &= f_0^{-1}f, & \hat{\phi} &= \frac{\phi'}{f_0UL}, & \hat{b} &= \frac{H}{f_0UL}b', \end{aligned} \quad (5.44)$$

where the hatted variables are nondimensional. The horizontal momentum and hydrostatic equations then become

$$Ro \frac{D\hat{\mathbf{u}}}{D\hat{t}} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi}, \quad (5.45)$$

and

$$\frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}. \quad (5.46)$$

The nondimensional mass conservation equation is simply

$$\nabla \cdot \hat{\mathbf{v}} = \left( \frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{\partial \hat{w}}{\partial \hat{z}} \right) = 0, \quad (5.47)$$

and the nondimensional thermodynamic equation is

$$\frac{f_0UL}{H} \frac{U}{L} \frac{D\hat{b}}{D\hat{t}} + \hat{N}^2 N_0^2 \frac{HU}{L} \hat{w} = 0, \quad (5.48)$$

or, re-arranging,

$$Ro \frac{D\hat{b}}{D\hat{t}} + \left( \frac{L_d}{L} \right)^2 \hat{N}^2 \hat{w} = 0. \quad (5.49)$$

The nondimensional equations are summarized in the box on the following page.

### Nondimensional Boussinesq Primitive Equations

The nondimensional, hydrostatic, Boussinesq equations in a rotating frame of reference are:

$$\text{Horizontal momentum:} \quad Ro \frac{D\hat{\mathbf{u}}}{Dt} + \hat{\mathbf{f}} \times \hat{\mathbf{u}} = -\nabla \hat{\phi}, \quad (\text{PE.1})$$

$$\text{Hydrostatic:} \quad \frac{\partial \hat{\phi}}{\partial \hat{z}} = \hat{b}, \quad (\text{PE.2})$$

$$\text{Mass continuity:} \quad \frac{\partial \hat{u}}{\partial \hat{x}} + \frac{\partial \hat{v}}{\partial \hat{y}} + \frac{\partial \hat{w}}{\partial \hat{z}} = 0, \quad (\text{PE.3})$$

$$\text{Thermodynamic:} \quad Ro \frac{D\hat{b}}{Dt} + \left( \frac{L_d}{L} \right)^2 \hat{N}^2 \hat{w} = 0. \quad (\text{PE.4})$$

## 5.4 PLANETARY-GEOSTROPHIC EQUATIONS FOR STRATIFIED FLUIDS

The *planetary-geostrophic equations* are appropriate for geostrophic flow at large horizontal scales. The specific assumptions we apply to the primitive equations are that:

$$(i) \quad Ro \ll 1,$$

$$(ii) \quad (L_d/L)^2 \ll 1. \text{ More specifically, } Ro(L/L_d)^2 = \mathcal{O}(1).$$

We also allow  $f$  to have its full variation with latitude, although we still use Cartesian coordinates. If we look at the equations in the shaded box above we see that, if we are to retain only the dominant terms, the momentum equation should simply be replaced by geostrophic balance, whereas hydrostatic and mass continuity equations are unaltered.

In the thermodynamic equation both terms are of order Rossby number, and therefore we retain both. This circumstance arises because, by assumption, we are dealing with large scales and on these scales the perturbation buoyancy varies as much as the mean buoyancy itself. The buoyancy equation reverts to the evolution of total buoyancy, namely

$$\frac{Db}{Dt} = 0, \quad (5.50)$$

where the material derivative is fully three-dimensional. This is the only evolution equation in the system. Thus, to summarize, the *planetary-geostrophic* equations of motion are, in dimensional form,

$$\begin{aligned} \mathbf{f} \times \mathbf{u} &= -\nabla \phi, & \frac{Db}{Dt} &= 0, & \nabla \cdot \mathbf{v} &= 0, \\ & & \frac{\partial \phi}{\partial z} &= b', & & \end{aligned} \quad (5.51)$$

### Potential vorticity

Manipulation of (5.51) reveals that we can equivalently write the equations as an evolution equation for potential vorticity. Thus, the evolution equations may be written as

$$\frac{DQ}{Dt} = 0, \quad Q = f \frac{\partial b}{\partial z}. \quad (5.52)$$

The inversion — i.e., the diagnosis of velocity, pressure and buoyancy — is carried out using the hydrostatic, geostrophic and mass conservation equations, and in a real situation the right-hand sides of the buoyancy and potential vorticity equations would have terms representing heating.

#### 5.4.1 Applicability to the Ocean and Atmosphere

In the atmosphere a typical deformation radius  $NH/f$  is about 1000 km. The constraint that the scale of motion be much larger than the deformation radius is thus quite hard to satisfy, since one quickly runs out of room on a planet whose equator-to-pole distance is 10 000 km. Thus, only the largest scales of motion can satisfy the planetary-geostrophic scaling in the atmosphere and we should then also properly write the equations in spherical coordinates. In the ocean the deformation radius is about 100 km, so there is lots of room for the planetary-geostrophic equations to hold, and indeed much of the theory of the large-scale structure of the ocean involves the planetary-geostrophic equations.

## 5.5 QUASI-GEOSTROPHIC EQUATIONS FOR STRATIFIED FLUIDS

Let us now consider the appropriate equations for geostrophic flow at scales of order the deformation radius.

### 5.5.1 Scaling and Assumptions

The nondimensionalization and scaling are as before and so the nondimensional equations are those in the shaded box on the facing page. The Coriolis parameter is given by

$$\mathbf{f} = (f_0 + \beta y) \hat{\mathbf{k}}. \quad (5.53)$$

The *variation* of the Coriolis parameter is now assumed to be small (this is a key difference between the quasi-geostrophic system and the planetary-geostrophic system), and in particular we assume that  $\beta y$  is approximately the size of the relative vorticity, and so is much smaller than  $f_0$ . The main assumptions needed to derive the QG system are then:

- (i) The Rossby number is small,  $Ro \ll 1$ .
- (ii) Length scales are of the same order as the deformation radius,  $L \sim L_d$  or  $L/L_d = \mathcal{O}(1)$ .
- (iii) Variations in Coriolis parameter are small, and specifically  $|\beta y| \sim Ro f_0$ .

Given these assumptions, we can write the horizontal velocity as the sum of a geostrophic component and an ageostrophic one:

$$\mathbf{u} = \mathbf{u}_g + \mathbf{u}_a, \quad \text{where} \quad f_0 \hat{\mathbf{k}} \times \mathbf{u}_g = -\nabla\phi \quad \text{and} \quad |\mathbf{u}_g| \gg |\mathbf{u}_a|. \quad (5.54)$$

Since the Coriolis parameter is constant in our definition of the geostrophic velocity the geostrophic divergence is zero; that is

$$\frac{\partial u_g}{\partial x} + \frac{\partial v_g}{\partial y} = 0. \quad (5.55)$$

The vertical velocity is thus given by the divergence of the *ageostrophic* velocity,

$$\frac{\partial w}{\partial z} = -\frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y}. \quad (5.56)$$

Since the ageostrophic velocity is small, the actual vertical velocity is smaller than the scaling suggested by the mass conservation equation in its original form. That is,

$$W \ll \frac{UH}{L}. \quad (5.57)$$

*The quasi-geostrophic equations are probably the most used set of equations in theoretical meteorology, certainly for midlatitude dynamics.*

### 5.5.2 Derivation of Stratified QG Equations

We begin by cross-differentiating the horizontal momentum equation to give, after a few lines of algebra, the vorticity equation:

$$\frac{D}{Dt}(\zeta + f) = -(\zeta + f) \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left( \frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right). \quad (5.58)$$

We now apply the quasi-geostrophic assumption, namely:

- (i) The geostrophic velocity and vorticity are much larger than their ageostrophic counterparts, and therefore we use geostrophic values for the terms on the left-hand side.
- (ii) On the right-hand side we keep the horizontal divergence (which is small) only where it is multiplied by the big term  $f$ . Furthermore, because  $f$  is nearly constant we replace it with  $f_0$ .
- (iii) The second term in large parentheses on the right-hand side is smaller than the advection terms on the left-hand side by the ratio  $[UW/(HL)]/[U^2/L^2] = [W/H]/[U/L] \ll 1$ , because  $w$  is small, as noted above. We thus neglect it.

Given the above, (5.58) becomes

$$\frac{D_g}{Dt}(\zeta_g + f) = -f_0 \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = f_0 \frac{\partial w}{\partial z}, \quad (5.59)$$

where the second equality uses mass continuity and  $D_g/Dt = \partial/\partial t + \mathbf{u}_g \cdot \nabla$  — note that only the (horizontal) geostrophic velocity does any advecting, because  $w$  is so small.

Now consider the three-dimensional thermodynamic equation. Since  $w$  is small it only advects the basic state, and the perturbation buoyancy is advected only by the geostrophic velocity. Thus, (5.33) becomes,

$$\frac{D_g b'}{Dt} + wN^2 = 0. \quad (5.60)$$

We now eliminate  $w$  between (5.59) and (5.60), which, after a little algebra, yields

$$\frac{D_g q}{Dt} = 0, \quad q = \zeta_g + f + \frac{\partial}{\partial z} \left( \frac{f_0 b'}{N^2} \right). \quad (5.61)$$

Hydrostatic and geostrophic wind balance enable us to write the geostrophic velocity, vorticity, and buoyancy in terms of streamfunction  $\psi$  [ $= p/(f_0 \rho_0)$ ]:

$$\mathbf{u}_g = \hat{\mathbf{k}} \times \nabla \psi, \quad \zeta_g = \nabla^2 \psi, \quad b' = f_0 \partial \psi / \partial z. \quad (5.62)$$

Thus, we have, now omitting the subscript  $g$ ,

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + f + f_0^2 \frac{\partial}{\partial z} \left( \frac{1}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (5.63a,b)$$

Only the variable part of  $f$  (i.e.,  $\beta y$ ) is relevant in the second term on the right-hand side of the expression for  $q$ . The material derivative may be expressed as

$$\frac{Dq}{Dt} = \frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = \frac{\partial q}{\partial t} + J(\psi, q). \quad (5.64)$$

where  $J(\psi, q) = \partial \psi / \partial x \partial q / \partial y - \partial \psi / \partial y \partial q / \partial x$ .

The quantity  $q$  is known as the *quasi-geostrophic potential vorticity*, and it is conserved when advected by the *horizontal* geostrophic flow. All the other dynamical variables may be obtained from potential vorticity as follows:

- (i) Streamfunction, using (5.63b).
- (ii) Velocity:  $\mathbf{u} = \hat{\mathbf{k}} \times \nabla \psi$  [ $\equiv \nabla^\perp \psi = -\nabla \times (\hat{\mathbf{k}} \psi)$ ].
- (iii) Relative vorticity:  $\zeta = \nabla^2 \psi$ .
- (iv) Perturbation pressure:  $\phi = f_0 \psi$ .
- (v) Perturbation buoyancy:  $b' = f_0 \partial \psi / \partial z$ .

By inspection of (5.63b) we see that a length scale  $L_d = NH/f_0$ , emerges naturally from the quasi-geostrophic dynamics. It is the scale at which buoyancy and relative vorticity effects contribute equally to the potential vorticity, and is called the *deformation radius*; it is analogous to the quantity  $\sqrt{gH}/f_0$  arising in shallow water theory. In the upper ocean, with  $N \approx 10^{-2} \text{ s}^{-1}$ ,  $H \approx 10^3 \text{ m}$  and  $f_0 \approx 10^{-4} \text{ s}^{-1}$ , then  $L_d \approx 100 \text{ km}$ . At high latitudes the ocean is much less stratified and  $f$  is somewhat larger, and the deformation radius may be as little as 20 km. In the atmosphere, with

$N \approx 10^{-2} \text{ s}^{-1}$ ,  $H \approx 10^4 \text{ m}$ , then  $L_d \approx 1000 \text{ km}$ . It is this order of magnitude difference in the deformation scales that accounts for a great deal of the quantitative difference in the dynamics of the ocean and the atmosphere. If we take the limit  $L_d \rightarrow \infty$  then the stratified quasi-geostrophic equations reduce to

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + f. \quad (5.65)$$

This is the two-dimensional vorticity equation; the high stratification of this limit has suppressed all vertical motion, and variations in the flow become confined to the horizontal plane.

Finally, if we allow density to vary in the vertical and carry through the quasi-geostrophic derivation we find that the quasi-geostrophic potential vorticity is given by

$$q = \nabla^2 \psi + f + \frac{f_0^2}{\bar{\rho}} \frac{\partial}{\partial z} \left( \frac{\bar{\rho}}{N^2} \frac{\partial \psi}{\partial z} \right), \quad (5.66)$$

where  $\bar{\rho}$  is a reference profile of density and a function of  $z$  only, typically decreasing approximately exponentially with height.

### 5.5.3 Upper and Lower Boundary Conditions and Buoyancy Advection

The solution of the elliptic equation in (5.63b) requires vertical boundary conditions on  $\psi$  at the ground and at some upper boundary (e.g., the tropopause), and these are given by use of the thermodynamic equation. For a flat, slippery, rigid surface the vertical velocity is zero so that the thermodynamic equation may be written as

$$\frac{Db'}{Dt} = 0, \quad b' = f_0 \frac{\partial \psi}{\partial z}, \quad z = 0, H. \quad (5.67)$$

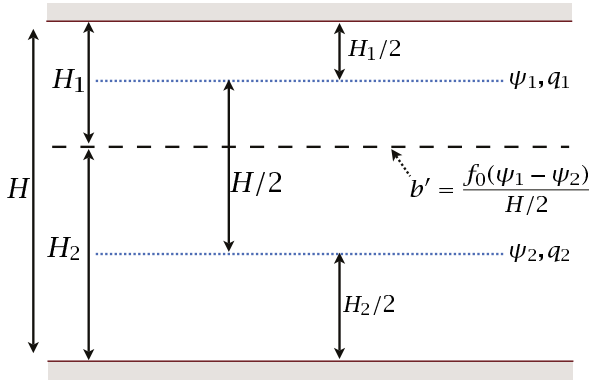
This equation provides the values of  $\partial \psi / \partial z$  at the boundary that are then used when solving (5.63b). A heating term could be added as needed, and if there is no upper boundary (as in the real atmosphere) we would apply a condition that the motion decays to zero with height, but we will not deal with that condition further. If the bottom boundary is not flat then the vertical velocity is non-zero and the thermodynamic equation becomes, at the bottom boundary,

$$\frac{D}{Dt} \left( \frac{\partial \psi}{\partial z} \right) + w N^2 = 0, \quad w = \mathbf{u} \cdot \nabla \eta_B, \quad (5.68)$$

where  $\eta_B$  is the height of the topography. In practice, the bottom boundary condition is often incorporated into the definition of potential vorticity, as we shall see in the two-level model.

## 5.6 THE TWO-LEVEL QUASI-GEOSTROPHIC EQUATIONS

The continuously-stratified quasi-geostrophic equations have, as their name suggests, a continuous variation in the vertical. It turns out to be



**Fig. 5.1:** A two-level quasi-geostrophic fluid system with a flat bottom and a rigid lid, with  $w = 0$  at both. Here the levels are allowed to be of unequal thicknesses, but in the text we take  $H_1 = H_2$ .

extraordinarily useful to constrain this variation to that of two levels, in which the velocity is defined at just two vertical levels, and the temperature at just one level in the middle, but keeping the full horizontal variation. The resulting equations are not only algebraically much simpler than the continuous equations but they also capture many of their important features. There are two ways to derive the equations. One is by considering, *ab initio*, the motion of two immiscible shallow layers of fluid, one on top of the other, giving the ‘two-layer’ quasi-geostrophic equations. The second is by finite-differencing the continuous equations, and this is the procedure we follow here. Both methods give identical answers and we use ‘two-layer’ and ‘two-level’ interchangeably through the book.

### 5.6.1 Deriving the Equations

The quasi-geostrophic potential vorticity may be written as

$$q = \nabla^2 \psi + \beta y + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (5.69)$$

We now apply this equation at the two levels, 1 and 2, in Fig. 5.1, taking  $N$  to be constant and  $H_1 = H_2 = H/2$  for simplicity (these restrictions may be easily relaxed). Using simple expressions for vertical finite differences gives

$$q_1 = \nabla^2 \psi_1 + \beta y + \frac{f_0^2}{N^2} \frac{1}{H/2} \left( \left( \frac{\partial \psi}{\partial z} \right)_{\text{top}} - \frac{\psi_1 - \psi_2}{H/2} \right). \quad (5.70)$$

In this equation  $(\partial \psi / \partial z)_{\text{top}}$  is the value of  $\partial \psi / \partial z$  at the top of the domain and  $2(\psi_1 - \psi_2)/H$  is the value of  $\partial \psi / \partial z$  in the middle of the domain. The value at the top is given by the thermodynamic equation, (5.67), and if there is no heating we set  $D/Dt (\partial \psi / \partial z)_{\text{top}} = 0$ . The potential vorticity equation for the upper level is then given by

$$\frac{Dq_1}{Dt} = 0, \quad q_1 = \nabla^2 \psi_1 + \beta y + \frac{k_d^2}{2} (\psi_2 - \psi_1), \quad (5.71a,b)$$

where  $k_d^2 = 8f_0^2/N^2H^2$ . In proceeding this way we have built the boundary conditions into the definition of potential vorticity. A thermody-

### Quasi-Geostrophic Equations

#### Continuously-stratified

The adiabatic quasi-geostrophic potential vorticity equation for a Boussinesq fluid on the  $\beta$ -plane, is

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + f\beta y + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right), \quad (\text{QG.1})$$

where  $\psi$  is the streamfunction. The horizontal velocities are given by  $(u, v) = (-\partial\psi/\partial y, \partial\psi/\partial x)$ . The boundary conditions at the top and bottom are given by the buoyancy equation,

$$\frac{Db}{Dt} = 0, \quad b = f_0 \frac{\partial \psi}{\partial z}. \quad (\text{QG.2})$$

#### Two-level

The two-level or two-layer quasi-geostrophic equations are

$$\frac{Dq_i}{Dt} = 0, \quad q_i = \nabla^2 \psi_i + \beta y + \frac{k_d^2}{2} (\psi_j - \psi_i), \quad (\text{QG.3})$$

where  $i = 1, 2$ , denoting the top and bottom levels respectively, and  $j = 3 - i$ .

Defining  $k_d$  so that  $k_d^2/2$  rather than just  $k_d^2$  is used in the two-level quasi-geostrophic potential vorticity is for later algebraic convenience rather than being of fundamental importance.

dynamic source and/or frictional terms may readily be added to the right-hand of (5.71a) as needed.

We apply an exactly analogous procedure to the lower level, and so obtain an expressions for the evolution of the lower layer potential vorticity, namely,

$$\frac{Dq_2}{Dt} = 0, \quad q_2 = \nabla^2 \psi_2 + \beta y + \frac{k_d^2}{2} (\psi_1 - \psi_2), \quad (5.72)$$

Once again the boundary conditions on buoyancy — the finite difference analogue of the buoyancy equations at the top and bottom — are *built in* to the definition of potential vorticity. If topography is present then it has an effect through the buoyancy equation at the bottom and it is incorporated into the definition of the lower level potential vorticity, which becomes

$$q_2 = \nabla^2 \psi_2 + \beta y + \frac{k_d^2}{2} (\psi_1 - \psi_2) + \frac{f_0 \eta_b}{H_2}. \quad (5.73)$$

This has a similar form to the shallow water potential vorticity with topography, as in (5.27). The two-level equations have proven to be enormously useful in the development of dynamical oceanography and meteorology and we will come back to them in later chapters. The quasi-geostrophic equations are summarized in the box above.



## 5.7 FRICTIONAL GEOSTROPHIC BALANCE AND EKMAN LAYERS

### 5.7.1 Preliminaries

Within a few hundred meters of the ground in the atmosphere, and in the upper and lower tens of metres in the ocean, frictional effects are important and geostrophic balance by itself is not the leading order balance in the momentum equations. Rather, a frictional term is important and the momentum equation becomes, for a Boussinesq fluid,

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + \mathbf{F}, \quad (5.74)$$

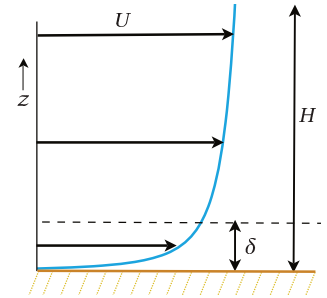
where  $\mathbf{F}$  is a frictional term and  $\nabla\phi$  is the horizontal gradient of pressure, at constant  $z$ . The friction ultimately comes from molecular viscosity and is of the form  $\mathbf{F} = \nu \nabla^2 \mathbf{u}$ . However, in geophysical settings the vertical derivative dominates and it is common to write (5.74) in the form

$$\mathbf{f} \times \mathbf{u} = -\nabla_z \phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad (5.75)$$

where  $\boldsymbol{\tau}$  is the stress, and since density is assumed constant we shall absorb it into the definition of  $\boldsymbol{\tau}$ . (The quantity  $\boldsymbol{\tau}/\rho_0$  is the ‘kinematic stress’, but we shall just refer to it as the stress and also denote it  $\boldsymbol{\tau}$ . Also, in some other fluid dynamical contexts the stress is a tensor but here  $\boldsymbol{\tau}$  is a vector.) Commonly, the regions where the stress term is important are *boundary layers* (see Fig. 5.2) and they exist because the fluid takes on values at the boundary that differ from the values in the fluid interior. Boundary layers are ubiquitous in fluids in many circumstances, and if the Coriolis term is important then the region is called an *Ekman layer*. As noted, the stress ultimately arises from molecular forces and  $\boldsymbol{\tau} = \nu \partial \mathbf{u} / \partial z$  where  $\nu$  is the coefficient of viscosity, again just including vertical derivatives. However, on the scale of the Ekman layer molecular effects are greatly amplified by the effects of small scale turbulence (as discussed more in Chapter 10) and we commonly replace  $\nu$  by a much larger ‘eddy viscosity’,  $A$ .

In the atmosphere we imagine that the flow is nearly geostrophic above an Ekman layer, with frictional effects coming from the need to bring the speed of flow down from its high geostrophic value to zero at the ground. In the ocean, on the other hand, the stress largely arises from the wind in the atmosphere blowing over the ocean surface. The stress then decays to zero with depth, and in the deeper ocean the flow again is geostrophic. The stress itself is continuous across the ocean–atmosphere interface, usually with a maximum value at the interface, decaying with height in the atmosphere and with depth in the ocean.

The equations of motion of the Ekman layer are completed by the mass continuity equation,  $\nabla \cdot \mathbf{v} = 0$ , and the hydrostatic equation,  $\partial \phi / \partial z = b$ . In order to treat the simplest case we take buoyancy to be constant, and in that case, and without any additional loss of generality, we take  $b = 0$ . The horizontal gradient of pressure is independent of height, which implies that, even though velocity may vary rapidly in the boundary layer the pressure gradient force does not.



**Fig. 5.2:** An idealized boundary layer in which fields vary rapidly in order to satisfy the boundary conditions, here that  $U = 0$ . The scale  $\delta$  is a measure of the thickness of the boundary layer and  $H$  is a typical scale of the variations away from the boundary.

*Fridtjof Nansen (1861–1930), the polar explorer and statesman, apparently wanted to understand the motion of pack ice and of his ship, the Fram, embedded in the ice and which seemed to move with neither the wind nor the surface current. He posed the problem to V. W. Ekman (1874–1954) who then calculated the direction of the flow in the upper ocean and published a paper on the matter in 1905.*

### 5.7.2 Properties of Ekman Layers

We shall now determine three important properties of Ekman layers:

- (i) The transport in an Ekman layer.
- (ii) The depth of the Ekman layer.
- (iii) The vertical velocity at the edge of the Ekman layer.

The importance of these will become clear as we go on.

#### Transport

We may write the Ekman layer equation (5.74) as

$$\mathbf{f} \times (\mathbf{u} - \mathbf{u}_g) = \frac{\partial \boldsymbol{\tau}}{\partial z} \quad \text{or equivalently} \quad \mathbf{f} \mathbf{u}_a = -\hat{\mathbf{k}} \times \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad (5.76a,b)$$

where  $\mathbf{u}_g$  is the geostrophic wind such that  $\mathbf{f} \times \mathbf{u}_g = -\nabla \phi$ ,  $\mathbf{u}_a = \mathbf{u} - \mathbf{u}_g$  is the ageostrophic wind and  $\hat{\mathbf{k}}$  is the unit vector in the vertical (so that  $\mathbf{f} = f\hat{\mathbf{k}}$ ). Consider an ocean with a stress that diminishes with depth from its surface values,  $\boldsymbol{\tau}_0$ . If we integrate (5.76) over the depth of the Ekman layer (i.e., down from the surface until the stress vanishes) we obtain

$$\int \mathbf{f} \mathbf{u}_a dz = -\hat{\mathbf{k}} \times \boldsymbol{\tau}_0, \quad \text{or} \quad \int f u_a dz = \tau_0^y, \quad \int f v_a dz = -\tau_0^x, \quad (5.77)$$

where the superscripts denote the  $x$  and  $y$  components of the stress. That is to say, *the integrated ageostrophic transport in the Ekman layer,  $\mathbf{U}_a = \int \mathbf{u}_a dz$ , is at right angles to the stress at the surface*, as in Fig. 5.3. This result is independent of the detailed form of the stress, and it gives a partial answer to Nansen's question — if the surface stress is in the direction of the wind, the water beneath it flows at an angle to it, as a consequence of the Coriolis force. More generally, the Ekman transport is the most immediate response of the ocean to the atmosphere and is thus of key importance in the ocean circulation. In the atmosphere we *live* in the Ekman layer, and Ekman layer effects are partly responsible for determining the surface wind.

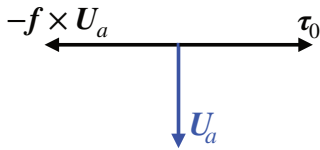
#### Depth

Although the transport in the Ekman layer is independent of the detailed form of the stress, the depth — namely the extent of the frictional influence of the surface — is not and to calculate it we use the Ekman-layer equation in the form

$$\mathbf{f} \times \mathbf{u} = -\nabla \phi + A \frac{\partial^2 \mathbf{u}}{\partial z^2}, \quad (5.78)$$

where  $A$  is a coefficient of viscosity. Suppose for the moment that the vertical scale of the flow is  $H$ ; the ratio of size of the frictional term to the Coriolis term is then  $(A/fH^2)$ , and this ratio defines the Ekman number,  $Ek$ :

$$Ek \equiv \left( \frac{A}{fH^2} \right). \quad (5.79)$$



**Fig. 5.3:** Vertically integrated transport in a Northern Hemisphere ocean Ekman layer. A stress  $\boldsymbol{\tau}_0$  at the surface induces an ageostrophic flow,  $\mathbf{U}_a$ . In order that the Coriolis force on the ageostrophic flow can balance the surface stress the ageostrophic flow must be at right angles to the stress, as illustrated.

In the fluid interior the Ekman number is usually small. In the Ekman layer itself, however, the viscous terms must be large (otherwise we are not in the Ekman layer), and the vertical scale must therefore be smaller than  $H$ . If the viscous and Coriolis terms in (5.78) are required to be the same magnitude we obtain an Ekman layer depth,  $\delta$ , of

$$\delta \sim \left( \frac{A}{f} \right)^{1/2}. \quad (5.80)$$

The Ekman layer depth decreases with viscosity, becoming a thin boundary layer as  $A \rightarrow 0$ . We also see that the Ekman number can be recast as the ratio of the Ekman layer depth to the vertical scale in the interior; that is, from (5.79) and (5.80),  $Ek = (\delta/H)^2$ .

### Vertical velocity

The vertical velocity at the edge of the Ekman layer may be calculated using the mass continuity equation,

$$\frac{\partial w}{\partial z} = - \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right). \quad (5.81)$$

The right-hand side may be calculated from (5.76) and, noting that divergence of the geostrophic velocity is given by  $\nabla \cdot \mathbf{v}_g = -\beta v_g/f$ , we find that

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -\frac{\beta v_g}{f} + \frac{\partial}{\partial z} \text{curl}_z \left( \frac{\boldsymbol{\tau}}{f} \right), \quad (5.82)$$

where  $\text{curl}_z(\boldsymbol{\tau}/f) = \partial_x(\tau^y/f) - \partial_y(\tau^x/f)$ . Consider an ocean Ekman layer in which the vertical velocity is zero at the surface and the stress is zero at the bottom of the layer. If we integrate (5.81) over the depth of that layer and use (5.82) we obtain

$$w_E = \text{curl}_z \frac{\boldsymbol{\tau}_0}{f} - \frac{\beta V_g}{f}. \quad (5.83)$$

where  $V_g = \int v_g dz$  is the integral of the meridional geostrophic velocity over the Ekman layer. Evidently, friction induces a vertical velocity at the edge of the Ekman layer, proportional to the curl of the stress at the surface, and this is perhaps the most used result in Ekman layer theory.

This result is particularly useful for the top Ekman layer in the ocean, where the stress can be regarded as a function of the overlying wind and is largely independent of the flow in the ocean; an equivalent result applies in the atmosphere, but here the surface stress is a function of the geostrophic wind in the free atmosphere. If the curl of the wind stress at the top of the ocean is anticyclonic (i.e., negative if  $f > 0$ , positive if  $f < 0$ ) then a downward velocity is induced at the base of the Ekman layer, a phenomenon known as *Ekman pumping*, and this is particularly important in setting the structure of the subtropical gyres in the ocean, discussed in Chapter 14.

### Notes and References

The first systematic derivation of the quasi-geostrophic equations was given by Charney (1948), although aspects of the concept, and the words ‘quasi-geostrophy’, appeared in Durst & Sutcliffe (1938) and Sutcliffe (1947). The various forms of geostrophic equations were brought together in a review article by Phillips (1963) and since then both the quasi-geostrophic and planetary-geostrophic equations have been staples of dynamical meteorology and dynamical oceanography.

### Problems

5.1 Do either or both:

- (a) Carry through the derivation of the quasi-geostrophic system starting with the anelastic equations and obtain (5.66).
- (b) Carry through the derivation of the quasi-geostrophic system in pressure coordinates.

In each case, state the differences between your results and the Boussinesq result.

5.2 (a) The shallow water *planetary geostrophic* equations may be derived by simply omitting  $\zeta$  in the equation

$$\frac{D}{Dt} \frac{\zeta + f}{h} = 0 \quad (\text{P5.1})$$

by invoking a small Rossby number, so that  $\zeta/f$  is small. We then relate the velocity field to the height field by hydrostatic balance and obtain:

$$\frac{D}{Dt} \left( \frac{f}{h} \right) = 0, \quad fu = -g \frac{\partial h}{\partial y}, \quad fv = g \frac{\partial h}{\partial x}. \quad (\text{P5.2})$$

The assumptions of hydrostatic balance and small Rossby number are the same as those used in deriving the quasi-geostrophic equations. Explain nevertheless how some of the assumptions used for quasi-geostrophy are in fact different from those used for planetary-geostrophy, and how the derivations and resulting systems differ from each other. Use any or all of the momentum and mass continuity equations, scaling, nondimensionalization and verbal explanations as needed.

- (b) Explain if and how your arguments in part (a) also apply to the stratified equations (using, for example, the Boussinesq equations or pressure coordinates).
- 5.3 In the derivation of the quasi-geostrophic equations, geostrophic balance leads to the lowest-order horizontal velocity being divergence-free — that is,  $\nabla \cdot \mathbf{u}_0 = 0$ . It seems that this can also be obtained from the mass conservation equation at lowest order. Is this a coincidence? Suppose that the Coriolis parameter varied, and that the momentum equation yielded  $\nabla \cdot \mathbf{u}_0 \neq 0$ . Would there be an inconsistency?
- 5.4 Consider a wind stress imposed by a mesoscale cyclonic storm (in the atmosphere) given by

$$\boldsymbol{\tau} = -Ae^{-(r/\lambda)^2} (y \hat{\mathbf{i}} - x \hat{\mathbf{j}}), \quad (\text{P5.3})$$

where  $r^2 = x^2 + y^2$ , and  $A$  and  $\lambda$  are constants. Also assume constant Coriolis gradient  $\beta = \partial f / \partial y$  and constant ocean depth  $H$ . In the ocean, find

(a) the Ekman transport, (b) the vertical velocity  $w_E(x, y, z)$  below the Ekman layer, (c) the northward velocity  $v(x, y, z)$  below the Ekman layer and (d) indicate how you would find the westward velocity  $u(x, y, z)$  below the Ekman layer.

5.5 In an atmospheric Ekman layer on the  $f$ -plane let us write the momentum equation as

$$\mathbf{f} \times \mathbf{u} = -\nabla\phi + \frac{1}{\rho_a} \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad (\text{P5.4})$$

where  $\boldsymbol{\tau} = A\rho_a \partial \mathbf{u} / \partial z$  and  $A$  is a constant eddy viscosity coefficient. An *independent* formula for the stress at the ground is  $\boldsymbol{\tau} = C\rho_a \mathbf{u}$ , where  $C$  is a constant. Let us take  $\rho_a = 1$ , and assume that in the free atmosphere the wind is geostrophic and zonal, with  $\mathbf{u}_g = U\hat{\mathbf{i}}$ .

- Find an expression for the wind vector at the ground. Discuss the limits  $C = 0$  and  $C = \infty$ . Show that when  $C = 0$  the frictionally-induced vertical velocity at the top of the Ekman layer is zero.
- Find the vertically integrated horizontal mass flux caused by the boundary layer.
- When the stress on the atmosphere is  $\boldsymbol{\tau}$ , the stress on the ocean beneath is also  $\boldsymbol{\tau}$ . Why? Show how this is consistent with Newton's third law.
- Determine the direction and strength of the surface current, and the mass flux in the oceanic Ekman layer, in terms of the geostrophic wind in the atmosphere, the oceanic Ekman depth and the ratio  $\rho_a / \rho_o$ , where  $\rho_o$  is the density of the seawater. Include a figure showing the directions of the various winds and currents. How does the boundary-layer mass flux in the ocean compare to that in the atmosphere? (Assume, as needed, that the stress in the ocean may be parameterized with an eddy viscosity.)

*Partial solution for (a):* A useful trick in Ekman layer problems is to write the velocity as a complex number,  $\hat{u} = u + iv$  and  $\hat{u}_g = u_g + iv_g$ . The fundamental Ekman layer equation may then be written as

$$A \frac{\partial^2 \hat{U}}{\partial z^2} = i f \hat{U}, \quad (\text{P5.5})$$

where  $\hat{U} = \hat{u} - \hat{u}_g$ . The solution to this is

$$\hat{u} - \hat{u}_g = [\hat{u}(0) - \hat{u}_g] \exp\left[-\frac{(1+i)z}{d}\right], \quad (\text{P5.6})$$

where  $d = \sqrt{2A/f}$  and the boundary condition of finiteness at infinity eliminates the exponentially growing solution. The boundary condition at  $z = 0$  is  $\partial \hat{u} / \partial z = (C/A)\hat{u}$ ; applying this gives  $[\hat{u}(0) - \hat{u}_g] \exp(i\pi/4) = -Cd\hat{u}(0)/(\sqrt{2}A)$ , from which we obtain  $\hat{u}(0)$ , and the rest of the solution follows.

# Rossby Waves

**W**AVES ARE FAMILIAR TO ALMOST EVERYONE. Gravity waves cover the ocean surface, sound waves allow us to talk and light waves enable us to see. This chapter provides an introduction to their properties, paying particular attention to a wave that is especially important to the large scale flow in both ocean and atmosphere — the Rossby wave. We start with an elementary introduction to wave kinematics, discussing such concepts as phase speed and group velocity. Then, beginning with Section 6.3, we discuss the dynamics of Rossby waves, and this part may be considered to be the natural follow-on from the geostrophic theory of the previous chapter. Rossby waves then reappear frequently in later chapters.

## 6.1 FUNDAMENTALS AND FORMALITIES

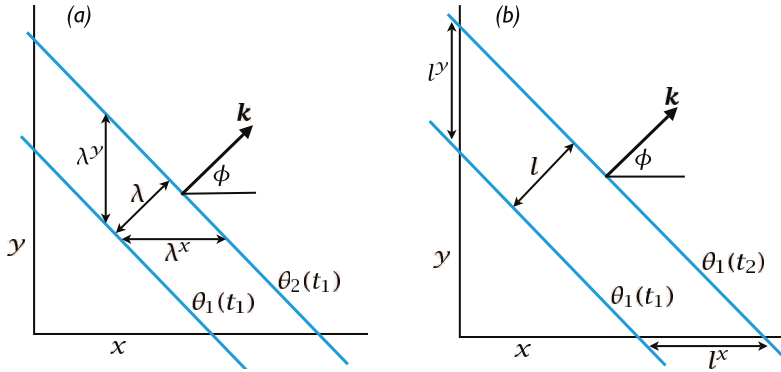
### 6.1.1 Definitions and Kinematics

A wave is more easily recognized than defined. Loosely speaking, a wave is a propagating disturbance that has a characteristic relationship between its frequency and size, called a *dispersion relation*. To see what all this means, and what a dispersion relation is, suppose that a disturbance,  $\psi(\mathbf{x}, t)$  (where  $\psi$  might be velocity, streamfunction, pressure, etc.), satisfies the equation

$$L(\psi) = 0, \quad (6.1)$$

where  $L$  is a linear operator, typically a polynomial in time and space derivatives; one example is  $L(\psi) = \partial \nabla^2 \psi / \partial t + \beta \partial \psi / \partial x$ . If (6.1) has constant coefficients (if  $\beta$  is constant in this example) then harmonic solutions may often be found that are a superposition of *plane waves*, each of which satisfy

$$\psi = \text{Re } \tilde{\psi} e^{i\theta(\mathbf{x}, t)} = \text{Re } \tilde{\psi} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (6.2)$$



**Fig. 6.1:** The propagation of a two-dimensional wave. (a) Two lines of constant phase (e.g., two wavecrests) at a time  $t_1$ . The wave is propagating in the direction  $\mathbf{k}$  with wavelength  $\lambda$ . (b) A line of constant phase at two successive times. The phase speed is the speed of advancement of the wavecrest in the direction of travel, and so  $c_p = l/(t_2 - t_1)$ . The phase speed in the  $x$ -direction is the speed of propagation of the wavecrest along the  $x$ -axis, and  $c_p^x = l^x/(t_2 - t_1) = c_p / \cos \phi$ .

where  $\tilde{\psi}$  is a complex constant,  $\theta$  is the phase,  $\omega$  is the wave frequency and  $\mathbf{k}$  is the vector wavenumber  $(k, l, m)$  (also written as  $(k^x, k^y, k^z)$  or, in subscript notation,  $k_i$ ). The prefix  $\text{Re}$  denotes the real part of the expression, but we will drop it if there is no ambiguity.

Waves are characterized by having a particular relationship between the frequency and wavevector known as the *dispersion relation*. This is an equation of the form

$$\omega = \Omega(\mathbf{k}), \quad (6.3)$$

where  $\Omega(\mathbf{k})$ , or  $\Omega(k_i)$ , and meaning  $\Omega(k, l, m)$ , is some function determined by the form of  $L$  in (6.1) and which thus depends on the particular type of wave — the function is different for sound waves, light waves and the Rossby waves and gravity waves we will encounter in this book. Unless it is necessary to explicitly distinguish the function  $\Omega$  from the frequency  $\omega$ , we often write  $\omega = \omega(\mathbf{k})$ .

### 6.1.2 Wave Propagation and Phase Speed

A common property of waves is that they propagate through space with some velocity, which in special cases might be zero. Waves in fluids may carry energy and momentum but do not necessarily transport fluid parcels themselves. Further, it turns out that the speed at which properties like energy are transported (the group speed) may be different from the speed at which the wave crests themselves move (the phase speed). Let's try to understand this statement, beginning with the phase speed. A summary of key results is given on page 107.

#### Phase speed

Consider the propagation of monochromatic plane waves, for that is all that is needed to introduce the phase speed. Given (6.2) a wave will propagate in the direction of  $\mathbf{k}$  (Fig. 6.1). At a given instant and location we can align our coordinate axis along this direction, and we write  $\mathbf{k} \cdot \mathbf{x} = Kx^*$ , where  $x^*$  increases in the direction of  $\mathbf{k}$  and  $K^2 = |\mathbf{k}|^2$  is the magnitude of the wavenumber. With this, we can write (6.2) as

$$\psi = \text{Re } \tilde{\psi} e^{i(Kx^* - \omega t)} = \text{Re } \tilde{\psi} e^{iK(x^* - ct)}, \quad (6.4)$$

where  $c = \omega/K$ . From this equation it is evident that the phase of the wave propagates at the speed  $c$  in the direction of  $\mathbf{k}$ , and we define the *phase speed* by

$$c_p \equiv \frac{\omega}{K}. \quad (6.5)$$

The wavelength of the wave,  $\lambda$ , is the distance between two wavecrests — that is, the distance between two locations along the line of travel whose phase differs by  $2\pi$  — and evidently this is given by

$$\lambda = \frac{2\pi}{K}. \quad (6.6)$$

In (for simplicity) a two-dimensional wave, and referring to Fig. 6.1, the wavelength and wave vectors in the  $x$ - and  $y$ -directions are given by,

$$\lambda^x = \frac{\lambda}{\cos \phi}, \quad \lambda^y = \frac{\lambda}{\sin \phi}, \quad k^x = K \cos \phi, \quad k^y = K \sin \phi. \quad (6.7)$$

In general, lines of constant phase intersect both the coordinate axes and propagate along them. The speed of propagation along these axes is given by

$$c_p^x = c_p \frac{l^x}{l} = \frac{c_p}{\cos \phi} = c_p \frac{K}{k^x} = \frac{\omega}{k^x}, \quad c_p^y = c_p \frac{l^y}{l} = \frac{c_p}{\sin \phi} = c_p \frac{K}{k^y} = \frac{\omega}{k^y}, \quad (6.8)$$

using (6.5) and (6.7), and again referring to Fig. 6.1 for notation. The speed of phase propagation along any one of the axes is in general *larger* than the phase speed in the primary direction of the wave. The phase speeds are clearly *not* components of a vector: for example,  $c_p^x \neq c_p \cos \phi$ . Analogously, the wavevector  $\mathbf{k}$  is a true vector, whereas the wavelength  $\lambda$  is not.

To summarize, the phase speed and its components are given by

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k^x}, \quad c_p^y = \frac{\omega}{k^y}. \quad (6.9)$$

### 6.1.3 The Dispersion Relation

The above description is kinematic, in that it applies to almost any disturbance that has a wavevector and a frequency. The particular *dynamics* of a wave are determined by the relationship between the wavevector and the frequency; that is, by the *dispersion relation*. Once the dispersion relation is known a great many of the properties of the wave follow in a more-or-less straightforward manner. Picking up from (6.3), the dispersion relation is a functional relationship between the frequency and the wavevector of the general form

$$\omega = \Omega(\mathbf{k}). \quad (6.10)$$

Perhaps the simplest example of a linear operator that gives rise to waves is the one-dimensional equation

$$\frac{\partial \psi}{\partial t} + c \frac{\partial \psi}{\partial x} = 0. \quad (6.11)$$



### Wave Fundamentals

- A wave is a propagating disturbance that has a characteristic relationship between its frequency and size, known as the dispersion relation. Waves typically arise as solutions to a linear problem of the form  $L(\psi) = 0$ , where  $L$  is, commonly, a linear operator in space and time. Two examples are

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0 \quad \text{and} \quad \frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0, \quad (\text{WF.1})$$

where the second example gives rise to Rossby waves.

- Solutions to the governing equation are often sought in the form of plane waves that have the form

$$\psi = \text{Re } A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad (\text{WF.2})$$

where  $A$  is the wave amplitude,  $\mathbf{k} = (k, l, m)$  is the wavevector, and  $\omega$  is the frequency.

- The dispersion relation connects the frequency and wavevector through an equation of the form  $\omega = \Omega(\mathbf{k})$  where  $\Omega$  is some function. The relation is normally derived by substituting a trial solution like (WF.2) into the governing equation. For the examples of (WF.1) we obtain  $\omega = c^2 K^2$  and  $\omega = -\beta k / K^2$  where  $K^2 = k^2 + l^2 + m^2$  or, in two dimensions,  $K^2 = k^2 + l^2$ .
- The phase speed is the speed at which the wave crests move. In the direction of propagation and in the  $x$ ,  $y$  and  $z$  directions the phase speeds are given by, respectively,

$$c_p = \frac{\omega}{K}, \quad c_p^x = \frac{\omega}{k}, \quad c_p^y = \frac{\omega}{l}, \quad c_p^z = \frac{\omega}{m}, \quad (\text{WF.3})$$

where  $K = 2\pi/\lambda$  and  $\lambda$  is the wavelength. The wave crests have both a speed ( $c_p$ ) and a direction of propagation (the direction of  $\mathbf{k}$ ), like a vector, but the components defined in (WF.3) are not the components of that vector.

- The group velocity is the velocity at which a wave packet or wave group moves. It is a vector and is given by

$$\mathbf{c}_g = \frac{\partial \omega}{\partial \mathbf{k}} \quad \text{with components} \quad c_g^x = \frac{\partial \omega}{\partial k}, \quad c_g^y = \frac{\partial \omega}{\partial l}, \quad c_g^z = \frac{\partial \omega}{\partial m}. \quad (\text{WF.4})$$

Most physical quantities of interest are transported at the group velocity.

Substituting a trial solution of the form  $\psi = \text{Re } A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$  into (6.11) we obtain  $(-i\omega + c i k)A = 0$ , giving the dispersion relation

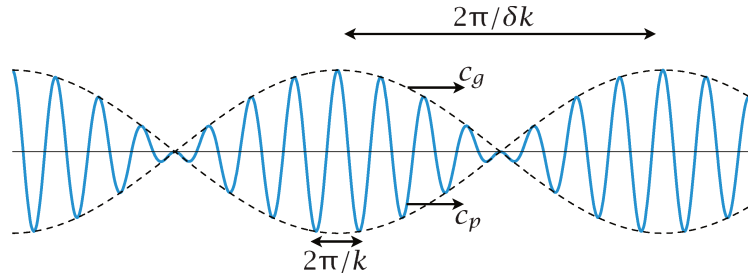
$$\omega = ck. \quad (6.12)$$

The phase speed of this wave is  $c_p = \omega/k = c$ . A couple of other examples of governing equations, dispersion relations and phase speeds are:

$$\frac{\partial^2 \psi}{\partial t^2} - c^2 \nabla^2 \psi = 0, \quad \omega^2 = c^2 K^2, \quad c_p = \pm c, \quad c_p^x = \pm \frac{cK}{k}, \quad c_p^y = \pm \frac{cK}{l}, \quad (6.13a)$$

$$\frac{\partial}{\partial t} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = 0, \quad \omega = -\frac{\beta k}{K^2}, \quad c_p = \frac{\omega}{K}, \quad c_p^x = -\frac{\beta}{K^2}, \quad c_p^y = -\frac{\beta k/l}{K^2}, \quad (6.13b)$$

**Fig. 6.2:** Superposition of two sinusoidal waves with wavenumbers  $k$  and  $k + \delta k$ , producing a wave (solid line) that is modulated by a slowly varying wave envelope or packet (dashed). The envelope moves at the group velocity,  $c_g = \partial\omega/\partial k$ , and the phase moves at the group speed,  $c_p = \omega/k$ .



where  $K^2 = k^2 + l^2$  and the examples are two-dimensional, with variation in  $x$  and  $y$  only.

A wave is said to be *nondispersive* if the phase speed is independent of the wavelength. This condition is satisfied for the simple example (6.11) but is manifestly not satisfied for (6.13b), and these waves (Rossby waves, in fact) are *dispersive*. Waves of different wavelengths then travel at different speeds so that a group of waves will spread out — disperse — even if the medium is homogeneous. When a wave is dispersive there is another characteristic speed at which the waves propagate, the group velocity, and we come to this shortly.

Most media are inhomogeneous, but if the medium varies sufficiently slowly in space and time — and in particular if the variations are slow compared to the wavelength and period — we may still have a *local* dispersion relation between frequency and wavevector,

$$\omega = \Omega(\mathbf{k}; \mathbf{x}, t), \quad (6.14)$$

where  $\mathbf{x}$  and  $t$  are slowly varying parameters. We resume our discussion of this topic in Section 6.5, but before that we introduce the group velocity.

## 6.2 GROUP VELOCITY

Information and energy do not, in general, propagate at the phase speed. Rather, most quantities of interest propagate at the *group velocity*, a quantity of enormous importance in wave theory. Roughly speaking, group velocity is the velocity at which a packet or a group of waves will travel, whereas the individual wave crests travel at the phase speed. To introduce the idea we will consider the superposition of plane waves, noting that a truly monochromatic plane wave already fills all space uniformly so that there can be no propagation of energy from place to place.

### 6.2.1 Superposition of Two Waves

Consider the linear superposition of two waves. Limiting attention to the one-dimensional case, consider a disturbance that is the sum of two waves,

$$\psi = \text{Re } \tilde{\psi}(e^{i(k_1 x - \omega_1 t)} + e^{i(k_2 x - \omega_2 t)}). \quad (6.15)$$

Group velocity seems to have been first articulated in about 1841 by the Irish mathematician and physicist William Rowan Hamilton (1806–1865), who is also remembered for his formulation of ‘Hamiltonian mechanics’. Hamilton was largely motivated by optics, and it was George Stokes, Osborne Reynolds and John Strutt (also known as Lord Rayleigh) who further developed and generalized the idea in fluid dynamics in the nineteenth and early twentieth centuries.

Let us further suppose that the two waves have similar wavenumbers and frequency, and, in particular, that  $k_1 = k + \Delta k$  and  $k_2 = k - \Delta k$ , and  $\omega_1 = \omega + \Delta\omega$  and  $\omega_2 = \omega - \Delta\omega$ . With this, (6.15) becomes

$$\begin{aligned}\psi &= \text{Re } \tilde{\psi} e^{i(kx - \omega t)} [e^{i(\Delta k x - \Delta\omega t)} + e^{-i(\Delta k x - \Delta\omega t)}] \\ &= 2 \text{Re } \tilde{\psi} e^{i(kx - \omega t)} \cos(\Delta k x - \Delta\omega t).\end{aligned}\quad (6.16)$$

The resulting disturbance, illustrated in Fig. 6.2 has two aspects: a rapidly varying component, with wavenumber  $k$  and frequency  $\omega$ , and a more slowly varying envelope, with wavenumber  $\Delta k$  and frequency  $\Delta\omega$ . The envelope modulates the fast oscillation, and moves with velocity  $\Delta\omega/\Delta k$ ; in the limit  $\Delta k \rightarrow 0$  and  $\Delta\omega \rightarrow 0$  this is the *group velocity*,  $c_g = \partial\omega/\partial k$ . Group velocity is equal to the phase speed,  $\omega/k$ , only when the frequency is a linear function of wavenumber. The energy in the disturbance moves at the group velocity — note that the node of the envelope moves at the speed of the envelope and no energy can cross the node. These concepts generalize to more than one dimension, and if the wavenumber is the three-dimensional vector  $\mathbf{k} = (k, l, m)$  then the three-dimensional envelope propagates at the group velocity given by

$$\mathbf{c}_g = \frac{\partial\omega}{\partial\mathbf{k}} \equiv \left( \frac{\partial\omega}{\partial k}, \frac{\partial\omega}{\partial l}, \frac{\partial\omega}{\partial m} \right). \quad (6.17)$$

The group velocity is also written as  $\mathbf{c}_g = \nabla_{\mathbf{k}}\omega$  or, in subscript notation,  $c_{gi} = \partial\omega/\partial k_i$ , with the subscript  $i$  denoting the component of a vector.

The above derivation can be generalized to apply to the superposition of many waves. It also applies when the medium through which the waves propagate is not homogeneous, provided that changes occur on a longer space scale than the wavelength of the waves. Energy and most other physically meaningful properties of waves travel at the group velocity, as explicitly shown for Rossby waves in Section 9.2.2.

### 6.3 ROSSBY WAVE ESSENTIALS

Rossby waves are the most prominent wave in the atmosphere and ocean on large scales, although gravity waves sometimes rival them. They can best be described using the quasi-geostrophic equations, as follows.

#### 6.3.1 The Linear Equation of Motion

The relevant equation of motion is the inviscid, adiabatic potential vorticity equation in the quasi-geostrophic system, as discussed in Chapter 5, namely

$$\frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q = 0, \quad (6.18)$$

where  $q(x, y, z, t)$  is the potential vorticity and  $\mathbf{u}(x, y, z, t)$  is the horizontal velocity. The velocity is in turn related to a streamfunction by  $u = -\partial\psi/\partial y$ ,  $v = \partial\psi/\partial x$ , and the potential vorticity is some function

The group velocity,  $\mathbf{c}_g$  is given by  $\mathbf{c}_g = \partial\omega/\partial\mathbf{k}$  or, in Cartesian components,  $\mathbf{c}_g = (\partial\omega/\partial k, \partial\omega/\partial l, \partial\omega/\partial m)$ . It is a vector, and it gives the velocity at which wave packets, and hence energy, are propagated. The phase speed,  $c_p = \omega/k$ , is the speed at which the phase of an individual wave moves. For some waves the group velocity and phase speed may be in opposite directions, so that the wave crests appear to move backwards through the wave packets.

*The Rossby wave, the most well-known wave in meteorology and oceanography, is named for Carl-Gustav Rossby who described the essential dynamics in Rossby (1939). The waves are in fact present in the solutions of Hough (1898) although in a rather oblique form and Rossby is normally given credit for having discovered them.*

of the streamfunction, which might differ from system to system. Two examples, one applying to a continuously-stratified system and the second to a single layer, are

$$q = f + \zeta + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) \quad \text{and} \quad q = \zeta + f - k_d^2 \psi, \quad (6.19a,b)$$

*Definitions of  $k_d$  and  $L_d$  differ from source to source (and book to book), often by factors of 2 or  $\pi$  or similar. Caveat lector.*

where  $\zeta = \nabla^2 \psi$  is the relative vorticity and  $k_d = 1/L_d$  is the inverse radius of deformation for a shallow water system.

We now *linearize* (6.18); that is, we suppose that the flow consists of a time-independent component (the ‘basic state’) plus a perturbation, with the perturbation being small compared with the mean flow. The basic state must satisfy the time-independent equation of motion, and it is common and useful to linearize about a zonal flow,  $\bar{u}(y, z)$ . The basic state is then purely a function of  $y$  and so we can write

$$q = \bar{q}(y, z) + q'(x, y, t), \quad \psi = \bar{\psi}(y, z) + \psi'(x, y, z, t), \quad (6.20)$$

with a similar notation for the other variables, and  $\bar{u} = -\partial \bar{\psi} / \partial y$  and  $\bar{v} = 0$ . Substituting into (6.18) gives, without approximation,

$$\frac{\partial q'}{\partial t} + \bar{\mathbf{u}} \cdot \nabla \bar{q} + \bar{\mathbf{u}} \cdot \nabla q' + \mathbf{u}' \cdot \nabla \bar{q} + \mathbf{u}' \cdot \nabla q' = 0. \quad (6.21)$$

The primed quantities are presumptively small so we neglect terms involving their products. Further, we are assuming that we are linearizing about a state that is a solution of the equations of motion, so that  $\bar{\mathbf{u}} \cdot \nabla \bar{q} = 0$ . Finally, since  $\bar{v} = 0$  (since  $\int \partial \bar{\psi} / \partial x \, dx = 0$ ) and  $\partial \bar{q} / \partial x = 0$  we obtain

$$\frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (6.22)$$

where  $U \equiv \bar{u}$ . This equation or one very similar appears very commonly in studies of Rossby waves. Let us first consider the simple example of waves in a single layer.

### 6.3.2 Waves in a Single Layer

Consider a system obeying (6.18) and (6.19b). The dynamics are more easily illustrated on a Cartesian  $\beta$ -plane for which  $f = f_0 + \beta y$ , and since  $f_0$  is a constant it does not appear in our subsequent derivations.

#### *Infinite deformation radius*

If the scale of motion is much less than the deformation scale then we make the approximation that  $k_d = 0$  and the vorticity equation may be written as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0. \quad (6.23)$$

We linearize about a constant zonal flow,  $U$ , by writing

$$\frac{\partial}{\partial t} \nabla^2 \psi' + U \frac{\partial \nabla^2 \psi'}{\partial x} + \beta \frac{\partial \psi'}{\partial x} = 0. \quad (6.24)$$

This equation is just a single-layer version of (6.22), with  $\partial \bar{q}/\partial y = \beta$ ,  $q' = \nabla^2 \psi'$  and  $v' = \partial \psi'/\partial x$ .

The coefficients in (6.24) are not functions of  $y$  or  $z$ ; this is not a requirement for wave motion to exist but it does enable solutions to be found more easily. Let us seek solutions in the form of a plane wave, namely

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly - \omega t)}, \quad (6.25)$$

where  $\tilde{\psi}$  is a complex constant. Solutions of this form are valid in a domain with doubly-periodic boundary conditions; solutions in a channel can be obtained using a meridional variation of  $\sin ly$ , with no essential changes to the dynamics. The amplitude of the oscillation is given by  $\tilde{\psi}$  and the phase by  $kx + ly - \omega t$ , where  $k$  and  $l$  are the  $x$ - and  $y$ -wavenumbers and  $\omega$  is the frequency of the oscillation.

Substituting (6.25) into (6.24) yields

$$[(-\omega + Uk)(-K^2) + \beta k] \tilde{\psi} = 0, \quad (6.26)$$

where  $K^2 = k^2 + l^2$ . For non-trivial solutions the above equation implies

$$\omega = Uk - \frac{\beta k}{K^2}, \quad (6.27)$$

and this is the *dispersion relation* for barotropic Rossby waves. Evidently the velocity  $U$  Doppler shifts the frequency by the amount  $Uk$ . The components of the phase speed and group velocity are given by, respectively,

$$c_p^x \equiv \frac{\omega}{k} = U - \frac{\beta}{K^2}, \quad c_p^y \equiv \frac{\omega}{l} = U \frac{k}{l} - \frac{\beta k}{K^2 l}, \quad (6.28a,b)$$

and

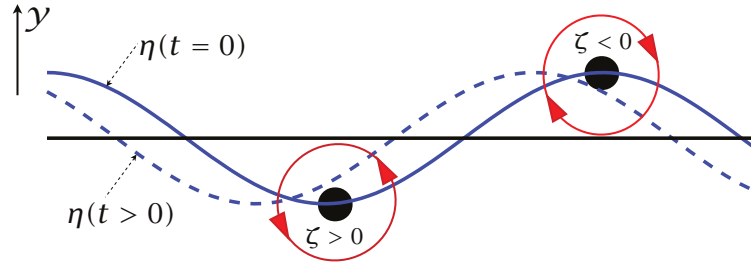
$$c_g^x \equiv \frac{\partial \omega}{\partial k} = U + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \quad c_g^y \equiv \frac{\partial \omega}{\partial l} = \frac{2\beta k l}{(k^2 + l^2)^2}. \quad (6.29a,b)$$

The phase speed in the absence of a mean flow is *westward*, with waves of longer wavelengths travelling more quickly, and the eastward current speed required to hold the waves of a particular wavenumber stationary (i.e.,  $c_p^x = 0$ ) is  $U = \beta/K^2$ . The background flow  $U$  evidently just provides a uniform shift to the phase speed, and (in this case) can be transformed away by a change of coordinate. The  $x$ -component of the group velocity may also be written as the sum of the phase speed plus a positive quantity, namely

$$c_g^x = c_p^x + \frac{2\beta k^2}{(k^2 + l^2)^2}. \quad (6.30)$$

This means that the zonal group velocity for Rossby wave packets moves eastward relative to its zonal phase speed. A stationary wave ( $c_p^x = 0$ ) has

**Fig. 6.3:** A two-dimensional ( $x$ - $y$ ) Rossby wave. An initial disturbance displaces a material line at constant latitude (the straight horizontal line) to the solid line marked  $\eta(t = 0)$ . Conservation of potential vorticity,  $\beta y + \zeta$ , leads to the production of relative vorticity,  $\zeta$ , as shown. The associated velocity field (arrows on the circles) then advects the fluid parcels, and the material line evolves into the dashed line with the phase propagating westward.



an eastward group velocity, and this has implications for the ‘downstream development’ of wave packets, but we will not pursue that topic here.

We see from (6.29) the group velocity is negative (westward) if the  $x$ -wavenumber is sufficiently small compared to the  $y$ -wavenumber. Essentially, *long waves move information westward and short waves move information eastward*, and this is a common property of Rossby waves. The  $x$ -component of the phase speed, on the other hand, is always westward relative to the mean flow.

### Finite deformation radius

For a finite deformation radius the basic state  $\Psi = -Uy$  is still a solution of the original equations of motion, but the potential vorticity corresponding to this state is  $q = Uyk_d^2 + \beta y$  and its gradient is  $\nabla q = (\beta + Uk_d^2)\hat{\mathbf{j}}$ . The linearized equation of motion is thus

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) (\nabla^2 \psi' - \psi' k_d^2) + (\beta + Uk_d^2) \frac{\partial \psi'}{\partial x} = 0. \quad (6.31)$$

Substituting  $\psi' = \tilde{\psi} e^{i(kx + ly - \omega t)}$  we obtain the dispersion relation,

$$\omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2} = Uk - k \frac{\beta + Uk_d^2}{K^2 + k_d^2}. \quad (6.32)$$

It is clear from the second form of the above equation that the uniform velocity field no longer provides just a simple Doppler shift of the frequency, nor does it provide a uniform addition to the phase speed. This is because the current does not just provide a uniform translation, but, if  $k_d$  is non-zero, it also modifies the basic potential vorticity gradient, as explored further in Problem 6.1.

### 6.3.3 The Mechanism of Rossby Waves

The fundamental mechanism underlying Rossby waves may be understood as follows. Consider a material line of stationary fluid parcels along a line of constant latitude, and suppose that some disturbance causes their displacement to the line marked  $\eta(t = 0)$  in Fig. 6.3. In the displacement, the potential vorticity of the fluid parcels is conserved, and in the simplest

### Essentials of Rossby Waves

- Rossby waves owe their existence to a gradient of potential vorticity in the fluid. If a fluid parcel is displaced, it conserves its potential vorticity and so its relative vorticity will in general change. The relative vorticity creates a velocity field that displaces neighbouring parcels, whose relative vorticity changes and so on.
- A common source of a potential vorticity gradient is differential rotation, or the  $\beta$ -effect, and in this case the associated Rossby waves are called *planetary waves*. In the presence of non-zero  $\beta$  the ambient potential vorticity increases northward and the phase of the Rossby waves propagates westward. Topography is another source of potential vorticity gradients. In general, Rossby waves propagate to the left of the direction of increasing potential vorticity.
- A common equation of motion for Rossby waves is

$$\frac{\partial \bar{q}'}{\partial t} + U \frac{\partial \bar{q}'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (\text{RW.1})$$

with an overbar denoting the basic state and a prime a perturbation. In the case of a single layer of fluid with no mean flow this equation becomes

$$\frac{\partial}{\partial t} (\nabla^2 + k_d^2) \psi' + \beta \frac{\partial \psi'}{\partial x} = 0, \quad (\text{RW.2})$$

with dispersion relation

$$\omega = \frac{-\beta k}{k^2 + l^2 + k_d^2}. \quad (\text{RW.3})$$

- In the absence of a mean flow (i.e.,  $U = 0$ ), the phase speed in the zonal direction ( $c_p^x = \omega/k$ ) is always negative, or westward, and is larger for large waves. For (RW.3) the components of the group velocity are given by

$$c_g^x = \frac{\beta(k^2 - l^2 - k_d^2)}{(k^2 + l^2 + k_d^2)^2}, \quad c_g^y = \frac{2\beta k l}{(k^2 + l^2 + k_d^2)^2}. \quad (\text{RW.4})$$

The group velocity is westward if the zonal wavenumber is sufficiently small, and eastward if the zonal wavenumber is sufficiently large.

case of barotropic flow on the  $\beta$ -plane the potential vorticity is the absolute vorticity,  $\beta y + \zeta$ . Thus, in either hemisphere, a northward displacement leads to the production of negative relative vorticity and a southward displacement leads to the production of positive relative vorticity. The relative vorticity gives rise to a velocity field which, in turn, advects the parcels in the material line in the manner shown, and the wave propagates westward.

In more complicated situations, such as flow in two layers, considered below, or in a continuously-stratified fluid, the mechanism is essentially the same. A displaced fluid parcel carries with it its potential vorticity and,

in the presence of a potential vorticity gradient in the basic state, a potential vorticity anomaly is produced. The potential vorticity anomaly produces a velocity field (an example of potential vorticity inversion) which further displaces the fluid parcels, leading to the formation of a Rossby wave. The vital ingredient is a basic state potential vorticity gradient, such as that provided by the change of the Coriolis parameter with latitude.

## 6.4 ROSSBY WAVES IN STRATIFIED QUASI-GEOSTROPHIC FLOW

We now consider the dynamics of linear waves in stratified quasi-geostrophic flow on a  $\beta$ -plane, with a resting basic state. The interior flow is governed by the potential vorticity equation, (5.63), and linearizing this about a constant mean zonal flow,  $U$ , gives

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \left[ \nabla^2 \psi' + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \beta \frac{\partial \psi'}{\partial x} = 0, \quad (6.33)$$

where, for simplicity, we suppose that  $f_0^2/N^2$  does not vary with  $z$ .

### 6.4.1 Dispersion Relation and Group Velocity

Deferring the issues of boundary conditions for a moment, let us seek solutions of the form

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx+ly+mz-\omega t)}, \quad (6.34)$$

where  $\tilde{\psi}$  is a constant. Substituting (6.34) into (6.33) gives, after a couple of lines of elementary algebra, the dispersion relation

$$\omega = Uk - \frac{\beta k}{k^2 + l^2 + m^2 f_0^2 / N^2}. \quad (6.35)$$

As in most wave problems the frequency is, by convention, a positive quantity.

In reality we usually have to satisfy a boundary condition at the top and bottom of the fluid and these are determined by the thermodynamic equation, (5.67), which after linearization becomes

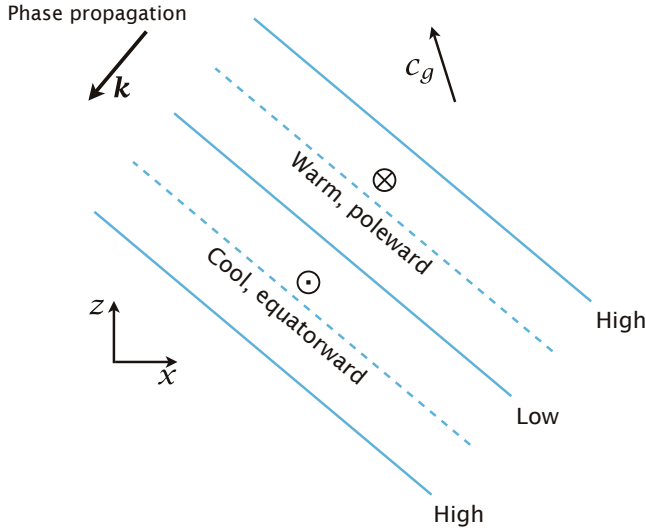
$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \left( \frac{\partial \psi'}{\partial z} \right) + N^2 w = 0. \quad (6.36)$$

If the boundaries are flat, rigid, surfaces then  $w = 0$  at those boundaries suggesting that instead of (6.34) we choose a streamfunction of the form

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx+ly-\omega t)} \cos\left(\frac{m_j \pi z}{H}\right), \quad m_j = 1, 2, \dots, \quad (6.37)$$

which satisfies  $\partial \psi' / \partial z = 0$  at  $z = 0, H$ . The dispersion relation is the same as (6.35) with  $m$  equal to  $m_j \pi / H$ . If the domain is finite in the horizontal direction also — as it nearly always is — then the horizontal wavenumbers are also quantized. For example, if the extent of the domain in the





**Fig. 6.4:** A schematic east-west section of an upwardly propagating Rossby wave. The slanting lines are lines of constant phase and 'high' and 'low' refer to the pressure or streamfunction values. Both  $k$  and  $m$  are negative so the phase lines are oriented up and to the west. The phase propagates westward and downward, but the group velocity is upward.

$x$ -direction is  $L_x$  and the flow is periodic in that direction, then  $k$  is restricted to the values  $k = 2\pi k_j / L_x$  where  $k_j$  is an integer, and similarly for  $l$  and  $l_j$ . (It is common albeit sometimes confusing to refer to both  $k$  and  $k_j$  as the wavenumber and also to drop the subscript  $j$  on  $k_j$  with the distinction then made clear by context, and similarly for  $l$  and  $l_j$ .)

Using (6.35), the three components of the group velocity for these waves are:

$$c_g^x = U + \frac{\beta[k^2 - (l^2 + m^2 f_0^2 / N^2)]}{(k^2 + l^2 + m^2 f_0^2 / N^2)^2}, \quad (6.38a)$$

$$c_g^y = \frac{2\beta k l}{(k^2 + l^2 + m^2 f_0^2 / N^2)^2}, \quad c_g^z = \frac{2\beta k m f_0^2 / N^2}{(k^2 + l^2 + m^2 f_0^2 / N^2)^2}. \quad (6.38b,c)$$

The propagation in the horizontal is similar to the propagation in a single-layer model. We see also that higher baroclinic modes (bigger  $m$ ) will have a more westward group velocity. The vertical group velocity is proportional to  $m$ , and for waves that propagate signals upward we choose  $m$  to have the same sign as  $k$  so that  $c_g^z$  is positive. If there is no mean flow then the zonal wavenumber  $k$  is negative (in order that frequency is positive) and  $m$  must then also be negative. Energy then propagates upward but the phase propagates downward! This case is illustrated in Fig. 6.4 and to understand it better let us look at the vertical propagation in more detail.

### 6.4.2 Vertical Propagation of Rossby Waves

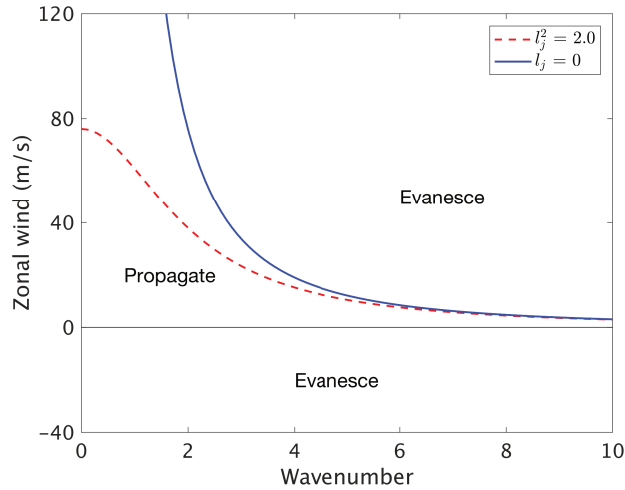
#### Conditions for wave propagation

With a little re-arrangement the dispersion relation, (6.35), may be written as

$$m^2 = \frac{N^2}{f_0^2} \left( \frac{\beta}{U - c} - (k^2 + l^2) \right), \quad (6.39)$$

*Vertical propagation of Rossby waves was not discussed by Rossby, who was mainly interested in their properties in a horizontal plane. Rather, it was Charney & Drazin (1961) who first considered the vertical propagation of the waves in a stratified fluid in any detail.*

**Fig. 6.5:** The boundary between propagating waves and evanescent waves as a function of zonal wind and wavenumber, calculated using (6.41). The abscissa is the  $x$ -wavenumber,  $k_j$ , and the results show the boundary with two values of meridional wavenumber,  $l_j$ . Small wavenumbers (large scales) can more easily propagate into the stratosphere, and propagation is inhibited if the zonal wind is too strong or is negative.



where  $c = \omega/k$ . For waves to propagate upwards we require that  $m^2 > 0$ : if  $m^2 < 0$  the wavenumber is imaginary and the wave amplitude either grows with height (which is unphysical) or is damped. This condition implies

$$0 < U - c < \frac{\beta}{k^2 + l^2}. \quad (6.40)$$

For waves of some given frequency ( $\omega = kc$ ) the above expression provides a condition on  $U$  for the vertical propagation of planetary waves. For stationary waves  $c = 0$  and the criterion is

$$0 < U < \frac{\beta}{k^2 + l^2}, \quad (6.41)$$

as illustrated in Fig. 6.5. That is to say, the vertical propagation of stationary Rossby waves occurs only in eastward winds, and winds that are weaker than some critical value,  $U_c = \beta/(k^2 + l^2)$  that depends on the scale of the wave. The lower limit, where  $U = c$  (and so  $U = 0$  if stationary) is known as a critical level and the upper limit is a ‘turning level’. Since (6.39) is just a form of the dispersion relation, for any given frequency there is a criterion for propagation given by (6.40), and for any given  $U$  there may be a frequency that allows propagation. However, the zero frequency case is often regarded as the most important.

The upper critical velocity,  $U_c = \beta/(k^2 + l^2)$ , is a function of wavenumber and increases with horizontal wavelength. Thus, for a given eastward flow long waves may penetrate vertically when short waves are trapped, an effect sometimes referred to as ‘Charney–Drazin filtering’. An important consequence is that the stratospheric motion is typically of larger scales than that of the troposphere, because Rossby waves tend to be excited first in the troposphere (by baroclinic instability and by flow over topography, among other things), but the shorter waves are trapped and

Stationary, vertically oscillatory modes can exist only for zonal flows that are eastward and that are less than some critical value  $U_c = \beta/(k^2 + l^2)$ .

only the longer ones reach the stratosphere. In the summer, the stratospheric winds are often westwards (because polar regions in the stratosphere are warmer than equatorial regions) and *all* waves are trapped in the troposphere; the eastward stratospheric winds that favour vertical penetration occur in the other three seasons, although very strong eastward winds can suppress penetration in mid-winter.

For westward flow, or for sufficiently strong eastward flow,  $m$  is imaginary and the waves decay exponentially in the vertical as  $\exp(-\alpha z)$  where

$$\alpha = \frac{N}{f_0} \left( k^2 + l^2 - \frac{\beta}{U} \right)^{1/2}, \quad (6.42)$$

and the decay scale is smaller for shorter waves.

### An interpretation

One physical way to interpret the propagation criterion is to write (6.41) as

$$0 < |Uk| < \frac{|\beta k|}{k^2 + l^2}, \quad (6.43)$$

allowing for the fact that  $k$  may be negative. Now, in a resting medium ( $U = 0$ ) the Rossby wave frequency has a maximum value when  $m = 0$  given by

$$\omega = \frac{|\beta k|}{k^2 + l^2}, \quad (6.44)$$

and the minimum frequency is zero. Let us suppose that the stationary Rossby waves are excited by flow of speed  $U$  moving over bottom topography. If we move into the frame of reference of the flow then the waves are forced by a moving topography, and the frequency of the forcing is just  $|Uk|$ . Thus, (6.43) is equivalent to saying that for oscillatory waves to exist *the forcing frequency,  $|Uk|$ , must lie within the frequency range of vertically propagating Rossby waves.*

### 6.4.3 Heat Transport and Vertical Propagation

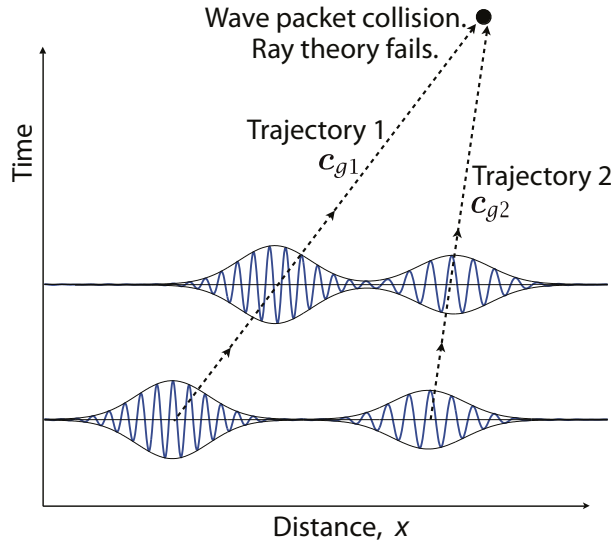
If the group velocity in the  $z$ -direction, given by (6.38) is to be positive, then we require the product  $km > 0$ . This has an important consequence for the heat transport. Remember that the buoyancy  $b$ , which is a proxy for temperature, is given by  $f_0 \partial \psi / \partial z$ , and the northward velocity is  $v = \partial \psi / \partial x$ . Thus, the northward flux of heat,  $H$  say, is given by

$$H = \overline{vb} = f_0 \overline{\frac{\partial \psi}{\partial z} \frac{\partial \psi}{\partial x}}, \quad (6.45)$$

where an overbar denotes a zonal average. To evaluate this expression it is simplest to suppose that  $\psi = \psi_0 \cos(kx + ly + mz)$ , where  $\psi_0$  is a real constant, in which case we obtain

$$H = f_0 k m \psi_0^2 \overline{\sin^2(kx + ly + mz)} = \frac{1}{2} f_0 \psi_0^2 k m, \quad (6.46)$$

**Fig. 6.6:** Idealised trajectory of two wavepackets, each with a different wavelength and moving with a different group velocity, as might be calculated using ray theory. If the wave packets collide ray theory must fail. Ray theory gives only the trajectory of the wave packet, not the detailed structure of the waves within a packet.



using the standard result that the average of  $\sin^2 x$  over a wavelength is equal to  $1/2$ . Thus, the heat flux, like the vertical component of the group velocity, is equal to  $km$  multiplied by a positive quantity, and we may conclude that an upward propagation of Rossby waves is associated with a polewards heat flux. A moment's thought will reveal that the same conclusion holds in the Southern Hemisphere, even though  $f_0$  is negative there.

## 6.5 RAY THEORY AND ROSSBY RAYS

In the real world most waves propagate in a medium that is inhomogeneous; for example, in the Earth's atmosphere and ocean the Coriolis parameter varies with latitude and the stratification with height. In these cases it can be hard to obtain the solution of a wave problem by Fourier methods, even approximately. Nonetheless, the idea of signals propagating at the group velocity is a robust one, and we can often obtain some of the information we want — and in particular the trajectory of a wave — using a recipe known as *ray theory*. If the background properties of the medium vary only slowly compared to the wavelength, we assume that the dispersion relation is satisfied *locally*. We then calculate the group velocity, and assume that the trajectory of the wave is along that group velocity.

To implement this recipe, we first obtain a dispersion relation from the governing equation. In the homogeneous case we obtained a dispersion relation  $\omega = f(\mathbf{k})$ ; that is the frequency is some function only of the wavenumber. In the inhomogeneous case we proceed the same way, and obtain a dispersion relation *as if* the parameters were fixed, but we then allow the parameters to vary in the dispersion relation; that is, we obtain a dispersion relation of the form  $\omega = f(\mathbf{k}; \mathbf{x}, t)$ . For example, for barotropic

Rossby waves, the dispersion relation might now be

$$\omega = U(y) - \frac{\beta(y)}{k^2 + l^2}. \quad (6.47)$$

This is the same as the usual dispersion relation except that  $U$  and  $\beta$  are now taken to be slowly varying functions of position. This procedure is valid provided that  $U$  and  $\beta$  vary more slowly than the wavelengths of the Rossby waves, which is essentially the same condition used in the WKB approximation, discussed in the appendix below.

The utility of ray theory is that it allows us to calculate the trajectory of a wave packet without fully solving the problem. Thus, suppose that a disturbance somewhere generates Rossby waves — perhaps a sea-surface temperature anomaly in the tropics. The waves propagate away from the disturbance following the group velocity, which we calculate using the dispersion relation. Thus, if the dispersion relation is given by (6.47) then the group velocity is given by

$$c_g^x = U + \frac{\beta(k^2 - l^2)}{(k^2 + l^2)^2}, \quad c_g^y = \frac{2kl\beta}{(k^2 + l^2)^2}, \quad (6.48a,b)$$

where  $\beta$  and  $U$  are functions of space. Thus, the position of the wave packet,  $\mathbf{x}$ , is given by solving (often numerically) the ray equations

$$\frac{dx}{dt} = c_g^x, \quad \frac{dy}{dt} = c_g^y, \quad (6.49)$$

where  $c_g^x$  and  $c_g^y$  are evaluated using (6.48). Figure 6.6 shows schematically how a wave packet might then propagate. In the Rossby-wave case here, the frequency is constant (determined by the source of the waves), as is the  $x$ -wavenumber,  $k$ , because  $U$  and  $\beta$  are only functions of  $y$ . As we move along the ray, we recalculate the  $y$ -wavenumber,  $l$ , at each point using the dispersion relation and then calculate the group velocity at each point, and thence the trajectory. When this is done on Earth, one sometimes can see Rossby waves propagating from the tropics to midlatitude that bring about long-range correlations in weather patterns known as ‘teleconnections’.

#### ♦ APPENDIX A: THE WKB APPROXIMATION

The WKB method is a way of finding approximate solutions to certain linear differential equations in which the term with the highest derivative is multiplied by a small parameter. In particular, WKB theory can be used to find approximate solutions to wave equations in which the coefficients vary slowly in space or time. Consider an equation of the form

$$\frac{d^2\xi}{dz^2} + m^2(z)\xi = 0. \quad (6.50)$$

Such an equation commonly arises in wave problems. If  $m^2$  is positive the equation has wavelike solutions, and if  $m$  is constant the solution has the

*In 1926, G. Wentzel, H. A. Kramers and L. Brillouin each presented a method to find approximate solutions of the Schrödinger equation in quantum mechanics, and the technique became known as the WKB method. It turns out that the technique had already been discovered by Harold Jeffreys, a mathematical geophysicist, a few years prior to that, and in fact the origins of the method go back to the early nineteenth century. Evidently, methods are named after the last people to discover them.*

harmonic form

$$\xi = \text{Re } A_0 e^{imz}, \quad (6.51)$$

where  $A_0$  is a complex constant. If  $m$  varies only slowly with  $z$  (meaning that the variations in  $m$  only occur on a scale much longer than  $1/m$ ) one might reasonably expect that the harmonic solution above would provide a decent first approximation; that is, we expect the solution to locally look like a plane wave with local wavenumber  $m(z)$ . However, we might also expect that the solution would not be *exactly* of the form  $\exp(im(z)z)$ , because the phase of  $\xi$  is  $\theta(z) = mz$ , so that  $d\theta/dz = m + z dm/dz \neq m$ . Thus, in (6.51)  $m$  is not the wavenumber unless  $m$  is constant.

The condition that variations in  $m$ , or in the wavelength  $\lambda \sim m^{-1}$ , occur only slowly may be variously expressed as

$$\lambda \left| \frac{\partial \lambda}{\partial z} \right| \ll \lambda \quad \text{or} \quad \left| \frac{\partial m^{-1}}{\partial z} \right| \ll 1 \quad \text{or} \quad \left| \frac{\partial m}{\partial z} \right| \ll m^2. \quad (6.52a,b,c)$$

This condition will generally be satisfied if variations in the background state, or in the medium, occur on a scale much longer than the wavelength.

### The Solution

Let us seek solutions of (6.50) in the form

$$\xi = A(z) e^{i\theta(z)}, \quad (6.53)$$

where  $A(z)$  and  $\theta(z)$  are both real. Using (6.53) in (6.50) yields

$$i \left[ 2 \frac{dA}{dz} \frac{d\theta}{dz} + A \frac{d^2\theta}{dz^2} \right] + \left[ A \left( \frac{d\theta}{dz} \right)^2 - \frac{d^2A}{dz^2} - m^2 A \right] = 0. \quad (6.54)$$

The terms in square brackets must each be zero. The WKB approximation is to assume that the amplitude varies sufficiently slowly that  $|A^{-1} d^2A/dz^2| \ll m^2$ , and hence that the term involving  $d^2A/dz^2$  may be neglected. The real and imaginary parts of (6.54) become

$$\left( \frac{d\theta}{dz} \right)^2 = m^2, \quad 2 \frac{dA}{dz} \frac{d\theta}{dz} + A \frac{d^2\theta}{dz^2} = 0. \quad (6.55a,b)$$

The solution of the first equation above is

$$\theta = \pm \int m dz, \quad (6.56)$$

and substituting this into (6.55b) gives

$$2 \frac{dA}{dz} m + A \frac{dm}{dz} = 0, \quad \text{with solution} \quad A = A_0 m^{-1/2}, \quad (6.57a,b)$$

where  $A_0$  is a constant. Using (6.56) and (6.57b) in (6.53) gives us

$$\xi(z) = A_0 m^{-1/2} \exp \left( \pm i \int m dz \right), \quad (6.58)$$

and this is the WKB solution to (6.50). In terms of real quantities the solution may be written

$$\xi(z) = B_0 m^{-1/2} \cos\left(\int m dz\right) + C_0 m^{-1/2} \sin\left(\int m dz\right), \quad (6.59)$$

where  $B_0$  and  $C_0$  are real constants.

Using (6.55a) and the real part of (6.54) we see that the condition for the validity of the approximation is that

$$\left| A^{-1} \frac{d^2 A}{dz^2} \right| \ll m^2, \quad (6.60a,b)$$

which using (6.57b) is

$$\left| \frac{1}{m^{-1/2}} \frac{d^2 m^{-1/2}}{dz^2} \right| \ll m^2. \quad (6.61)$$

Equation (6.52) expresses a similar condition to (6.61).

## Notes and References

Many books discuss waves in fluids with the ones by LeBlond & Mysak (1980), Pedlosky (2003) and Gill (1982) having a geophysical emphasis. Accessible introductions to WKB theory can be found in Simmonds & Mann (1998) and Holmes (2013). An example of the application of WKB theory to the atmosphere is given in Hoskins & Karoly (1981).

## Problems

- 6.1 *Rossby waves and Galilean invariance in shallow water.* Consider the flat-bottomed shallow-water quasi-geostrophic equations in standard notation,

$$\frac{D}{Dt} \left( \zeta + \beta y - \frac{f_0 \eta}{H} \right) = 0. \quad (P6.1)$$

- How is  $\zeta$  related to  $\eta$ ? Express  $u, v, \eta$  and  $\zeta$  in terms of a streamfunction.
  - Linearize (P6.1) about a state of rest, and show that the resulting system supports two-dimensional Rossby waves. Discuss the limits in which the wavelength is much shorter or much longer than the deformation radius.
  - Now linearize (P6.1) about a *geostrophically balanced state* that is translating uniformly eastwards. This means that:  $u = U + u'$  and  $\eta = \bar{\eta}(y) + \eta'$ , where  $\eta(y)$  is in geostrophic balance with  $U$ . Obtain an expression for the form of  $\bar{\eta}(y)$ . Obtain the dispersion relation for Rossby waves in this system. Show that their speed is different from that obtained by adding a constant  $U$  to the speed of Rossby waves in part (b). The problem is therefore not *Galilean invariant* — why?
- 6.2 *Horizontal propagation of Rossby waves.* Consider barotropic Rossby waves obeying the dispersion relation  $\omega = Uk - \beta k / (k^2 + l^2)$ , where  $U$  and/or  $\beta$  vary slowly with latitude.

- (i) By re-arranging this expression to obtain an expression for the meridional wavenumber, or otherwise, show that the Rossby waves can only propagate in the horizontal if

$$0 < U - c < \frac{\beta}{k^2}, \quad (\text{P6.2})$$

where  $c = \omega/k$ .

- (ii) If the waves approach a latitude where  $U = c$  (a ‘critical latitude’) show that the meridional wavenumber  $l$  becomes large but the group velocity in the  $y$ -direction becomes small. Show that Rossby waves generated in the atmosphere in midlatitudes are unlikely to propagate into the tropics (where the mean flow is westward).
- (iii) Suppose that the Rossby waves approach a latitude where  $U - c = \beta/k^2$ . Calculate the  $x$ - and  $y$ -components of the group velocity in this limit and infer that a wave will turn away from such a latitude.
- 6.3 Consider barotropic Rossby waves with an infinite deformation radius. Obtain an expression for the group velocity in the presence of a uniform eastward mean flow. Show that for stationary waves the group velocity is eastward relative to Earth’s surface, and hence deduce that energy propagation is downstream of topographic sources. Is this still true if the deformation radius is finite? Is it true if the mean flow is westward?
- 6.4 *Rossby waves in the two-level model.*
- (a) Consider a two-layer (or two-level) quasi-geostrophic system on a  $\beta$ -plane, and you may consider the layers to be of equal depth. Linearize the equations about a state of rest, and show that they may be transformed into two uncoupled equations. Give a physical interpretation of what these equations represent.
- (b) Obtain the dispersion relations for this system (there are two). Show that one of the dispersion relations corresponds to the synchronous, depth-independent motion in the two layers. What does the other one correspond to? Obtain the phases and group velocities of these waves and determine the conditions under which they are eastward or westward.
- 6.5 In realistic calculations of the vertical propagation of Rossby waves one must take into account the vertical variation of density. Carry through the calculation leading to the Charney–Drazin condition, either using pressure (or log-pressure) coordinates or using the anelastic version of quasi-geostrophy, using (5.66) with  $\bar{\rho} = \rho_0 \exp(-z/H)$ . Show that the condition for propagation analogous to (6.41) becomes

$$0 < U < \frac{\beta}{k^2 + l^2 + \gamma^2}, \quad (\text{P6.3})$$

and obtain an expression for  $\gamma$ .



## Gravity Waves

**W**AVES ARISE WHEN A SYSTEM IS PERTURBED and a restoring force tries to bring the system back to equilibrium; the system then overshoots and oscillations ensue. Gravity waves are waves in a fluid in which gravity provides the restoring force. For gravity to have an effect the fluid density must vary, and thus the waves must either exist at a fluid interface or in a stratified fluid — and a fluid interface is just an abrupt form of stratification. It is thus common to think of gravity waves as being either internal waves or surface waves: the former being in the interior of a fluid where the density changes may be continuous and the latter at a fluid interface, and naturally enough the two waves have many similarities. We considered surface waves in the hydrostatic, shallow water case in Chapter 4; now we consider internal waves in the continuously-stratified equations.

*Gravity waves are those waves that exist in a fluid for which gravity provides the restoring force. Gravitational waves are a disturbance in the fabric of spacetime caused by accelerating massive bodies, as predicted by the general theory of relativity.*

### 7.1 INTERNAL WAVES IN A CONTINUOUSLY-STRATIFIED FLUID

*Internal gravity waves* are waves that are internal to a stratified fluid and that owe their existence to the restoring force of gravity. In this section we will consider the simplest and most fundamental case, that of internal waves in a Boussinesq fluid with constant stratification and no background rotation. To this end, consider a fluid, initially at rest, in which the background buoyancy varies only with height and so the buoyancy frequency,  $N$ , is a function only of  $z$ . The system satisfies the Boussinesq equations (Section 2.5) and linearizing those equations of motion about this basic state gives the linear momentum equations,

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi', \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (7.1a,b)$$

and the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.1c,d)$$

Our notation is such that  $\mathbf{u} \equiv u\hat{\mathbf{i}} + v\hat{\mathbf{j}}$ ,  $\mathbf{v} \equiv u\hat{\mathbf{i}} + v\hat{\mathbf{j}} + w\hat{\mathbf{k}}$ , where  $(\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$  are the unit vectors in the  $x$ ,  $y$  and  $z$  directions, and the gradient operator is horizontal unless noted. Thus,  $\nabla \equiv \hat{\mathbf{i}}\partial_x + \hat{\mathbf{j}}\partial_y$  and  $\nabla_3 \equiv \hat{\mathbf{i}}\partial_x + \hat{\mathbf{j}}\partial_y + \hat{\mathbf{k}}\partial_z$ .

A little algebra gives a single equation for  $w'$ ,

$$\left[ \frac{\partial^2}{\partial t^2} \left( \nabla^2 + \frac{\partial^2}{\partial z^2} \right) + N^2 \nabla^2 \right] w' = 0. \quad (7.2)$$

This equation is evidently *not* isotropic. If  $N^2$  is a constant — that is, if the background buoyancy varies linearly with  $z$  — then the coefficients of each term are constant, and we may then seek solutions of the form

$$w' = \text{Re } \tilde{w} e^{i(kx+ly+mz-\omega t)}, \quad (7.3)$$

where  $\text{Re}$  denotes the real part, a denotation that will frequently be dropped unless ambiguity arises, and other variables oscillate in a similar fashion. Using (7.3) in (7.2) yields the dispersion relation:

$$\omega^2 = \frac{(k^2 + l^2)N^2}{k^2 + l^2 + m^2} = \frac{K^2 N^2}{K_3^2}, \quad (7.4)$$

where  $K^2 = k^2 + l^2$  and  $K_3^2 = k^2 + l^2 + m^2$ . The frequency (see Fig. 7.1) is thus always less than  $N$ , approaching  $N$  for small horizontal scales,  $K^2 \gg m^2$ . If we neglect pressure perturbations, as in the parcel argument of Section 3.4, then the two equations,

$$\frac{\partial w'}{\partial t} = b', \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (7.5)$$

form a closed set, and give  $\omega^2 = N^2$ .

If the basic state density increases with height then  $N^2 < 0$  and the basic state is unstable. The disturbance grows exponentially according to  $\exp(\sigma t)$  where

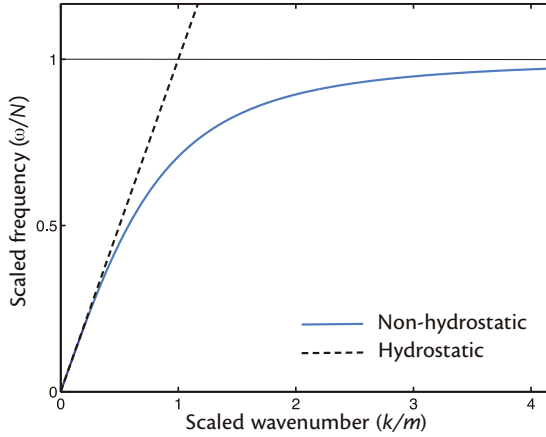
$$\sigma = i\omega = \pm \frac{K\tilde{N}}{K_3}, \quad (7.6)$$

where  $\tilde{N}^2 \equiv -N^2$  and  $K_3 = \sqrt{K^2}$ . Most convective activity in the ocean and atmosphere is, ultimately, related to an instability of this form, although of course there are many complicating issues — water vapour in the atmosphere, salt in the ocean, the effects of rotation and so forth.

### 7.1.1 Hydrostatic Internal Waves

Let us now suppose that the fluid satisfies the hydrostatic Boussinesq equations. The linearized two-dimensional equations of motion become

$$\frac{\partial \mathbf{u}'}{\partial t} = -\nabla \phi', \quad 0 = -\frac{\partial \phi'}{\partial z} + b', \quad (7.7a)$$



**Fig. 7.1:** Scaled frequency,  $\omega/N$ , plotted as a function of scaled horizontal wavenumber,  $k/m$ , using the full dispersion relation of (7.4) with  $l = 0$  (solid line, asymptoting to unit value for large  $k/m$ ), and with the hydrostatic dispersion relation (7.8) (dashed line, tending to  $\infty$  for large  $k/m$ ).

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0, \quad (7.7b)$$

where these are the horizontal and vertical momentum equations, the mass continuity equation and the thermodynamic equation respectively. A little algebra gives the dispersion relation,

$$\omega^2 = \frac{(k^2 + l^2)N^2}{m^2}. \quad (7.8)$$

The frequency and, if  $N^2$  is negative, the growth rate, are unbounded as  $K^2/m^2 \rightarrow \infty$ , and the hydrostatic approximation thus has quite unphysical behaviour for small horizontal scales. Many numerical models of the large-scale circulation in the atmosphere and ocean do make the hydrostatic approximation. In these models convection must be *parameterized*; otherwise, it would simply occur at the smallest scale available, namely the size of the numerical grid, and this type of unphysical behaviour should be avoided. In nonhydrostatic models convection must also be parameterized if the horizontal resolution of the model is too coarse to properly resolve the convective scales.

## 7.2 PROPERTIES OF INTERNAL WAVES

Internal waves have a number of interesting and counter-intuitive properties — let's discuss them.

### 7.2.1 The Dispersion Relation

We can write the dispersion relation, (7.4), as

$$\omega = \pm N \cos \vartheta, \quad (7.9)$$

where  $\cos^2 \vartheta = K^2/(K^2 + m^2)$  so that  $\vartheta$  is the angle between the three-dimensional wave-vector,  $\mathbf{k} = k\hat{\mathbf{i}} + l\hat{\mathbf{j}} + m\hat{\mathbf{k}}$ , and the horizontal. The frequency is evidently a function only of  $N$  and  $\vartheta$ , and, if this is given, the frequency is not a function of wavelength. This has some interesting consequences for wave reflection, as we see below.

We can also write the dispersion relation, (7.4), as

$$\frac{\omega^2}{N^2 - \omega^2} = \frac{K^2}{m^2}. \quad (7.10)$$

Thus, and consistently with our first point, given the wave frequency the ratio of the vertical to the horizontal wavenumber is fixed.

### 7.2.2 Polarization Relations

The oscillations of pressure, velocity and buoyancy are, naturally, connected, and we can obtain the relations between them with some simple manipulations. If the pressure field is oscillating like  $\phi' = \tilde{\phi} \exp[i(\mathbf{k} \cdot \mathbf{x} - \omega t)] = \tilde{\phi} \exp[i(kx + ly + mz - \omega t)]$  then, using (7.1a), the horizontal velocity components satisfy

$$(\tilde{u}, \tilde{v}) = (k, l) \omega^{-1} \tilde{\phi}. \quad (7.11)$$

Evidently, since the frequency is real, the velocities are in phase with the pressure. We can obtain similar relations for the other variables and, since all the fields are real, it is convenient to express the relations in terms of sines and cosines. If we choose pressure to vary as a cosine then after some algebra we obtain

$$\phi = \Phi_0 \cos(kx + ly + mz - \omega t), \quad (7.12a)$$

$$(u, v) = (k, l) \frac{\Phi_0}{\omega} \cos(kx + ly + mz - \omega t), \quad (7.12b)$$

$$w = \frac{-K^2}{m\omega} \Phi_0 \cos(kx + ly + mz - \omega t), \quad (7.12c)$$

$$b = \frac{N^2 K^2}{m\omega^2} \Phi_0 \sin(kx + ly + mz - \omega t), \quad (7.12d)$$

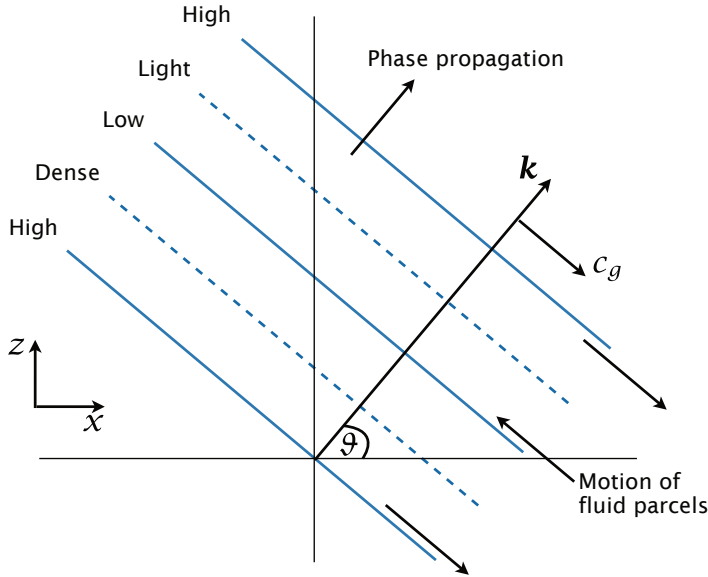
where  $\Phi_0$  is a constant. The vertical velocity is thus in phase with the pressure perturbation, and for regions of positive  $m$  (and so with upward phase propagation) regions of high relative pressure are associated with downward fluid motion. The above relations between pressure, buoyancy and velocity are known as *polarization relations*.

### 7.2.3 Relation between Wave Vector and Velocity

On multiplying (7.12b) and (7.12c) by  $(k, l)$  and  $m$ , respectively, we see that

$$\mathbf{k} \cdot \tilde{\mathbf{v}} = 0, \quad (7.13)$$

where  $\mathbf{k}$  and  $\tilde{\mathbf{v}}$  are three-dimensional vectors. This means that, at any instant, the wave vector is perpendicular to the velocity vector, and the velocity is therefore aligned *along* the direction of the troughs and crests,



**Fig. 7.2:** An internal wave propagating in the direction  $\mathbf{k}$ . Both  $k$  and  $m$  are positive for the wave shown. The solid lines show crests and troughs of constant pressure, and the dashed lines the corresponding crests and troughs of buoyancy (or density). The motion of the fluid parcels is along the lines of constant phase, as shown, and is parallel to the group velocity and perpendicular to the phase speed.

along which there is no pressure gradient. If the wave vector is purely horizontal (i.e.,  $m = 0$ ), then the motion is purely vertical and  $\omega = N$ .

The vertical and horizontal velocities are related to the wavenumbers. If (for simplicity, and with no loss of generality) the motion is in the  $x$ - $y$  plane with  $v = l = 0$ , then it is a corollary of (7.13) that

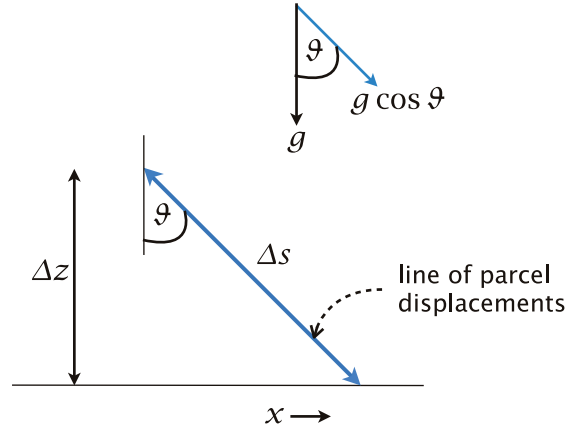
$$\frac{\tilde{u}}{\tilde{w}} = -\frac{m}{k}. \quad (7.14)$$

Furthermore, from (7.3) with  $l = 0$ , at any given instant all of the perturbation quantities in the wave are constant along the lines  $kx + mz = \text{constant}$ . Thus, *all fluid parcel motions are parallel to the wave fronts*. Now, since the wave frequency is related to the background buoyancy frequency by  $\omega = \pm N \cos \vartheta$ , it follows that the fluid parcels oscillate along lines that are at an angle  $\vartheta = \cos^{-1}(\omega/N)$  to the vertical. The polarization relations and the group and phase velocities are illustrated in Fig. 7.2. Let us now discuss the wave properties in a little more detail.

#### 7.2.4 A Parcel Argument and Physical Interpretation

Let us consider first the dispersion relation itself and try to derive it more physically, or at least heuristically. Let us suppose there is a wave propagating in the  $(x, z)$  plane at some angle  $\vartheta$  to the horizontal, with fluid parcels moving parallel to the troughs and crests, as in Fig. 7.2. In general the restoring force on a parcel is due to both the pressure gradient and gravity, but along the crests there is no pressure gradient. Referring to Fig. 7.3, for a total displacement  $\Delta s$  the restoring force,  $F_{\text{res}}$ , in the direc-

**Fig. 7.3:** Parcel displacements and associated forces in an internal gravity wave in which the parcel displacements are occurring at an angle  $\vartheta$  to the vertical, as in Fig. 7.2.



tion of the particle displacement is

$$F_{\text{res}} = g \cos \vartheta \Delta \rho = g \cos \vartheta \frac{\partial \rho}{\partial z} \Delta z = g \cos \vartheta \frac{\partial \rho}{\partial z} \Delta s \cos \vartheta = \rho_0 \frac{\partial b}{\partial z} \cos^2 \vartheta \Delta s, \quad (7.15)$$

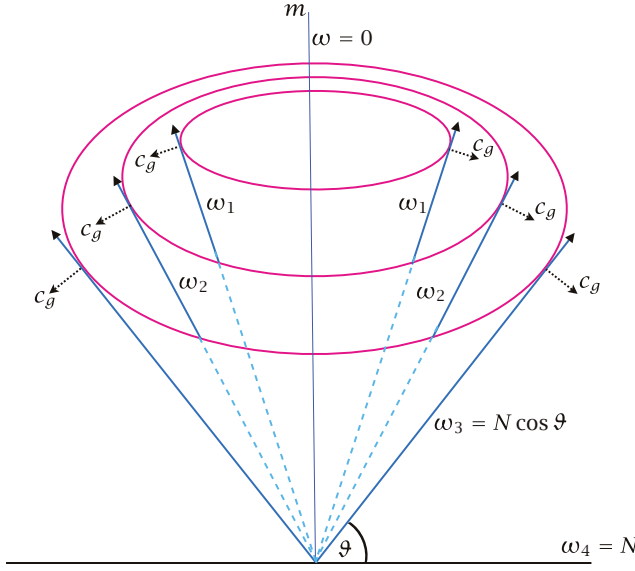
noting that  $\Delta z = \cos \vartheta \Delta s$ . The equation of motion of a parcel moving along a trough or crest is therefore

$$\rho_0 \frac{d^2 \Delta s}{dt^2} = -\rho_0 N^2 \cos^2 \vartheta \Delta s, \quad (7.16)$$

which implies a frequency  $\omega = N \cos \vartheta$ , as in (7.9). One of the  $\cos \vartheta$  factors in (7.16) comes from the fact that the parcel displacement is at an angle to the direction of gravity, and the other comes from the fact that the restoring force that a parcel experiences is proportional to  $N \cos \vartheta$ . (The reader may also wish to refer ahead to Fig. 7.6 and Section 7.3.1 for a similar argument.)

Now consider the wave illustrated in Fig. 7.2. For this wave both  $k$  and  $m$  are positive, and the frequency is assumed positive by convention to avoid duplicative solutions. The slanting solid and dashed lines are lines of constant phase, and from (7.12) the buoyancy and pressure are  $1/4$  of a wavelength out of phase. When  $k$  and  $m$  are both positive the extrema in the buoyancy field lag the extrema in the vertical velocity by  $\pi/2$ , as illustrated. The perturbation velocities are zero along the lines of extreme buoyancy. This follows because the velocities are in phase with the pressure, which as we noted is out of phase with the buoyancy.

Given the direction of the fluid parcel displacement in Fig. 7.2, the direction of the phase propagation  $c_p$  up and to the right may be deduced from the following argument. Buoyancy perturbations arise because of vertical advection of the background stratification,  $w' \partial b_0 / \partial z = w' N^2$ . A local maximum in rising motion, and therefore a tendency to increase the fluid density, is present along the 'Low' line  $1/4$  wavelength upward and to the right of the 'Dense' phase line. Thus, the density of fluid along the 'Low' phase line increases and the 'Dense' phase line moves upward and to



**Fig. 7.4:** Internal wave cones. The surfaces of constant frequency are cones, defined by the surface that has a constant angle to the horizontal. The wave vector, and so the phase velocity, point along the cone away from the origin, and the frequency of any wave with a wave vector in the cone is  $N \cos \vartheta$ . The group velocity is at right angles to the cone and pointed in the direction of increasing frequency, as indicated by the arrows on the dashed lines. In the vertical direction the phase speed and group velocity have opposite signs.

the right. If the fluid parcel motion were reversed the pattern of ‘High–Dense–Low–Light–High’ in Fig. 7.2 would remain the same. However, the downward fluid motion along the ‘Low’ line would cause the fluid to lose density, and so the phase lines would propagate downward and to the left. Evidently, the wave fronts, or the lines of constant phase, move at right angles to the fluid-parcel trajectories. In the figure we see that the group velocity is denoted as being at right angles to the phase speed, so let’s discuss this.

### 7.2.5 Group Velocity and Phase Speed

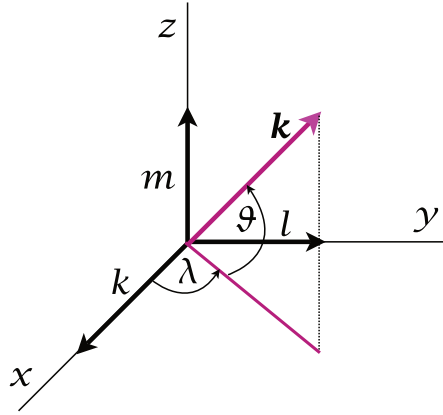
As we noted above, the frequency of internal waves is given by  $\omega = N \cos \vartheta$ , where  $\vartheta$  is the angle the wave vector makes with the horizontal. This means that the surfaces of constant frequency are *cones*, as illustrated in Fig. 7.4. To evaluate phase and group velocities in a useful way it is convenient to use spherical polar coordinates, as in Fig. 7.5, in which

$$k = K_3 \cos \vartheta \cos \lambda, \quad l = K_3 \cos \vartheta \sin \lambda, \quad m = K_3 \sin \vartheta, \quad (7.17)$$

so that  $\mathbf{k} = K_3(\cos \vartheta \cos \lambda, \cos \vartheta \sin \lambda, \sin \vartheta)$ . The angles are  $\vartheta$ , the angle of the wave vector with the horizontal and  $\lambda$ , which determines the orientation in the horizontal plane. (The notation is similar to the spherical coordinates of Chapter 2 — see Fig. 2.2 — although here  $\vartheta$  is the angle with the horizontal, not the angle with the equatorial plane. Note also that  $\lambda$  is also used in this chapter for wavelength.) We also note that

$$\sin^2 \vartheta = \frac{m^2}{k^2 + l^2 + m^2}, \quad \cos^2 \vartheta = \frac{K_3^2}{K_3^2} = \frac{k^2 + l^2}{k^2 + l^2 + m^2}, \quad \tan \lambda = \frac{l}{k}. \quad (7.18)$$

**Fig. 7.5:** The spherical coordinates used to describe internal waves, as in (7.17). The angle  $\vartheta$  is the angle of the wave vector with the horizontal, and  $\lambda$  determines the orientation in the horizontal plane. The wave vector  $\mathbf{k}$  is given by  $\mathbf{k} = (k, l, m)$ , these being the wavenumbers in the direction of increasing  $(x, y, z)$ , respectively.



In many problems we can align the direction of the wave propagation with the  $x$ -axis and take  $l = 0$  and  $\tan \lambda = 0$ .

The phase speed of the internal waves in the direction of the wave vector (sometimes referred to as the phase velocity) is given by

$$c_p = \frac{\omega}{K_3} = \frac{N}{K_3} \cos \vartheta = \frac{NK}{K_3^2}. \quad (7.19)$$

The phase speeds (as conventionally-defined) in the  $x, y$  and  $z$  directions are

$$c_p^x \equiv \frac{\omega}{k} = \frac{N}{k} \cos \vartheta, \quad c_p^y \equiv \frac{\omega}{l} = \frac{N}{l} \cos \vartheta, \quad c_p^z \equiv \frac{\omega}{m} = \frac{N}{m} \cos \vartheta. \quad (7.20a,b,c)$$

As noted in Section 6.1.2, these quantities are the speed of propagation of the wave crests in the respective directions. In general, each speed is *larger* than the phase speed in the direction perpendicular to the wave crests (that is, in the direction of the wave vector), but no information is transmitted at these speeds.

The group velocity is given by

$$\mathbf{c}_g = \left( \frac{\partial \omega}{\partial k}, \frac{\partial \omega}{\partial l}, \frac{\partial \omega}{\partial m} \right). \quad (7.21)$$

Using (7.4) we find

$$c_g^x = \frac{\partial \omega}{\partial k} = \frac{Nm}{K_3^2} \frac{km}{KK_3} = \left( \frac{N}{K_3} \sin \vartheta \right) \cos \lambda \sin \vartheta, \quad (7.22a)$$

$$c_g^y = \frac{\partial \omega}{\partial l} = \frac{Nm}{K_3^2} \frac{lm}{KK_3} = \left( \frac{N}{K_3} \sin \vartheta \right) \sin \lambda \sin \vartheta, \quad (7.22b)$$

$$c_g^z = \frac{\partial \omega}{\partial m} = -\frac{Nm}{K_3^2} \frac{K}{K_3} = -\left( \frac{N}{K_3} \sin \vartheta \right) \cos \vartheta. \quad (7.22c)$$

The magnitude of the group velocity is evidently

$$|c_g| = \frac{N}{K_3} \sin \vartheta, \quad (7.23)$$



and the group velocity vector is directed at an angle  $\vartheta$  to the vertical, as in Fig. 7.4. This angle is perpendicular to the cone itself; that is, the group velocity is perpendicular to the wave vector, as may be verified by taking the dot product of (7.17) and (7.22) which gives

$$\mathbf{k} \cdot \mathbf{c}_g = 0. \quad (7.24)$$

The group velocity is therefore parallel to the motion of the fluid parcels, as illustrated in Fig. 7.2. Furthermore, because energy propagates with the group velocity, and the latter is *parallel* to lines of constant phase, energy propagates perpendicular to the direction of phase propagation — very different from most other wave types. In the vertical direction we see from (7.20c) and (7.22c) that

$$\frac{\omega}{m} \frac{\partial \omega}{\partial m} = -\frac{N^2}{K^2} \cos^2 \vartheta < 0. \quad (7.25)$$

That is, the phase speed and the group velocity have opposite signs, meaning that if the wave crests move downward the group moves upward!

### 7.2.6 Effect of a Mean Flow

Suppose that there is a mean flow,  $U$ , in the  $x$ -direction, as is common in both atmosphere and ocean. The dispersion relation, (7.4), simply becomes

$$(\omega - Uk)^2 = \frac{K^2 N^2}{K^2 + m^2}. \quad (7.26)$$

The frequency is Doppler shifted, as expected, but the upward propagation of waves is affected in an interesting way. From (7.26) we find that the vertical component of the group velocity may be written as

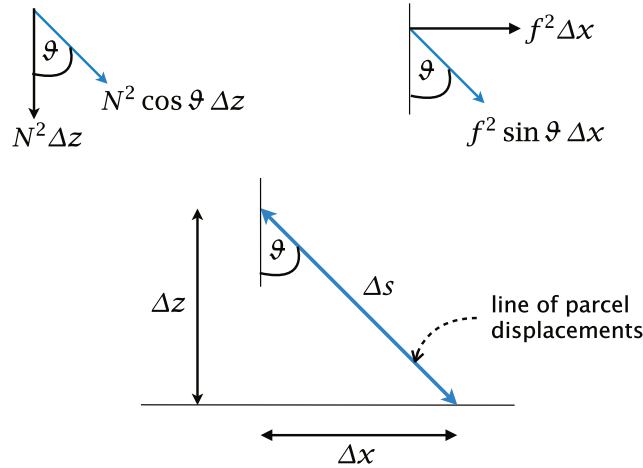
$$\frac{\partial \omega}{\partial m} = \frac{-m(\omega - Uk)}{K^2 + m^2} = \frac{-mk(c - U)}{K^2 + m^2}, \quad (7.27)$$

where  $c = \omega/k$  is the phase speed in the  $x$ -direction. If  $U$  is not constant but is varying slowly with  $z$  then (7.27) still holds, although  $m$  itself will also vary slowly with  $z$ . We see that the group velocity goes to zero at the location where  $U = c$ , known as a ‘critical line’, and the wave stalls.

The physical consequence of group velocity going to zero approaching a critical line is that any dissipation that may be present has more time to act. That is, we can expect a wave to be preferentially dissipated near a critical line, giving up its momentum to the mean flow and its energy to create mixing — the former being important in the atmosphere (for this is the mechanism producing the ‘quasi-biennial oscillation’) and the latter in the ocean. The mixing caused by internal wave breaking is in part responsible for the global scale meridional overturning circulation in the ocean, which we discuss more in Chapter 15.

*The quasi-biennial oscillation, or QBO, is a quasi-periodic reversal of the zonal wind in the equatorial stratosphere, with a period of about 26 months. It is caused by the deposition of momentum in the stratosphere by upward-propagating Rossby and gravity waves.*

**Fig. 7.6:** Parcel displacements and associated forces in an inertia-gravity wave in which the parcel displacements are occurring at an angle  $\vartheta$  to the vertical. Coriolis and buoyancy forces are present, and  $\Delta s = \Delta z / \cos \vartheta = \Delta x / \sin \vartheta$ .



### 7.3 ♦ INTERNAL WAVES IN A ROTATING FRAME OF REFERENCE

In the presence of both a Coriolis force and stratification a displaced fluid will feel two restoring forces — one due to gravity and the other to rotation. The first gives rise to gravity waves, as we have discussed, and the second to inertial waves. When the two forces both occur the resulting waves are called *inertia-gravity waves*. The algebra describing them can be complicated so we begin with a simple parcel argument to lay bare the basic dynamics; refer to Section 7.2.4 as needed.

#### 7.3.1 A Parcel Argument

Consider a parcel that is displaced along a slantwise path in the  $x$ - $z$  plane, as shown in Fig. 7.6, with a horizontal displacement of  $\Delta x$  and a vertical displacement of  $\Delta z$ . Let us suppose that the fluid is Boussinesq and that there is a stable and uniform stratification given by  $N^2 = -g\rho_0^{-1}\partial\rho_0/\partial z = \partial b/\partial z$ . Referring to (7.15) as needed, the component of the restoring buoyancy force,  $F_b$  say, in the direction of the parcel oscillation is given by (7.15),

$$F_b = -N^2 \cos \vartheta \Delta z = -N^2 \cos^2 \vartheta \Delta s. \quad (7.28)$$

The parcel also experiences a restoring Coriolis force,  $F_C$ , and the component of this in the direction of the parcel displacement is

$$F_C = -f^2 \sin \vartheta \Delta x = -f^2 \sin^2 \vartheta \Delta s. \quad (7.29)$$

Here, and for the rest of the chapter, we denote the Coriolis parameter by  $f$ . It should be regarded as a constant in any given problem (so there are no Rossby waves), but its value varies with latitude. Using (7.28) and (7.29) the (Lagrangian) equation of motion for a displaced parcel is

$$\frac{d^2 \Delta s}{dt^2} = -(N^2 \cos^2 \vartheta + f^2 \sin^2 \vartheta) \Delta s, \quad (7.30)$$

and hence the frequency is given by

$$\omega^2 = N^2 \cos^2 \vartheta + f^2 \sin^2 \vartheta. \quad (7.31)$$

Now, nearly everywhere in both atmosphere and ocean,  $N^2 > f^2$ . From (7.31) we then see that the frequency lies in the interval  $N^2 > \omega^2 > f^2$ . (To see this, put  $N = f$  or  $f = N$  in (7.31), and use  $\sin^2 \vartheta + \cos^2 \vartheta = 1$ . See also (7.43) below.) If the parcel displacements approach the vertical then the Coriolis force diminishes and  $\omega \rightarrow N$ , and similarly  $\omega \rightarrow f$  as the displacements become horizontal. The ensuing waves are then pure inertial waves.

We can write (7.31) in terms of wavenumbers since, for motion in the  $x$ - $z$  plane,

$$\cos^2 \vartheta = \frac{k^2}{k^2 + m^2}, \quad \sin^2 \vartheta = \frac{m^2}{k^2 + m^2}, \quad (7.32)$$

where  $k$  and  $m$  are the horizontal and vertical wavenumbers and  $l = 0$ . The dispersion relation becomes

$$\omega^2 = \frac{N^2 k^2 + f^2 m^2}{k^2 + m^2}. \quad (7.33)$$

Let's now move on to a discussion using the linearized equations of motion.

### 7.3.2 Equations of Motion

In a rotating frame of reference, specifically on an  $f$ -plane, the linearized equations of motion are the momentum equations

$$\frac{\partial \mathbf{u}'}{\partial t} + \mathbf{f}_0 \times \mathbf{u}' = -\nabla \phi', \quad \frac{\partial w'}{\partial t} = -\frac{\partial \phi'}{\partial z} + b', \quad (7.34a,b)$$

and the mass continuity and thermodynamic equations,

$$\frac{\partial u'}{\partial x} + \frac{\partial v'}{\partial y} + \frac{\partial w'}{\partial z} = 0, \quad \frac{\partial b'}{\partial t} + w' N^2 = 0. \quad (7.34c,d)$$

These are similar to (7.1), with the addition of a Coriolis term in the horizontal momentum equations.

To obtain a single equation for  $w'$  we take the horizontal divergence of (7.34a) and use the continuity equation to give

$$\frac{\partial}{\partial t} \left( \frac{\partial w'}{\partial z} \right) + f \zeta' = \nabla^2 \phi', \quad (7.35)$$

where  $\zeta' \equiv (\partial v' / \partial x - \partial u' / \partial y)$  is the vertical component of the vorticity. We may obtain an evolution equation for that vorticity by taking the curl of (7.34a), giving

$$\frac{\partial \zeta'}{\partial t} = f \frac{\partial w'}{\partial z}. \quad (7.36)$$

Eliminating vorticity between these equations gives

$$\left( \frac{\partial^2}{\partial t^2} + f^2 \right) \frac{\partial w'}{\partial z} = \frac{\partial}{\partial t} \nabla^2 \phi'. \quad (7.37)$$

We may obtain another equation linking pressure and vertical velocity by eliminating the buoyancy between (7.34b) and (7.34d), so giving

$$\frac{\partial^2 w'}{\partial t^2} + N^2 w' = - \frac{\partial}{\partial t} \frac{\partial \phi'}{\partial z}. \quad (7.38)$$

Eliminating  $\phi'$  between (7.37) and (7.38) gives a single equation for  $w'$  analogous to (7.2), namely

$$\left[ \frac{\partial^2}{\partial t^2} \left( \nabla^2 + \frac{\partial^2}{\partial z^2} \right) + f^2 \frac{\partial^2}{\partial z^2} + N^2 \nabla^2 \right] w' = 0. \quad (7.39)$$

If we assume a time dependence of the form  $w' = \tilde{w} e^{-i\omega t}$ , this equation may be written in the sometimes useful form,

$$\frac{\partial^2 \tilde{w}}{\partial z^2} = \left( \frac{N^2 - \omega^2}{\omega^2 - f^2} \right) \nabla^2 \tilde{w}. \quad (7.40)$$

### 7.3.3 Dispersion Relation

If we assume wave solutions to (7.39) of the standard harmonic form, that is  $w' = \tilde{w} \exp[i(kx + ly + mz - \omega t)]$ , we obtain the dispersion relation

$$\omega^2 = \frac{f^2 m^2 + (k^2 + l^2) N^2}{k^2 + l^2 + m^2}, \quad (7.41)$$

which is a minor generalization of (7.33). We can also write the dispersion relation as

$$\omega^2 = f^2 \sin^2 \vartheta + N^2 \cos^2 \vartheta, \quad (7.42)$$

as we have derived already, and which we may write as

$$\omega^2 = f^2 - (f^2 - N^2) \cos^2 \vartheta, \quad \text{or} \quad \omega^2 = N^2 - (N^2 - f^2) \sin^2 \vartheta, \quad (7.43)$$

where  $\vartheta$  is the angle of the wavevector with the horizontal, and as noted earlier the frequency therefore lies between  $N$  and  $f$ . The waves satisfying (7.41) are called inertia-gravity waves and are analogous to surface gravity waves in a rotating frame — that is, Poincaré waves — discussed in Section 4.3.2.

In many atmospheric and oceanic situations  $f \ll N$  (in fact typically  $N/f \sim 100$ , the main exception being weakly stratified near-surface mixed layers and the deep abyss in the ocean) and  $f < \omega < N$ . From (7.42) the frequency is dependent only on the angle the wavevector makes with the horizontal, and the surfaces of constant frequency again form cones in wavenumber space, although depending on the values of  $f$  and  $\omega$  the frequency does not necessarily decrease monotonically with  $\vartheta$  as in the non-rotating case.

From the dispersion relation, (7.43), we can see that the frequency lies between  $N$  and  $f$ , no matter which is larger.

### A few limits

Internal gravity waves with rotation have a few interesting special cases:

- (i) A purely horizontal wave vector. In this case  $m = 0$  and  $\omega = N$ . The waves are then unaffected by the Earth's rotation. This is because the Coriolis force is (in the  $f$ -plane approximation) due to the product of the Coriolis parameter and the horizontal component of the velocity. If the wave vector is horizontal, the fluid velocities are purely vertical and so the Coriolis force vanishes.
- (ii) A purely vertical wave vector. In this case  $\omega = f$ , the fluid velocities are horizontal and the fluid parcels do not feel the stratification. The oscillations are then known as *inertial waves*, although they are not inertial in the sense of there being no implied force in an inertial frame of reference. This case and the previous one show that when  $\omega = N$  or  $\omega = f$  the group velocity is zero.
- (iii) In the limit  $N \rightarrow 0$  we have pure inertial waves with a frequency  $0 < \omega < f$ , and specifically  $\omega = f \sin \vartheta$ . Similarly, as  $f \rightarrow 0$  we have pure internal waves, as discussed previously, with  $\omega = N \cos \vartheta$ .
- (iv) The hydrostatic limit. Hydrostasy occurs in the limit of large horizontal scales,  $k, l \ll m$ . If we therefore neglect  $k^2$  and  $l^2$  where they appear with  $m^2$  in (7.41) we obtain

$$\omega^2 = f^2 + N^2 \frac{K^2}{m^2} = f^2 + N^2 \cos^2 \vartheta. \quad (7.44)$$

In this case the frequency of the waves is small compared to the buoyancy frequency, or  $\omega^2 N^2$ , because  $K^2/m^2 \ll 1$ . Put another way, low frequencies tend to have a small aspect ratio and be hydrostatic.

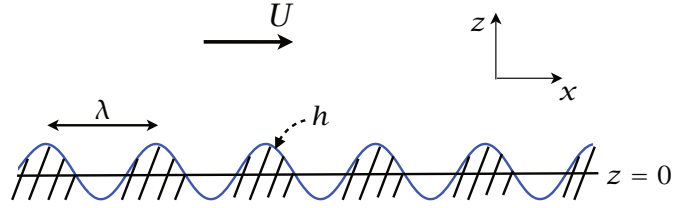
## 7.4♦ TOPOGRAPHIC GENERATION OF INTERNAL WAVES

How are internal waves generated? One way that is important in both the ocean and atmosphere is by way of a horizontal flow, such as a mean wind or, in the ocean, a tide or a mesoscale eddy, passing over a topographic feature. This forces the fluid to move up and/or down, so generating an internal wave, commonly known as a mountain wave. In this section we illustrate the mechanism with simple examples of steady, uniform flow with constant stratification over idealized topography.

### 7.4.1 Sinusoidal Mountain Waves

For simplicity we ignore the effects of the Earth's rotation and pose the problem in two dimensions,  $x$  and  $z$ , using the Boussinesq approximation. Our goal is to calculate the response to a steady, uniform flow of magnitude  $U$  over a sinusoidally varying boundary  $h = \bar{h} \cos kx$  at  $z = 0$ , as in Fig. 7.7 with  $k = 2\pi/\lambda$ . The topographic variations are assumed small, so allowing the dynamics to be linearized, which enables an arbi-

**Fig. 7.7:** Uniform flow,  $U$ , in the  $x$ -direction flowing over sinusoidal topography,  $h$ . The vertical coordinate is stretched, and in reality  $|h| \ll \lambda$ .



trarily shaped boundary to be considered by appropriately summing over Fourier modes.

### Equation of motion and dispersion relation

The momentum equations, the buoyancy equation, and the mass continuity equation are, respectively,

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) u' = -\frac{\partial \phi}{\partial x}, \quad \left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) w' = -\frac{\partial \phi}{\partial z} + b', \quad (7.45a,b)$$

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) u' + w' N^2 = 0, \quad \frac{\partial u'}{\partial x} + \frac{\partial w'}{\partial z} = 0. \quad (7.45c,d)$$

We henceforth drop the primes on the perturbation quantities.

The dispersion relation is obtained by noting that the equations are the same as with no mean flow, save that  $\partial/\partial t$  is replaced by  $\partial/\partial t + U\partial/\partial x$ . Thus, similarly to (7.26), the dispersion relation is

$$(\omega - Uk)^2 = \frac{k^2 N^2}{k^2 + m^2} \quad \text{or} \quad \omega = Uk \pm \frac{kN}{(k^2 + m^2)^{1/2}}. \quad (7.46a,b)$$

The horizontal and vertical components of the group velocity are then given by

$$c_g^z = \pm \frac{Nm k}{(k^2 + m^2)^{3/2}}, \quad c_g^x = U \pm \frac{Nm^2}{(k^2 + m^2)^{3/2}}. \quad (7.47a,b)$$

Steady waves have  $\omega = 0$  and if  $U > 0$  such waves are associated with the negative root in (7.46b), the positive root in (7.47a) and the negative root in (7.47b). Energy will propagate upwards, and away from the mountain, if  $c_g^z$  is positive.

### The solution

If we are looking for steady solutions we can follow the recipe of Section 7.1 but with  $U\partial/\partial x$  replacing the time derivative. By analogy to (7.2) we find a single equation for  $w$  namely

$$\left[ U \frac{\partial^2}{\partial x^2} \left( \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2} \right) + N^2 \frac{\partial^2}{\partial x^2} \right] w = 0. \quad (7.48)$$

We take the topography to have the harmonic form  $h = \text{Re } h_0 \exp(ikx)$  and its presence is felt via a lower boundary condition,  $w = U\partial h/\partial x =$

$\text{Re } U h_0 i k \exp(ikx)$  at  $z = 0$ . We thus seek solutions to (7.47) of the form

$$w = \text{Re } U h_0 i k e^{i(kx+mz)}. \quad (7.49)$$

The harmonic dependence in  $z$  is valid only because we take  $N$  to be constant. The value of  $m$  is given by the dispersion relation (7.46) with  $\omega = 0$ , which gives

$$m^2 = \left(\frac{N}{U}\right)^2 - k^2. \quad (7.50)$$

We see that  $m^2$  may be negative, and so  $m$  imaginary, if  $N^2 < k^2 U^2$ , so evidently there is a qualitative difference between short waves, with  $k^2 > (N/U)^2$ , and long waves, with  $k^2 < (N/U)^2$ . In the atmosphere we might take  $U \sim 10 \text{ m s}^{-1}$  and  $N \sim 10^{-2} \text{ s}^{-1}$ , in which case  $U/N \sim 1 \text{ km}$ . In the ocean below the thermocline we might take  $U \sim 0.1 \text{ m s}^{-1}$  and  $N \sim 10^{-3} \text{ s}^{-1}$ , whence  $U/N \sim 100 \text{ m}$ .

A useful interpretation of (7.50) arises if, since the mean flow is constant, we consider the problem in the frame of reference of that mean flow. In this case the topography has the form  $h = \text{Re } h_0 \exp ik(x - Ut)$ , and the solution in the moving frame is

$$w = \text{Re } U h_0 i k e^{i(kx+mz+Ukt)}. \quad (7.51)$$

That is, the fluid now oscillates with a frequency  $\omega = -Uk$ . The condition for propagation,  $N^2 < k^2 U^2$  is just the same as  $\omega^2 < N^2$ , which is just the condition for gravity waves to oscillate in a stratified fluid.

Given the solution for  $w$  we can use the polarization relations of Section 7.2 to obtain the solutions for perturbation horizontal velocity and pressure. One way to proceed is to pose the problem in the moving frame, with  $\omega = -Uk$ , and directly use (7.12). Then back in the stationary frame we obtain the solutions

$$w = w_0 e^{i(kx+mz)} = iUk h_0 e^{imz} e^{ikx}, \quad (7.52a)$$

$$u = u_0 e^{i(kx+mz)} = -imU h_0 e^{imz} e^{ikx}, \quad (7.52b)$$

$$\phi = \phi_0 e^{i(kx+mz)} = imU^2 h_0 e^{imz} e^{ikx}, \quad (7.52c)$$

where  $m$  is given by (7.50). Let us see what the solutions mean, and if and how waves propagate.

#### 7.4.2 Energy Propagation

The direction of energy propagation is given by the group velocity. For steady waves ( $\omega = 0$ ) we have, using (7.47a),

$$c_g^z = \frac{Nkm}{(k^2 + m^2)^{3/2}} = \frac{Ukm}{k^2 + m^2}. \quad (7.53a,b)$$

This means that for positive  $U$  an upward group velocity, and hence upward energy propagation, occur when  $k$  and  $m$  have the same sign. This property may also be deduced by evaluating the vertical energy flux,  $\overline{w\phi}$ , where the overbar denotes a zonal average, using (7.52), with appropriate care to take the real part of the fields. The phase speed,  $c_p^z = \omega/m$ , is zero in the stationary frame and it is  $-Uk/m$  in the translating frame.

*Flow over topography is an important source of gravity waves in both atmosphere and ocean, with the flow coming from weather systems in the atmosphere or mesoscale eddies and tides in the atmosphere. Convection is also a significant source of gravity waves in the tropical atmosphere.*

*Short, trapped waves*

If the undulations on the boundary are sufficiently short then  $k^2 > (N/U)^2$  and  $m^2$  is negative and  $m$  is purely imaginary. Writing  $m = is$ , so that  $s^2 = k^2 - (N/U)^2$ , the solutions have the form

$$w = \text{Re } w_0 e^{ikx - sz}. \quad (7.54)$$

We must choose the solution with  $s > 0$  in order that the solution decays away from the mountain, and internal waves are not propagated into the interior. (If there were a rigid lid or a density discontinuity at the top of the fluid, as at the top of the ocean, then the possibility of reflection would arise and we would seek to satisfy the upper boundary condition with a combination of decaying and amplifying modes.) The above result is entirely consistent with the dispersion relation for internal waves, namely  $\omega = N \cos \vartheta$ : because  $\cos \vartheta < 1$  the frequency  $\omega$  must be less than  $N$  so that if the forcing frequency is higher than  $N$  no internal waves will be generated.

Because the waves are trapped waves we do not expect energy to propagate away from the mountains. To verify this, from the polarization relation (7.52) we have

$$w = \frac{k}{mU} \phi = \frac{-ik}{sU} \phi. \quad (7.55)$$

The pressure and the vertical velocity are therefore out of phase by  $\pi/2$ , and the vertical energy flux,  $\overline{w\phi}$  is identically zero. This is consistent with the fact that the energy flux is in the direction of the group velocity; the group velocity is given by (7.53a) and for an imaginary  $m$  the real part is zero. A solution to the problem in the short wave limit is shown in the top panels of Fig. 7.8, with  $s = 1$  and  $m = i$ . (Here and in Fig. 7.9 the lengths and wavenumbers are in units such that the length of the domain is  $2\pi$  and its height is that shown.)

*Long, propagating waves*

Suppose now that  $k^2 < (N/U)^2$  so that  $\omega^2 < N^2$ . From (7.50)  $m$  is now real and the solution has propagating waves of the form

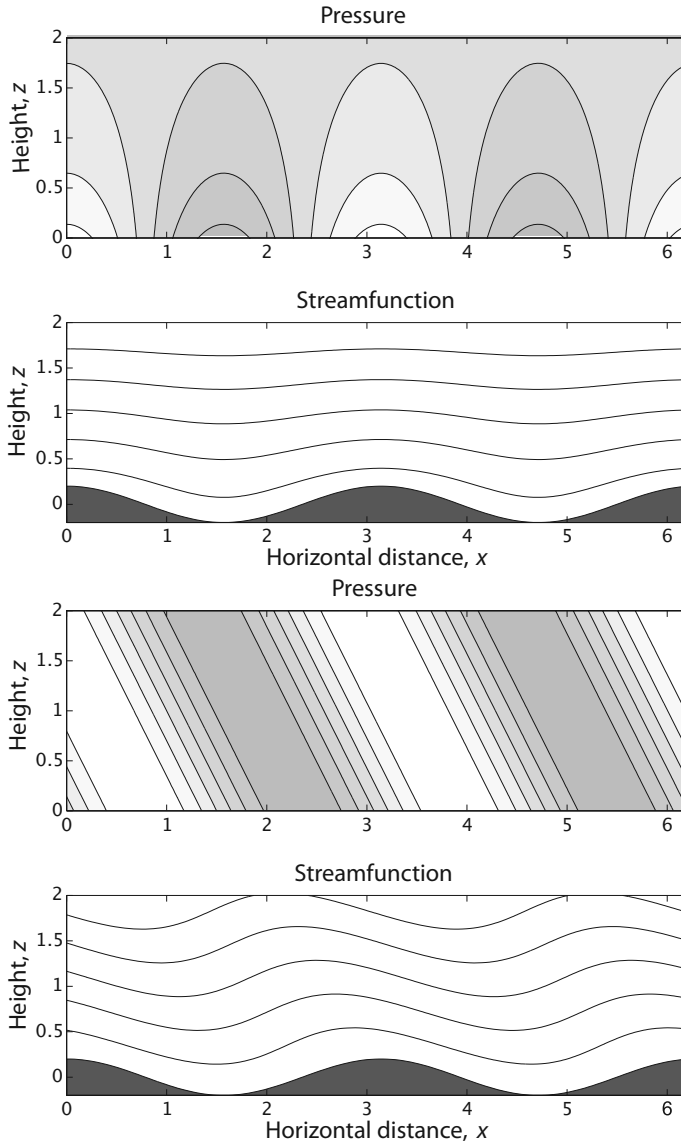
$$w = w_0 e^{i(kx + mz)}, \quad m^2 = \left(\frac{N}{U}\right)^2 - k^2. \quad (7.56)$$

Vertical propagation is occurring because the forcing frequency is less than the buoyancy frequency. The angle at which fluid parcel oscillations occur is then slanted off the vertical at an angle  $\vartheta$  such that the forcing frequency is equal to the natural frequency of oscillations at that angle, namely

$$\vartheta = \cos^{-1} \left( \frac{Uk}{N} \right). \quad (7.57)$$

The angle  $\vartheta$  is also the angle between the wavevector  $\mathbf{k}$  and the horizontal, as in (7.9), because the wavevector is at right angles to the parcel oscillations. If  $Uk = N$  then the fluid parcel oscillations are vertical and, using





**Fig. 7.8:** Flow over a sinusoidal ridge in the shortwave limit (top two panels) and the longwave limit (bottom two panels), in a nondimensional domain of length  $2\pi$  and height 2.

The top two panels show the solutions for the flow over a sinusoidal ridge, using (7.52), in the short wave limit ( $Uk > N$ ) and with  $m = i$ . The top panel shows the pressure, with darker grey indicating higher pressure. The second panel shows contours of the total streamfunction,  $\psi - Uz$ , with flow coming in from the left, and the topography itself (solid). The perturbation amplitude decreases exponentially with height.

The bottom two panels show solutions in the long wave limit ( $Uk < N$ ) with  $m = 1$ . The pressure is high on the windward side of the topography, and phase lines tilt upstream with height for both pressure and streamfunction.

(7.50),  $m = 0$ . Thus, although the group velocity is directed vertically, parallel to the fluid parcel oscillations, its magnitude is zero, from (7.53).

Solutions for flow over topography in the long wave limit and  $m = 1$  are shown in the lower two panels of Fig. 7.8. The flow is coming in from the left, and the phase lines evidently tilt upstream with height. Lines of constant phase follow  $kx + mz = \text{constant}$ , and in the solution shown both  $k$  and  $m$  are positive ( $k = m = 1$ ). Thus, the lines slope back at a slope  $x/z = -m/k$ , and energy propagates up and to the left. The phase propagation is actually downward in this example, as the reader may confirm. The pressure is high on the upstream side of the mountain, and this provides a drag on the flow — a so-called topographic form drag.

### Topographically Generated Gravity Waves (Mountain Waves)

- In both atmosphere and ocean an important mechanism for the generation of gravity waves is flow over bottom topography, and the ensuing waves are sometimes called mountain waves. A canonical case is that of a uniform flow over a sinusoidal topography, with constant stratification. If the flow is in the  $x$ -direction and there is no  $y$ -variation then the boundary condition is

$$w(x, z = 0) = U \frac{\partial h}{\partial x} = -iUk\tilde{h}. \quad (\text{MW.1})$$

Solutions of the problem may be found in the form  $w(x, z, t) = w_0 \exp[i(kx + mz - \omega t)]$ , where the boundary condition at  $z = 0$  is given by (MW.1), the frequency is given by the internal wave dispersion relation, and the other dynamical fields are obtained using the polarization relations.

- One way to solve the problem is to transform into a frame moving with the background flow,  $U$ , provided  $U$  is uniform. The topography then appears to oscillate with a frequency  $-Uk$ , and this in turn becomes the frequency of the gravity waves.
- Propagating gravity waves can only be supported if the frequency is less than  $N$ , meaning that  $Uk < N$ . That is, the waves must be sufficiently long and therefore the topography must be of sufficiently large scale.
- When propagating waves exist, energy is propagated upward away from the topography. The topography also exerts a drag on the background flow.
- If the waves are too short they are evanescent, decaying exponentially with height. That is, they are trapped near the topography
- In the presence of rotation the wave frequency must lie between the buoyancy frequency  $N$  and the inertial frequency  $f$ . If the flow is constant, waves can thus radiate upward if

$$f < Uk < N. \quad (\text{MW.2})$$

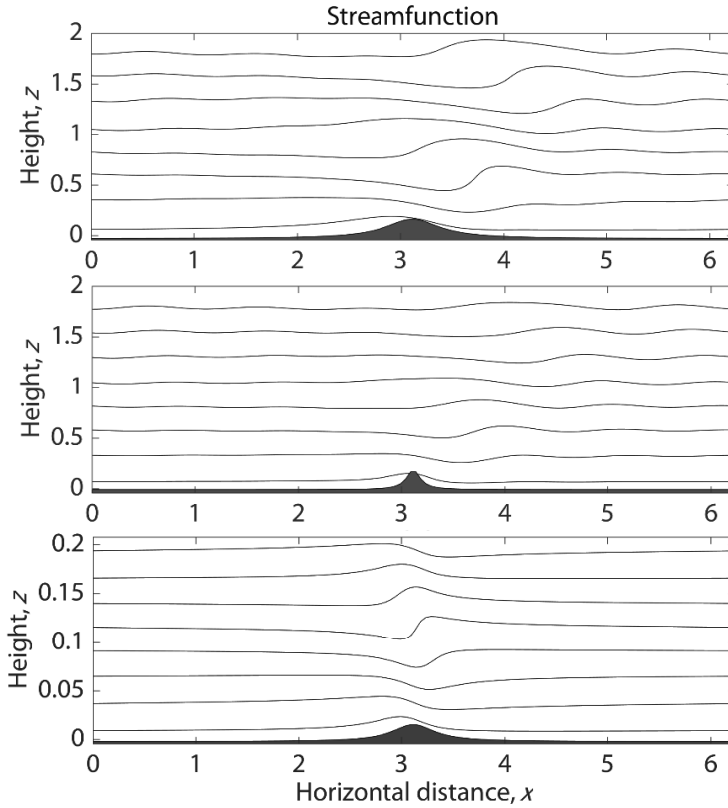
Thus, both very long waves and very short waves are evanescent.

## 7.5 ♦ FLOW OVER AN ISOLATED RIDGE

Most mountains are of course not perfect sinusoids, but we can construct a solution for any given topography using a superposition of Fourier modes. In this section we will illustrate the solution for a mountain consisting of a single ridge; the actual solution must usually be obtained numerically, and we will sketch the method and show some results.

### 7.5.1 Sketch of the Methodology

The methodology to compute a solution is as follows. Consider a topographic profile,  $h(x)$ , and let us suppose that it is periodic in  $x$  over some



**Fig. 7.9:** The streamfunction for three solutions for uniform flow over isolated topography. The top panel shows flow over a wide hump, resulting in waves propagating vertically and downstream. The middle panel shows flow over a narrow hump and most of the waves evanesce away from the hump. The bottom panel shows the solution in a hydrostatic regime, very common in the atmosphere and ocean; here the group velocity is purely vertical and the disturbance is directly above the mountain.

distance  $L$ . Such a profile can (nearly always) be decomposed into a sum of Fourier coefficients, meaning that we can write

$$h(x) = \sum_k \tilde{h}_k e^{ikx}, \quad (7.58)$$

where  $\tilde{h}_k$  are the Fourier coefficients. We can obtain the set of  $\tilde{h}_k$  by multiplying (7.58) by  $e^{-ikx}$  and integrating over the domain from  $x = 0$  to  $x = L$ , a procedure known as taking the discrete Fourier transform of  $h(x)$ , and there are standard computer algorithms for doing this efficiently. Once we have obtained the values of  $\tilde{h}_k$  we essentially solve the problem separately for each  $k$  in precisely the same manner as we did in the previous section. For *each*  $k$  there is a vertical wavenumber given by (7.50), so that for each wavenumber we obtain a solution for pressure of the form  $\tilde{\phi}_k(z)$ , and similarly for the other variables. Once we have the solution for each wavenumber, then at each level we sum over all the wavenumbers to obtain the solution in real space; that is, we evaluate

$$\phi(x, z) = \sum_k \tilde{\phi}_k(z) e^{ikx}, \quad (7.59)$$

that is to say, we take the inverse discrete Fourier transform.

### 7.5.2 A Witch's Brew

For specificity let us consider the bell-shaped topographic profile

$$h(x) = \frac{h_0 a^2}{a^2 + x^2}, \quad (7.60)$$

*The profile is named for Maria Agnesi, 1718–1799, an Italian mathematician and later a theologian who discussed the properties of the curve. The term 'witch' (in 'Witch of Agnesi') seems to be a mistranslation from Italian, or a deliberate and possibly unkind play on words, of versiera, which refers to a curve and not an adversary of God or a she-devil (which would be avversiera). Maria Agnesi, evidently a remarkable person, may have been the first woman appointed to a professorship of mathematics and that was at the University of Bologna.*

sometimes called the Witch of Agnesi. Such a profile is composed of *many* (in fact an infinite number of) Fourier coefficients of differing amplitudes. If the profile is narrow (meaning  $a$  is small, in a sense made clearer below) then there will be a great many significant coefficients at high wavenumbers. In fact, in the limiting case of an infinitely thin ridge (a delta function) all wavenumbers are present with equal weight, so there are certainly more large wavenumbers than small wavenumbers. However, if  $a$  is large, then the contributing wavenumbers are predominantly small.

In the problem of flow over topography the natural horizontal scale of the flow is  $U/N$ . If  $a \gg U/N$  then the dominant wavenumbers are small and the solution consists of waves propagating upward with little loss of amplitude and phase lines tilting upstream, as illustrated in the top panel of Fig. 7.9. In the case of a narrow ridge, as illustrated in the middle panel of that figure, the perturbation is largely trapped near to the mountain and the perturbation fields largely decay exponentially with height. Nevertheless, because the ridge does contain *some* small wavenumbers, some weak, propagating large-scale disturbances are generated. The fluid acts as a low-pass filter, and the perturbation aloft consists only of large scales.

#### A hydrostatic solution

If the ridge is sufficiently wide then the solution is essentially hydrostatic, with little dependence of the vertical structure on the horizontal wavenumber; that is, using (7.50) at large scales,  $m^2 \approx (N/U)^2$ . Furthermore, the  $x$ -component of the group velocity is zero, which can be seen from (7.47b) using  $m = N/U$  (since  $m$  is positive for upward wave propagation and  $m \gg k$  by hydrostasy). Explicitly we have

$$c_g^x = U - \frac{Nm^2}{(k^2 + m^2)^{3/2}} \approx U - \frac{N}{m} = U - U = 0. \quad (7.61)$$

Thus, the disturbance appears directly over the mountain, with no downstream propagation, as in the bottom panel of Fig. 7.9. The pattern therefore repeats itself in the vertical at intervals of  $2\pi/m$  or  $2\pi U/N$ , with neither a downstream nor upstream influence. This solution is in fact the most atmospherically relevant one, for it is that produced by atmospheric flow over mountains that have horizontal scales larger than a few kilometres (i.e.  $a \gg U/N$ ). It is also oceanographically relevant for flow over features of greater than a few kilometres, except possibly in regions of very weak stratification and large abyssal currents. Scales that are much larger are filtered by the Coriolis effect, because the frequency must lie between  $f$  and  $N$ .

## Notes and References

The books by Gill (1982) and Sutherland (2010) provide more discussion of gravity waves in general, and Durran (2015) provides more discussion of mountain waves.

## Problems

- 7.1 In the atmosphere gravity waves may be generated by flow over mountains, for example by westerly winds flowing over the Rockies. Suppose that the wind is  $10 \text{ m s}^{-1}$ . Under what circumstances of stratification and width of the mountain range will the disturbance (a) propagate downstream; (b) upstream; (c) appear directly above the mountain; (d) evanesce? Be as quantitative as possible, and comment on the realism and/or likelihood of each case.
- 7.2 In the ocean, gravity waves may be generated by flow over bottom topography, with the flow generated either by tides, which have a frequency of order hours, or by mesoscale eddies, which have a timescale of weeks and which may have a different strength. Discuss the properties of the gravity waves generated by these two effects. (This is an open-ended question that will require research into the literature.)
- 7.3 As shown in Section 7.1, the frequency of hydrostatic gravity waves increases without limit as the horizontal scale becomes smaller and smaller.
  - (a) What is the physical reason behind this effect. (That is, give an argument in words as to why this effect occurs, and why it does not in the nonhydrostatic case.)
  - (b) In the atmosphere, at what scale does this effect become a practical concern? Suppose we are building a numerical model of the atmosphere; at what grid resolution do we need to be concerned as to whether we have a nonhydrostatic or hydrostatic model?
- 7.4 *Co-existence of gravity waves and Rossby waves.* Consider the linear shallow-water equations in a rotating frame of reference:

$$\frac{\partial u}{\partial t} - fv = -\frac{\partial \phi}{\partial x}, \quad \frac{\partial v}{\partial t} + fu = -\frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial t} + c^2 \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0, \quad (\text{P7.1})$$

where  $\phi = g'\eta$  and  $c^2 = g'H$ ,  $\eta$  is the free surface height,  $H$  is the reference depth of the fluid and  $g'$  is the reduced gravity. Show that these equations reduce to a single equation for  $v$ , namely

$$\frac{1}{c^2} \frac{\partial^3 v}{\partial t^3} + \frac{f^2}{c^2} \frac{\partial v}{\partial t} - \frac{\partial}{\partial t} \nabla^2 v - \beta \frac{\partial v}{\partial x} = 0, \quad (\text{P7.2})$$

where  $\beta = \partial f / \partial y$ . Now suppose that  $f$  and  $\beta$  are both constant in the above equation and derive the dispersion relation

$$\omega^2 - \frac{\beta kc^2}{\omega} = f_0^2 + c^2(k^2 + l^2). \quad (\text{P7.3})$$

By considering appropriate limits of this equation, or otherwise, show that the system contains both Rossby and gravity waves. Plot the dispersion relation and show that with sensible values of parameters there is a frequency gap between the two kinds of waves.

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# Instability

**W**HAT HYDRODYNAMIC STATES OCCUR IN NATURE? Any flow must of course be a solution of the equations of motion, subject to the relevant initial and boundary conditions. There are many steady solutions to the equations of motion — certain purely zonal flows, for example — but the flows we experience are unsteady, time-dependent solutions, not steady solutions. Why should this be? It is because for any steady flow to persist it must be stable to those small perturbations that inevitably arise, but all the steady solutions that are known for the large-scale flow in the Earth's atmosphere and ocean have been found to be unstable.

Our focus in this chapter is on barotropic and baroclinic instability. *Baroclinic instability* (and we will define the term more precisely later on) is an instability that arises in rotating, stratified fluids that are subject to a horizontal temperature gradient. It is the instability that gives rise to the large- and mesoscale motion in the atmosphere and ocean — it produces atmospheric weather systems, for example — and so is, perhaps, the form of hydrodynamic instability that most affects the human condition. *Barotropic instability* is an instability that arises because of the shear in a flow, and may occur in fluids of constant density. It is important to us for two reasons: first, in its own right as an instability mechanism for jets and vortices and as an important process in turbulence; second, many problems in barotropic and baroclinic instability are very similar, so that the solutions and insight we obtain in the often simpler problems in barotropic instability may be useful in the baroclinic problem.

## 8.1 KELVIN–HELMHOLTZ INSTABILITY

We first consider what is perhaps the simplest physically interesting instance of a fluid-dynamical instability — that of a constant-density flow with a shear perpendicular to the fluid's mean velocity, this being an ex-

ample of a *Kelvin–Helmholtz instability*. Specifically, we consider two fluid masses of equal density, with an interface at  $y = 0$ , moving with velocities  $-U$  and  $+U$  in the  $x$ -direction, respectively, as in Fig. 8.1. There is no variation in the basic flow in the  $z$ -direction (normal to the page), and we will assume this is also true for the instability. This flow is clearly a solution of the Euler equations.

What happens if the flow is perturbed slightly? If the perturbation is initially small then even if it grows we can, for small times after the onset of instability, neglect the nonlinear interactions in the governing equations because these are the squares of small quantities. The equations determining the evolution of the initial perturbation are then the Euler equations linearized about the steady solution. Thus, denoting perturbation quantities with a prime and basic state variables with capital letters, for  $y > 0$  the perturbation satisfies

$$\frac{\partial \mathbf{u}'}{\partial t} + U \frac{\partial \mathbf{u}'}{\partial x} = -\nabla p', \quad \nabla \cdot \mathbf{u}' = 0, \quad (8.1a,b)$$

and a similar equation holds for  $y < 0$ , but with  $U$  replaced by  $-U$ . Given periodic boundary conditions in the  $x$ -direction, we may seek solutions of the form

$$\phi'(x, y, t) = \text{Re} \sum_k \tilde{\phi}_k(y) \exp[ik(x - ct)], \quad (8.2)$$

where  $\phi$  is any field variable (e.g., pressure or velocity), and  $\text{Re}$  denotes that only the real part should be taken. (Typically we use tildes over variables to denote Fourier-like modes, and we often omit the marker ‘Re’.) Because (8.1a) is linear, the Fourier modes do not interact and we may confine attention to just one. Taking the divergence of (8.1a), the left-hand side vanishes and the pressure satisfies Laplace’s equation

$$\nabla^2 p' = 0. \quad (8.3)$$

This has solutions in the form

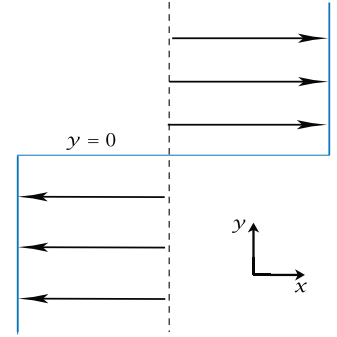
$$p' = \begin{cases} \tilde{p}_1 e^{ikx - ky} e^{\sigma t} & y > 0, \\ \tilde{p}_2 e^{ikx + ky} e^{\sigma t} & y < 0, \end{cases} \quad (8.4)$$

where, anticipating the possibility of growing solutions, we have written the time variation in terms of a growth rate,  $\sigma = -ikc$ . In general  $\sigma$  is complex: if it has a positive real component, the amplitude of the perturbation will grow and there is an instability; if  $\sigma$  has a non-zero imaginary component, then there will be oscillatory motion, and there may be both oscillatory motion *and* an instability. To obtain the dispersion relationship, we consider the  $y$ -component of (8.1a), namely (for  $y > 0$ )

$$\frac{\partial v'_1}{\partial t} + U \frac{\partial v'_1}{\partial x} = -\frac{\partial p'_1}{\partial y}. \quad (8.5)$$

Substituting a solution of the form  $v'_1 = \tilde{v}_1 \exp(ikx + \sigma t)$  yields, with (8.4),

$$(\sigma + ikU)\tilde{v}_1 = k\tilde{p}_1. \quad (8.6)$$



**Fig. 8.1:** A simple basic state giving rise to shear-flow instability. The velocity profile is discontinuous and the density is uniform. In more general problems in Kelvin–Helmholtz instability the density may also vary.

But the velocity normal to the interface is, at the interface, nothing but the rate of change of the position of the interface itself; that is, at  $y = +0$

$$v_1 = \frac{\partial \eta'}{\partial t} + U \frac{\partial \eta'}{\partial x}, \quad (8.7)$$

or

$$\tilde{v}_1 = (\sigma + ikU)\tilde{\eta}, \quad (8.8)$$

where  $\eta'$  is the displacement of the interface from its equilibrium position. Using this in (8.6) gives

$$(\sigma + ikU)^2 \tilde{\eta} = k \tilde{p}_1. \quad (8.9)$$

The above few equations pertain to motion on the  $y > 0$  side of the interface. Similar reasoning on the other side gives (at  $y = -0$ )

$$(\sigma - ikU)^2 \tilde{\eta} = -k \tilde{p}_2. \quad (8.10)$$

But at the interface  $p_1 = p_2$ , because pressure must be continuous. The dispersion relationship then emerges from (8.9) and (8.10), giving

$$\sigma^2 = k^2 U^2. \quad (8.11)$$

This equation has two roots, one of which is positive. Thus, the amplitude of the perturbation grows exponentially, like  $e^{\sigma t}$ , and the flow is *unstable*. The instability itself can be seen in the natural world when billow clouds appear wrapped up into spirals: the clouds are acting as tracers of fluid flow, and are a manifestation of the instability at finite amplitude, as seen later in Fig. 8.2.

## 8.2 INSTABILITY OF PARALLEL SHEAR FLOW

We now consider a little more systematically the instability of parallel shear flows. This is a classic problem in hydrodynamic stability theory, and there are two particular reasons for our own interest:

- (i) The instability is an example of *barotropic instability*, which abounds in the ocean and atmosphere. Roughly speaking, barotropic instability arises when a flow is unstable by virtue of its horizontal shear, with gravitational and buoyancy effects being secondary.
- (ii) The instability is in many ways analogous to *baroclinic instability*, which is the main instability giving rise to weather systems in the atmosphere and similar phenomena in the ocean.

We restrict attention to two-dimensional, incompressible flow; this illustrates the physical mechanisms in the most transparent way, in part because it allows for the introduction of a streamfunction and the automatic satisfaction of the mass continuity equation. In fact, for parallel two-dimensional shear flows the most unstable disturbances are two-dimensional ones.



The vorticity equation for incompressible two-dimensional flow is just

$$\frac{D\zeta}{Dt} = 0. \quad (8.12)$$

We suppose the basic state to be a parallel flow in the  $x$ -direction that may vary in the  $y$ -direction. That is

$$\bar{\mathbf{u}} = U(y)\hat{\mathbf{i}}. \quad (8.13)$$

The linearized vorticity equation is then

$$\frac{\partial \zeta'}{\partial t} + U \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \zeta}{\partial y} = 0, \quad (8.14)$$

where  $\zeta = -\partial_y U$ . Because the mass continuity equation has the simple form  $\partial u'/\partial x + \partial v'/\partial y = 0$ , we may introduce a streamfunction  $\psi$  such that  $u' = -\partial \psi'/\partial y$ ,  $v' = \partial \psi'/\partial x$  and  $\zeta' = \nabla^2 \psi'$ . The linear vorticity equation becomes

$$\frac{\partial \nabla^2 \psi'}{\partial t} + U \frac{\partial \nabla^2 \psi'}{\partial x} + \frac{\partial \zeta}{\partial y} \frac{\partial \psi'}{\partial x} = 0. \quad (8.15)$$

The coefficients of the  $x$ -derivatives are not themselves functions of  $x$ ; thus, we may seek solutions that are harmonic functions (sines and cosines) in the  $x$ -direction, but the  $y$  dependence must remain arbitrary at this stage and we write

$$\psi' = \text{Re } \tilde{\psi}(y) e^{ik(x-ct)}. \quad (8.16)$$

The full solution is a superposition of all wavenumbers, but since the problem is linear the waves do not interact and it suffices to consider them separately. If  $c$  is purely real then  $c$  is the phase speed of the wave; if  $c$  has a positive imaginary component then the wave will grow exponentially and is thus *unstable*.

From (8.16) we have

$$u' = \tilde{u}(y) e^{ik(x-ct)} = -\tilde{\psi}_y e^{ik(x-ct)}, \quad (8.17a)$$

$$v' = \tilde{v}(y) e^{ik(x-ct)} = ik\tilde{\psi} e^{ik(x-ct)}, \quad (8.17b)$$

$$\zeta' = \tilde{\zeta}(y) e^{ik(x-ct)} = (-k^2\tilde{\psi} + \tilde{\psi}_{yy}) e^{ik(x-ct)}, \quad (8.17c)$$

where the  $y$  subscript denotes a derivative. Using (8.17) in (8.14) gives

$$(U - c)(\tilde{\psi}_{yy} - k^2\tilde{\psi}) - U_{yy}\tilde{\psi} = 0, \quad (8.18)$$

which is known as *Rayleigh's equation*. It is the linear vorticity equation for disturbances to parallel shear flow, and in the presence of a  $\beta$ -effect it generalizes slightly to

$$(U - c)(\tilde{\psi}_{yy} - k^2\tilde{\psi}) + (\beta - U_{yy})\tilde{\psi} = 0, \quad (8.19)$$

which is known as the Rayleigh–Kuo equation.

*John Strutt (1842–1919) became 3rd Baron Rayleigh on the death of his father in 1873 and, in a testament to the enduring British class system, is almost universally known as Lord Rayleigh. He made major contributions in many areas of physics, among them fluid mechanics (including the theory of sound and instability theory), the analysis of the composition of gases (leading to the discovery of argon), and scattering theory.*

### 8.2.1 Piecewise Linear Flows

Although Rayleigh's equation is linear and has a simple form, it is nevertheless quite difficult to analytically solve for an arbitrary smoothly varying profile. It is simpler to consider *piecewise linear* flows, in which  $U_y$  is constant over some interval, with  $U$  or  $U_y$  changing abruptly to another value at a line of discontinuity, as for example in Fig. 8.1. The curvature,  $U_{yy}$ , is accounted for through the satisfaction of matching conditions, analogous to boundary conditions, at the lines of discontinuity (as in Section 8.1), and solutions in each interval are then exponential functions.

#### Jump or matching conditions

The idea, then, is to solve the linearized vorticity equation separately in the continuous intervals in which vorticity is constant, matching the solution with that in the adjacent regions. The matching conditions arise from two physical conditions:

- (i) That normal stress should be continuous across the interface. For an inviscid fluid this implies that pressure be continuous.
- (ii) That the normal velocity of the fluid on either side of the interface should be consistent with the motion of the interface itself.

Let us consider the implications of these two conditions.

#### (i) Continuity of pressure

The linearized momentum equation in the direction along the interface is:

$$\frac{\partial u'}{\partial t} + U \frac{\partial u'}{\partial x} + v' \frac{\partial U}{\partial y} = -\frac{\partial p'}{\partial x}. \quad (8.20)$$

For normal modes,  $u' = -\tilde{\psi}_y e^{ik(x-ct)}$ ,  $v' = ik\tilde{\psi} e^{ik(x-ct)}$  and  $p' = \tilde{p} e^{ik(x-ct)}$ , and (8.20) becomes

$$ik(U - c)\tilde{\psi}_y - ik\tilde{\psi}U_y = -ik\tilde{p}. \quad (8.21)$$

Because pressure is continuous across the interface we have the first *matching or jump condition*,

$$\Delta[(U - c)\tilde{\psi}_y - \tilde{\psi}U_y] = 0, \quad (8.22)$$

where the operator  $\Delta$  denotes the difference in the values of the argument (in square brackets) across the interface. That is, the quantity  $(U - c)\tilde{\psi}_y - \tilde{\psi}U_y$  is continuous.

We can obtain this condition directly from Rayleigh's equation, (8.19), written in the form

$$[(U - c)\tilde{\psi}_y - U_y\tilde{\psi}]_y + [\beta - k^2(U - c)]\tilde{\psi} = 0. \quad (8.23)$$

Integrating across the interface gives (8.22).

(ii) *Material interface condition*

At the interface, the normal velocity  $v$  is given by the kinematic condition

$$v = \frac{D\eta}{Dt}, \quad (8.24)$$

where  $\eta$  is the interface displacement. The linear version of (8.24) is

$$\frac{\partial \eta'}{\partial t} + U \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'}{\partial x}. \quad (8.25)$$

If the fluid itself is continuous then this equation must hold at either side of the interface, giving two equations and their normal-mode counterparts, namely,

$$\frac{\partial \eta'}{\partial t} + U_1 \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'_1}{\partial x} \longrightarrow (U_1 - c)\tilde{\eta} = \tilde{\psi}_1, \quad (8.26)$$

$$\frac{\partial \eta'}{\partial t} + U_2 \frac{\partial \eta'}{\partial x} = \frac{\partial \psi'_2}{\partial x} \longrightarrow (U_2 - c)\tilde{\eta} = \tilde{\psi}_2. \quad (8.27)$$

Material continuity at the interface thus gives the second jump condition

$$\Delta \left[ \frac{\tilde{\psi}}{U - c} \right] = 0. \quad (8.28)$$

That is,  $\tilde{\psi}/(U - c)$  is continuous at the interface. Note that if  $U$  is continuous across the interface the condition becomes one of continuity of the normal velocity.

### 8.2.2 Kelvin–Helmholtz Instability, Revisited

We now use Rayleigh's equation and the jump conditions to consider the situation illustrated in Fig. 8.1; that is, vorticity is everywhere zero except in a thin sheet at  $y = 0$ . On either side of the interface, Rayleigh's equation is simply

$$(U_i - c)(\partial_{yy}\tilde{\psi}_i - k^2\tilde{\psi}_i) = 0, \quad i = 1, 2 \quad (8.29)$$

or, assuming that  $U_i \neq c$ ,  $\partial_{yy}\tilde{\psi}_i - k^2\tilde{\psi}_i = 0$ . (This is just Laplace's equation, coming from  $\nabla^2\psi' = \zeta'$ , with  $\zeta' = 0$  everywhere except at the interface.) Solutions of this that decay away on either side of the interface are

$$y > 0 : \quad \tilde{\psi}_1 = \Psi_1 e^{-ky}, \quad (8.30a)$$

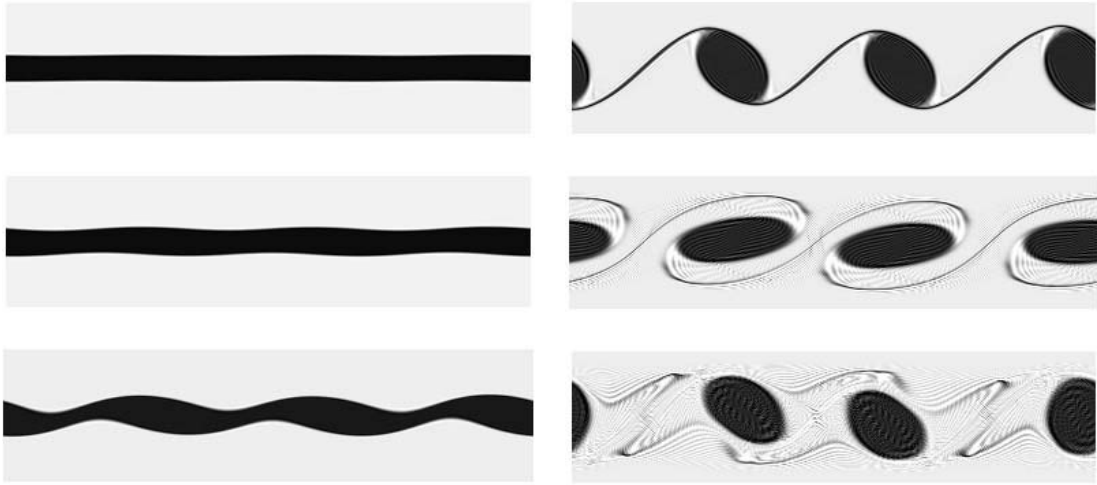
$$y < 0 : \quad \tilde{\psi}_2 = \Psi_2 e^{ky}, \quad (8.30b)$$

where  $\Psi_1$  and  $\Psi_2$  are constants. The boundary condition (8.22) gives

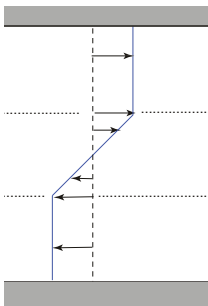
$$(U_1 - c)(-k)\Psi_1 = (U_2 - c)(k)\Psi_2, \quad (8.31)$$

and (8.28) gives

$$\frac{\Psi_1}{(U_1 - c)} = \frac{\Psi_2}{(U_2 - c)}. \quad (8.32)$$



**Fig. 8.2:** A sequence of plots of the vorticity, at equal time intervals, from a numerical solution of the nonlinear vorticity equation (8.12), with initial conditions of an unstable shear flow, as in the figure below. Time increases first down the left column and then down the right column. The instability first grows exponentially, as in the left column, and then nonlinearities cause vortices to form, as in the right column.



**Fig. 8.3:** An unstable shear flow, providing initial conditions for the evolution shown in Fig. 8.2.

The last two equations combine to give  $(U_1 - c)^2 = -(U_2 - c)^2$ , which, supposing that  $U = U_1 = -U_2$  gives  $c^2 = -U^2$ . Thus, since  $U$  is purely real,  $c = \pm iU$ , and the disturbance grows exponentially as  $\exp(kU_1 t)$ , just as we obtained in Section 8.1, with all wavelengths being unstable. An example of a numerically calculated instability is given in Fig. 8.2, although here the initial vorticity is not just confined to a sheet but spread out over a finite width, as in Fig. 8.3, resulting in a preferred instability scale at about nondimensional wavenumber 3 (i.e., with 3 wavelengths spanning the domain).

### 8.3 NECESSARY CONDITIONS FOR INSTABILITY

#### 8.3.1 Rayleigh's Criterion

For simple profiles it may be possible to calculate, or even intuit, the instability properties, but for continuous profiles of  $U(y)$  this is often impossible and it would be nice to have some general guidelines as to when a profile might be unstable. To this end, we derive a *necessary* condition for instability, or *sufficient* conditions for stability, that will at least tell us if a flow *might* be unstable.

Consider the vorticity equation, linearized about a parallel shear flow (cf. (8.14) with a  $\beta$ -term),

$$\frac{\partial \zeta}{\partial t} + U \frac{\partial \zeta}{\partial x} + v \left( \frac{\partial Z}{\partial y} + \beta \right) = 0, \quad (8.33)$$

where  $Z = -\partial U / \partial y$  is the vorticity of the mean state and we have dropped the primes on the perturbation quantities. If we multiply by  $\zeta$  and divide by  $\beta + \partial_y Z$  we obtain

$$\frac{\partial}{\partial t} \left( \frac{\zeta^2}{Q_y} \right) + \frac{U}{Q_y} \frac{\partial \zeta^2}{\partial x} + 2v\zeta = 0, \quad (8.34)$$

where  $Q_y = \partial_y Q = \beta + \partial_y Z$  is the gradient of the absolute vorticity, here equal to the gradient of the potential vorticity (hence the symbol  $Q$ ). Integrating the above equation with respect to  $x$  gives

$$\frac{\partial}{\partial t} \int \left( \frac{\zeta^2}{Q_y} \right) dx = -2 \int v \zeta dx. \quad (8.35)$$

Now, using  $\nabla \cdot \mathbf{u} = 0$ , the vorticity flux may be written as

$$v \zeta = -\frac{\partial}{\partial y} (uv) + \frac{1}{2} \frac{\partial}{\partial x} (v^2 - u^2). \quad (8.36)$$

That is, the flux of vorticity is the divergence of some quantity. Its integral therefore vanishes provided there are no contributions from the boundary, and integrating (8.35) with respect to  $y$  gives

$$\frac{d}{dt} \int \left( \frac{\zeta^2}{Q_y} \right) dx dy = 0. \quad (8.37)$$

If there is to be an instability  $\zeta$  must grow, but the integral is identically zero. These two conditions can only be simultaneously satisfied if  $Q_y$ , or equivalently  $\beta - U_{yy}$ , is zero somewhere in the domain. Thus,

*A necessary condition for instability is that the expression*

$$\beta - U_{yy}$$

*change sign somewhere in the domain.*

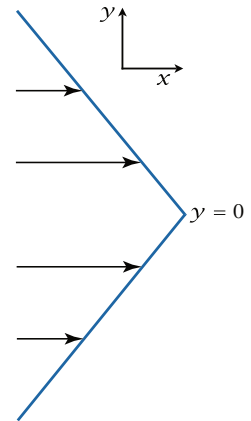
Equivalently, a sufficient criterion for stability is that  $\beta - U_{yy}$  does not vanish in the domain interior. This condition is known as Rayleigh's inflection-point criterion, or when  $\beta \neq 0$ , the Rayleigh–Kuo inflection point criterion. The quantity  $\zeta^2/Q_y$  is an example of a *wave-activity density* — a wave activity being a conserved quantity, quadratic in the amplitude of the wave. Such quantities play an important role in instabilities, and they crop up again in Chapter 9.

Interestingly, the  $\beta$ -effect can be either stabilizing or destabilizing: it can stabilize the middle two profiles of Fig. 8.5, because if it is large enough  $\beta - U_{yy}$  will be one-signed. However, the  $\beta$ -effect will destabilize a westward point jet,  $U(y) = -(1 - |y|)$  (the negative of the jet in Fig. 8.4), because  $\beta - U_{yy}$  is negative at  $y = 0$  and positive elsewhere. An eastward point jet is stable, with or without  $\beta$ . Finally, we emphasize that Rayleigh's criterion is a necessary condition for instability, not a sufficient one. In some circumstances a flow may satisfy the criterion for instability yet still be stable to infinitesimal perturbations, and this is found to be the case in the last row of Fig. 8.5.

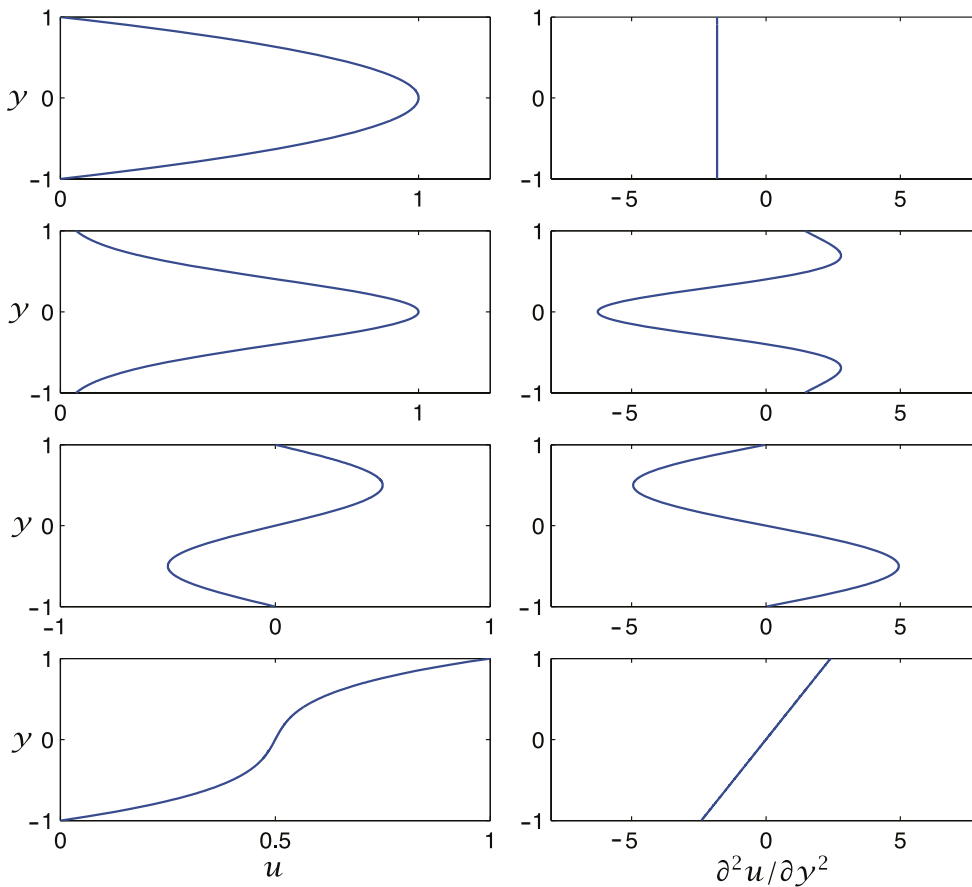
## 8.4 BAROCLINIC INSTABILITY

Baroclinic instability is an instability that occurs in stably stratified, rotating fluids, and it is ubiquitous in the Earth's atmosphere and oceans, and almost certainly occurs in other planetary atmospheres. It gives rise to weather, and so is perhaps the form of hydrodynamic instability that affects us most, and that certainly we talk about most.

*The classical stability criterion for shear flow was written-down by Rayleigh (1880) and extended to a beta-plane by Kuo (1949) and then to a baroclinic flow (Section 8.7) by Charney & Stern (1962) and Pedlosky (1964).*



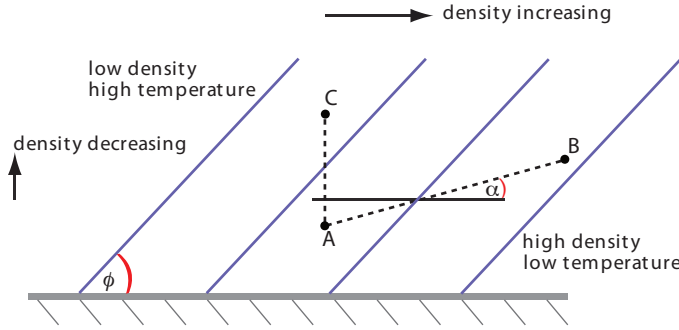
**Fig. 8.4:** Velocity profile of a point jet, in which vorticity is concentrated at a point. Although the vorticity is discontinuous, a small perturbation gives rise only to *edge waves* centred at  $y = 0$ , and the jet is stable



**Fig. 8.5:** Example parallel velocity profiles (left column) and their second derivatives (right column). From the top: Poiseuille flow ( $u = 1 - y^2$ ); a Gaussian jet; a sinusoidal profile; a polynomial profile. By Rayleigh's criterion, the top profile is stable, whereas the lower three are potentially unstable. The bottom profile is in fact stable (although we do not demonstrate that here). If the  $\beta$ -effect were present and large enough it would stabilize the middle two profiles.

#### 8.4.1 A Physical Picture

We first draw a picture of baroclinic instability as a form of 'sloping convection' in which the fluid, although statically stable, is able to release available potential energy when parcels move along a sloping path. To this end, let us first ask: what is the basic state that is baroclinically unstable? In a stably stratified fluid potential density decreases with height; we can also easily imagine a state in which the basic state temperature decreases, and the potential density increases, polewards. (We couch most of our discussion in terms of the Boussinesq equations and drop the qualifier 'potential' from density.) Can we construct a steady solution from these two conditions? The answer is yes, provided the fluid is also rotating; rotation is necessary because the meridional temperature gradient generally implies a meridional pressure gradient; there is nothing to balance this in the absence of rotation, and a fluid parcel would therefore accelerate. In a rotating fluid this pressure gradient can be balanced by the Coriolis force and a steady solution can be maintained even in the absence of viscosity. Consider a stably stratified Boussinesq fluid in geostrophic and hydrostatic balance on an  $f$ -plane, with buoyancy decreasing uniformly



**Fig. 8.6:** A steady basic state giving rise to baroclinic instability. Potential density decreases upwards and equatorwards, and the associated horizontal pressure gradient is balanced by the Coriolis force. Parcel 'A' is heavier than 'C', and so statically stable, but it is lighter than 'B'. Hence, if 'A' and 'B' are interchanged there is a release of potential energy.

polewards. Then  $fu = -\partial\phi/\partial y$  and  $\partial\phi/\partial z = b$ , where  $b = -g\delta\rho/\rho_0$  is the buoyancy. These together give the thermal wind relation,  $\partial u/\partial z = \partial b/\partial y$ . If there is no variation of these fields in the zonal direction, then, for any variation of  $b$  with  $y$ , this is a steady solution to the primitive equations of motion, with  $v = w = 0$ .

The density structure corresponding to a uniform increase of density in the meridional direction is illustrated in Fig. 8.6. Is this structure stable to perturbations? The answer is no, although the perturbations must be a little special. Suppose the particle at 'A' is displaced upwards; then, since the fluid is (by assumption) stably stratified it will be denser than its surroundings and hence experience a restoring force, and similarly if displaced downwards. Suppose, however, we interchange the two parcels at positions 'A' and 'B'. Parcel 'A' finds itself surrounded by parcels of higher density than itself, and it is therefore buoyant; it is also higher than where it started. Parcel 'B' is negatively buoyant, and at a lower altitude than where it started. Thus, overall, the centre of gravity of the fluid has been lowered, and so its overall potential energy lowered. This loss in potential energy (PE) of the basic state must be accompanied by a gain in kinetic energy of the perturbation. Thus, the perturbation amplifies and converts potential energy into kinetic energy.

The loss of potential energy is easily calculated. Since

$$PE = \int \rho g \, dz, \quad (8.38)$$

the change in potential energy due to the interchange is

$$\Delta PE = g(\rho_A z_A + \rho_B z_B - \rho_A z_B - \rho_B z_A) = g(z_A - z_B)(\rho_A - \rho_B) = g \Delta \rho \Delta z. \quad (8.39)$$

If both  $\rho_B > \rho_A$  and  $z_B > z_A$  then the initial potential energy is larger than the final one, energy is released and the state is unstable. If the slope of the isopycnals is  $\phi$  [so that  $\phi = -(\partial_y \rho)/(\partial_z \rho)$ ] and the slope of the displacements is  $\alpha$ , then for a displacement of horizontal distance  $L$  the change in potential energy is given by

$$\Delta PE = g \Delta \rho \Delta z = g \left( L \frac{\partial \rho}{\partial y} + L \alpha \frac{\partial \rho}{\partial z} \right) \alpha L = g L^2 \alpha \frac{\partial \rho}{\partial y} \left( 1 - \frac{\alpha}{\phi} \right), \quad (8.40)$$

if  $\alpha$  and  $\phi$  are small. If  $0 < \alpha < \phi$  then energy is released by the perturbation, and it is maximized when  $\alpha = \phi/2$ . For the atmosphere the actual slope of the isotherms is about  $10^{-3}$ , so that the slope and potential parcel trajectories are indeed shallow.

Although intuitively appealing, the thermodynamic arguments presented in this section pay no attention to satisfying the dynamical constraints of the equations of motion, and we now turn our attention to that.

### 8.4.2 Linearized Quasi-Geostrophic Equations

To explore the dynamics of baroclinic instability we use the quasi-geostrophic equations, specifically a potential vorticity equation for the fluid interior and a buoyancy or temperature equation at two vertical boundaries, one representing the ground and the other the tropopause. (The tropopause is the boundary between the troposphere and stratosphere at about 10 km; it is not a true rigid surface, but the higher static stability of the stratosphere does inhibit vertical motion.) For a Boussinesq fluid, the potential vorticity equation is

$$\begin{aligned} \frac{\partial q}{\partial t} + \mathbf{u} \cdot \nabla q &= 0, & 0 < z < H, \\ q &= \nabla^2 \psi + \beta y + \frac{\partial}{\partial z} \left( F \frac{\partial \psi}{\partial z} \right), \end{aligned} \quad (8.41)$$

where  $F = f_0^2/N^2$ , and the buoyancy equation, with  $w = 0$ , is

$$\begin{aligned} \frac{\partial b}{\partial t} + \mathbf{u} \cdot \nabla b &= 0, & z = 0, H, \\ b &= f_0 \frac{\partial \psi}{\partial z}. \end{aligned} \quad (8.42)$$

A solution of these equations is a purely zonal flow,  $\mathbf{u} = U(y, z)\hat{\mathbf{i}}$  with a corresponding temperature field given by thermal wind balance. The potential vorticity of this basic state is

$$Q = \beta y - \frac{\partial U}{\partial y} + \frac{\partial}{\partial z} F \frac{\partial \Psi}{\partial z} = \beta y + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial}{\partial z} F \frac{\partial \Psi}{\partial z}, \quad (8.43)$$

where  $\Psi$  is the streamfunction of the basic state, related to  $U$  by  $U = -\partial \Psi / \partial y$ . Linearizing (8.41) about this zonal flow gives the potential vorticity equation for the interior,

$$\frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial Q}{\partial y} = 0, \quad 0 < z < H, \quad (8.44)$$

where  $q' = \nabla^2 \psi' + \partial_z (F \partial_z \psi')$  and  $v' = \partial_x \psi'$ . Similarly, the linearized buoyancy equation at the boundary is

$$\frac{\partial b'}{\partial t} + U \frac{\partial b'}{\partial x} + v' \frac{\partial B}{\partial y} = 0, \quad z = 0, H, \quad (8.45)$$



where  $b' = f_0 \partial_z \psi'$  and  $\partial_y B = \partial_y (f_0 \partial_z \Psi) = -f_0 \partial U / \partial z$ .

Just as for the barotropic problem, a standard way of proceeding is to seek normal-mode solutions. Since the coefficients of (8.44) and (8.45) are functions of  $y$  and  $z$ , but not of  $x$ , we seek solutions of the form

$$\psi'(x, y, z, t) = \text{Re } \tilde{\psi}(y, z) e^{ik(x-ct)}, \quad (8.46)$$

and similarly for the derived quantities  $u', v', b'$  and  $q'$ . In particular,

$$\tilde{q} = \frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{\partial}{\partial z} F \frac{\partial \tilde{\psi}}{\partial z} - k^2 \tilde{\psi}. \quad (8.47)$$

Using (8.47) and (8.46) in (8.44) and (8.45) gives, with subscripts  $y$  and  $z$  denoting derivatives,

$$(U - c) (\tilde{\psi}_{yy} + (F \tilde{\psi}_z)_z - k^2 \tilde{\psi}) + Q_y \tilde{\psi} = 0 \quad 0 < z < H, \quad (8.48a)$$

$$(U - c) \tilde{\psi}_z - U_z \tilde{\psi} = 0 \quad z = 0, H. \quad (8.48b)$$

These equations are analogous to Rayleigh's equations for parallel shear flow, and illustrate the similarity between the baroclinic instability problem and that of a parallel shear flow.

## 8.5 THE EADY PROBLEM

We now proceed to explicitly calculate the stability properties of a particular configuration that has become known as the *Eady problem*. This was one of the first two mathematical descriptions of baroclinic instability, the other being the *Charney problem*. The Charney problem is in some respects more complete (for example in allowing a  $\beta$ -effect), but the Eady problem displays the instability in a more transparent form. The  $\beta$ -effect can be incorporated relatively simply in the two-layer model of the next section.

To begin, let us make the following simplifying assumptions:

- (i) The motion is on the  $f$ -plane ( $\beta = 0$ ). This assumption, although not particularly realistic for the Earth's atmosphere, greatly simplifies the analysis.
- (ii) The fluid is uniformly stratified; that is,  $N^2$  is a constant. This is a decent approximation for the atmosphere below the tropopause, but less so for the ocean where stratification varies considerably, being much larger in the upper ocean.
- (iii) The basic state has uniform shear; that is,  $U_0(z) = \Lambda z = Uz/H$ , where  $\Lambda$  is the (constant) shear and  $U$  is the zonal velocity at  $z = H$ , where  $H$  is the domain depth. This profile is more appropriate for the atmosphere than the ocean — below the thermocline the ocean is relatively quiescent and the shear is small.

*Eric Eady (1915–1966) and Jule Charney (1917–1981) were the two great pioneers of the mathematical theory of baroclinic instability. They worked independently in the late 1940s, each formulating and solving a problem that became the (largely unsupervised) PhD thesis of its respective author, Eady at Imperial College London and Charney at UCLA. Their achievements in this problem rank among the most significant in all the physical sciences in the twentieth century.*

- (iv) The motion is contained between two rigid, flat horizontal surfaces. In the atmosphere this corresponds to the ground and a 'lid' at a constant-height tropopause.

Although, apart from (i), these assumptions are more appropriate for the atmosphere than the ocean, the same qualitative nature of baroclinic instability carries through to the ocean.

### 8.5.1 The Linearized Problem

With a basic state streamfunction of  $\Psi = -\Lambda z y$ , the basic state potential vorticity,  $Q$ , is

$$Q = \nabla^2 \Psi + \frac{H^2}{L_d^2} \frac{\partial}{\partial z} \left( \frac{\partial \Psi}{\partial z} \right) = 0. \quad (8.49)$$

The fact that  $Q = 0$  makes the Eady problem a special case, albeit an illuminating one. The linearized potential vorticity equation is

$$\left( \frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \left( \nabla^2 \psi' + \frac{H^2}{L_d^2} \frac{\partial^2 \psi'}{\partial z^2} \right) = 0. \quad (8.50)$$

This equation has no  $x$ -dependent coefficients and in a periodic channel we may seek solutions of the form  $\psi'(x, y, z, t) = \text{Re } \tilde{\psi}(y, z) e^{ik(x-ct)}$ , yielding

$$(\Lambda z - c) \left( \frac{\partial^2 \tilde{\psi}}{\partial y^2} + \frac{H^2}{L_d^2} \frac{\partial^2 \tilde{\psi}}{\partial z^2} - k^2 \tilde{\psi} \right) = 0. \quad (8.51)$$

This equation is (8.48a) applied to the Eady problem.

#### Boundary conditions

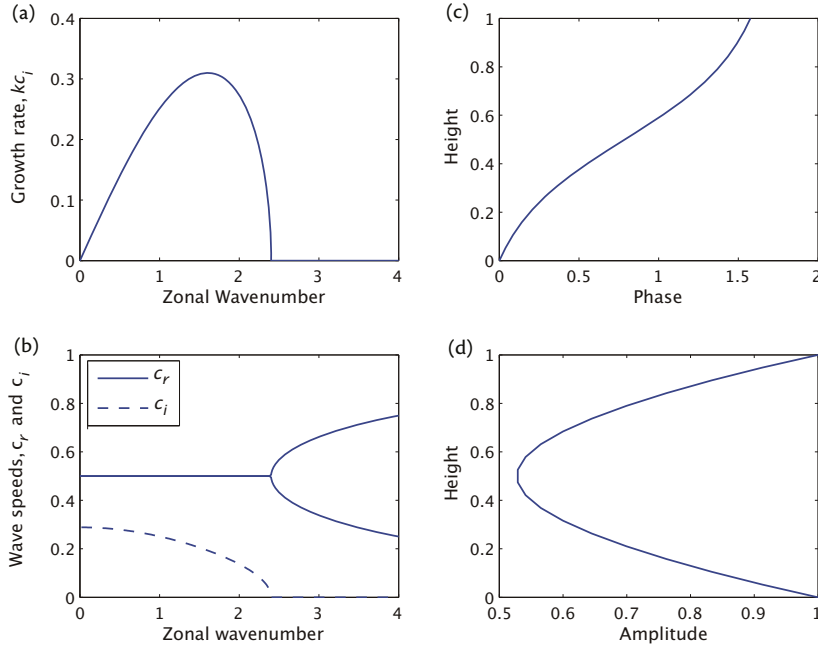
There are two sets of boundary conditions to satisfy, the vertical boundary conditions at  $z = 0$  and  $z = H$  and the lateral boundary conditions. In the horizontal plane we may either consider the flow to be in a channel, periodic in  $x$  and confined between two meridional walls, or, with a slightly greater degree of idealization but with little change to the essential dynamics, we may suppose that the domain is doubly-periodic. Either case is dealt with easily enough by the choice of geometric basis function; we choose a channel of width  $L$  and impose  $\psi = 0$  at  $y = +L/2$  and  $y = -L/2$  and, to satisfy this, seek solutions of the form  $\Psi = \Phi(z) \sin ly$  or, using (8.46)

$$\psi'(x, y, z, t) = \text{Re } \Phi(z) \sin ly e^{ik(x-ct)}, \quad (8.52)$$

where  $l = n\pi/L$ , with  $n$  being a positive integer.

The vertical boundary conditions are that  $w = 0$  at  $z = 0$  and  $z = H$ . We follow the procedure of Section 8.4.2 and from (8.45) we obtain

$$\left( \frac{\partial}{\partial t} + \Lambda z \frac{\partial}{\partial x} \right) \frac{\partial \psi'}{\partial z} - \Lambda \frac{\partial \psi'}{\partial x} = 0, \quad \text{at } z = 0, H. \quad (8.53)$$



**Fig. 8.7:** Solution of the Eady problem, in nondimensional units. (a) The growth rate,  $kc_i$ , as a function of scaled wavenumber  $\mu$ , from (8.62) with  $\Lambda = H = 1$  and for the largest meridional mode. (b) The real (solid) and imaginary (dashed) wave speeds of those modes, as a function of horizontal wavenumber. (c) The phase of the single most unstable mode as a function of height. (d) The amplitude of that mode as a function of height. To obtain dimensional values, multiply the growth rate by  $\Lambda H/L_d$  and the wavenumber by  $1/L_d$ .

### Solutions

Substituting (8.52) into (8.51) gives the interior potential vorticity equation

$$(\Lambda z - c) \left[ \frac{H^2}{L_d^2} \frac{\partial^2 \Phi}{\partial z^2} - (k^2 + l^2) \Phi \right] = 0, \quad (8.54)$$

and substituting (8.52) into (8.53) gives, at  $z = 0$  and  $z = H$ ,

$$c \frac{d\Phi}{dz} + \Lambda \Phi = 0 \quad \text{and} \quad (c - \Lambda H) \frac{d\Phi}{dz} + \Lambda \Phi = 0. \quad (8.55a,b)$$

These are equivalent to (8.48b) applied to the Eady problem. If  $\Lambda z \neq c$  then (8.54) becomes

$$H^2 \frac{d^2 \Phi}{dz^2} - \mu^2 \Phi = 0, \quad (8.56)$$

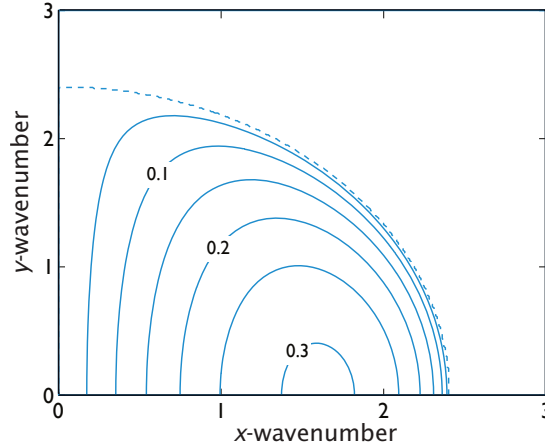
where  $\mu^2 = L_d^2(k^2 + l^2)$ . The nondimensional parameter  $\mu$  is a horizontal wavenumber, scaled by the inverse of the Rossby radius of deformation. Solutions of (8.56) are

$$\Phi(z) = A \cosh \mu \hat{z} + B \sinh \mu \hat{z}, \quad (8.57)$$

where  $\hat{z} = z/H$ ; thus,  $\mu$  determines the vertical structure of the solution. The boundary conditions (8.55) are satisfied if

$$\begin{aligned} A [\Lambda H] + B [\mu c] &= 0, \\ A [(c - \Lambda H) \mu \sinh \mu + \Lambda H \cosh \mu] + B [(c - \Lambda H) \mu \cosh \mu + \Lambda H \sinh \mu] &= 0. \end{aligned} \quad (8.58)$$

**Fig. 8.8:** Contours of the growth rate,  $\sigma$ , in the Eady problem, in the  $k$ - $l$  plane using (8.62), nondimensionalized as in Fig. 8.7. The growth rate peaks near the deformation scale, and for any given zonal wavenumber the most unstable wavenumber is that with the largest meridional scale.



Equations (8.58) are two coupled homogeneous equations in the two unknowns  $A$  and  $B$ . Non-trivial solutions only exist if the determinant of their coefficients (the terms in square brackets) vanishes, and this leads to

$$c^2 - Uc + U^2(\mu^{-1} \coth \mu - \mu^{-2}) = 0, \quad (8.59)$$

where  $U \equiv \Lambda H$  and  $\coth \mu = \cosh \mu / \sinh \mu$ . The solution of (8.59) is

$$c = \frac{U}{2} \pm \frac{U}{\mu} \left[ \left( \frac{\mu}{2} - \coth \frac{\mu}{2} \right) \left( \frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (8.60)$$

The waves, being proportional to  $\exp(-ikct)$ , will grow exponentially if  $c$  has an imaginary part. Since  $\mu/2 > \tanh(\mu/2)$  for all  $\mu$ , for an instability we require that

$$\frac{\mu}{2} < \coth \frac{\mu}{2}, \quad (8.61)$$

which is satisfied when  $\mu < \mu_c$  where  $\mu_c = 2.399$ . The growth rates of the instabilities themselves are given by the imaginary part of (8.60), multiplied by the  $x$ -wavenumber; that is

$$\sigma = kc_i = k \frac{U}{\mu} \left[ \left( \coth \frac{\mu}{2} - \frac{\mu}{2} \right) \left( \frac{\mu}{2} - \tanh \frac{\mu}{2} \right) \right]^{1/2}. \quad (8.62)$$

These solutions suggest a natural nondimensionalization: scale length by  $L_d$ , height by  $H$  and time by  $L_d/U = L_d/(\Lambda H)$ . The growth rate scales as the inverse of the time scaling and so by  $U/L_d$ . The timescale is also usefully written as

$$T_E = \frac{L_d}{U} = \frac{NH}{f_0 U} = \frac{N}{f_0 \Lambda} = \frac{1}{Fr f_0} = \frac{\sqrt{Ri}}{f_0}, \quad (8.63)$$

where  $Fr = U/(NH)$  and  $Ri = N^2/\Lambda^2$  are the Froude and Richardson numbers for this problem.

From (8.62) we can (with a little work) determine that the maximum growth rate occurs when  $\mu = \mu_m = 1.61$ . For any given  $x$ -wavenumber,

the most unstable wavenumber has the largest meridional scale, which here is  $n = 1$ , and we may further consider a wide channel so that  $l^2 \ll k^2$ . The maximum growth rate,  $\sigma_E$ , is then given by

$$\sigma_E = \frac{0.31U}{L_d} = \frac{0.31\Lambda H}{L_d} = \frac{0.31\Lambda f}{N}, \quad (8.64)$$

and this is known as the *Eady growth rate*. We have removed the subscript 0 from the Coriolis parameter here. Although  $f$  is taken as constant in the quasi-geostrophic derivation, we might wish to calculate the Eady growth rate at various locations around the globe, in which case we should use the local value of the Coriolis parameter and deformation radius. Evidently, the growth rate is proportional to the shear times the Prandtl ratio,  $f/N$ . The associated phase speed is the real part of  $c$  and is given by  $c_r = 0.5U$ .

For small  $l$  the unstable  $x$ -wavenumbers and corresponding wavelengths occur for

$$k < k_c = \frac{\mu_c}{L_d} = \frac{2.4}{L_d}, \quad \lambda > \lambda_c = \frac{2\pi L_d}{\mu_c} = 2.6L_d. \quad (8.65a,b)$$

The wavenumber and wavelength at which the instability is greatest are:

$$k_m = \frac{1.6}{L_d}, \quad \lambda_m = \frac{2\pi L_d}{\mu_m} = 3.9L_d. \quad (8.66a,b)$$

These properties can be seen in the left-hand panels of Fig. 8.7 and in Fig. 8.8.

Given  $c$ , we may use (8.58) to determine the vertical structure of the Eady wave and this is, to within an arbitrary constant factor,

$$\Phi(\tilde{z}) = \cosh \mu \tilde{z} - \frac{U}{\mu c} \sinh \mu \tilde{z} = \left( \cosh \mu \tilde{z} - \frac{U c_r \sinh \mu \tilde{z}}{\mu |c^2|} + \frac{i U c_i \sinh \mu \tilde{z}}{\mu |c^2|} \right). \quad (8.67)$$

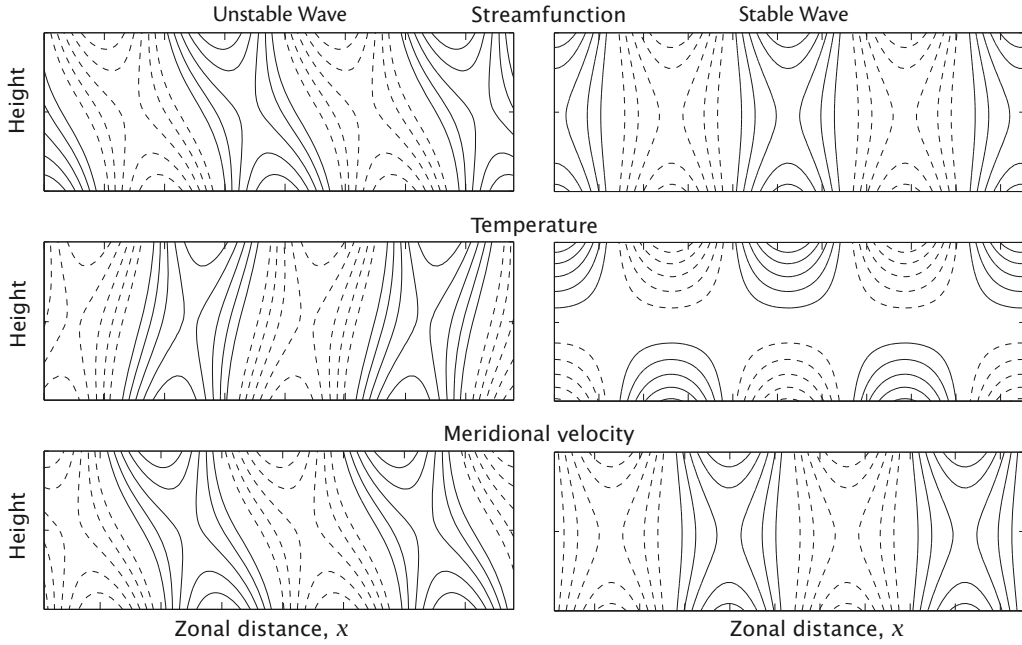
The wave therefore has a phase,  $\theta(z)$ , given by

$$\theta(\tilde{z}) = \tan^{-1} \left( \frac{U c_i \sinh \mu \tilde{z}}{\mu |c^2| \cosh \mu \tilde{z} - U c_r \sinh \mu \tilde{z}} \right). \quad (8.68)$$

The phase and amplitude of the Eady waves are plotted in the right panels of Fig. 8.7, and their overall structure in Fig. 8.9, where we see the unstable wave tilting into the shear.

### 8.5.2 Atmospheric and Oceanic Parameters

To get a qualitative sense of the nature of the instability we choose some typical parameters, as follows:



**Fig. 8.9:** Left column: vertical structure of the most unstable Eady mode. Top: contours of streamfunction. Middle: temperature,  $\partial\psi/\partial z$ . Bottom: meridional velocity,  $\partial\psi/\partial x$ . Negative contours are dashed, and two complete wavelengths are present in the horizontal direction. Polewards flowing (positive  $v$ ) air is generally warmer than equatorwards flowing air. Right column: the same, but now for a wave just beyond the short-wave cut-off.

#### For the atmosphere

Let us choose

$$H \sim 10 \text{ km}, \quad U \sim 10 \text{ m s}^{-1}, \quad N \sim 10^{-2} \text{ s}^{-1}.$$

We then obtain:

$$\begin{aligned} \text{deformation radius:} \quad L_d &= \frac{NH}{f} \approx \frac{10^{-2} 10^4}{10^{-4}} \approx 1000 \text{ km}, \\ \text{scale of max. instability:} \quad L_{\max} &\approx 3.9 L_d \approx 4000 \text{ km}, \\ \text{growth rate:} \quad \sigma &\approx 0.3 \frac{U}{L_d} \approx \frac{0.3 \times 10}{10^6} \text{ s}^{-1} \approx 0.26 \text{ day}^{-1}. \end{aligned}$$

#### For the ocean

For the main thermocline in the ocean let us choose

$$H \sim 1 \text{ km}, \quad U \approx 0.1 \text{ m s}^{-1}, \quad N \sim 10^{-2} \text{ s}^{-1}.$$

We then obtain:

$$\begin{aligned} \text{deformation radius:} \quad L_d &= \frac{NH}{f} \approx \frac{10^{-2} \times 1000}{10^{-4}} = 100 \text{ km}, \\ \text{scale of max. instability:} \quad L_{\max} &\approx 3.9 L_d \approx 400 \text{ km}, \end{aligned}$$

$$\text{growth rate: } \sigma \approx 0.3 \frac{U}{L_d} \approx \frac{0.3 \times 0.1}{10^5} \text{ s}^{-1} \approx 0.026 \text{ day}^{-1}.$$

In the ocean, the Eady problem is not quantitatively applicable because of the non-uniformity of the stratification. Nevertheless, the above estimates give a qualitative sense of the scale and growth rate of the instability relative to the corresponding values in the atmosphere. A summary of the main points of the Eady problem is given in the shaded box on the next page.

## 8.6 TWO-LEVEL BAROCLINIC INSTABILITY

The eigenfunctions displaying the largest growth rates in the Eady problem have a relatively simple vertical structure. This suggests that an even simpler mathematical model of baroclinic instability might be constructed in which the vertical structure is a priori restricted to a very simple form, namely the two-layer or two-level quasi-geostrophic (QG) model of Section 5.6. One notable advantage over the Eady model is that it is possible to include the  $\beta$ -effect in a simple way.

*This two-level instability problem is often called the 'Phillips problem,' after Norman Phillips who formulated it in 1954. He also made a great many other contributions to dynamical meteorology and was one of the pioneers of numerical weather prediction.*

### 8.6.1 Posing the Problem

For two layers or two levels of equal thickness, we write the potential vorticity equations in the dimensional form,

$$\frac{D}{Dt} \left[ \zeta_i + \beta y + \frac{k_d^2}{2} (\psi_j - \psi_i) \right] = 0, \quad i = 1, 2, \quad j = 3 - i, \quad (8.69)$$

where, using two-level notation for definiteness,

$$\frac{k_d^2}{2} = \left( \frac{2f_0}{NH} \right)^2 \rightarrow k_d = \frac{\sqrt{8}}{L_d}, \quad (8.70)$$

where  $H$  is the total depth of the domain, as in the Eady problem. The basic state we choose is

$$\Psi_1 = -U_1 y, \quad \Psi_2 = -U_2 y = +U_1 y. \quad (8.71)$$

It is possible to choose  $U_2 = -U_1$  without loss of generality because there is no topography and the system is Galilean invariant. The basic state potential vorticity gradient is then given by

$$Q_1 = \beta y + k_d^2 U y, \quad Q_2 = \beta y - k_d^2 U y, \quad (8.72)$$

where  $U = U_1$ . (Note that  $U$  differs by a constant multiplicative factor from the  $U$  in the Eady problem.) Even in the absence of  $\beta$  there is a non-zero potential vorticity gradient. Why should this be different from the Eady problem? — after all, the shear is uniform in both problems. The difference arises from the vertical boundary conditions. In the two-layered formulation the temperature gradient at the boundary is *absorbed into the definition of the potential vorticity in the interior*. This results in a

### Some Results in Baroclinic Instability

#### Eady Problem

- The length scales of the instability are characterized by the deformation scale. The most unstable scale has a wavelength a few times larger than the deformation radius  $L_d$ , where  $L_d = NH/f$ .
- The growth rate of the instability is approximately

$$\sigma_E \approx \frac{0.3U}{L_d} = 0.3\Lambda \frac{f}{N}. \quad (\text{B.1})$$

That is, it is proportional to the shear scaled by the Prandtl ratio  $f/N$ . The value  $\sigma_E$  is known as the *Eady growth rate*.

- There is a *short-wave cutoff* beyond which (i.e., at higher wavenumber than) there is no instability. This occurs near the deformation radius.
- The two-layer (Phillips) problem with zero beta captures many of the results of the Eady model, including the short-wave cutoff.

#### Effects of beta

- Potential vorticity may now change sign because of the relation between the surface temperature gradient and the interior potential vorticity gradient. Thus, in the continuously-stratified problem, unstable modes do not need to extend the full depth of the atmosphere.
- There is a long-wave cut-off to the main instability branch. At scales larger than this the instabilities are slowly growing, and absent in the two-layer (Phillips) problem.
- In the continuously-stratified problem there is no short-wave cut-off. These modes are slowly growing and shallow, and in the two-layer model they are absent.
- In the two-layer model with beta, there is a minimum shear,  $\Lambda_c$ , for instability given by

$$\Lambda_c = \frac{\beta H}{2} \frac{N^2}{f^2}. \quad (\text{B.2})$$

This shear does not arise in the continuous problem, but it may be a useful criterion for the onset of rapidly growing deep modes.

non-zero interior potential vorticity gradient at the two levels adjacent to the boundary (the only layers in the two-layer problem), but with isothermal boundary conditions. In the Eady problem we have a zero interior gradient of potential vorticity but a temperature gradient at the top and bottom boundary.

The linearized potential vorticity equation is, for each layer,

$$\frac{\partial q'_i}{\partial t} + U_i \frac{\partial q'_i}{\partial x} + v'_i \frac{\partial Q_i}{\partial y} = 0, \quad i = 1, 2, \quad (8.73)$$



or, more explicitly,

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \left[ \nabla^2 \psi'_1 + \frac{k_d^2}{2} (\psi'_2 - \psi'_1) \right] + \frac{\partial \psi'_1}{\partial x} (\beta + k_d^2 U) = 0, \quad (8.74a)$$

$$\left( \frac{\partial}{\partial t} - U \frac{\partial}{\partial x} \right) \left[ \nabla^2 \psi'_2 + \frac{k_d^2}{2} (\psi'_1 - \psi'_2) \right] + \frac{\partial \psi'_2}{\partial x} (\beta - k_d^2 U) = 0. \quad (8.74b)$$

For simplicity let us set the problem in a square, doubly periodic domain, and so seek solutions of the form,

$$\psi'_i = \text{Re } \tilde{\psi}_i e^{i(kx + ly - \omega t)} = \text{Re } \tilde{\psi}_i e^{ik(x - ct)} e^{ily}, \quad i = 1, 2. \quad (8.75)$$

Here,  $k$  and  $l$  are the  $x$ - and  $y$ -wavenumbers, and  $(k, l) = (2\pi/L)(m, n)$ , where  $L$  is the size of the domain, and  $m$  and  $n$  are integers. The constant  $\tilde{\psi}_i$  is the complex amplitude.

### 8.6.2 The Solution

Substituting (8.75) into (8.74) we obtain

$$[ik(U - c)] [-K^2 \tilde{\psi}_1 + k_d^2 (\tilde{\psi}_2 - \tilde{\psi}_1)/2] + ik \tilde{\psi}_1 (\beta + k_d^2 U) = 0, \quad (8.76a)$$

$$[-ik(U + c)] [-K^2 \tilde{\psi}_2 + k_d^2 (\tilde{\psi}_1 - \tilde{\psi}_2)/2] + ik \tilde{\psi}_2 (\beta - k_d^2 U) = 0, \quad (8.76b)$$

where  $K^2 = k^2 + l^2$ . A little algebra turns these equations into a quadratic equation for  $c$ , and solving that gives

$$c = -\frac{\beta}{K^2 + k_d^2} \left\{ 1 + \frac{k_d^2}{2K^2} \pm \frac{k_d^2}{2K^2} \left[ 1 + \frac{4K^4(K^4 - k_d^4)}{k_\beta^4 k_d^4} \right]^{1/2} \right\}, \quad (8.77)$$

where  $K^4 = (k^2 + l^2)^2$  and  $k_\beta = \sqrt{\beta/U}$  (its inverse is known as the Kuo scale). We may nondimensionalize this equation using the deformation radius  $L_d$  as the length scale and the shear velocity  $U$  as the velocity scale. Then, denoting nondimensional parameters with hats, we have

$$k = \frac{\hat{k}}{L_d}, \quad c = \hat{c} U, \quad t = \frac{L_d}{U} \hat{t}, \quad (8.78)$$

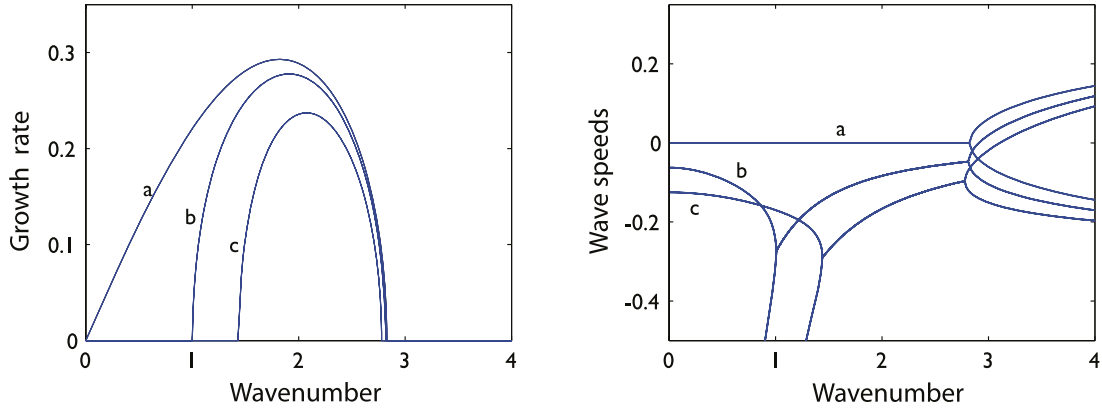
and the nondimensional form of (8.77) is just

$$\hat{c} = -\frac{\hat{k}_\beta^2}{\hat{K}^2 + \hat{k}_d^2} \left\{ 1 + \frac{\hat{k}_d^2}{2\hat{K}^2} \pm \frac{\hat{k}_d^2}{2\hat{K}^2} \left[ 1 + \frac{4\hat{K}^4(\hat{K}^4 - \hat{k}_d^4)}{\hat{k}_\beta^4 \hat{k}_d^4} \right]^{1/2} \right\}, \quad (8.79)$$

where  $\hat{k}_\beta = k_\beta L_d$  and  $\hat{k}_d = \sqrt{8}$ , as in (8.70). The nondimensional parameter

$$\gamma = \frac{1}{4} \hat{k}_\beta^2 = \frac{\beta L_d^2}{4U}, \quad (8.80)$$

is often useful as a measure of the importance of  $\beta$ ; it is proportional to the ratio of the deformation radius to the Kuo scale  $\sqrt{U/\beta}$ , squared. Let us look at two special cases first, before considering the general solution to these equations.



**Fig. 8.10:** Growth rates and wave speeds for the two-layer baroclinic instability problem, from (8.79), with three (nondimensional) values of  $\beta$  as labelled: a,  $\gamma = 0$  ( $\hat{k}_\beta = 0$ ); b,  $\gamma = 0.5$  ( $\hat{k}_\beta = \sqrt{2}$ ); c,  $\gamma = 1$  ( $\hat{k}_\beta = 2$ ). As  $\beta$  increases, so does the low-wavenumber cut-off to instability, but the high-wavenumber cut-off is little changed. The solutions are obtained from (8.79), with  $\hat{k}_d = \sqrt{8}$  and  $U_1 = -U_2 = 1/4$ .

### I. Zero shear, non-zero $\beta$

If there is no shear (i.e.,  $U = 0$ ) then the phase speeds are purely real and are given by

$$c = -\frac{\beta}{K^2} \quad \text{and} \quad c = -\frac{\beta}{K^2 + k_d^2}. \quad (8.81)$$

The first of these is the dispersion relationship for Rossby waves in a purely barotropic flow, and corresponds to the eigenfunction  $\tilde{\psi}_1 = \tilde{\psi}_2$ . The second solution corresponds to the baroclinic eigenfunction  $\tilde{\psi}_1 + \tilde{\psi}_2 = 0$ .

### II. Zero $\beta$ , non-zero shear

If  $\beta = 0$ , then (8.76) yields, after a little algebra,

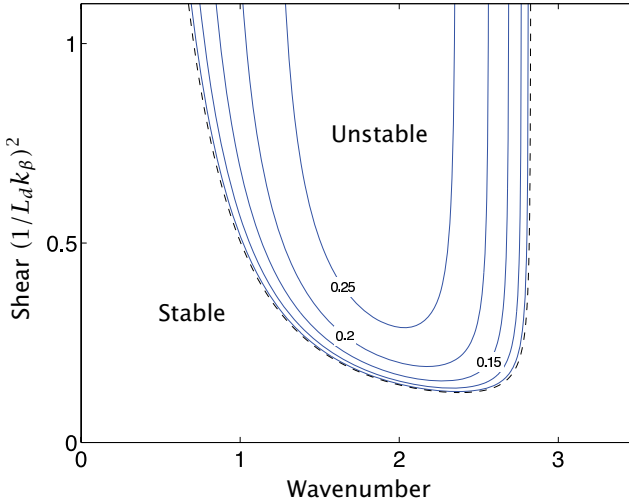
$$c = \pm U \left( \frac{K^2 - k_d^2}{K^2 + k_d^2} \right)^{1/2} \quad \text{or} \quad \sigma = Uk \left( \frac{k_d^2 - K^2}{K^2 + k_d^2} \right)^{1/2}, \quad (8.82)$$

where  $\sigma = -i\omega$  is the growth rate. These expressions are similar to those in the Eady problem. We note the following:

- There is an instability for *all* values of  $U$ .
- There is a high-wavenumber cut-off, at a scale proportional to the radius of deformation. For the two-layer model, if  $K > k_d = 2.82/L_d$  there is no growth. For the Eady problem, the high wavenumber cut-off occurs at  $2.4/L_d$ .
- There is no low wavenumber cut-off.
- For any given  $k$ , the highest growth rate occurs for  $l = 0$ . In the two-layer model, from (8.82), for  $l = 0$  the maximum growth rate occurs when  $k = 0.634k_d = 1.79/L_d$ . For the Eady problem, the maximum growth rate occurs at  $1.61/L_d$ .

### Solution with non-zero $\beta$

Using (8.79), the growth rate and wave speeds as a function of wavenumber are plotted in Fig. 8.10. We observe that there still appears to be a



**Fig. 8.11:** Contours of growth rate in the two-layer baroclinic instability problem. The dashed line is the neutral stability curve, (8.84), and the other curves are contours of growth rates obtained from (8.79). The wavenumber is scaled by  $1/L_d$  (i.e., by  $k_d/\sqrt{8}$ ) and growth rates are scaled by the inverse of the Eady time scale (i.e., by  $U/L_d$ ). Thus, for  $L_d = 1000$  km and  $U = 10$  m s $^{-1}$ , a nondimensional growth rate of 0.25 corresponds to a dimensional growth rate of  $0.25 \times 10^{-5}$  s $^{-1} = 0.216$  day $^{-1}$ .

high-wavenumber cut-off and, for  $\beta = 0$ , there is a low-wavenumber cut-off. A little analysis elucidates the origin of these features:

#### The neutral curve

For instability, there must be an imaginary component to the phase speed in (8.77); that is, we require

$$k_\beta^4 k_d^4 + 4K^4(K^4 - k_d^4) < 0. \quad (8.83)$$

This is a quadratic equation in  $K^4$  for the value of  $K$ ,  $K_c$  say, at which the growth rate is zero. Solving, we find

$$K_c^4 = \frac{1}{2} k_d^4 \left( 1 \pm \sqrt{1 - k_\beta^4/k_d^4} \right), \quad (8.84)$$

and this is plotted in Fig. 8.11.

#### Minimum shear for instability

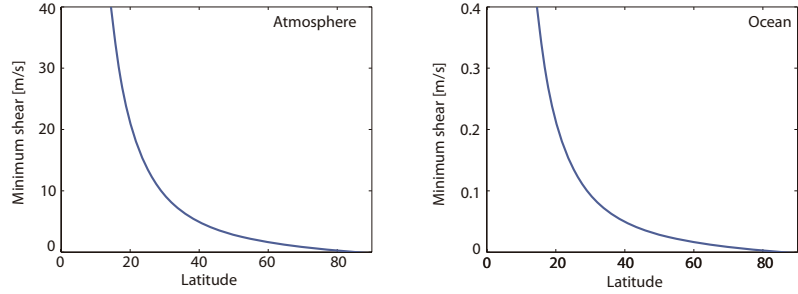
From (8.83), instability arises when  $\beta^2 k_d^4/U^2 < 4K^4(k_d^4 - K^4)$ . The maximum value of the right-hand side of this expression arises when  $K^4 = \tilde{k}_d^4/2$ ; thus, instability arises only when

$$\frac{\beta^2 k_d^4}{U^2} < 4 \frac{k_d^4}{2} \frac{k_d^4}{2} \quad \text{or} \quad \kappa_\beta < k_d. \quad (8.85)$$

That is, *instability only arises if the deformation radius is sufficiently smaller than the Kuo scale,  $\sqrt{U/\beta}$ . The critical velocity difference required for instability is then*

$$U_1 - U_2 > U_c = \frac{2\beta}{k_d^2} = \frac{1}{4} \beta L_d^2, \quad (8.86)$$

**Fig. 8.12:** The minimum shear (the velocity difference  $U_1 - U_2$ ) required for baroclinic instability in a two-layer model, calculated using (8.86), i.e.,  $U_{\min} = \beta L_d^2/4$  where  $\beta = 2\Omega a^{-1} \cos \vartheta$  and  $L_d = NH/f$ , with  $f = 2\Omega \sin \vartheta$ . The left panel uses  $H = 10$  km and  $N = 10^{-2} \text{ s}^{-1}$ , and the right panel uses parameters representative of the main thermocline,  $H = 1$  km and  $N = 10^{-2} \text{ s}^{-1}$ . The results are not quantitatively accurate, but the implications that the minimum shear is much less for the ocean, and that in both the atmosphere and the ocean the shear increases rapidly at low latitudes, are robust.



recalling our notation that  $U_1 - U_2 = 2U$  and that  $k_d^2 = 8/L_d^2$ , as in (8.70). The critical shear for instability is

$$\Lambda_c = \frac{\beta H N^2}{2 f^2}, \quad (8.87a,b)$$

where the shear  $\Lambda$  is defined by  $(U_1 - U_2)/(0.5H)$ , where  $H$  is the total depth of the domain. Note the relationship of the minimum shear to the basic state potential vorticity gradient in the respective layers. In the upper and lower layers the potential vorticity gradients are given by, respectively,

$$\frac{\partial Q_1}{\partial y} = \beta + k_d^2 U, \quad \frac{\partial Q_2}{\partial y} = \beta - k_d^2 U. \quad (8.88a,b)$$

Thus, the requirement for instability is exactly that which causes the potential vorticity gradient to change sign somewhere in the domain, in this case becoming negative in the lower layer.

Figure 8.12 sketches how the critical shear varies — very approximately — with latitude in the atmosphere and ocean because of the variation of the Coriolis parameter. (In any given quasi-geostrophic calculation  $f$  is held constant, but if we wish to see how the critical shear varies with latitude we vary  $f$  accordingly.)

#### High-wavenumber cut-off

Instability can only arise when, from (8.83),

$$4K^4(k_d^4 - K^4) > k_\beta^4 k_d^4, \quad (8.89)$$

so that a necessary condition for instability is

$$k_d^2 > K^2. \quad (8.90)$$

Thus, waves shorter than the deformation radius are always stable, no matter what the value of  $\beta$ .

#### Low-wavenumber cut-off

Suppose that  $K \ll k_d$ . Then (8.83) simplifies to  $k_\beta^4 < 4K^4$ . That is,

for instability we require

$$K^2 > \frac{1}{2} k_\beta^2 = \frac{\beta}{2U}. \quad (8.91)$$

Thus, using (8.90) and (8.91) the unstable waves lie approximately in the interval  $\beta/(\sqrt{2}U) < K < k_d$ .

### 8.7♦ NECESSARY CONDITIONS FOR BAROCLINIC INSTABILITY

As in the two-dimensional case discussed in Section 8.3, we can derive a necessary condition for instability, or a sufficient condition for stability, for small perturbations. The procedure is entirely analogous to that of Section 8.3 but now we must take into account boundary conditions at the top and bottom and this complicates the algebra. Trusting readers may skip to the results.

Starting with the linear potential vorticity equation, (8.44), we multiply by  $q'$  and divide by  $Q_y$  to give (dropping the primes on perturbed quantities to avoid clutter)

$$\frac{\partial}{\partial t} \frac{q^2}{Q_y} + \frac{U}{Q_y} \frac{\partial q^2}{\partial x} + 2vq = 0, \quad 0 < z < H, \quad (8.92)$$

where, as before,  $Q$  is the background potential vorticity and  $Q_y$  its meridional gradient. The above equation holds in the fluid interior, and at the top and bottom boundary we use (8.45) to similarly obtain

$$\frac{\partial b^2}{\partial t} + \frac{U}{B_y} \frac{\partial b^2}{\partial x} + 2vb = 0, \quad z = 0, H, \quad (8.93)$$

where  $B$  is buoyancy at the boundary and  $B_y$  its meridional gradient.

Now, the poleward eddy flux of potential vorticity is

$$\overline{vq} = -\frac{\partial}{\partial y} \overline{uv} + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \overline{vb} \right), \quad (8.94)$$

and integrating this expression with respect to both  $y$  and  $z$  gives

$$\int_A \overline{vq} \, dy \, dz = \left[ \frac{f_0}{N^2} \overline{vb} \right]_0^H, \quad (8.95)$$

assuming that the meridional boundaries are at quiescent latitudes. Integrating (8.92) over  $y$  and  $z$  and using (8.95) gives

$$\frac{d}{dt} \iint \frac{1}{2} \frac{\overline{q^2}}{Q_y} \, dy \, dz = - \left[ \frac{f_0}{N^2} \overline{v'b'} \right]_0^H. \quad (8.96)$$

Using (8.93) to eliminate  $\overline{vb}$  finally gives

$$\frac{d}{dt} \left\{ \iint \frac{1}{2} \frac{\overline{q'^2}}{Q_y} \, dy \, dz - \int \left[ \frac{1}{2} \frac{f_0}{N^2} \frac{\overline{b^2}}{B_y} \right]_0^H \, dy \right\} = 0. \quad (8.97)$$

The perturbation can only grow if the expression within curly brackets has both positive and negative terms, because (8.97) tells us that the expression is a constant. Thus, If the expression is positive or negative definite the perturbation cannot grow and the basic state is stable. Stability thus depends on the meridional gradient of potential vorticity in the interior, and the meridional gradient of buoyancy at the boundary. If  $Q_y$  changes sign in the interior, or  $B_y$  changes sign at the boundary, we have the potential for instability. If these are both one signed, then various possibilities exist, and using the thermal wind relation ( $f_0 \partial U / \partial z = -\partial B / \partial y$ ) we obtain the following:

I. *A stable case:*

$$\frac{\partial Q}{\partial y} > 0 \text{ and } \left. \frac{\partial U}{\partial z} \right|_{z=0} < 0 \text{ and } \left. \frac{\partial U}{\partial z} \right|_{z=H} > 0 \Rightarrow \text{stability. (8.98)}$$

Stability also ensues if all inequalities are switched.

II. *Instability via interior–surface interactions:*

$$\frac{\partial Q}{\partial y} > 0 \text{ and } \left. \frac{\partial U}{\partial z} \right|_{z=0} > 0 \text{ or } \left. \frac{\partial U}{\partial z} \right|_{z=H} < 0 \Rightarrow \text{potential instability. (8.99)}$$

The condition  $\partial Q / \partial y > 0$  and  $(\partial U / \partial z)_{z=0} > 0$  (or  $f_0^{-1} \partial B / \partial y < 0$ ) is the most common criterion for instability that is met in the atmosphere. Stability is then determined by the interior potential vorticity gradient and the surface buoyancy gradient.

III. *Instability via interaction between edge waves at top and bottom:*

$$\left. \frac{\partial U}{\partial z} \right|_{z=0} > 0 \text{ and } \left. \frac{\partial U}{\partial z} \right|_{z=H} > 0 \Rightarrow \text{potential instability. (8.100)}$$

(And similarly, with both inequalities switched.) Such an instability may occur where the troposphere acts like a lid, as for example in the Eady problem. If  $Q_y = 0$  and there is no lid at  $z = H$  (e.g., the Eady problem with no lid) then the instability disappears.

Readers should explore the consequences of these conditions for themselves, and in particular how they are consistent with the instability results from the Eady problem and the two-level problem calculated directly. The necessary conditions above are in some ways more powerful than the explicit calculations for they do not rely on the instability being of normal mode form, although they do not supply the actual growth rates.

## Notes and References

The book by Drazin & Reid (1981) discusses many classical problems in hydrodynamic stability, although its treatment of baroclinic instability is rather brief. The continuously stratified baroclinic instability problem with non-zero beta, namely the Charney problem, is discussed by Pedlosky (1987a).

## Problems

- 8.1 Using values of physical parameters obtained from the literature, estimate the baroclinic growth rates on Venus, Mars and Jupiter, using the Eady growth rate and/or the growth rate determined from the two-level model. (Make sensible choices as to the physical parameters on these planets, but there is no need to do a detailed calculation.) Given these results, discuss whether baroclinic instability is likely to be a significant source of atmospheric motion on those planets.
- 8.2 Show, using the two-layer model (or otherwise) that the presence of  $\beta$  reduces the efficiency of baroclinic instability. For example, show that it makes the meridional velocity slightly out of phase with the temperature.
- 8.3 Following the same procedure used in Sections 8.3 and 8.7, obtain the necessary conditions for instability in the two-level quasi-geostrophic model in the case with uniform shear. Show that these conditions are consistent with the conditions for instability calculated directly with a normal-mode approach.
- 8.4 Suppose that the equator to pole temperature gradient on Earth and all other parameters remain fixed, except that Earth's rotation is slowed. At what value of rotation will baroclinic instability cease, and why? (You may use the Eady model, or the two-layer model with beta, or another of your choice.) Suppose now that the rotation rate is increased. Will baroclinic instability ever cease to be important, and why? Briefly discuss and justify your answers in both cases.
- 8.5 *Stability conditions in the continuously stratified QG model*  
Consider the modified Eady problem (Boussinesq, uniform stratification, flow contained between two flat horizontal surfaces at 0 and  $H$ ), but instead of a uniform shear suppose that the basic state is given by

$$U = -U_0 \cos(\pi m z / H), \quad (\text{P8.1})$$

where  $U_0 > 0$ , and allow  $\beta$  to be non-zero.

- (a) Show there is a critical shear, and that this diminishes as  $m$  increases.  
(b) Compare the results to those of the original Eady model and two-layer model with  $\beta$ .

*Sketch of solution to (a).*

The basic state has no temperature gradient at the boundary, and so stability is assured if  $\partial Q / \partial y$  in the interior is positive everywhere. The basic state potential vorticity is  $Q = \nabla^2 \Psi + (f_0^2 / N^2) \partial^2 \Psi / \partial z^2 + \beta y$ , so that

$$\frac{dQ}{dy} = \beta - k_d^2 \pi^2 m^2 U_0 \cos(\pi p z / H), \quad (\text{P8.2})$$

where  $k_d^2 = f_0^2 H^2 / N^2$  and  $U = -\partial \Psi / \partial y$ . Thus, stability is assured if  $U_0 < (\beta / k_d^2 \pi^2 m^2)$ .

## Waves and Mean-Flows

**L**INEAR DYNAMICS is mainly concerned with waves and instabilities that live on a pre-determined background flow. But the real world isn't quite like that. Rather, the mean state is the *result* of the combined effects of thermal and mechanical forcing (by radiation from the sun and, for the ocean, the winds) *plus the action of the waves and instabilities themselves*. In this chapter we explore the geophysical fluid dynamics underlying such wave–mean-flow interactions. We try to keep our discussion as elementary as possible by staying within the comfortable bounds of the quasi-geostrophic approximation and considering only zonal averages. Nevertheless the subject is often regarded as an advanced one, and all the sections in this chapter may be considered to be implicitly marked with a diamond, ♦.

### 9.1 QUASI-GEOSTROPHIC WAVE–MEAN-FLOW INTERACTION

#### 9.1.1 Preliminaries

The Jacobian operator for any two quantities  $a$  and  $b$  is given by  $J(a, b) = \partial_x a \partial_y b - \partial_y a \partial_x b$ . When  $a$  is the streamfunction the Jacobian gives the horizontal advection of  $b$ , namely  $\mathbf{u} \cdot \nabla b$ .

To fix our dynamical system and notation, we write down the Boussinesq quasi-geostrophic potential vorticity equation

$$\frac{\partial q}{\partial t} + J(\psi, q) = D, \quad (9.1)$$

where  $J(\psi, q) = \partial\psi/\partial x \partial q/\partial y - \partial\psi/\partial y \partial q/\partial x$  and  $D$  represents any non-conservative terms. The potential vorticity in a Boussinesq system is

$$q = \beta y + \zeta + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} b \right), \quad (9.2)$$

where  $\zeta$  is the relative vorticity and  $b$  is the buoyancy perturbation from a background state characterized by  $N^2$ . (Nearly all the derivations in this chapter could be done in pressure coordinates with minor modifications.)



We refer to lines of constant  $b$  as isentropes. In terms of the streamfunction, the variables are

$$\zeta = \nabla^2 \psi, \quad b = f_0 \frac{\partial \psi}{\partial z}, \quad q = \beta y + \left[ \nabla^2 + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \psi, \quad (9.3)$$

where  $\nabla^2 \equiv (\partial_x^2 + \partial_y^2)$ . The potential vorticity equation holds in the fluid interior; the boundary conditions on (9.3) are provided by the thermodynamic equation

$$\frac{\partial b}{\partial t} + J(\psi, b) + wN^2 = S, \quad (9.4)$$

where  $S$  represents heating terms. The vertical velocity at the boundary,  $w$ , is zero in the absence of topography and Ekman friction so that the boundary condition is just

$$\frac{\partial b}{\partial t} + J(\psi, b) = S. \quad (9.5)$$

Equations (9.1) and (9.5) are the evolution equations for the system, and if both  $D$  and  $S$  are zero they conserve both the total energy,  $\hat{E}$  and the total enstrophy,  $\hat{Z}$ :

$$\begin{aligned} \frac{d\hat{E}}{dt} &= 0, & \hat{E} &= \frac{1}{2} \int_V (\nabla \psi)^2 + \frac{f_0^2}{N^2} \left( \frac{\partial \psi}{\partial z} \right)^2 dV, \\ \frac{d\hat{Z}}{dt} &= 0, & \hat{Z} &= \frac{1}{2} \int_V q^2 dV, \end{aligned} \quad (9.6)$$

where  $V$  is a volume bounded by surfaces at which the normal velocity is zero, or that has periodic boundary conditions. The enstrophy is also conserved layerwise; that is, the horizontal integral of  $q^2$  is conserved at every level.

### 9.1.2 Potential Vorticity Flux in the Linear Equations

Let us decompose the fields into a mean (to be denoted with an overbar) plus a perturbation (denoted with a prime), and let us suppose the perturbation fields are of small amplitude. (In linear problems, such as those considered in Chapter 8, we decomposed the flow into a 'basic state' plus a perturbation, with the basic state fixed in time. Our approach here is similar, but soon we will allow the mean state to evolve.) The linearized quasi-geostrophic potential vorticity equation is then

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + u' \frac{\partial \bar{q}}{\partial x} + \bar{v} \frac{\partial q'}{\partial y} + v' \frac{\partial \bar{q}}{\partial y} = D', \quad (9.7)$$

where  $D'$  represents eddy forcing and dissipation and, in terms of streamfunction,

$$(u'(x, y, z, t), v'(x, y, z, t)) = \left( -\frac{\partial \psi'}{\partial y}, \frac{\partial \psi'}{\partial x} \right), \quad (9.8a)$$

$$q'(x, y, z, t) = \nabla^2 \psi' + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial z} \right). \quad (9.8b)$$

If the mean is a zonal mean then  $\partial \bar{q} / \partial x = 0$  and  $\bar{v} = 0$  (because  $v$  is purely geostrophic) and (9.7) simplifies to

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = D', \quad (9.9)$$

where

$$\bar{q} = \beta y - \frac{\partial \bar{u}}{\partial y} + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \bar{b} \right), \quad \frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right), \quad (9.10a,b)$$

having used the thermal wind relation,

$$f_0 \frac{\partial \bar{u}}{\partial z} = - \frac{\partial \bar{b}}{\partial y}. \quad (9.11)$$

Multiplying (9.9) by  $q'$  and zonally averaging gives the enstrophy equation:

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{q'^2} = - \overline{v' q' \frac{\partial \bar{q}}{\partial y}} + \overline{D' q'}. \quad (9.12)$$

The quantity  $\overline{v' q'}$  is the meridional flux of potential vorticity; this is downgradient (by definition) when the first term on the right-hand side is positive (i.e.,  $\overline{v' q' \partial \bar{q} / \partial y} < 0$ ), and it then acts to increase the variance of the perturbation. (This occurs, for example, when the flux is diffusive so that  $\overline{v' q'} = -\kappa \partial \bar{q} / \partial y$ , where  $\kappa$  may vary but is everywhere positive.) This argument may be inverted: for unforced, inviscid flow ( $D = 0$ ), if the waves are growing, as for example in the canonical models of baroclinic instability discussed in Chapter 8, then *the potential vorticity flux is downgradient*.

If the second term on the right-hand side of (9.12) is negative, as it will be if  $D'$  is a dissipative process (e.g., if  $D' = A \nabla^2 q'$  or if  $D' = -r q'$ , where  $A$  and  $r$  are positive) then a statistical balance can be achieved between enstrophy production via downgradient transport, and dissipation. If the waves are steady (by which we mean statistically steady, neither growing nor decaying in amplitude) and conservative (i.e.,  $D' = 0$ ) then we must have

$$\overline{v' q'} = 0. \quad (9.13)$$

Similar results follow for the buoyancy at the boundary; we start by linearizing the thermodynamic equation (9.5) to give

$$\frac{\partial b'}{\partial t} + \bar{u} \frac{\partial b'}{\partial x} + v' \frac{\partial \bar{b}}{\partial y} = S', \quad (9.14)$$

where  $S'$  is a diabatic source term. Multiplying (9.14) by  $b'$  and averaging gives

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{b'^2} = - \overline{v' b' \frac{\partial \bar{b}}{\partial y}} + \overline{S' b'}. \quad (9.15)$$

If the flow is adiabatic ( $S' = 0$ ) then growing waves have a downgradient flux of buoyancy at the boundary. In the Eady problem there is no interior gradient of basic-state potential vorticity and all the terms in (9.12) are zero, but the perturbation grows at the boundary. If the waves are steady and adiabatic then, analogously to (9.13),  $\overline{v'q'} = 0$ . In models with discrete vertical layers or a finite number of levels it is common practice to absorb the boundary conditions into the definition of potential vorticity at top and bottom, as in the two level model of Section 5.6.

### 9.1.3 Wave–Mean-Flow Interaction

In linear problems we usually suppose that the mean-flow is fixed and that the zonal mean terms,  $\bar{u}$  and  $\bar{q}$  in (9.9), are functions only of  $y$  and  $z$ . However, in reality we might expect that the mean-flow would change because of momentum and heat flux convergences arising from the eddy–eddy interactions. To calculate these changes we begin with the potential vorticity equation (9.1) and, in the usual way, express the variables as a zonal mean plus an eddy term and obtain

$$\frac{\partial \bar{q}}{\partial t} + \bar{v} \frac{\partial \bar{q}}{\partial y} + \frac{\partial}{\partial y} (\overline{v'q'}) = \bar{D}. \quad (9.16)$$

Now,  $\bar{v} = 0$  (since the flow is geostrophic) and the mean-flow thus evolves according to

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial}{\partial y} \overline{v'q'} = \bar{D}. \quad (9.17)$$

Similarly, at the boundary the mean buoyancy evolution equation is

$$\frac{\partial \bar{b}}{\partial t} + \frac{\partial}{\partial y} \overline{v'b'} = \bar{S}. \quad (9.18)$$

To obtain  $\bar{u}$  from  $\bar{q}$  and  $\bar{b}$  we use thermal wind balance, (9.11), to define a streamfunction  $\Psi$ . That is, since  $f_0 \partial \bar{u} / \partial z = -\partial \bar{b} / \partial y$ , then

$$\left( \bar{u}, \frac{1}{f_0} \bar{b} \right) = \left( -\frac{\partial \Psi}{\partial y}, \frac{\partial \Psi}{\partial z} \right), \quad (9.19)$$

whence, using (9.10a), the zonal mean potential vorticity is

$$\bar{q}(y, z, t) - \beta y = \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \Psi}{\partial z} \right) + \frac{\partial^2 \Psi}{\partial y^2}. \quad (9.20)$$

If  $\bar{q}$  is known in the interior from (9.18), and  $\bar{b}$  (i.e.,  $f_0 \partial \Psi / \partial z$ ) is known at the boundaries, then  $\bar{u}$  and  $\bar{b}$  in the interior may be obtained using (9.20) and (9.19b). The equations are also summarized in the shaded box on page 176.

To close the system we suppose that the eddy terms themselves evolve according to (9.9) and (9.14). If in those equations we were to include the eddy–eddy interaction terms we would simply recover the full system, so

in neglecting those terms we have constructed an eddy–mean-flow system, commonly called a *wave–mean-flow* system because by eliminating the nonlinear terms in the perturbation equation the eddies will often be wavelike. It is important to realise that such systems do differ from linear ones in which we regard the mean flow as fixed; we have gone one step further and allowed the mean flow to evolve because of the effects of eddies, but we do not allow the eddies to interact with themselves.

We now consider some more properties of the waves themselves — how they propagate and what they conserve — beginning with a discussion of the potential vorticity flux and its relative, the so-called Eliassen–Palm flux.

## 9.2 POTENTIAL VORTICITY FLUX

The eddy flux of potential vorticity may be expressed in terms of vorticity and buoyancy fluxes as

$$v'q' = v'\zeta' + f_0v' \frac{\partial}{\partial z} \left( \frac{b'}{N^2} \right). \quad (9.21)$$

The second term on the right-hand side can be written as

$$\begin{aligned} f_0v' \frac{\partial}{\partial z} \left( \frac{b'}{N^2} \right) &= f_0 \frac{\partial}{\partial z} \left( \frac{v'b'}{N^2} \right) - f_0 \frac{\partial v'}{\partial z} \frac{b'}{N^2} \\ &= f_0 \frac{\partial}{\partial z} \left( \frac{v'b'}{N^2} \right) - f_0 \frac{\partial}{\partial x} \left( \frac{\partial \psi'}{\partial z} \right) \frac{b'}{N^2} \\ &= f_0 \frac{\partial}{\partial z} \left( \frac{v'b'}{N^2} \right) - \frac{f_0^2}{2N^2} \frac{\partial}{\partial x} \left( \frac{\partial \psi'}{\partial z} \right)^2, \end{aligned} \quad (9.22)$$

using  $b' = f_0 \partial \psi' / \partial z$ .

Similarly, the flux of relative vorticity can be written

$$v'\zeta' = -\frac{\partial}{\partial y}(u'v') + \frac{1}{2} \frac{\partial}{\partial x}(v'^2 - u'^2), \quad (9.23)$$

and using (9.22) and (9.23), (9.21) becomes

$$v'q' = -\frac{\partial}{\partial y}(u'v') + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} v'b' \right) + \frac{1}{2} \frac{\partial}{\partial x} \left( (v'^2 - u'^2) - \frac{b'^2}{N^2} \right). \quad (9.24)$$

Thus the meridional potential vorticity flux, in the quasi-geostrophic approximation, can be written as the divergence of a vector:  $v'q' = \nabla \cdot \mathcal{E}$  where

$$\mathcal{E} \equiv \frac{1}{2} \left( (v'^2 - u'^2) - \frac{b'^2}{N^2} \right) \hat{\mathbf{i}} - (u'v') \hat{\mathbf{j}} + \left( \frac{f_0}{N^2} v'b' \right) \hat{\mathbf{k}}. \quad (9.25)$$

A particularly useful form of this arises after zonally averaging, for then (9.24) becomes

$$\overline{v'q'} = -\frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \overline{v'b'} \right). \quad (9.26)$$

The vector defined by

$$\mathcal{F} \equiv -\overline{u'v'} \hat{\mathbf{j}} + \frac{f_0}{N^2} \overline{v'b'} \hat{\mathbf{k}} \quad (9.27)$$

is the wave activity flux, often called the (quasi-geostrophic) Eliassen–Palm (EP) flux, and its divergence, given by (9.26), gives the poleward flux of potential vorticity:

$$\overline{v'q'} = \nabla_x \cdot \mathcal{F}, \quad (9.28)$$

where  $\nabla_x \cdot \equiv (\partial/\partial y, \partial/\partial z) \cdot$  is the divergence in the meridional-vertical plane, at constant  $x$ . Unless the meaning is unclear, the subscript  $x$  will be dropped.

### 9.2.1 The Eliassen–Palm Relation

On dividing by  $\partial \bar{q}/\partial y$  and using (9.28), the enstrophy equation (9.12) becomes

$$\frac{\partial \mathcal{P}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (9.29)$$

where

$$\mathcal{P} = \frac{\overline{q'^2}}{2\partial \bar{q}/\partial y}, \quad \mathcal{D} = \frac{\overline{D'q'}}{\partial \bar{q}/\partial y}, \quad (9.30a,b)$$

and  $\mathcal{F}$  is given by (9.27). Equation (9.29) is known as the *Eliassen–Palm relation*, and it is a conservation law (when  $\mathcal{D} = 0$ ) for the *pseudomomentum*  $\mathcal{P}$ . The conservation law is exact (in the linear approximation) if the mean-flow is constant in time; it is a good approximation if  $\partial \bar{q}/\partial y$  varies slowly compared to the variation of  $q'^2$ .

If we integrate (9.29) over a meridional area  $A$  bounded by walls where the eddy activity vanishes, and if  $\mathcal{D} = 0$ , we obtain

$$\frac{d}{dt} \int_A \mathcal{P} dA = 0. \quad (9.31)$$

The integral is a ‘wave activity’ — a quantity that is quadratic in the amplitude of the perturbation and that is conserved in the absence of forcing and dissipation. The quantity  $\mathcal{P}$  is an example of a ‘wave activity density’, generically denoted  $\mathcal{A}$ ; other kinds of wave activity density exist — the pseudoenergy for example, but we do not consider them here. If there is no ambiguity we drop the word density and also refer to  $\mathcal{A}$  and  $\mathcal{P}$  as wave activities. Note that neither the perturbation energy nor the perturbation enstrophy are wave activities of the linearized equations, because there can be an exchange of energy or enstrophy between mean and perturbation — indeed, this is how a perturbation grows in baroclinic or barotropic instability! This is already evident from (9.12), or in general take (9.7) with  $D' = 0$  and multiply by  $q'$  to give the enstrophy equation,

$$\frac{1}{2} \frac{\partial q'^2}{\partial t} + \frac{1}{2} \bar{\mathbf{u}} \cdot \nabla q'^2 + \mathbf{u}' q' \cdot \nabla \bar{q} = 0, \quad (9.32)$$

*The Eliassen–Palm flux and Eliassen–Palm relation are named for quantities that appeared in Eliassen & Palm (1961). That paper was mainly concerned with the transfer of energy in mountain waves rather than with matters related to potential vorticity flux and large-scale flow. ‘Wave activity flux’ is a descriptive alternative name for the EP flux.*

### Quasi-Geostrophic Wave–Mean-Flow Interaction

The inviscid and unforced Boussinesq quasi-geostrophic set of wave–mean-flow equations is

$$\frac{\partial q'}{\partial t} + \bar{u} \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (\text{WM.1a})$$

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial}{\partial y} \overline{v' q'} = 0, \quad (\text{WM.1b})$$

along with similar equations as needed for buoyancy at the boundary (see main text). The eddy terms are

$$q' = \left[ \nabla^2 + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \psi', \quad (u', v') = \left( -\frac{\partial \psi'}{\partial y}, \frac{\partial \psi'}{\partial x} \right). \quad (\text{WM.2a,b})$$

The mean-flow terms are

$$\bar{q}(y, t) = \beta y - \frac{\partial \bar{u}}{\partial y} + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \bar{b} \right), \quad (\text{WM.3})$$

and

$$\frac{\partial \bar{q}}{\partial y} = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} + \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \frac{\partial \bar{b}}{\partial y} \right) = \beta - \frac{\partial^2 \bar{u}}{\partial y^2} - \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \bar{u}}{\partial z} \right), \quad (\text{WM.4})$$

using thermal wind. To solve for the mean-flow we may define a streamfunction  $\Psi$  such that

$$\left( \bar{u}, \frac{1}{f_0} \bar{b} \right) = \left( -\frac{\partial \Psi}{\partial y}, \frac{\partial \Psi}{\partial z} \right), \quad (\text{WM.5})$$

whence

$$\bar{q}(y, t) - \beta y = \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \Psi}{\partial z} \right) + \frac{\partial^2 \Psi}{\partial y^2}. \quad (\text{WM.6})$$

Given  $\bar{q}$  from (WM.1b) we solve (WM.6) to give  $\Psi$  and thence  $\bar{u}$  and  $\bar{b}$ . Equivalently, we may derive a single equation for the zonal wind by differentiating (WM.1b) with respect to  $y$  and, using (WM.4), we obtain

$$\left[ \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \frac{\partial \bar{u}}{\partial t} = \frac{\partial^2}{\partial y^2} \overline{v' q'}. \quad (\text{WM.7})$$

The evolution of the mean-flow may also be written in TEM form as described on page 186.

where here the overbar is an average (although it need not be a zonal average). Integrating this over a volume  $V$  gives

$$\frac{d\hat{Z}'}{dt} \equiv \frac{d}{dt} \int_V \frac{1}{2} q'^2 dV = - \int_V \mathbf{u}' q' \cdot \nabla \bar{q} dV. \quad (9.33)$$

The right-hand side does not, in general, vanish and so  $\hat{Z}'$  is not in general conserved.

### 9.2.2 ♦ The Group Velocity Property for Rossby Waves

The vector  $\mathcal{F}$  describes how the wave activity propagates. We stated in Chapter 6 that in a wave most physical quantities of interest propagate at the group velocity, not the phase velocity. We now show that this is true for the wave activity for Rossby waves; in particular we show that  $\mathcal{F} = \mathbf{c}_g \mathcal{P}$ , where  $\mathbf{c}_g$  is the group velocity and  $\mathcal{P}$  the pseudomomentum.

The Boussinesq quasi-geostrophic equation on the  $\beta$ -plane, linearized around a constant zonal flow and with constant static stability, is

$$\frac{\partial q'}{\partial t} + U \frac{\partial q'}{\partial x} + v' \frac{\partial \bar{q}}{\partial y} = 0, \quad (9.34)$$

where  $q' = [\nabla^2 + (f_0^2/N^2)\partial^2/\partial z^2]\psi'$  and, if  $U$  is constant,  $\partial \bar{q}/\partial y = \beta$ . Thus we have

$$\left( \frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) \left[ \nabla^2 \psi' + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \psi'}{\partial z} \right) \right] + \beta \frac{\partial \psi'}{\partial x} = 0. \quad (9.35)$$

Seeking solutions of the form

$$\psi' = \text{Re } \tilde{\psi} e^{i(kx + ly + mz - \omega t)}, \quad (9.36)$$

we find the dispersion relation,

$$\omega = Uk - \frac{\beta k}{\kappa^2}, \quad (9.37)$$

where  $\kappa^2 = (k^2 + l^2 + m^2 f_0^2/N^2)$ , and the group velocity components:

$$c_g^y = \frac{2\beta k l}{\kappa^4}, \quad c_g^z = \frac{2\beta k m f_0^2/N^2}{\kappa^4}. \quad (9.38)$$

Also, if  $u' = \text{Re } \tilde{u} \exp[i(kx + ly + mz - \omega t)]$ , and similarly for the other fields, then

$$\begin{aligned} \tilde{u} &= -\text{Re } i l \tilde{\psi}, & \tilde{v} &= \text{Re } i k \tilde{\psi}, \\ \tilde{b} &= \text{Re } i m f_0 \tilde{\psi}, & \tilde{q} &= -\text{Re } \kappa^2 \tilde{\psi}. \end{aligned} \quad (9.39)$$

The wave activity density is then

$$\mathcal{P} = \frac{1}{2} \frac{\overline{q'^2}}{\beta} = \frac{\kappa^4}{4\beta} |\tilde{\psi}|^2, \quad (9.40)$$

where the additional factor of 2 in the denominator arises from the averaging. Using (9.39) the EP flux, (9.27), is

$$\mathcal{F}^y = -\overline{u'v'} = \frac{1}{2} k l |\tilde{\psi}|^2, \quad \mathcal{F}^z = \frac{f_0}{N^2} \overline{v'b'} = \frac{f_0^2}{2N^2} k m |\tilde{\psi}|^2. \quad (9.41)$$

Using (9.38), (9.40) and (9.41) we obtain

$$\mathcal{F} = (\mathcal{F}^y, \mathcal{F}^z) = \mathbf{c}_g \mathcal{P}. \quad (9.42)$$

Equation (9.29) may then be written

$$\frac{\partial \mathcal{P}}{\partial t} + \nabla \cdot (\mathcal{P} \mathbf{c}_g) = 0, \quad (9.43)$$

completing our demonstration. The equation essentially tells us how the disturbance associated with Rossby wave propagates, and is of practical as well as fundamental importance.

### 9.3 THE TRANSFORMED EULERIAN MEAN

The so-called *transformed Eulerian mean*, or TEM, is a transformation of the equations of motion that provides a useful framework for discussing eddy effects under a wide range of conditions. It is useful because, as we shall see, it is equivalent to a natural form of averaging the equations that serves to eliminate eddy fluxes in the thermodynamic equation and collect them together, in a simple form, in the momentum equation, and in so doing it highlights the role of potential vorticity fluxes. The TEM also provides a useful separation between diabatic and adiabatic effects or between advective and diffusive fluxes and a pleasing simplification of the equations. However, the TEM does have some of its own difficulties, for example in the implementation of proper boundary conditions on the velocity.

#### 9.3.1 Quasi-Geostrophic Form

The Boussinesq zonally-averaged Eulerian mean equations for the zonally-averaged zonal velocity and buoyancy may be written as

$$\frac{\partial \bar{u}}{\partial t} + \bar{v} \frac{\partial \bar{u}}{\partial y} + \bar{w} \frac{\partial \bar{u}}{\partial z} - f \bar{v} = -\frac{\partial}{\partial y} \overline{u'v'} - \frac{\partial}{\partial z} \overline{u'w'} + \bar{F}, \quad (9.44a)$$

$$\frac{\partial \bar{b}}{\partial t} + \bar{v} \frac{\partial \bar{b}}{\partial y} + \bar{w} \frac{\partial \bar{b}}{\partial z} = -\frac{\partial}{\partial y} \overline{v'b'} - \frac{\partial}{\partial z} \overline{w'b'} + \bar{S}, \quad (9.44b)$$

where  $\bar{F}$  and  $\bar{S}$  represent frictional and heating terms, respectively, and the meridional velocity,  $\bar{v}$ , is purely ageostrophic. Using quasi-geostrophic scaling we neglect the vertical eddy flux divergences and all ageostrophic velocities except when multiplied by  $f_0$  or  $N^2$ , which we take to be equal to  $\partial \bar{b} / \partial z$ . The above equations then become

$$\frac{\partial \bar{u}}{\partial t} = f_0 \bar{v} - \frac{\partial}{\partial y} \overline{u'v'} + \bar{F}, \quad (9.45a)$$

$$\frac{\partial \bar{b}}{\partial t} = -N^2 \bar{w} - \frac{\partial}{\partial y} \overline{v'b'} + \bar{S}. \quad (9.45b)$$

These two equations are connected by the thermal wind relation,  $f_0 \partial \bar{u} / \partial z = -\partial \bar{b} / \partial y$ , which itself is a combination of the geostrophic  $v$ -momentum equation ( $f_0 \bar{u} = -\partial \bar{\phi} / \partial y$ ) and hydrostasy ( $\partial \bar{\phi} / \partial z = \bar{b}$ ). One



less than ideal aspect of (9.45) is that in the extratropics the dominant balance is usually between the first two terms on the right-hand sides of each equation, even in time-dependent cases. Thus, the Coriolis force closely balances the divergence of the eddy momentum fluxes, and the advection of the mean stratification ( $N^2\bar{w}$ , or ‘adiabatic cooling’) often balances the divergence of eddy heat flux, with heating being a small residual. This may lead to an underestimation of the importance of diabatic heating, as this is ultimately responsible for the mean meridional circulation. Furthermore, the link between  $\bar{u}$  and  $\bar{b}$  via thermal wind dynamically couples buoyancy and momentum, and obscures the understanding of how the eddy fluxes influence these fields — is it through the eddy heat fluxes or momentum fluxes, or some combination?

To address this issue we combine the terms  $N^2\bar{w}$  and the eddy flux in (9.45b) into a single total or *residual* (so recognizing the cancellation between the mean and eddy terms) heat transport term that in a steady state is balanced by the diabatic term  $\bar{S}$ . To do this, we first note that because  $\bar{v}$  and  $\bar{w}$  are related by mass conservation we can define a mean meridional streamfunction  $\psi_m$  such that

$$(\bar{v}, \bar{w}) = \left( -\frac{\partial \psi_m}{\partial z}, \frac{\partial \psi_m}{\partial y} \right). \quad (9.46)$$

The velocities then satisfy  $\partial \bar{v} / \partial y + \partial \bar{w} / \partial z = 0$  automatically. If we define a *residual streamfunction* by

$$\psi^* \equiv \psi_m + \frac{1}{N^2} \overline{v' b'}, \quad (9.47a)$$

the components of the *residual mean meridional circulation* are then given by

$$(\bar{v}^*, \bar{w}^*) = \left( -\frac{\partial \psi^*}{\partial z}, \frac{\partial \psi^*}{\partial y} \right), \quad (9.47b)$$

and

$$\bar{v}^* = \bar{v} - \frac{\partial}{\partial z} \left( \frac{1}{N^2} \overline{v' b'} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left( \frac{1}{N^2} \overline{v' b'} \right). \quad (9.48)$$

Note that by construction, the residual overturning circulation satisfies

$$\frac{\partial \bar{v}^*}{\partial y} + \frac{\partial \bar{w}^*}{\partial z} = 0. \quad (9.49)$$

Substituting (9.48) into (9.45a) and (9.45b) the zonal momentum and buoyancy equations then take the simple forms

$$\begin{aligned} \frac{\partial \bar{u}}{\partial t} &= f_0 \bar{v}^* + \overline{v' q'} + \bar{F}, \\ \frac{\partial \bar{b}}{\partial t} &= -N^2 \bar{w}^* + \bar{S}, \end{aligned} \quad (9.50a,b)$$

There are no explicit eddy fluxes in the buoyancy budget; the only eddy term is the flux of potential vorticity, and this is the divergence of the Eliassen–Palm flux; that is  $\overline{v' q'} = \nabla_x \cdot \mathcal{F}$ .

which are known as the (quasi-geostrophic) *transformed Eulerian mean equations*, or TEM equations. The potential vorticity flux,  $\overline{v'q'}$ , is given in terms of the heat and vorticity fluxes by (9.26), and is equal to the divergence of the Eliassen–Palm flux as in (9.28).

The TEM equations make it apparent that we may consider the potential vorticity fluxes, rather than the separate contributions of the vorticity and heat fluxes, to force the circulation. If we know the potential vorticity flux as well as  $\overline{F}$  and  $\overline{S}$ , then (9.49) and (9.50), along with thermal wind balance, (9.11), form a complete set. The meridional overturning circulation is obtained by eliminating time derivatives from (9.50) using thermal wind to give

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = f_0 \frac{\partial \overline{v'q'}}{\partial z} + f_0 \frac{\partial \overline{F}}{\partial z} + \frac{\partial \overline{S}}{\partial y}. \quad (9.51)$$

Thus, the residual or net overturning circulation is driven by the (vertical derivative of the) potential vorticity fluxes and the diabatic terms — ‘driven’ in the sense that if we know those terms we can calculate the overturning circulation, although of course the fluxes themselves ultimately depend on the circulation. Note that this equation applies at every instant, even if the equations are not in a steady state.

Use of the equations in TEM form is particularly useful when the eddy potential vorticity flux arises from wave activity, for example from Rossby waves. The potential vorticity flux is the convergence of the EP flux  $\mathcal{F}$ , as in (9.28), and if the eddies satisfy a dispersion relation the components of the EP flux are equal to the group velocity multiplied by the wave activity density  $\mathcal{P}$ , as in (9.42). Thus, knowing the group velocity tells us a great deal about how momentum is transported by waves.

### Connection to potential vorticity and wave–mean-flow interaction

If we cross-differentiate (9.50) then, after using the residual mass continuity equation (9.49), we recover the zonally-averaged potential vorticity equation, namely

$$\frac{\partial \overline{q}}{\partial t} = -\frac{\partial}{\partial y} \overline{v'q'} - \frac{\partial \overline{F}}{\partial y}, \quad \text{where} \quad \overline{q}(y, t) = \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \overline{b} \right) - \frac{\partial \overline{u}}{\partial y}. \quad (9.52a,b)$$

These equations are equivalent to (9.17) and (9.20) (since we may fold  $\beta y$  into  $\overline{q}$  as it is constant in time).

The corresponding equation for the evolution of eddy potential vorticity is, in its inviscid form,

$$\left( \frac{\partial}{\partial t} + \overline{u}(y, t) \frac{\partial}{\partial x} \right) q' + v' \frac{\partial \overline{q}}{\partial y} = 0, \quad (9.53)$$

as in (9.7). Equations (9.52) and (9.53) are a closed set of quasi-linear equations, and we have recovered the wave–mean-flow system described in Section 9.1.3.

### 9.3.2 The TEM in the Shallow Water Equations

The residual circulation has an illuminating interpretation if we think of the fluid as comprising one or more layers of shallow water. In a single layer system the momentum and mass conservation equation can be written, in the usual way, as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} - f\mathbf{v} = \mathbf{F}, \quad \frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = S, \quad (9.54a,b)$$

where  $S$  is a thickness source term. With quasi-geostrophic scaling, so that variations in Coriolis parameter and layer thickness are small, zonally averaging gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v' \zeta'} + \bar{F}, \quad \frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}}{\partial y} = -\frac{\partial}{\partial y} \overline{v' h'} + \bar{S}, \quad (9.55a,b)$$

where the meridional velocity is purely ageostrophic. By analogy with (9.48), we define the residual circulation by

$$\bar{v}^* \equiv \bar{v} + \frac{1}{H} \overline{v' h'}, \quad (9.56)$$

where  $H$  is the reference thickness of the layer. Using (9.56) in (9.55) gives

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \overline{v' q'} + \bar{F}, \quad \frac{\partial \bar{h}}{\partial t} + H \frac{\partial \bar{v}^*}{\partial y} = \bar{S}, \quad (9.57a,b)$$

where

$$\overline{v' q'} = \overline{v' \zeta'} - \frac{f_0}{H} \overline{v' h'}, \quad (9.58)$$

is the meridional potential vorticity flux in a shallow water system. From (9.56) we see that the residual velocity is a measure of the *total meridional thickness flux*, eddy plus mean, in an isentropic layer. This is often a more useful quantity than the Eulerian velocity  $\bar{v}$  because it is generally the former, not the latter, that is constrained by the external forcing. We have effectively used a thickness-weighted mean in (9.54b); to see this explicitly, define the thickness-weighted mean by

$$\bar{v}_* \equiv \frac{\overline{h v}}{\bar{h}}. \quad (9.59)$$

From (9.59) we have

$$\bar{v}_* = \bar{v} + \frac{1}{\bar{h}} \overline{v' h'}, \quad (9.60)$$

then the zonal average of (9.54b) is just

$$\frac{\partial \bar{h}}{\partial t} + \frac{\partial}{\partial y} (\bar{h} \bar{v}_*) = \bar{S}, \quad (9.61)$$

which is the same as (9.57b) if we take  $H = \bar{h}$ .

The value of this approach becomes more apparent when we consider multiple layers of fluid, or equivalently if we express the continuous system in isentropic coordinates, in which the thickness of isentropic layer of fluid is used as one of the state variables. The residual velocity in the TEM approach is the same as that which arises from a thickness-weighted average, and this velocity represents the actual average flow of fluid parcels more truthfully than does a conventional Eulerian average at a fixed height. The interested reader may pursue this topic in the references given at the end of the chapter.

## 9.4 THE NON-ACCELERATION RESULT

In this section we derive an important result in wave-mean-flow dynamics, the so-called non-acceleration condition. Under certain conditions, to be made precise below, we can show that waves have no net effect on the mean-flow, an important and somewhat counter-intuitive result.

### 9.4.1 A Derivation from the Potential Vorticity Equation

Consider how the potential vorticity fluxes affect the mean fields. The unforced and inviscid zonally-averaged potential vorticity equation is

$$\frac{\partial \bar{q}}{\partial t} + \frac{\partial \overline{v'q'}}{\partial y} = 0. \quad (9.62)$$

Now, in quasi-geostrophic theory the geostrophically balanced velocity and buoyancy can be determined from the potential vorticity via an elliptic equation, as in (9.20), namely

$$\bar{q} - \beta y = \frac{\partial^2 \bar{\psi}}{\partial y^2} + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial \bar{\psi}}{\partial z} \right), \quad (9.63)$$

where  $\bar{\psi}$  is such that  $(\bar{u}, \bar{b}/f_0) = (-\partial \bar{\psi}/\partial y, \partial \bar{\psi}/\partial z)$ . Differentiating (9.62) with respect to  $y$  we obtain

$$\left[ \frac{\partial^2}{\partial y^2} + \frac{\partial}{\partial z} \left( \frac{f_0^2}{N^2} \frac{\partial}{\partial z} \right) \right] \frac{\partial \bar{u}}{\partial t} = (\nabla \cdot \mathcal{F})_{yy}, \quad (9.64)$$

where  $\nabla \cdot \mathcal{F} = \overline{v'q'}$  is the divergence of the EP flux (in the  $y$ - $z$  plane, i.e.,  $\nabla_x \cdot \mathcal{F}$ ). This is determined using the wave activity equation for pseudomomentum which, from (9.29), is

$$\frac{\partial \mathcal{P}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad (9.65)$$

where  $\mathcal{P}$  is the pseudomomentum. If the waves are statistically steady (i.e.,  $\partial \mathcal{P}/\partial t = 0$ ) and have no dissipation ( $\mathcal{D} = 0$ ) then evidently  $\nabla \cdot \mathcal{F} = 0$ . If there is no acceleration at the boundaries then the solution of (9.64) is

$$\frac{\partial \bar{u}}{\partial t} = 0. \quad (9.66)$$

This is a *non-acceleration result*. That is to say, under certain conditions the tendency of the mean fields, and in particular of the zonally-averaged zonal flow, are independent of the waves. To be explicit, those conditions are:

- (i) The waves are steady (so that, using the wave activity equation  $\mathcal{P}$  does not vary).
- (ii) The waves are conservative; that is,  $\mathcal{D} = 0$  in (9.29). Given this and item (i), the Eliassen–Palm relation implies that  $\nabla \cdot \mathcal{F} = 0$  and the potential vorticity flux is zero.
- (iii) The waves are of small amplitude (all of our analysis has neglected terms that are cubic in perturbation amplitude).
- (iv) The waves do not affect the boundary conditions (so there are no boundary contributions to the acceleration).

Given the way we have derived it, the result does not seem too surprising; however, it can be powerful and counter-intuitive, for it means that steady waves (i.e., those whose amplitude does not vary) do not affect the zonal flow. However, they *do* affect the Eulerian meridional overturning circulation, and the relative vorticity flux may also be non-zero. In fact, the non-acceleration theorem is telling us that the changes in the vorticity flux are exactly compensated for by changes in the meridional circulation, and there is no net effect on the zonally-averaged zonal flow. It is *irreversibility*, often manifested by the breaking of waves, that leads to permanent changes in the mean-flow.

The derivation of this result by way of the momentum equation, which one might expect to be more natural, is rather awkward because one must consider momentum and buoyancy fluxes separately. Nevertheless, we *can* proceed that way if we use the transformed Eulerian mean, as follows.

#### 9.4.2 Using TEM to Give the Non-Acceleration Result

We may use the TEM formalism to obtain the non-acceleration result. We begin with a purely two-dimensional case as here the derivation is very simple.

##### *A two-dimensional case*

Consider two-dimensional incompressible flow on the  $\beta$ -plane, for which there is no buoyancy flux. The linearized vorticity equation is

$$\frac{\partial \zeta'}{\partial t} + \bar{u} \frac{\partial \zeta'}{\partial x} + v' \frac{\partial \bar{\zeta}}{\partial y} = D', \quad (9.67)$$

from which we derive, analogously to (9.29), the Eliassen–Palm relation

$$\frac{\partial \mathcal{P}}{\partial t} + \frac{\partial \mathcal{F}}{\partial y} = \mathcal{D}, \quad (9.68)$$

where  $\mathcal{F} = -\overline{u'v'}$ ,  $\mathcal{D}$  represents non-conservative forces, and

$$\mathcal{P} = \frac{\overline{\zeta'^2}}{2\partial_y \bar{\zeta}}. \quad (9.69)$$

Now consider the  $x$ -momentum equation, namely

$$\frac{\partial u}{\partial t} = -\frac{\partial u^2}{\partial x} - \frac{\partial uv}{\partial y} - \frac{\partial \phi}{\partial x} + fv. \quad (9.70)$$

Zonally averaging, noting that  $\bar{v} = 0$ , gives

$$\frac{\partial \bar{u}}{\partial t} = -\frac{\partial \bar{u}\bar{v}}{\partial y} = \overline{v'\zeta'} = \frac{\partial \mathcal{F}}{\partial y}. \quad (9.71)$$

Finally, combining (9.68) and (9.71) gives

$$\frac{\partial}{\partial t} (\bar{u} + \mathcal{P}) = \mathcal{D}. \quad (9.72)$$

In the absence of non-conservative terms (i.e., if  $\mathcal{D} = 0$ ) the quantity  $\bar{u} + \mathcal{P}$  is constant. Further, if the waves are steady and conservative then  $\mathcal{P}$  is constant and, therefore, so is  $\bar{u}$ . This is the non-acceleration result.

### The stratified case

Beginning with (9.50a), and using the relation  $\overline{v'q'} = \nabla \cdot \mathcal{F}$ , the unforced zonally-averaged zonal momentum equation becomes

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \mathcal{F}. \quad (9.73)$$

Using the Eliassen–Palm relation, (9.29) the above expression may be written as

$$\frac{\partial}{\partial t} (\bar{u} + \mathcal{P}) - f_0 \bar{v}^* = \mathcal{D}, \quad (9.74)$$

again illustrating how  $\mathcal{P}$  is related to the momentum of the flow (and why it is called the pseudomomentum). If the waves are steady ( $\partial \mathcal{P} / \partial t = 0$ ) and conservative ( $\mathcal{D} = 0$ ), then  $\partial \bar{u} / \partial t - f_0 \bar{v}^* = 0$ . However, under these same conditions the residual circulation is also zero, as we now show.

The residual meridional circulation ( $\bar{v}^*$ ,  $\bar{w}^*$ ) arises via the necessity to keep the temperature and velocity fields in thermal wind balance and is determined by an elliptic equation, namely (9.51). If the waves are steady and adiabatic then, since  $\overline{v'q'} = 0$  in the fluid interior, the right-hand side of the equation is zero and (9.51) becomes

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = 0. \quad (9.75)$$

If  $\psi^* = 0$  at the boundaries, then the unique solution of this is  $\psi^* = 0$  everywhere. At the meridional boundaries we may certainly suppose that

$\psi^*$  vanishes if these are quiescent latitudes. At the horizontal boundaries the buoyancy flux will vanish if the waves there are steady, because from (9.15) we have

$$\overline{v'b'} \frac{\partial \bar{b}}{\partial y} = -\frac{1}{2} \frac{\partial}{\partial t} \overline{b'^2} = 0. \quad (9.76)$$

Now, the residual streamfunction is related to the Eulerian streamfunction by (9.47a), and if the eddy buoyancy flux and the Eulerian streamfunction are zero at the boundary, then so is the residual streamfunction. Under these circumstances, then, the residual meridional circulation vanishes in the interior and, from (9.73), the mean-flow is steady, and this is the non-acceleration result.

Compare (9.73) with the momentum equation in conventional Eulerian form, namely

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v} = \overline{v'\zeta'}. \quad (9.77)$$

There is no reason that the vorticity flux should vanish when waves are present, even if they are steady. Indeed, such a flux is (under non-acceleration conditions) precisely compensated by the meridional circulation  $f_0 \bar{v}$ , something that is hard to infer or intuit directly from (9.77), and even when non-acceleration conditions do not apply there is significant cancellation between the Coriolis and eddy terms. The difficulty boils down to the fact that, in contrast to  $\overline{v'q'}$ ,  $\overline{v'\zeta'}$  is not the flux of a wave activity.

Various results regarding the TEM and non-acceleration are summarized in the shaded box on the next page.

### 9.4.3 ♦ The EP Flux and Form Drag

It may seem a little magical that the zonal flow is driven by the Eliassen–Palm flux via (9.73). The poleward vorticity flux is clearly related to the momentum flux convergence, but why should a poleward buoyancy flux affect the momentum? The TEM form of the momentum equation may be written as

$$\frac{\partial \bar{u}}{\partial t} = \frac{\partial}{\partial z} \left( \frac{f_0}{N^2} \overline{v'b'} \right) + F_m, \quad (9.78)$$

where  $F_m = \overline{v'\zeta'} + f_0 \bar{v}^*$  represents forces from the momentum flux and Coriolis force. The first term on the right-hand side certainly does not look like a force; however, it turns out to be directly proportional to the *form drag* between isentropic layers.

The pressure drag, or the ‘form drag’,  $\tau_d$ , at an interface between two isentropic layers of fluid is given by the product of the interface displacement,  $\eta'$ , and the pressure gradient:

$$\tau_d = -\eta' \frac{\partial \phi'}{\partial x}, \quad (9.79)$$

and the interface displacement is just proportional to the buoyancy perturbation,

$$\eta' = -b'/N^2, \quad (9.80)$$

Some readers may find (9.79) and (9.80) obvious, some will find them mysterious. They may be derived with a little effort, or the reader may consult the references given at the end of the chapter, or take them on trust. The purpose of this section is simply to make it plausible that a buoyancy flux affects the momentum.

### TEM, Residual Velocities and Non-Acceleration

For a Boussinesq quasi-geostrophic system, the TEM form of the unforced momentum equation and the thermodynamic equation are:

$$\frac{\partial \bar{u}}{\partial t} - f_0 \bar{v}^* = \nabla \cdot \mathcal{F}, \quad \frac{\partial \bar{b}}{\partial t} + \bar{w}^* N^2 = \bar{S}, \quad (\text{T.1})$$

where  $N^2 = \partial \bar{b}_0 / \partial z$ ,  $\bar{S}$  represents diabatic effects,  $\mathcal{F}$  is the Eliassen–Palm (EP) flux and its divergence is the potential vorticity flux; thus,  $\nabla \cdot \mathcal{F} = \nabla_x \cdot \mathcal{F} = \overline{v'q'}$ . The residual velocities are

$$\bar{v}^* = \bar{v} - \frac{\partial}{\partial z} \left( \frac{1}{N^2} \overline{v'b'} \right), \quad \bar{w}^* = \bar{w} + \frac{\partial}{\partial y} \left( \frac{1}{N^2} \overline{v'b'} \right). \quad (\text{T.2})$$

We may define a meridional overturning streamfunction such that  $(\bar{v}^*, \bar{w}^*) = (-\partial \psi^* / \partial z, \partial \psi^* / \partial y)$ , and using thermal wind to eliminate time-derivatives in (T.1) we obtain

$$f_0^2 \frac{\partial^2 \psi^*}{\partial z^2} + N^2 \frac{\partial^2 \psi^*}{\partial y^2} = f_0 \frac{\partial}{\partial z} \overline{v'q'} + \frac{\partial \bar{S}}{\partial y}. \quad (\text{T.3})$$

The manipulations (given in the main text) that lead to the above equations may seem formal, in that they simply transform the momentum and thermodynamic equations from one form to another. However, the resulting equations have two potential advantages over the untransformed ones:

- (i) The residual meridional velocity is approximately equal to the average thickness-weighted velocity between two neighbouring isentropic surfaces, and so is a measure of the total (Eulerian mean plus eddy) meridional transport of thickness or buoyancy.
- (ii) The EP flux is directly related to certain conservation properties of waves. The divergence of the EP flux is the meridional flux of potential vorticity:

$$\mathcal{F} = -(\overline{u'v'}) \hat{\mathbf{j}} + \left( \frac{f_0}{N^2} \overline{v'b'} \right) \hat{\mathbf{k}}, \quad \nabla \cdot \mathcal{F} = \overline{v'q'}, \quad (\text{T.4})$$

and the *pseudomomentum*,  $\mathcal{P}$ , satisfies

$$\frac{\partial \mathcal{P}}{\partial t} + \nabla \cdot \mathcal{F} = \mathcal{D}, \quad \text{where} \quad \mathcal{P} = \frac{\overline{q'^2}}{2\partial \bar{q} / \partial y}, \quad \mathcal{D} = \frac{\overline{D'q'}}{\partial \bar{q} / \partial y}. \quad (\text{T.5})$$

The quantity  $\mathcal{P}$  is a *wave activity density*, and  $\mathcal{D}$  is its dissipation. For nearly plane waves,  $\mathcal{P}$  and  $\mathcal{F}$  are connected by the *group velocity property*,

$$\mathcal{F} = (\mathcal{F}^y, \mathcal{F}^z) = \mathbf{c}_g \mathcal{P}, \quad (\text{T.6})$$

where  $\mathbf{c}_g$  is the group velocity of the waves. If the waves are steady ( $\partial \mathcal{P} / \partial t = 0$ ) and dissipationless ( $\mathcal{D} = 0$ ) then  $\nabla \cdot \mathcal{F} = 0$  and using (T.1) and (T.3) there is no wave-induced acceleration of the mean-flow; this is the ‘non-acceleration’ result. Commonly there is enstrophy dissipation, or wave-breaking, and  $\nabla \cdot \mathcal{F} < 0$ ; such *wave drag* leads to flow deceleration and/or a poleward residual meridional velocity, discussed more in Chapter 12.



and using (9.79), (9.80) and geostrophic balance gives

$$\tau_d = \frac{f_0}{N^2} \overline{v' b'}. \quad (9.81)$$

Thus, the vertical component of the EP flux (i.e., the meridional buoyancy flux) is in fact a real stress acting on a fluid layer and equal to the momentum flux caused by the wavy interface. One might say that the vertical component of the EP flux is a force in drag, masquerading as a buoyancy flux.

### Notes and References

The TEM and the non-acceleration corollary were introduced by Boyd (1976) and Andrews & McIntyre (1976). A precursor is the paper of Riehl & Fultz (1957), who noted the shortcomings of zonal averaging in uncovering the meaning of indirect cells in laboratory experiments, and by extension the atmosphere. An advanced text on wave–mean-flow is that of Bühler (2009).

# Turbulence

*Horace Lamb (1849–1934) was a British applied mathematician who published the classic text Hydrodynamics in 1895, probably the oldest book on fluid mechanics still in print. Werner Heisenberg (1901–1976) was a German physicist most famous for the matrix formulation of quantum mechanics and the uncertainty principle. His doctoral thesis of 1923 was, however, concerned with turbulence, and he returned to the subject in the late 1940s.*

**A** **N APOCRYPHAL STORY** that has been attributed to both Horace Lamb and Werner Heisenberg goes as follows. ‘When I die and go to Heaven,’ they are each said to have predicted, ‘I would like to ask my Maker to explain two things, namely turbulence and quantum electrodynamics. About the latter I am hopeful of getting an answer.’ Aside from the confidence of these two men as to where they were headed, the story speaks to the inherent difficulty of turbulence. But they may have been more likely to get an answer had their lives been more dissolute, for it is said that turbulence is the invention of the Devil, put on Earth to torment us.

Putting aside these metaphysical issues, in this chapter we will give an introduction to three concrete aspects of turbulence: (i) turbulent diffusion; (ii) the classical spectral scaling theory of turbulence in two and three dimensions; and (iii) the theory of geostrophic turbulence. Before all that we’ll describe what the ‘problem of turbulence’ actually is.

## 10.1 THE PROBLEM OF TURBULENCE

What is turbulence? Roughly speaking, turbulence is high Reynolds number fluid flow with both spatial and temporal disorder, and a couple of beautiful sketches of what Leonardo da Vinci called *turbolenza* are shown in Fig. 10.1.

Traditionally, turbulent flow has often been thought of as occurring at small scales but in fact a turbulent flow has, as a consequence of being so disordered, a *range* of scales from large to small. The weather itself is an example of a turbulent flow — the great storms sweeping across the midlatitudes contain many scales of motion within them and, as we know from experience, are very hard to predict. Still, turbulence in general and weather in particular *do* have predictable aspects — we know that next winter will be colder than this summer, and that any given month in



**Fig. 10.1:** Two sketches of turbulent flows from Leonardo's notebooks, from around the year 1500.

Spain will be warmer than the same month in the UK, and we know that if a storm is approaching from the west it is likely to be windier and rainier than normal, even if we do not know exactly when or where. We might like to be able to predict the *average* flow over a wide area or over a period of a time without necessarily predicting every detail. However, the details may be important — if not in themselves but because they have an effect on the large scale by transporting and mixing properties of the fluid; thus, a turbulent fluid will become well-mixed much more quickly than a laminar fluid. If we drop some ink into a glass of water then we can speed up the mixing by stirring the water, creating turbulent eddies that mix the ink into the water. However, to go beyond this picture and to properly understand the effect of the small scales on the large ones is very difficult because of the ‘closure problem’, as we now see.

*Leonardo da Vinci (1452–1519) is now most famous for his paintings, in particular Mona Lisa and The Last Supper, but he was also a sculptor, architect, inventor and scientist — the embodiment of a ‘renaissance man’. He made a number of drawings of eddying flow and coined the word turbolenza to describe it.*

### 10.1.1 The Closure Problem

Let us suppose that a flow has a mean component and a fluctuating component, so that the velocity is given by

$$\mathbf{v} = \bar{\mathbf{v}} + \mathbf{v}'. \quad (10.1)$$

Here  $\bar{\mathbf{v}}$  is the mean velocity field, and  $\mathbf{v}'$  is the deviation from that mean. The mean might be a time average, in which case  $\bar{\mathbf{v}}$  is a function only of space and not of time, or it might be a time mean over a finite period (e.g., a season if we are dealing with the weather), or it might be some form

of ensemble mean. The average of the deviation is, by definition, zero; that is  $\overline{v'} = 0$ . We would like to predict the evolution of the mean flow,  $\bar{v}$ , without predicting the evolution of the eddying flow and to do this we might substitute (10.1) into the momentum equation and try to obtain a closed equation for the mean quantity  $\bar{v}$ . To keep the algebra simple, and to avoid dealing with the full Navier–Stokes equations, let us carry out this program for a model nonlinear system that obeys

$$\frac{du}{dt} + uu + ru = 0, \quad (10.2)$$

where  $r$  is a constant. The average of this equation is:

$$\frac{d\bar{u}}{dt} + \overline{uu} + r\bar{u} = 0. \quad (10.3)$$

The value of the term  $\overline{uu}$  (i.e.,  $\overline{u^2}$ ) is not deducible simply by knowing  $\bar{u}$ , since it involves correlations between eddy quantities, namely  $u'u'$ . That is,  $\overline{uu} = \bar{u}\bar{u} + \overline{u'u'} \neq \bar{u}\bar{u}$ . We can go to the next order to try (vainly!) to obtain an equation for  $\overline{u'u'}$ . First multiply (10.2) by  $u$  to obtain an equation for  $u^2$ , and then average it to yield

$$\frac{1}{2} \frac{d\overline{u^2}}{dt} + \overline{uuu} + r\overline{u^2} = 0. \quad (10.4)$$

This equation contains the undetermined cubic term  $\overline{uuu}$ . An equation determining this would contain a quartic term, and so on in an unclosed hierarchy. Many methods of closing the hierarchy make assumptions about the relationship of  $(n + 1)$ th order terms to  $n$ th order terms, for example by supposing that

$$\overline{uuuu} = \alpha \bar{u}\bar{u}\bar{u}\bar{u} + \beta \overline{uuuu}, \quad (10.5)$$

where  $\alpha$  and  $\beta$  are parameters to be determined, one may hope, by a theory. If we know that the variables are distributed normally then such closures can be made exact, but this is not generally true in turbulence and all closures that have been proposed so far are, at best, approximations.

This same closure problem arises in the Navier–Stokes equations. If density is constant (as we shall assume in this chapter) the  $x$ -momentum equation for an averaged flow is

$$\frac{\partial \bar{u}}{\partial t} + (\bar{v} \cdot \nabla) \bar{u} = -\frac{\partial \bar{\phi}}{\partial x} - \nabla \cdot \overline{v'u'}. \quad (10.6)$$

Written out in full in Cartesian coordinates, the last term is

$$\nabla \cdot \overline{v'u'} = \frac{\partial}{\partial x} \overline{u'u'} + \frac{\partial}{\partial y} \overline{u'v'} + \frac{\partial}{\partial z} \overline{u'w'}. \quad (10.7)$$

These terms, and the similar ones in the  $y$ - and  $z$ -momentum equations, represent the effects of eddies on the mean flow and are known as *Reynolds stress* terms. The ‘closure problem’ of turbulence may be thought

of as finding a representation of the Reynolds stresses in terms of mean flow quantities. Nobody has been able to usefully close the system without introducing physical assumptions not directly deducible from the equations of motion themselves. Indeed it is not clear that in general a useful closed-form solution even exists.

## 10.2 TURBULENT DIFFUSION

The most widely used recipe to address the closure problem is by way of turbulent diffusion, or eddy diffusion. The idea comes by way of an analogy with molecular diffusion, and is roughly as follows. Suppose that a fluid carries with it a tracer,  $\varphi$ , that satisfies an equation like

$$\frac{D\varphi}{Dt} = \kappa \nabla^2 \varphi, \quad (10.8)$$

where  $\kappa$  is a molecular diffusivity. For simplicity we suppose the flow is two-dimensional and incompressible, and that the flow and the tracer have both a mean and a fluctuating component. The mean component of (10.8) may then be written as

$$\frac{\partial \bar{\varphi}}{\partial t} + \frac{\partial \bar{u}\bar{\varphi}}{\partial x} + \frac{\partial \bar{v}\bar{\varphi}}{\partial y} = -\frac{\partial \overline{u'\varphi'}}{\partial x} - \frac{\partial \overline{v'\varphi'}}{\partial y} + \kappa \nabla^2 \bar{\varphi}, \quad (10.9)$$

Now, consider a fluctuating parcel of fluid that, on average, carries its value of  $\varphi$  with it a certain distance  $\ell$ , a 'mixing length', before mixing with its surroundings. If there is a mean gradient of  $\varphi$  in the direction of movement (the  $y$ -direction, say) then the value of  $\varphi'$  is given by

$$\varphi' = -\ell \frac{\partial \varphi}{\partial y}. \quad (10.10)$$

If the dominant eddies have a typical speed  $v'$  then the eddy transport is given by

$$\overline{v'\varphi'} = -K \frac{\partial \bar{\varphi}}{\partial y}, \quad \text{where} \quad K = \overline{v'\ell}. \quad (10.11)$$

In this expression,  $K$  is an *eddy diffusivity*, the product of the velocity and length scale of the dominant eddies in the system. If we assume that a similar process occurs in the  $x$ -direction then (10.1) becomes

$$\frac{\partial \bar{\varphi}}{\partial t} + \frac{\partial \bar{u}\bar{\varphi}}{\partial x} + \frac{\partial \bar{v}\bar{\varphi}}{\partial y} = K \nabla^2 \bar{\varphi} + \kappa \nabla^2 \bar{\varphi}. \quad (10.12)$$

In most turbulent flows the eddy diffusivity is much larger than the molecular diffusivity  $\kappa$  because the mixing length is orders of magnitude larger than the corresponding molecular mixing length, which is the average distance that a molecule goes before interacting with another molecule. Thus, we commonly neglect the last term on the right-hand side in (10.12). However, the presence of a molecular viscosity is important in that it allows mixing to take place in the first instance; the turbulence amplifies the molecular mixing enormously, but that mixing must be present.

*To suppose that turbulence acts like an enhanced diffusion is probably the most widely used parameterization of turbulence in practical situations. It is easily implemented and has a rational basis. On the other hand, there are many circumstances when it does not work, and momentum in large-scale atmospheric flows is typically not diffused downgradient, or diffused at all.*

*The theory of turbulent diffusion stems from work by G. I. Taylor and L. Prandtl, two great figures in fluid dynamics in the early twentieth century.*

Equation (10.12) is a practical recipe, and no more, for treating the enhanced transport associated with a turbulent flow. It says that if we are unable to explicitly model the small scales of a turbulent flow, perhaps because we don't know what is happening at those scales, then we might be able to approximately simulate the effects of the small scales using a turbulent diffusion. The idea is rather ad hoc, because we don't have a good theory for the magnitude and structure of the eddy diffusion coefficient  $K$ , but it is often better than doing nothing.

### 10.2.1 Homogenization and Lack of Extrema

Now consider a tracer that is advected and diffused. The diffusion might be molecular or, if the effects of turbulence on a tracer are indeed diffusive, there might be an eddy diffusion. An important consequence of this is that, in the absence of additional forcing, there can be no extreme values of the tracer in the interior of the fluid and the diffusion acts to homogenize values of the tracer in broad regions.

Consider a tracer that obeys the equation

$$\frac{D\varphi}{Dt} = \nabla \cdot (\kappa \nabla \varphi), \quad (10.13)$$

where  $\kappa > 0$  and the advecting velocity is divergence-free. Given an extremum, there will then be a surrounding surface (in three dimensions), or a surrounding contour (in two), connecting constant values of  $\varphi$ . For definiteness consider three-dimensional incompressible flow which in the steady state flow satisfies

$$\nabla \cdot (\mathbf{v}\varphi) = \nabla \cdot (\kappa \nabla \varphi). \quad (10.14)$$

Integrating the left-hand side over the volume,  $V$ , enclosed by an iso-surface,  $A$ , of  $\varphi$ , and applying the divergence theorem, gives

$$\iiint_V \nabla \cdot (\mathbf{v}\varphi) dV = \iint_A (\mathbf{v}\varphi) \cdot \mathbf{n} dA = \varphi \iint_A \mathbf{v} \cdot \mathbf{n} dA = \varphi \iiint_V \nabla \cdot \mathbf{v} dV = 0, \quad (10.15)$$

where  $\mathbf{n}$  is a unit vector normal to the bounding surface. But the integral of the right-hand side of (10.14) over the same area is non-zero; that is

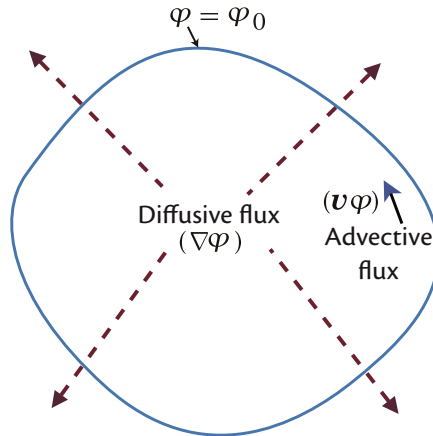
$$\iiint_V \nabla \cdot (\kappa \nabla \varphi) dV = \iint_A \kappa \nabla \varphi \cdot \mathbf{n} dA \neq 0, \quad (10.16)$$

if the integral surrounds an extremum. This is a contradiction for steady flow. Hence, there can be no isolated extrema of a conserved quantity in the interior of a fluid, if there is any diffusion at all. The result is kinematic, in that  $\varphi$  can be any tracer at all, active or passive.

#### *Interpretation and consequences*

The physical essence of the result is that the integrated effects of diffusion are non-zero surrounding an extremum, and cannot be balanced by





**Fig. 10.2:** If an extremum of a tracer  $\varphi$  exists in the fluid interior, then diffusion will provide a downgradient tracer flux. But over an area bounded by an isosurface of  $\varphi$  the integrated advective flux is zero. Thus, the diffusion cannot be balanced by advection and so in a steady state no extrema can exist.

advection (Fig. 10.2). Thus, if the initial conditions contain an extremum, diffusion will smooth away the extremum until it no longer exists. Thus, in a steady state, there can be no extrema in the interior of a fluid. One consequence of this is that fluid properties tend to become homogenized in regions of closed contours. Suppose, for example, that the flow is steady and circulates around some point. If that point is initially an extremum, diffusion will effectively *expel* gradients of  $\varphi$  to the enclosing circulation contours, forming a plateau of  $\varphi$ -values.

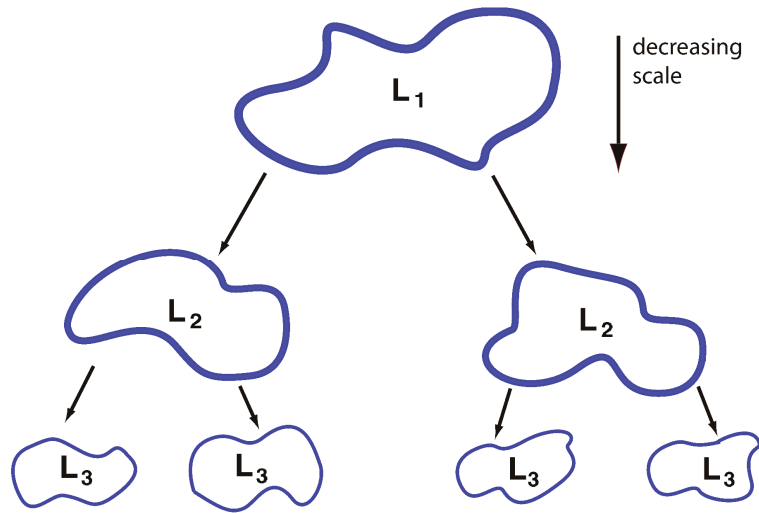
This result only applies to diffusive steady flows, or to turbulent flows in which the turbulent transport can be parameterized by an eddy diffusivity. This is not always the case and momentum, in particular, is not diffused by turbulence and local maxima of momentum do form on Earth and other planets. However, where such maxima do form the result implies that the flow is not steady, and this is important when studying *super-rotation*, when an atmosphere rotates faster than the planet beneath it: a steady atmosphere cannot super-rotate. However, when turbulence does act as a diffusion mechanism the consequences can be striking, and it appears that potential vorticity is indeed homogenized over large swaths of the ocean, a likely consequence of turbulent diffusion.

### 10.3 SPECTRAL THEORIES OF TURBULENCE

If we can't close the turbulent equations exactly just what *can* we say about a turbulent flow? One theory that, although it too is not exact, has proven remarkably resilient and that has been shown to agree with experiments in a variety of settings is the spectral theory of Kolmogorov, first proposed in 1941 and often known as K41 theory. The idea is as follows.

Suppose that we stir a fluid at large spatial scale,  $L_0$  say, and thereby we put energy into the fluid at that scale at a rate  $\varepsilon$ , where  $\varepsilon$  has units of energy/(time  $\times$  mass), or  $\text{m}^2 \text{s}^{-3}$  or more generically  $\text{L}^2 \text{T}^{-3}$ . The flow becomes turbulent, creating scales of motion much smaller than  $L_0$ . In other words, energy is transferred to smaller scales, and thence to smaller scales still, rather like the ditty in the margin on the following page. Eventually,

**Fig. 10.3:** The passage of energy to smaller scales: eddies at large scale break up into ones at smaller scale, thereby transferring energy to smaller scales. (The eddies in reality are embedded within each other.) If the passage occurs between eddies of similar sizes (i.e., if it is spectrally local) the transfer is said to be a cascade.



and no matter how small the molecular viscosity (if not zero), the energy reaches as a scale,  $L_v$  say, at which the Reynolds number,  $UL_v/\nu$ , is order one and viscous effects are important. If  $\nu = 1 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$  and  $U = 1 \text{ m s}^{-1}$  then this viscous scale is given by

$$L_v \sim \frac{\nu}{U} = 10^{-7} \text{ m}. \quad (10.17)$$

This estimate is rather rough and ready, because the velocity itself is likely to be smaller than  $1 \text{ m s}^{-1}$  at the viscous scale, but nevertheless it is clear that the viscous scale is a small one. We thus have a picture of energy being put into the fluid at large scales, being transferred across scales in an inviscid fashion, and finally being removed by viscosity at very small scales (where the kinetic energy turns into heat). At these intermediate scales the energy transfer from scale to scale must be constant. To describe all this it is useful to deal with Fourier transforms of quantities, so let us have a brief lesson on that.

### 10.3.1 A Spectral Diversion

Since we are dealing with scales of motion it is convenient to express the velocity field as Fourier components so that we write

$$u(x, y, z, t) = \sum_{\mathbf{k}} \tilde{u}(\mathbf{k}, t) e^{i(k^x x + k^y y + k^z z)}, \quad (10.18)$$

where  $\tilde{u}$  is the Fourier transformed field of  $u$ , with similar identities for  $v$  and  $w$ , and  $\mathbf{k} = (k^x, k^y, k^z)$ . The sum is a triple sum over all wavenumbers  $(k^x, k^y, k^z)$ , and in a finite domain these wavenumbers are quantized. Using Parseval's theorem the energy in the fluid is given by

$$\frac{1}{V} \int_V E \, dV = \frac{1}{2V} \int_V (u^2 + v^2 + w^2) \, dV = \frac{1}{2} \sum_{\mathbf{k}} (|\tilde{u}|^2 + |\tilde{v}|^2 + |\tilde{w}|^2) \equiv \sum_{\mathbf{k}} E_{\mathbf{k}}, \quad (10.19)$$

*Big whorls have little whorls,  
that feed on their vorticity;  
And little whorls  
have lesser whorls,  
and so on to viscosity.*  
L. F. Richardson, 1922  
(slightly modified).



where  $E$  is the energy density per unit mass,  $V$  is the volume of the domain, and the last equality serves to define the discrete energy spectrum  $\mathcal{E}_k$ . We now assume that the turbulence is isotropic, and that the domain is sufficiently large that the sums in the above equations may be replaced by integrals. We may then write

$$\bar{E} = \frac{1}{V} \hat{E} = \frac{1}{2V} \int_V \mathbf{v}^2 dV = \int \mathcal{E}(k) dk, \quad (10.20)$$

where  $\bar{E}$  is the average energy,  $\hat{E}$  is the total energy and  $\mathcal{E}(k)$  is the energy spectral density, or the energy spectrum, so that  $\mathcal{E}(k) \delta k$  is the energy in the small wavenumber interval  $\delta k$ . Because of the assumed isotropy, the energy is a function only of the scalar wavenumber  $k$ , where  $k^2 = \mathbf{k} \cdot \mathbf{k} = k_x^2 + k_y^2 + k_z^2$ . The units of  $\mathcal{E}(k)$  are  $L^3/T^2$  and the units of  $\bar{E}$  are  $L^2/T^2$ .

### 10.3.2 Inertial-Range Theory

We now suppose that the fluid is stirred at large scales and that this energy is transferred to small scales where it is dissipated by viscosity. The key assumption is to suppose that, if the forcing scale is sufficiently larger than the dissipation scale, there exists a range of scales that is intermediate between the large scale and the dissipation scale and where neither forcing nor dissipation are explicitly important to the dynamics. This assumption, known as the *locality hypothesis*, depends on the nonlinear transfer of energy being sufficiently local (in spectral space). This intermediate range is known as the *inertial range*, because the inertial terms and not forcing or dissipation dominate in the momentum balance. If the rate of energy input per unit volume by stirring is equal to  $\epsilon$ , then if we are in a steady state there must be a flux of energy from large to small scales that is also equal to  $\epsilon$ , and an energy dissipation rate, also  $\epsilon$ .

Now, we have no general theory for the energy spectrum of a turbulent fluid, but we might suppose it takes the general form

$$\mathcal{E}(k) = f(\epsilon, k, k_0, k_\nu), \quad (10.21)$$

where the right-hand side denotes a function of the spectral energy flux or cascade rate  $\epsilon$ , the wavenumber  $k$ , the forcing wavenumber  $k_0$  and the wavenumber at which dissipation acts,  $k_\nu$  (and  $k_\nu \sim L_\nu^{-1}$ ). In general, the function  $f$  depends on the particular nature of the forcing. Now, the locality hypothesis essentially says that at some scale within the inertial range the flux of energy to smaller scales depends only on processes occurring at or near that scale. That is to say, the energy flux is only a function of  $\mathcal{E}$  and  $k$ , or equivalently that the energy spectrum can be a function *only* of the energy flux  $\epsilon$  and the wavenumber itself. From a physical point of view, as energy cascades to smaller scales the details of the forcing are forgotten but the effects of viscosity are not yet apparent, and the energy spectrum takes the form,

$$\mathcal{E}(k) = g(\epsilon, k). \quad (10.22)$$

The function  $g$  is assumed to be *universal*, the same for every turbulent flow.

*The theory described in this section is not an exact theory of turbulence. It relies on assumptions of spectral locality and the near constancy of the energy transfer across scales, and these assumptions are not exactly satisfied. Nevertheless, the theory has been enormously useful and is one of the enduring foundations of the field.*

### Dimensions and the Kolmogorov Spectrum

| Quantity                          | Dimension       |
|-----------------------------------|-----------------|
| Wavenumber, $k$                   | $1/L$           |
| Energy per unit mass, $E$         | $U^2 = L^2/T^2$ |
| Energy spectrum, $\mathcal{E}(k)$ | $EL = L^3/T^2$  |
| Energy flux, $\varepsilon$        | $E/T = L^2/T^3$ |

If  $\mathcal{E} = g(\varepsilon, k)$  then the only dimensionally consistent relation for the energy spectrum is

$$\mathcal{E} = \mathcal{K} \varepsilon^{2/3} k^{-5/3},$$

where  $\mathcal{K}$  is a dimensionless constant.

Let us now use dimensional analysis to give us the form of the function  $g(\varepsilon, k)$  (see the shaded box above). In (10.22), the left-hand side has dimensions  $L^3/T^2$ ; the factor  $T^{-2}$  can only be balanced by  $\varepsilon^{2/3}$  because  $k$  has no time dependence; that is, (10.22), and its dimensions, must take the form

$$\mathcal{E}(k) = \varepsilon^{2/3} g(k), \quad (10.23a)$$

$$\frac{L^3}{T^2} \sim \frac{L^{4/3}}{T^2} g(k), \quad (10.23b)$$

where  $g(k)$  is some function. Evidently  $g(k)$  must have dimensions  $L^{5/3}$ , and the functional relationship we must have, if the physical assumptions are right, is

$$\mathcal{E}(k) = \mathcal{K} \varepsilon^{2/3} k^{-5/3}. \quad (10.24)$$

This is the famous ‘Kolmogorov -5/3 spectrum’, sketched in Fig. 10.4, enshrined as one of the cornerstones of turbulence theory and very well (if not exactly) supported by experiment. The parameter  $\mathcal{K}$  is a dimensionless constant, undetermined by this theory; it is known as Kolmogorov’s constant and experimentally its value is found to be about 1.5.

An equivalent, and revealing, way to derive this result is to first define an eddy turnover time  $\tau_k$ , which is the time taken for a parcel with velocity  $v_k$  to move a distance  $1/k$ ,  $v_k$  being the velocity associated with the (inverse) scale  $k$ . On dimensional considerations  $v_k = [\mathcal{E}(k)k]^{1/2}$  so that

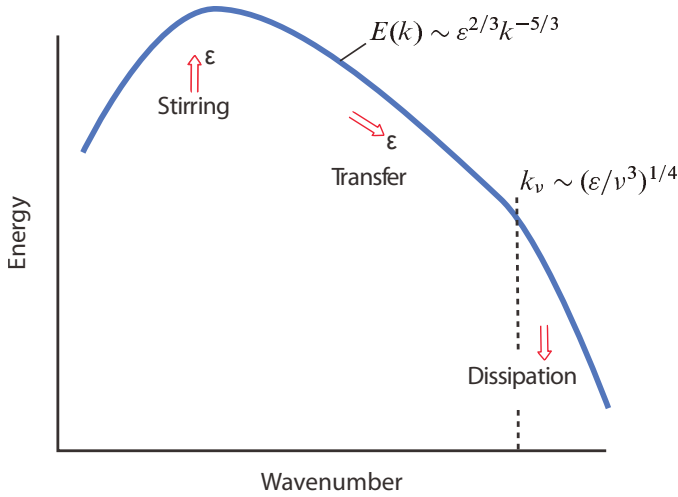
$$\tau_k = [k^3 \mathcal{E}(k)]^{-1/2}. \quad (10.25)$$

Kolmogorov’s assumptions are then equivalent to setting

$$\varepsilon \sim \frac{v_k^2}{\tau_k} = \frac{k \mathcal{E}(k)}{\tau_k}. \quad (10.26)$$

If we demand that  $\varepsilon$  be a constant then (10.25) and (10.26) yield (10.24).

*The spectral theory of turbulence as presented here was formulated by the Russians A. N. Kolmogorov and A. M. Obukhov in 1941. Prior to that, L. F. Richardson had adumbrated some of its essential qualities in 1920, encapsulated in his ditty on page 194.*



**Fig. 10.4:** The energy spectrum in three-dimensional turbulence, in the theory of Kolmogorov. Energy is supplied at some rate  $\varepsilon$ ; it is cascaded to small scales, where it is ultimately dissipated by viscosity. There is no systematic energy transfer to scales larger than the forcing scale, so here the energy falls off.

### The viscous scale and energy dissipation

At some small length scale we should expect viscosity to become important and the scaling theory we have just set up will fail. What is that scale? In the inertial range friction is unimportant because the time scales on which it acts are too long for it to be important and dynamical effects dominate. In the momentum equation the viscous term is  $\nu \nabla^2 u$  so that a viscous or dissipation time scale at a scale  $k^{-1}$ ,  $\tau_k^\nu$ , is

$$\tau_k^\nu \sim \frac{1}{k^2 \nu}, \quad (10.27)$$

so that the viscous time scale decreases as  $k$  increases and the length scale decreases. The eddy turnover, or the inertial, timescale,  $\tau_k$ , in the Kolmogorov spectrum is

$$\tau_k = \varepsilon^{-1/3} k^{-2/3}. \quad (10.28)$$

The wavenumber at which dissipation becomes important is then given by equating the above two time scales, yielding the dissipation wavenumber,  $k_\nu$  and the associated length scale,  $L_\nu$ ,

$$k_\nu \sim \left( \frac{\varepsilon}{\nu^3} \right)^{1/4}, \quad L_\nu \sim \left( \frac{\nu^3}{\varepsilon} \right)^{1/4}. \quad (10.29a,b)$$

$L_\nu$  is called the *Kolmogorov scale*. It is the *only* quantity which can be created from the quantities  $\nu$  and  $\varepsilon$  that has the dimensions of length.

An interesting consequence of the theory is this. The rate of energy dissipation by viscous effects is *independent of the magnitude of the coefficient of viscosity*. This might seem strange, since surely there must be less dissipation if the viscosity is smaller? Not so; if we reduce the viscosity, the cascade proceeds to smaller scales but the total dissipation stays the same, for it is equal to the energy input  $\varepsilon$ . Back in physical space, let us consider the viscous term  $\nu \nabla^2 u$ . If  $\nu$  decreases the scale at which dissipation takes place also decreases and so the second derivative,  $\nabla^2$  becomes

correspondingly larger in just such a way as to keep energy dissipation constant.

How big is  $L_v$  in the atmosphere? A crude estimate comes from noting that  $\varepsilon$  has units of  $U^3/L$ , and that at length scales of the order of 100 m in the atmospheric boundary layer (where there might be a three-dimensional energy cascade to small scales) velocity fluctuations are of the order of  $1 \text{ cm s}^{-1}$ , giving  $\varepsilon \approx 10^{-8} \text{ m}^2 \text{ s}^{-3}$ . Using (10.29b) we then find the dissipation scale to be of the order of a millimetre or so. In the ocean the dissipation scale is also of the order of millimetres. Various inertial range properties, in both three and two dimensions, are summarized in the shaded box on page 203.

## 10.4 TWO-DIMENSIONAL TURBULENCE

Two-dimensional turbulence behaves in a profoundly different way from three-dimensional turbulence, largely because of the presence of another quadratic invariant, the enstrophy, defined below. In two dimensions, the vorticity equation for incompressible flow is:

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta = F + \nu \nabla^2 \zeta, \quad (10.30)$$

where  $\mathbf{u} = u\mathbf{i} + v\mathbf{j}$  and  $\zeta = \mathbf{k} \cdot \nabla \times \mathbf{u}$  and  $F$  is a stirring term. In terms of a streamfunction,  $u = -\partial\psi/\partial y$ ,  $v = \partial\psi/\partial x$ , and  $\zeta = \nabla^2\psi$ , and (10.30) may be written as

$$\frac{\partial \nabla^2 \psi}{\partial t} + J(\psi, \nabla^2 \psi) = F + \nu \nabla^4 \psi. \quad (10.31)$$

We obtain an energy equation by multiplying by  $-\psi$  and integrating over the domain, and an enstrophy equation by multiplying by  $\zeta$  and integrating. When  $F = \nu = 0$  we find

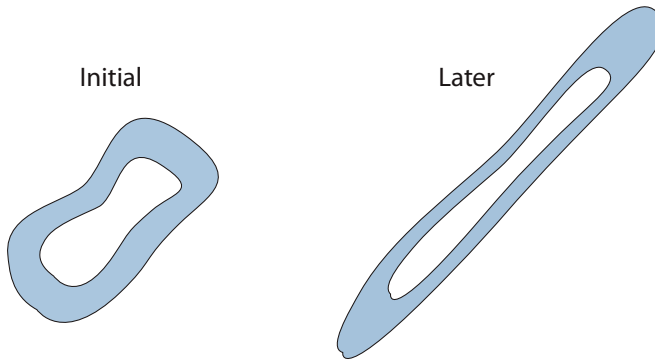
$$\hat{E} = \frac{1}{2} \int_A (u^2 + v^2) dA = \frac{1}{2} \int_A (\nabla \psi)^2 dA, \quad \frac{d\hat{E}}{dt} = 0, \quad (10.32a)$$

$$\hat{Z} = \frac{1}{2} \int_A \zeta^2 dA = \frac{1}{2} \int_A (\nabla^2 \psi)^2 dA, \quad \frac{d\hat{Z}}{dt} = 0, \quad (10.32b)$$

where the integral is over a finite area with either no-normal flow or periodic boundary conditions. The quantity  $\hat{E}$  is the energy, and  $\hat{Z}$  is known as the *enstrophy*. The enstrophy invariant arises because the vortex stretching term, so important in three-dimensional turbulence, vanishes identically in two dimensions. In fact, because vorticity is conserved on parcels it is clear that the integral of *any* function of vorticity is zero when integrated over  $A$ ; that is, from (10.30)

$$\frac{Df(\zeta)}{Dt} = 0 \quad \text{and} \quad \frac{d}{dt} \int_A f(\zeta) dA = 0, \quad (10.33)$$

where  $f(\zeta)$  is an arbitrary function. Of this infinity of conservation properties, enstrophy conservation (with  $f(\zeta) = \zeta^2$ ) in particular has been found to have enormous consequences to the flow of energy between scales, as we soon discover.



**Fig. 10.5:** In two-dimensional incompressible flow, a band of fluid is elongated, but its area is preserved. Elongation is followed by folding and more elongation, but still area-preserving in the absence of viscosity. As vorticity is tied to fluid parcels, the values of the vorticity in the shaded area (and in the hole) are maintained; thus, vorticity gradients increase and the enstrophy is thereby, on average, moved to smaller scales until it finally reaches the viscous scale.

### 10.4.1 Energy and Enstrophy Transfer

In three-dimensional turbulence we posited that energy is cascaded to small scales via vortex stretching. In two dimensions that mechanism is absent, and there is reason to expect energy to be transferred to *larger* scales. This counter-intuitive behaviour arises from the twin integral constraints of energy and enstrophy conservation, and the following two arguments illustrate why this should be so.

#### I. Vorticity elongation

Consider a band or a patch of vorticity, as in Fig. 10.5, in a nearly inviscid fluid. The vorticity of each element of fluid is conserved as the fluid moves. Now, we should expect that the quasi-random motion of the fluid will act to elongate the band but, as its area must be preserved, the band narrows and so vorticity gradients will increase. This is equivalent to the enstrophy moving to smaller scales. Now, the energy in the fluid is

$$\hat{E} = -\frac{1}{2} \int \psi \zeta \, dA, \quad (10.34)$$

where the streamfunction is obtained by solving the Poisson equation  $\nabla^2 \psi = \zeta$ . If the vorticity is locally elongated primarily only in one direction (as it must be to preserve area), the integration involved in solving the Poisson equation leads to the scale of the streamfunction becoming larger in the direction of stretching, but virtually no smaller in the perpendicular direction. Because stretching occurs, on average, in all directions, the overall scale of the streamfunction increases in all directions, and the cascade of enstrophy to small scales is accompanied by a transfer of energy to large scales.

#### II. An energy–enstrophy conservation argument

A moment's thought will reveal that the distributions of energy and enstrophy in wavenumber space are respectively analogous to the distribution of mass and moment of inertia of a lever, with wavenumber playing the role of distance from the fulcrum. Any rearrangement of mass such

that its distribution also becomes wider must be such that the centre of mass moves toward the fulcrum. Thus, analogously, any rearrangement of a flow that preserves both energy and enstrophy, and that causes the distribution to spread out in wavenumber space, will tend to move energy to small wavenumbers and enstrophy to large. To prove this we begin with expressions for the average energy and enstrophy:

$$\bar{E} = \int \mathcal{E}(k) dk, \quad \bar{Z} = \int \mathcal{Z}(k) dk = \int k^2 \mathcal{E}(k) dk, \quad (10.35)$$

where  $\mathcal{E}(k)$  and  $\mathcal{Z}(k)$  are the energy and enstrophy spectra. A wavenumber characterizing the spectral location of the energy is the centroid,

$$k_e = \frac{\int k \mathcal{E}(k) dk}{\int \mathcal{E}(k) dk}, \quad (10.36)$$

and, for simplicity, we normalize units so that the denominator is unity. The spreading out of the energy distribution is formalized by setting

$$I \equiv \int (k - k_e)^2 \mathcal{E}(k) dk, \quad \frac{dI}{dt} > 0. \quad (10.37)$$

Here,  $I$  measures the width of the energy distribution, and this is assumed to increase. Expanding out the integral gives

$$\begin{aligned} I &= \int k^2 \mathcal{E}(k) dk - 2k_e \int k \mathcal{E}(k) dk + k_e^2 \int \mathcal{E}(k) dk \\ &= \int k^2 \mathcal{E}(k) dk - k_e^2 \int \mathcal{E}(k) dk, \end{aligned} \quad (10.38)$$

where the last equation follows because  $k_e = \int k \mathcal{E}(k) dk$  is, from (10.36), the energy-weighted centroid. Because both energy and enstrophy are conserved, (10.38) gives

$$\frac{dk_e^2}{dt} = -\frac{1}{\bar{E}} \frac{dI}{dt} < 0. \quad (10.39)$$

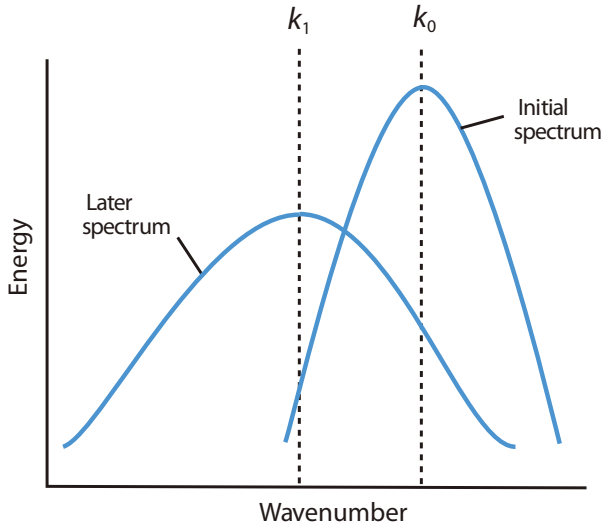
Thus, the centroid of the distribution moves to smaller wavenumbers and to larger scales (see Fig. 10.6).

An appropriately defined measure of the centre of the enstrophy distribution, on the other hand, moves to higher wavenumbers. The demonstration follows easily if we work with the inverse wavenumber, which is a direct measure of length. Let  $q = 1/k$  and assume that the enstrophy distribution spreads out by nonlinear interactions, so that, analogously to (10.37),

$$J \equiv \int (q - q_e)^2 \mathcal{Y}(q) dq, \quad \frac{dJ}{dt} > 0, \quad (10.40)$$

where  $\mathcal{Y}(q)$  is such that the enstrophy is  $\int \mathcal{Y}(q) dq$  and

$$q_e = \frac{\int q \mathcal{Y}(q) dq}{\int \mathcal{Y}(q) dq}. \quad (10.41)$$



**Fig. 10.6:** In two-dimensional flow, the centroid of the energy spectrum moves to large scales (smaller wavenumber) provided that the width of the distribution increases — as can be expected in a nonlinear, eddying flow.

Expanding the integrand in (10.40) and using (10.41) gives

$$J = \int q^2 \mathcal{Y}(q) dq - q_e^2 \int \mathcal{Y}(q) dq. \quad (10.42)$$

But  $\int q^2 \mathcal{Y}(q) dq$  is conserved, because this is the energy. Thus,

$$\frac{dJ}{dt} = -\frac{d}{dt} q_e^2 \int \mathcal{Y}(q) dq, \quad (10.43)$$

whence

$$\frac{dq_e^2}{dt} = -\frac{1}{Z} \frac{dJ}{dt} < 0. \quad (10.44)$$

Thus, the length scale characterizing the enstrophy distribution gets smaller, and the corresponding wavenumber gets larger.

### 10.4.2 Inertial Ranges in Two-Dimensional Turbulence

The above results suggest that, in a forced-dissipative two-dimensional fluid, energy is transferred to larger scales and enstrophy is transferred to small scales. We thus expect that two inertial ranges may form — an *energy inertial range* carrying energy to larger scales, and an *enstrophy inertial range* carrying enstrophy to small scales (Fig. 10.7). Let us calculate their properties.

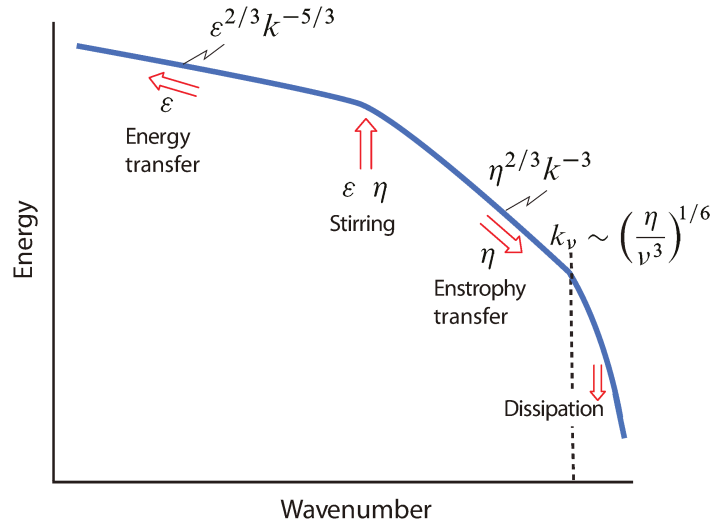
#### The enstrophy inertial range

In the enstrophy inertial range the enstrophy cascade rate  $\eta$ , equal to the rate at which enstrophy is supplied by stirring, is assumed constant. By analogy with (10.26) we may assume that this rate is given by

$$\eta \sim \frac{k^3 \mathcal{E}(k)}{\tau_k}. \quad (10.45)$$

In two-dimensional turbulence energy is generally transferred to large scales whereas enstrophy is transferred to small scales. If these transfers are spectrally local they are referred to as *cascades*. In an upscale energy cascade, meaning a cascade to larger scales, the energy spectrum varies as  $k^{-5/3}$  and in an enstrophy spectrum it varies as  $k^{-3}$ .

**Fig. 10.7:** The energy spectrum of two-dimensional turbulence. (Compare with Fig. 10.4.) Energy supplied at some rate  $\varepsilon$  is transferred to large scales, whereas enstrophy supplied at some rate  $\eta$  is transferred to small scales, where it may be dissipated by viscosity. If the forcing is localized at a scale  $k_f^{-1}$  then  $\eta \approx k_f^2 \varepsilon$ .



With  $\tau_k$  (still) given by (10.25) we obtain

$$\mathcal{E}(k) = \mathcal{K}_\eta \eta^{2/3} k^{-3}, \quad (10.46)$$

where  $\mathcal{K}_\eta$  is, we presume, a universal constant, analogous to the Kolmogorov constant of (10.24). This, and various other properties in two- and three-dimensional turbulence, are summarized in the shaded box on the facing page.

The velocity and time at a particular wavenumber then scale as

$$v_k \sim \eta^{1/3} k^{-1}, \quad t_k \sim l_k / v_k \sim 1 / (k v_k) \sim \eta^{-1/3}. \quad (10.47a,b)$$

We may also obtain (10.47) by substituting (10.46) into (10.25). Thus, *the eddy turnover time in the enstrophy range of two-dimensional turbulence is length-scale invariant*. The appropriate viscous scale is given by equating the inertial and viscous terms in (10.30). Using (10.47a) we obtain, analogously to (10.29a), the viscous wavenumber

$$k_v \sim \left( \frac{\eta^{1/3}}{\nu} \right)^{1/2}. \quad (10.48)$$

The enstrophy dissipation goes to a finite limit given by

$$\frac{d}{dt} \bar{Z} = \nu \int_A \zeta \nabla^2 \zeta \, dA \sim \nu k_v^4 v_{k_v}^2 \sim \eta, \quad (10.49)$$

using (10.47a) and (10.48). Thus, the enstrophy dissipation in two-dimensional turbulence is (at least according to this theory) independent of the viscosity, just as energy dissipation is independent of the viscosity in three-dimensional turbulence.



### Inertial Range Properties in 3D and 2D Turbulence

A few inertial range properties are listed below, omitting nondimensional constants.

|                        | 3D energy range               | 2D enstrophy range   |       |
|------------------------|-------------------------------|----------------------|-------|
| Energy spectrum        | $\varepsilon^{2/3} k^{-5/3}$  | $\eta^{2/3} k^{-3}$  | (T.1) |
| Turnover time          | $\varepsilon^{-1/3} k^{-2/3}$ | $\eta^{-1/3}$        | (T.2) |
| Viscous scale, $L_\nu$ | $(\nu^3/\varepsilon)^{1/4}$   | $(\nu^3/\eta)^{1/6}$ | (T.3) |

In these expressions:

$\nu$  = viscosity,  $k$  = wavenumber,  $\varepsilon$  = energy cascade rate,  
 $\eta$  = enstrophy cascade rate,

#### Energy inertial range

The energy inertial range of two-dimensional turbulence is quite similar to that of three-dimensional turbulence, except in one major respect: the energy flows from smaller to larger scales! Because the atmosphere and ocean behave in some ways as two-dimensional fluids, this has profound consequences on their behaviour, and is something we return to in the next section. The upscale energy flow is known as the *inverse cascade*, and the associated energy spectrum is, as in the three-dimensional case,

$$\mathcal{E}(k) = \mathcal{K}_\varepsilon \varepsilon^{2/3} k^{-5/3}, \quad (10.50)$$

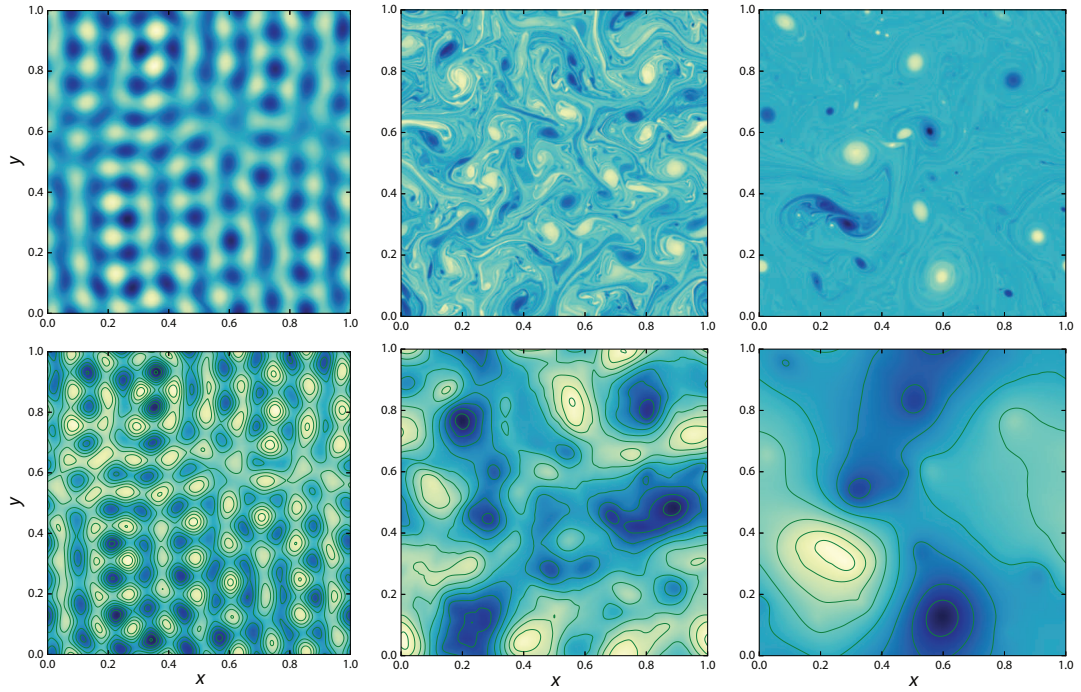
where  $\mathcal{K}_\varepsilon$  is a nondimensional constant — sometimes called the Kolmogorov–Kraichnan constant, and not necessarily equal to  $\mathcal{K}$  in (10.24) — and  $\varepsilon$  is the rate of energy transfer to larger scales, where it is removed by boundary-layer friction.

The atmosphere itself is not observed to have an inverse  $-5/3$  spectrum at large scales; indeed there is no well-defined inverse energy cascade in the sense described above. The atmosphere does have an approximate  $-3$  cascade but whether we should attribute this to a classical forward enstrophy cascade is not settled. A numerical simulation of two-dimensional turbulence, illustrating both the energy and enstrophy cascade, is shown in Fig. 10.8.

## 10.5 ♦ GEOSTROPHIC TURBULENCE

The atmosphere is not two-dimensional but, because of the effects of stratification and rotation, it is ‘quasi-two-dimensional’ and approximately satisfies the quasi-geostrophic equations. Let us therefore extend our two-dimensional theory to include the effects of stratification.

*Robert Kraichnan (1928–2008) made a host of significant contributions to the theory of turbulence in the second half of the twentieth century, working independently first in New Hampshire and later in New Mexico. Among other things he is largely responsible for the inertial-range theory of two-dimensional turbulence described here.*



**Fig. 10.8:** Evolution of vorticity (top) and streamfunction (bottom) in a numerical simulation of two-dimensional turbulence with no forcing. The initial conditions (left) have just a few non-zero Fourier modes (around wavenumber 9) with randomly generated phases. Time proceeds left to right, with like-signed vortices merging (with an example in the top right panel) and enstrophy cascading to smaller scales between the vortices. The scale of the streamfunction grows larger, reflecting the transfer of energy to larger scales. (See Fig. 10.11 to see jets form because of the effects of rotation.)

### 10.5.1 An Analogue to Two-Dimensional Flow

Consider a quasi-geostrophic atmosphere in which the unforced, inviscid governing equations are

$$\frac{Dq}{Dt} = 0, \quad q = \nabla^2 \psi + Pr^2 \frac{\partial^2 \psi}{\partial z^2}, \quad (10.51a)$$

where  $Pr = f_0/N$  is the *Prandtl ratio* (and  $Pr/H$  is the inverse of the deformation radius) and  $D/Dt = \partial/\partial t + \mathbf{u} \cdot \nabla$  is the two-dimensional material derivative. The vertical boundary conditions are

$$\frac{D}{Dt} \left( \frac{\partial \psi}{\partial z} \right) = 0, \quad \text{at } z = 0, H. \quad (10.51b)$$

These equations are analogous to the equations of motion for purely two-dimensional flow. In particular, with periodic lateral boundary conditions, or conditions of no-normal flow, there are two quadratic invariants of the motion, the energy and the enstrophy, which are obtained by multiplying (10.51a) by  $-\psi$  and  $q$  and integrating over the domain. The conserved energy is

$$\frac{d\hat{E}}{dt} = 0, \quad \hat{E} = \frac{1}{2} \int_V \left[ (\nabla \psi)^2 + Pr^2 \left( \frac{\partial \psi}{\partial z} \right)^2 \right] dV, \quad (10.52)$$

where the integral is over a *three-dimensional* domain. The enstrophy is conserved at each vertical level, and of course the volume integral is also

conserved, namely

$$\frac{d\widehat{Z}}{dt} = 0, \quad \widehat{Z} = \frac{1}{2} \int_V q^2 dV = \frac{1}{2} \int_V \left[ \nabla^2 \psi + Pr^2 \left( \frac{\partial^2 \psi}{\partial z^2} \right) \right]^2 dV. \quad (10.53)$$

The analogy with two-dimensional flow is even more transparent if we further rescale the vertical coordinate by  $1/Pr$ , and so let  $z' = z/Pr$ . Then the energy and enstrophy invariants are:

$$\widehat{E} = \int (\nabla_3 \psi)^2 dV, \quad \widehat{Z} = \int q^2 dV = \int (\nabla_3^2 \psi)^2 dV, \quad (10.54)$$

where  $\nabla_3 = \hat{i} \partial / \partial x + \hat{j} \partial / \partial y + \hat{k} \partial / \partial z'$ . The invariants then have almost the same form as the two-dimensional invariants, but with a three-dimensional Laplacian operator instead of a two-dimensional one.

Given these invariants, we expect that any dynamical behaviour that occurs in the two-dimensional equations *that depends solely on the energy/enstrophy constraints* should have an analogue in quasi-geostrophic flow. In particular, the transfer of energy to large-scales and enstrophy to small scales will also occur in quasi-geostrophic flow with, in so far as these transfers are effected by a local cascade, corresponding spectra of  $k^{-5/3}$  and  $k^{-3}$ . However, in the quasi-geostrophic case, it is the *three-dimensional* wavenumber that is relevant, with the vertical component scaled by the Prandtl ratio. As a consequence, the energy cascade to larger horizontal scales is generally accompanied by a cascade to larger vertical scales — a *barotropization* of the flow. Still, the analogy between two-dimensional and quasi-geostrophic cascades should not be taken too far, because in the latter the potential vorticity is advected only by the horizontal flow. Thus, the dynamics of quasi-geostrophic turbulence are *not* in general isotropic in three-dimensional wavenumber. To examine these dynamics more fully we turn to a simpler model, that of two-layer flow.

Geostrophic turbulence is similar to two-dimensional turbulence in that energy is transferred to large scales. The transfer occurs in both the vertical and horizontal directions, and the transfer to large scales leads to the *barotropization* of the flow.

### 10.5.2 Two-Layer Geostrophic Turbulence

We consider flow in two layers of equal depth, governed by the two-level quasi-geostrophic equations as described in Section 5.6. Taking  $\beta = 0$  the equations of motion are

$$\frac{\partial q_i}{\partial t} + J(\psi_i, q_i) = 0, \quad i = 1, 2, \quad (10.55)$$

where, for any  $a$  and  $b$ ,  $J(a, b) = \partial a / \partial x \partial b / \partial y - \partial a / \partial y \partial b / \partial x$  and

$$q_1 = \nabla^2 \psi_1 + \frac{1}{2} k_d^2 (\psi_2 - \psi_1), \quad q_2 = \nabla^2 \psi_2 + \frac{1}{2} k_d^2 (\psi_1 - \psi_2), \quad (10.56a, b)$$

$$\frac{1}{2} k_d^2 = \frac{4 f_0^2}{N^2 H^2}, \quad (10.56c)$$

and  $k_d$  is inversely proportional to the baroclinic deformation radius.

It turns out to be convenient to rewrite the equations in terms of the sum and difference of the streamfunctions in each layer, and to this end

we define the barotropic and baroclinic streamfunctions by

$$\hat{\psi} \equiv \frac{1}{2}(\psi_1 + \psi_2), \quad \tau \equiv \frac{1}{2}(\psi_1 - \psi_2), \quad (10.57)$$

so that  $\hat{\psi}$  is the vertically averaged streamfunction and  $\tau$  is proportional to the buoyancy (since  $b = f_0 \partial \psi / \partial z$ ; think ‘tau for temperature’). The potential vorticities for each layer are then

$$q_1 = \nabla^2 \hat{\psi} + (\nabla^2 - k_d^2)\tau, \quad q_2 = \nabla^2 \hat{\psi} - (\nabla^2 - k_d^2)\tau, \quad (10.58a,b)$$

and the equations of motion may be rewritten as the following evolution equations for  $\hat{\psi}$  and  $\tau$ :

$$\frac{\partial}{\partial t} \nabla^2 \hat{\psi} + J(\hat{\psi}, \nabla^2 \hat{\psi}) + J(\tau, (\nabla^2 - k_d^2)\tau) = 0, \quad (10.59a)$$

$$\frac{\partial}{\partial t} (\nabla^2 - k_d^2)\tau + J(\tau, \nabla^2 \hat{\psi}) + J(\hat{\psi}, (\nabla^2 - k_d^2)\tau) = 0. \quad (10.59b)$$

With these new variables a little algebra reveals that the conservation of energy takes the form

$$\frac{d\hat{E}}{dt} = 0, \quad \hat{E} = \frac{1}{2} \int_A [(\nabla \hat{\psi})^2 + (\nabla \tau)^2 + k_d^2 \tau^2] dA, \quad (10.60)$$

and the conservation of total enstrophy takes the form

$$\frac{d\hat{Z}}{dt} = 0, \quad \hat{Z} = \int_A (\nabla^2 \hat{\psi})^2 + [(\nabla^2 - k_d^2)\tau]^2 dA. \quad (10.61)$$

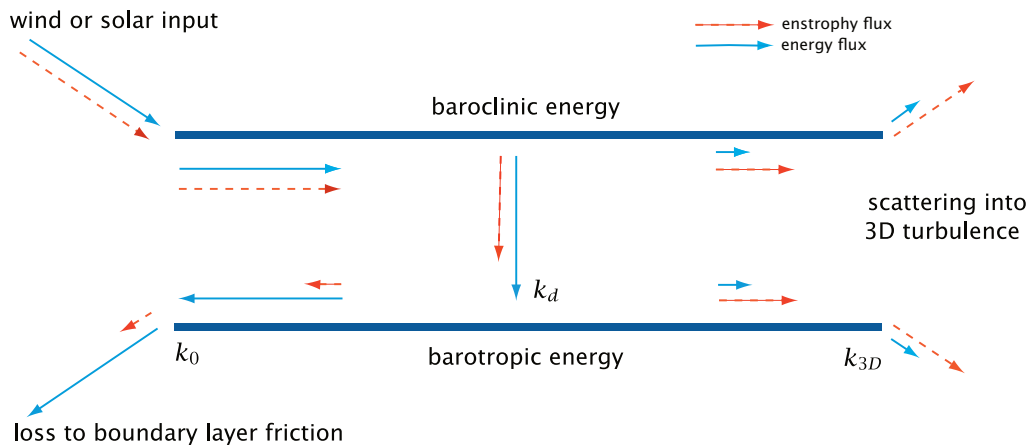
In addition the enstrophy in each layer is conserved. The nonlinear interactions in (10.59) will lead to a transfer of energy between scales, as in two-dimensional turbulence, and to better understand that we note:

- (i)  $\hat{\psi}$  and  $\tau$  are analogous to Fourier modes in the vertical. That is,  $\hat{\psi}$  is a *barotropic mode* with a ‘vertical wavenumber’ of zero, and  $\tau$  is a *baroclinic mode* with a vertical wavenumber of one.
- (ii) From (10.59) we see there are two distinct kinds of nonlinear interaction, namely

$$(i) \quad (\hat{\psi}, \hat{\psi}) \rightarrow \hat{\psi}, \quad (ii) \quad (\tau, \tau) \rightarrow \hat{\psi} \quad \text{or} \quad (\hat{\psi}, \tau) \rightarrow \tau. \quad (10.62)$$

The first is a *barotropic interaction*, involving only the barotropic mode. The other two are examples of mixed barotropic–baroclinic interactions: two baroclinic modes giving rise to a barotropic mode, or a barotropic mode interacting with a baroclinic mode. There are no interactions that involve only the baroclinic mode.

- (iii) Wherever the Laplacian operator acts on  $\tau$  in the evolution equations, it is accompanied by  $-k_d^2$ . That is, it is as if the effective wavenumber (squared) of  $\tau$  is shifted, so that  $k^2 \rightarrow k^2 + k_d^2$ . Similarly, the energy and enstrophy invariants bear the same relation to each other that they do in two-dimensional turbulence, except that the relevant wavenumber for the baroclinic mode is  $k^2 + k_d^2$  instead of  $k^2$ . Effectively, at any given horizontal wavenumber, the baroclinic modes have a higher ‘total’ wavenumber than the barotropic ones.



### Phenomenology of Geostrophic Turbulence

Putting together the above considerations with our experience of two-dimensional turbulence leads to the following picture of geostrophic turbulence, summarized in Fig. 10.9. In general, nonlinear interactions among the various modes will lead to a transfer of energy to the largest possible scales, or the smallest possible total wavenumber, but since the effective wavenumber of the baroclinic mode is larger than that of the barotropic mode, energy will also be transferred to the barotropic mode (this is the barotropization process referred to earlier). That is, the energy will be transferred to the largest possible vertical scale as well as the largest horizontal scale. The conversion from baroclinic to barotropic energy tends to occur at the scale of the radius of deformation, since this process is, essentially, baroclinic instability.

The overall energy cycle thus goes as follows. At large horizontal scales we may imagine some source of baroclinic energy, which in the atmosphere might be the differential heating between pole and equator, or in the ocean might be the wind and surface heat fluxes. Baroclinic instability effects a non-local transfer of energy to the deformation scale, where both baroclinic and barotropic modes are excited. Energy is then transferred back to large scales in the barotropic mode (an inverse cascade), and ultimately it is dissipated by bottom friction. At the same time, there is an enstrophy cascade in each layer to smaller and smaller scales, until the wavenumber is large enough (denoted by  $k_{3D}$  in Fig. 10.9) that non-geostrophic effects become important and enstrophy is scattered by three-dimensional effects and dissipated.

This cycle of energy occurs in baroclinic eddies in both atmosphere and ocean. Higher baroclinic modes, not captured in the two-layer system, can sometimes be excited (especially in the ocean) but the two-layer system does capture the main features of the real systems. Indeed, in the atmosphere the great mid-latitude weather systems can be succinctly characterized by the phrase ‘baroclinic growth and barotropic decay’, which is essentially the process illustrated in Fig. 10.9.

**Fig. 10.9:** Two-layer baroclinic turbulence. The horizontal axis represents horizontal wavenumber, and the vertical variation is decomposed into two vertical modes — the barotropic and first baroclinic. Energy transfer is shown by solid arrows and enstrophy transfer by dashed arrows. Large-scale forcing provides energy to the baroclinic mode at large scales. Energy is transferred to smaller baroclinic scales, and then into barotropic energy, and finally transferred to larger barotropic scales in an upscale cascade.

## 10.6 BETA-PLANE TURBULENCE

We now turn to looking at the effects of a variable Coriolis parameter — the beta effect — on geostrophic turbulence. To display the effects in their most austere fashion we restrict attention to two-dimensional turbulence on the beta-plane, and we can write the equation of motion in the various equivalent forms,

$$\frac{Dq}{Dt} = \frac{D}{Dt}(\zeta + f) = F - D, \quad \text{or} \quad \frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla(\zeta + f) = F - D, \quad (10.63a,b)$$

or

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta + \beta y) = F - D \quad \text{or} \quad \frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta v = F - D. \quad (10.63c,d)$$

In these equations  $q = \zeta + f$  is the potential vorticity (equal to the absolute vorticity here),  $\zeta = \nabla^2 \psi$  is the relative vorticity,  $J(\psi, \zeta) = \psi_x \zeta_y - \psi_y \zeta_x$  is the Jacobian,  $f = f_0 + \beta y$ , and  $F$  and  $D$  represent the forcing and dissipation.

The form of (10.63d) suggests that the presence of  $\beta$  will lead to the production of zonal flow, since in a statistically steady state we can write

$$\beta v = \text{forcing and nonlinear terms.} \quad (10.64)$$

The forcing is ultimately responsible for the flow, and if we imagine increasing  $\beta$  while keeping the forcing constant then  $v$  (the meridional flow) must diminish or the equation simply cannot balance. However, the zonal flow,  $u$ , will remain strong and the vorticity equation can then display a three-way balance between nonlinearity,  $\mathbf{u} \cdot \nabla \zeta$ , the beta term,  $\beta v$ , and the forcing,  $F$ . Still, the argument is a little heuristic and the precise balance depends on the scale of the motion, so let us look at the dynamics a little more closely.

### 10.6.1 The Wave–Turbulence Cross-over

Omitting the forcing term we write the equation of motion as

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla \zeta + \beta v = 0. \quad (10.65)$$

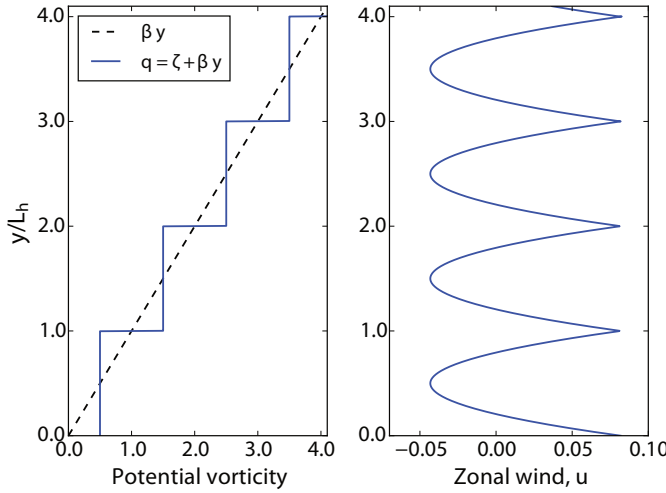
If  $\zeta \sim U/L$  and if  $t \sim T$  then the respective terms in this equation scale as

$$\frac{U}{LT} \quad \frac{U^2}{L^2} \quad \beta U, \quad (10.66)$$

and for now we do not differentiate between  $U$  and  $V$ . If the nonlinear term dominates, as (10.66) suggests it will at small scales, then the flow will be turbulent and display all the characteristics of two-dimensional turbulence. If the beta term dominates then Rossby waves will dominate and the flow will be more linear, and this behaviour will occur at large scales. The cross-over scale, denoted  $L_R$ , is called the *Rhines scale* and is given by

$$L_R \sim \left( \frac{U}{\beta} \right)^{1/2}. \quad (10.67)$$





**Fig. 10.10:** An idealized potential vorticity staircase on a  $\beta$ -plane with  $\beta = 1$ . The left panel shows homogeneous regions of potential vorticity of meridional extent  $L_h$  separated by jumps, as calculated using (10.71). The right-hand panel shows the corresponding zonal flow, calculated using (10.75) with  $u_0$  chosen to be such that the average zonal flow is zero.

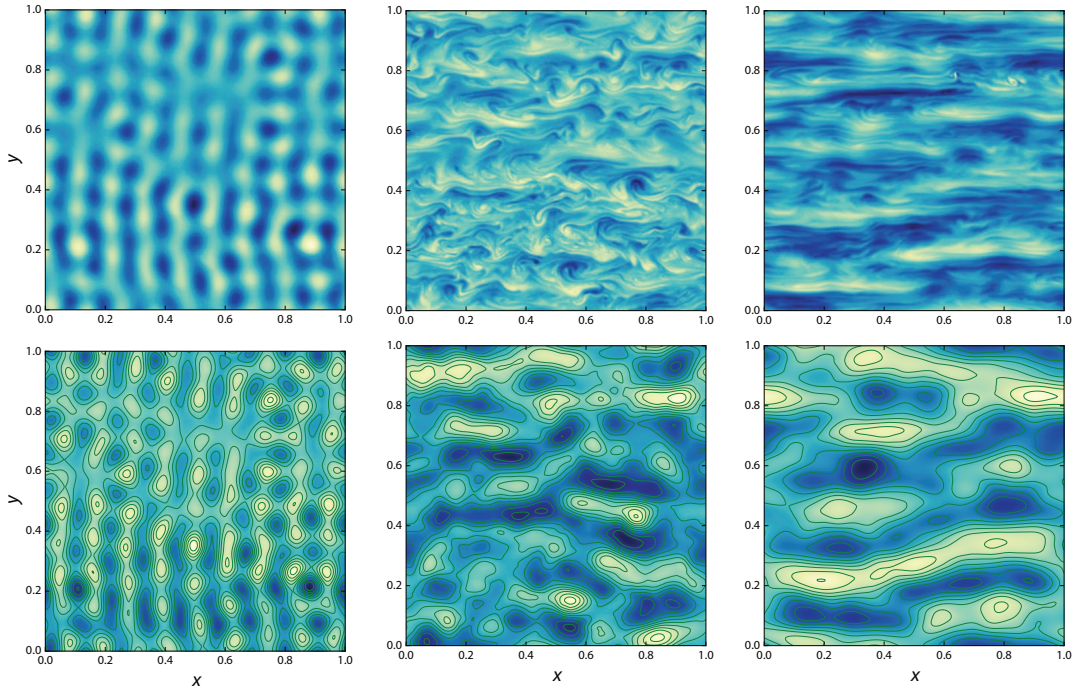
At scales larger than  $L_R$  Rossby waves are likely to dominate whereas at smaller scales advection and turbulence dominate and the flow is more isotropic. (The  $U$  in (10.67) is best interpreted as the root-mean-square velocity at the energy containing scales, not a mean or translational velocity.) At these large scales the Rossby waves interact with the turbulence and produce a zonal flow, as discussed in Problem 10.3 and in Chapter 12. However, here we shall give a complementary argument and show that the tendency of the flow to homogenize potential vorticity leads to jet production on the beta-plane.

### 10.6.2 Jets and Potential Vorticity Staircases

The gist of the argument is as follows. Suppose that a turbulent flow exists on a background state that has a potential vorticity gradient, such as might be caused by the beta-effect. The turbulent eddies seek to homogenize the potential vorticity but, in general, they are unable to do so completely. Instead, the flow becomes partially homogenized, with regions of homogeneous potential vorticity separated by jumps, as illustrated in Fig. 10.10. Such a structure is called a *potential vorticity staircase*, and the staircase implies the existence of zonal jets.

To go into more detail recall that, as discussed in Section 10.2.1, the gradients of a scalar that is both advected and diffused become smeared out as much as possible and, as a consequence, the scalar may become homogenized. Potential vorticity is such a scalar; however, potential vorticity is not a *passive* scalar and the process of homogenization affects the flow itself, often preventing complete homogenization both because of the need to satisfy boundary conditions and because the process of homogenization may require more energy than the flow contains. To see this, consider freely-evolving flow on a beta-plane obeying the barotropic vorticity equation

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta + \beta y) = \nu \nabla^2 \zeta, \quad (10.68)$$



**Fig. 10.11:** Evolution of vorticity (top) and streamfunction (bottom) in a numerical simulation on the  $\beta$ -plane, obeying (10.65) with the addition of a weak viscous term on the right-hand side. The initial conditions are the same as for Fig. 10.8, and time proceeds from left to right. Compared to Fig. 10.8, vortex formation is inhibited and there is a tendency toward zonal flow

where  $v$  is sufficiently small that energy is well conserved over the timescales of interest. If potential vorticity,  $q = \zeta + \beta y$ , is to be homogenized over some meridional scale  $L_h$  then, in the homogenized region,  $\zeta = -\beta y$ , taking the value of  $q$  to be zero for simplicity. If the flow is predominantly zonal then the zonal flow will have a magnitude  $U \sim \beta y^2$  and the energy in the region is, very approximately,

$$\frac{1}{2} \int u^2 dx dy \approx \int (\beta y^2)^2 dx dy, \quad (10.69)$$

giving the estimate  $U^2 L_h \sim \beta^2 L_h^5$ . Solving the above estimate for  $L_h$  suggests that the potential vorticity can become homogenized over the scale

$$L_h \sim \left( \frac{U}{\beta} \right)^{1/2}. \quad (10.70)$$

This is the same as the Rhines scale given in (10.67), as it has to be by dimensional analysis, although now it is supposed that much of the energy lies in the zonal flow. Although the scale is the same, this way of looking at the problem adds something to the physical picture — *an asymmetry between eastward and westward flow*, as we now show.

In the idealized staircase of Fig. 10.10 potential vorticity is piecewise continuous and given by

$$q = \zeta + \beta y = q_0, \quad 0 < y < L_h, \quad (10.71a)$$

$$q = \zeta + \beta y = q_0 + \beta L_h \equiv q_1, \quad L_h < y < 2L_h, \quad \text{and so on.} \quad (10.71b)$$



In any one of the homogenized regions the flow is given by solving

$$\frac{\partial u}{\partial y} = \beta y - q_n \quad \text{giving} \quad u = \frac{1}{2}\beta y^2 - q_n y + \text{constant}, \quad (10.72)$$

where  $q_n = \beta L_h/2 + n\beta L_h$ . The constants in each region may be determined by requiring continuity of  $u$  across the regions, and we then obtain

$$u = \frac{1}{2}\beta(y - L_h/2)^2 + u_0, \quad 0 < y < L_h, \quad (10.73)$$

$$u = \frac{1}{2}\beta(y - 3L_h/2)^2 + u_0, \quad L_h < y < 2L_h, \quad (10.74)$$

where  $u_0$  is a constant. In general we have

$$u = \frac{1}{2}\beta(y - (n - 1/2)L_h)^2 + u_0, \quad (n - 1)L_h < y < nL_h, \quad n = 1, 2, 3 \dots \quad (10.75)$$

Relative to  $u_0$ , the flow is weakly westward in the homogenized regions, whereas in the transition regions the flow has sharp eastward peaks, as shown in Fig. 10.10.

The staircase structure implies that potential vorticity is (obviously) well mixed in the regions of constant potential vorticity, but that there is no mixing across the jumps. The jumps are known as *mixing barriers* and, once formed, are found to be very persistent in numerical simulations. The edge of the stratospheric polar vortex, and the boundaries of ocean gyres, are examples of such barriers. Numerical simulations do show a tendency to produce zonal flow on the beta-plane, as seen in Fig. 10.11. However, the subject continues to evolve and the reader should consult the references below and the recent primary literature to learn more.

## Notes and References

The study of turbulence itself goes back to, at least, Leonardo da Vinci, and blossomed in the twentieth century. There are so many relevant papers on the subject that the reader should refer to a book such as Pope (2000) or Davidson (2015) for a discussion of the classical theory and a bibliography. The phenomenology of two-dimensional turbulence was put forward by Kraichnan (1967, 1971), Leith (1968) and Batchelor (1969). Geostrophic turbulence was introduced by Charney (1971) with subsequent work by Salmon (1980) and Rhines (1977). The interaction of Rossby waves and turbulence and the generation of zonal flows on a beta-plane comes from Rhines (1975) and Vallis & Maltrud (1993). A potential vorticity staircase was proposed by Marcus (1993), his motivation being the jets of Jupiter.

## Problems

- 10.1 *Predictability.* The eddy turnover time of three-dimensional turbulence at a wavenumber  $k$  is given by  $\tau_k = \varepsilon^{-1/3} k^{-2/3}$ , and that of two-dimensional turbulence by  $\tau_k = \eta^{-1/3}$  where  $k$  is the wavenumber and  $\varepsilon$  and  $\eta$  are constants (the energy and enstrophy cascade rates, respectively). Suppose that in weather prediction the error is confined to small scales and that the time

taken for the error to contaminate the next largest scale (in a logarithmic sense) is  $\tau_{k_i}$  so that the time taken for an error at a small scale  $k_s$  to reach the large scales  $k_l$  is given by

$$T = \int_{k_s}^{k_l} \tau_k dk. \quad (\text{P10.1})$$

If the inertial range extends indefinitely show that this time is infinite for classical two-dimensional turbulence and finite for three-dimensional turbulence, and discuss the implications for weather predictability and weather forecasting. If the atmosphere is two-dimensional down to a scale of 50 km, estimate a limit to weather predictability (e.g., a timescale in days), making sensible (and clearly stated) assumptions about the magnitude of the flow.

- 10.2 *Heat transport.* Considering large-scale turbulence in the atmosphere, suppose that the mixing length is given by the radius of deformation and a mixing velocity is given by a typical synoptic velocity. Estimate a value for the eddy diffusivity provided by weather systems, and hence estimate the polewards heat transport by these systems. Compare with the values of the heat transport in the atmosphere given by observations, and discuss the realism, or otherwise, of your calculation.

- 10.3 *Jets.* The Rossby wave frequency is given by  $\omega_R = -\beta k/(k^2 + l^2)$ , and this is anisotropic in  $(k, l)$  space. Suppose that a ‘turbulence frequency’, in two dimensions, is given by  $\omega_t = Uk$ , where  $U$  is a constant.

- (a) By equating the above two expressions, obtain an expression for the line in wavenumber space on which the turbulence frequency and Rossby-wave frequency are the same. Plot the curve; it should look a little like a dumbbell, with the Rossby-wave frequency higher than the turbulence frequency inside the dumbbell.

*Partial solution.* The crossover  $x$ -wavenumber is

$$k^x = (\beta/U)^{1/2} \cos^{3/2} \theta \quad \text{where} \quad \theta = \tan^{-1}(k^y/k^x).$$

What is  $k^y$ ?

- (b) Energy is transferred from small scales to large in two-dimensional turbulence; however, it will be unable to excite Rossby waves if they have a higher frequency than the turbulence. Hence argue that the cascade will excite zonal motion, and make a rough estimate for the meridional scale of the zonal flow.
- (c) Repeat the above but now with a turbulence frequency given by  $\omega_t = \varepsilon^{1/3} k^{2/3}$ , which is the inverse of the eddy turnover time in the energy inertial range, given (as in 3D turbulence) by (10.28).

- 10.4 Suppose that the inverse cascade of two-dimensional turbulence is halted by a linear drag. (That is, the vorticity equation is  $D\zeta/Dt = F - r\zeta$ , where  $F$  is the stirring and  $r$  is a constant.)

- (a) By equating the drag timescale to that of the turbulence obtain an estimate for the scale at which the cascade is halted, and an estimate for the velocity at that scale.
- (b) Show that an estimate of the horizontal eddy diffusivity,  $\mathcal{K}$ , is then  $\mathcal{K} \sim \varepsilon/r^2$ , where  $\varepsilon$  is the energy cascade rate.

Part II

## **ATMOSPHERES**



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## The Tropical Atmosphere

**I**N THIS CHAPTER AND THE TWO FOLLOWING we discuss the structure and circulation of planetary atmospheres. This chapter and the next focus mainly on Earth, first on the tropical circulation and then on the midlatitudes and the stratosphere. Then, in Chapter 13, we look at planetary atmospheres a little more generally. We begin with a brief observational overview of Earth's atmosphere as a whole.

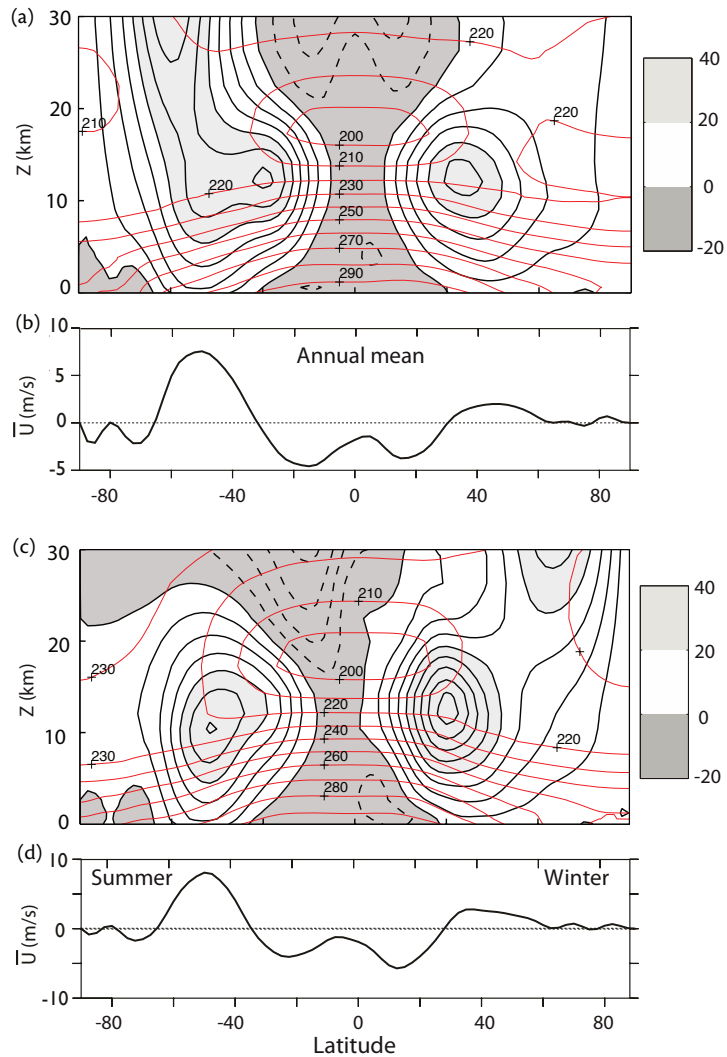
### 11.1 AN OBSERVATIONAL OVERVIEW

Many of the main zonally- and/or time-averaged features of Earth's atmosphere can be seen in Figs. 11.1–11.3. The most prominent features are:

- (i) The temperature falls monotonically with height to an altitude of about 16 km (in the tropics) or 8 km at high latitudes, before increasing with height. The lower region is called the *troposphere*, above which lies the *stratosphere*, and the boundary between the two is the *tropopause*.
- (ii) The temperature also falls monotonically from equator to pole, at the surface falling from about 300 K to 240 K at the pole. The tropopause temperature is more uniform, varying from about 230 K to 210 K.
- (iii) The surface winds are easterly in low latitudes (from about 30°S to 30°N), westerly in mid- and high latitudes, with weak polar easterlies in some seasons.
- (iv) The winds increase in height, especially in midlatitudes, with pronounced westerly jets in both hemispheres centred at about 40° latitude in both hemispheres.

**Fig. 11.1:** (a) Annual mean, zonally-averaged zonal wind (heavy contours and shading) and the zonally-averaged temperature (red, thinner contours). (b) Annual mean, zonally averaged zonal winds at the surface. (c) and (d) Same as (a) and (b), except for northern hemisphere winter (December–January–February, or DJF).

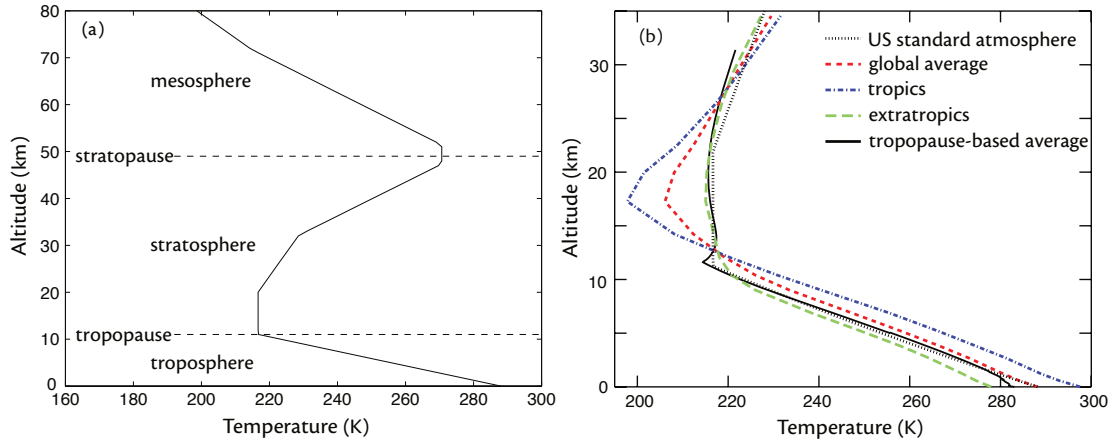
The wind contours are at intervals of  $5 \text{ m s}^{-1}$  with dark shading for eastward winds above  $20 \text{ m s}^{-1}$  and light shading for all westward winds, and the temperature contours are labelled.



(v) In the meridional ( $y$ - $z$ ) plane the circulation in each hemisphere (Fig. 11.3) is characterized by:

- (i) A 'direct' *Hadley Cell*, with warm air rising near the equator and sinking in the tropics. The winter Hadley Cell is much stronger and has greater latitudinal extent than the summer one, with the warm air rising at low latitudes in the summer hemisphere, sinking in the tropics in the winter hemisphere.
- (ii) An 'indirect' *Ferrel Cell*, with cool air apparently rising around  $60^\circ$  and sinking in the tropics.

A direct cell is one that is thermally driven, with warm, buoyant fluid rising and cold fluid descending. An indirect cell may be mechanically driven, as we discuss later.



## 11.2 AN IDEAL HADLEY CIRCULATION

To make matters as simple as possible we imagine the Earth to be a sphere with a uniform surface (so no mountains and no oceans), and that there are no seasons (unlike the case in Fig. 11.3). Because of the differential solar heating the air is warmer at low latitudes than at high, so we may reasonably imagine that the warm air rises and moves polewards before cooling and sinking at some high latitude, perhaps near the pole, and returning near the ground. Indeed such a concept was envisioned by George Hadley over 300 hundred years ago. However, observations tell us that the air does not go all the way to the pole; rather, it sinks in the subtropics at about 25–30°. There are two reasons why it must sink, one related to thermodynamic constraints and the other to hydrodynamic instabilities. Both are related to the properties of the air as it moves, conserving its angular momentum.

### 11.2.1 Zonally-Symmetric Equations of Motion

In Chapter 2 we wrote down the equations of motion on a sphere. If the flow is zonally symmetric (no longitudinal variation) then, with a little manipulation, the zonal momentum equation may be written in the form

$$\frac{\partial u}{\partial t} - (f + \zeta)v + w \frac{\partial u}{\partial z} = 0. \quad (11.1)$$

The variables in this equation are functions of latitude and height ( $\vartheta$  and  $z$ ) only, and not longitude,  $\lambda$ , and  $\zeta = -(a \cos \vartheta)^{-1} \partial_\vartheta(u \cos \vartheta)$ . If the vertical advection is small then a steady solution obeys

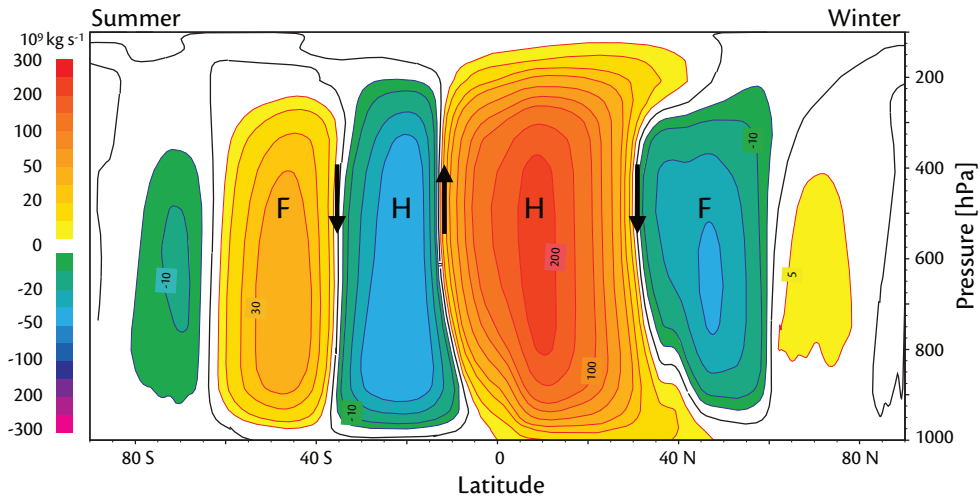
$$(f + \zeta)v = 0. \quad (11.2)$$

Presuming that the meridional flow  $v$  is non-zero then  $f + \zeta = 0$ , or equivalently, on the sphere,

$$2\Omega \sin \vartheta = \frac{1}{a} \frac{\partial u}{\partial \vartheta} - \frac{u \tan \vartheta}{a}. \quad (11.3)$$

**Fig. 11.2:** (a) The temperature profile of the so-called US standard atmosphere. (b) Observed, annually averaged profiles of temperature in the atmosphere, where the ordinate is log-pressure. Tropics here is the average from 30°S to 30°N, and extratropics is the average over the rest of the globe. (The ‘tropopause-based average’ uses the tropopause itself as the origin of the height scale, set to be 11 km, but we shall not discuss it further.)

*The Hadley Cell is named for George Hadley (1685–1768), a British meteorologist who put forward perhaps the first scientific model of Earth’s overturning circulation, in which the air rose near the equator and sank near the pole.*



**Fig. 11.3:** The observed meridional overturning circulation (MOC) of the atmosphere ( $\text{kg s}^{-1}$ ) averaged over December–January–February. Note the direct *Hadley Cells* (H) particularly strong in winter with rising motion near the equator, descending motion in the subtropics, and the weaker, indirect, *Ferrel Cells* (F) in midlatitudes. (From an ECMWF reanalysis.)

At the equator we may assume that  $u = 0$ , because here parcels have risen from the surface where the flow is weak. Equation (11.3) then has a solution of

$$u = \Omega a \frac{\sin^2 \vartheta}{\cos \vartheta}. \quad (11.4)$$

This gives the zonal velocity of the poleward moving air in the upper branch of the (model) Hadley Cell, above the frictional boundary layer. Evidently the zonal velocity increases rapidly with latitude: at  $20^\circ$  and  $40^\circ$  the values of  $u$  are about  $59$  and  $256 \text{ m s}^{-1}$  respectively (look ahead to Fig. 11.6 or Fig. 11.7), becoming far larger than the observed values at midlatitudes.

### Angular momentum conservation

An instructive interpretation of (11.4) comes by way of the conservation of angular momentum,  $m$ , of a ring of air at a latitude  $\vartheta$ , as in Fig. 11.4. The angular momentum per unit mass of a parcel of air with zonal velocity  $u$  is

$$m = (u + \Omega a \cos \vartheta) a \cos \vartheta, \quad (11.5)$$

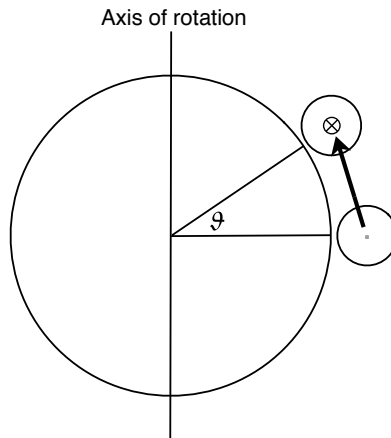
and if  $u = 0$  at  $\vartheta = 0$  and if  $m$  is conserved on a poleward moving parcel, then (11.5) leads directly to (11.4).

The air returning to the equator close to the surface has a small zonal velocity, meaning that there is a large thermal wind, which we may calculate using the thermal wind expression

$$2\Omega \sin \vartheta \frac{\partial u}{\partial z} = -\frac{1}{a} \frac{\partial b}{\partial \vartheta}, \quad (11.6)$$

where  $b = g \delta \theta / \theta_0$  and  $\delta \theta$  is the deviation of potential temperature from a constant reference value  $\theta_0$ . (Be reminded that  $\theta$  is potential temperature,





**Fig. 11.4:** If a ring of air at the equator moves poleward it moves closer to the axis of rotation. If the parcels in the ring conserve their angular momentum their zonal velocity must increase; thus, if  $m = (u + \Omega a \cos \vartheta)a \cos \vartheta$  is preserved and  $u = 0$  at  $\vartheta = 0$  we recover (11.4).

whereas  $\vartheta$  is latitude.) Vertically integrating from the ground to the height  $H$  where the outflow occurs and substituting (11.4) for  $u$  yields

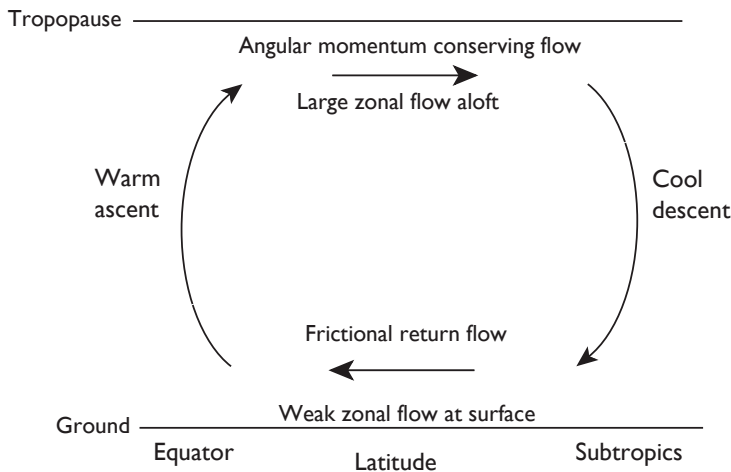
$$\frac{1}{a\theta_0} \frac{\partial \theta}{\partial \vartheta} = -\frac{2\Omega^2 a \sin^3 \vartheta}{gH \cos \vartheta}, \quad (11.7)$$

where  $\theta = H^{-1} \int_0^H \delta \theta dz$  is the vertically averaged potential temperature.

If the latitudinal extent of the Hadley Cell is not too great we can make a small-angle approximation, and replace  $\sin \vartheta$  by  $\vartheta$  and  $\cos \vartheta$  by one, and then integrating (11.7) gives

$$\theta = \theta(0) - \frac{\theta_0 \Omega^2 y^4}{2gHa^2}, \quad (11.8)$$

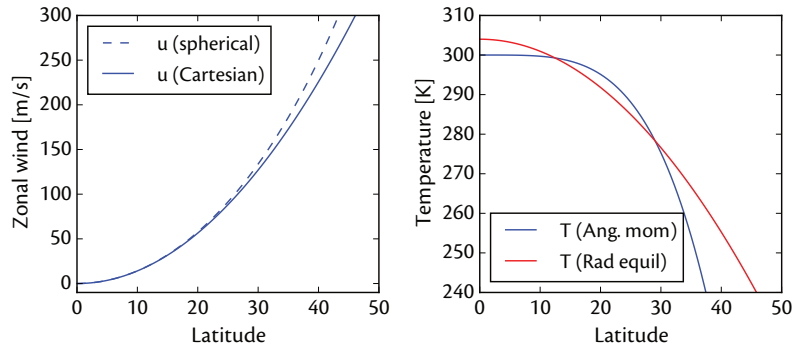
where  $y = a\vartheta$  and  $\theta(0)$  is the potential temperature at the equator. Evidently, the temperature in an angular momentum conserving Hadley Cell



**Fig. 11.5:** A simple model of the Hadley Cell. Rising air near the equator moves poleward near the tropopause, descending in the subtropics and returning.

The poleward moving air conserves its angular momentum, leading to a shear of the zonal wind that increases away from the equator. By thermal wind the temperature of the air falls as it moves poleward, and to satisfy the thermodynamic budget it sinks in the subtropics.

**Fig. 11.6:** Left: Upper tropospheric zonal wind computed for an angular momentum conserving Hadley Cell. The solid line makes the small-angle approximation. Right: The blue line shows temperature of the angular momentum conserving wind, computed using (11.8). The red curve shows the radiative equilibrium temperature, using (11.9). The Hadley Cell extends to the latitude of the second crossing, at about 30°.



drops rapidly away from the equator — if (11.8) were satisfied the temperature would drop by over 80 K between the equator and 40° latitude, and would drop to absolute zero at around 50°. Plainly, the Hadley Cell cannot extend to the pole and remain angular momentum conserving. What happens? Essentially, the air becomes so cold that it sinks in order to satisfy the thermodynamic constraints, as in Fig. 11.5 and as we now explain.

### 11.2.2 Thermodynamic Constraints

In the absence of any motion the atmosphere will evolve to a radiative-equilibrium temperature, in which the warming by solar radiation exactly balances the cooling by infra-red radiation. A difficult calculation is needed to determine the true radiative equilibrium temperature, but an approximate one is easier to come by. Essentially, at each latitude the net incoming solar radiation must be balanced by the outgoing infra-red radiation, which is an increasing function of temperature in the upper troposphere. Since the incoming solar radiation is higher at low latitudes then so is the radiative equilibrium temperature, and a simple estimate of the radiative equilibrium potential temperature in the upper troposphere is

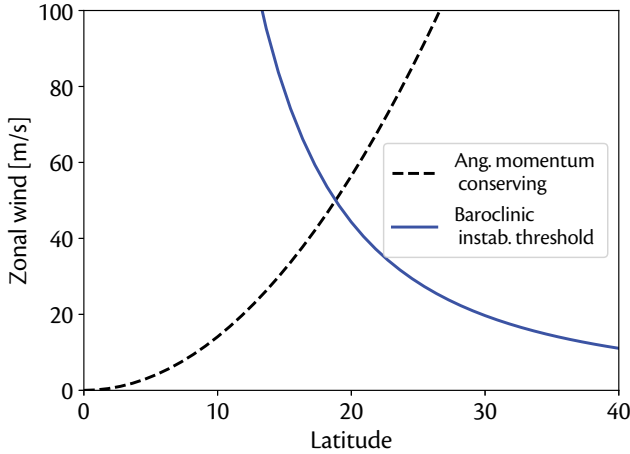
$$\theta_E = \theta_{E0} - \Delta\theta \left( \frac{y}{a} \right)^2, \quad (11.9)$$

where  $\theta_{E0}$  is the equilibrium temperature at the equator and  $\Delta\theta$  determines the equator–pole radiative-equilibrium temperature difference.

Now, the Hadley Cell itself carries heat poleward. If its temperature is given by (11.8) then it will not extend polewards beyond the latitude at which its temperature becomes equal to the radiative equilibrium temperature. Although the radiative equilibrium temperature at the equator,  $\theta_{E0}$ , is a little higher than the actual temperature it is not significantly so and we can take  $\theta(0) \approx \theta_{E0}$ . Equating (11.8) with (11.9) and solving for  $y$  gives the latitudinal extent of the Hadley Cell,  $y_M$ , or its equivalent latitude  $\vartheta_M$ , namely

$$\vartheta_M = \frac{y_M}{a} = \left( \frac{2\Delta\theta gH}{\Omega^2 a^2 \theta_0} \right)^{1/2}, \quad (11.10)$$

The more accurate calculation shown in the right-hand panel of Fig. 11.6 computes the equatorial temperature using the requirement that the circulation in the Hadley Cell does not create energy, so that the areas between the red and blue lines within the Hadley Cell sum to zero. This consideration does not significantly change (11.10).



**Fig. 11.7:** Upper tropospheric zonal winds. The dashed curve shows the approximate angular-momentum conserving wind with  $u = 0$  at the equator using (11.12). The solid curve shows the threshold for baroclinic instability of a zonal flow using a two-layer quasi-geostrophic calculation, (11.11). Polewards of about  $20^\circ$  the angular momentum conserving flow is, in this calculation, baroclinically unstable.

with a corresponding latitude  $\vartheta_M = y_M/a$ . This is the latitude at which the temperature in the outflowing branch of the Hadley Cell equals the radiative equilibrium temperature, at which point the Hadley Cell must terminate. (A more accurate calculation that does not set  $\theta(0) = \theta_{E0}$ , as shown in Fig. 11.6, makes a small quantitative difference but does not change the scaling.)

The main point here is a qualitative one: an ideal Hadley Cell cannot fill the hemisphere; rather, it descends at some latitude, beyond which the air will take on the radiative equilibrium temperature and have no meridional motion. In reality, the Hadley Cell is also baroclinically unstable, as we are about to see.

### 11.3 INSTABILITY OF THE IDEAL HADLEY CIRCULATION

If the air flowing poleward in the Hadley Cell conserves its angular momentum, but the flow at lower levels is smaller, then there will be a large shear that increases with latitude, and eventually this will become baroclinically unstable. A simple model of this uses the two-level quasi-geostrophic equations for which the critical velocity difference between the two layers is given by

$$\Delta U_C \equiv (U_1 - U_2)_C = \frac{1}{4} \beta L_d^2 = \frac{H^2 N^2 a}{8 \Omega y^2}, \quad (11.11)$$

where, using the small angle approximation,  $\beta = 2\Omega/a$ ,  $f = 2\Omega y/a$  and  $L_d = NH/f$  with  $H$  being the depth of the domain and  $N$  the buoyancy frequency. The velocity difference over the entire depth of the troposphere is roughly twice this value. The angular momentum conserving solution gives a velocity difference over depth of the troposphere of

$$\Delta U_M = \frac{\Omega y^2}{a}, \quad (11.12)$$

and equating twice (11.11) with (11.12) gives a cross-over latitude of

$$\vartheta_C = \frac{y_C}{a} = \left( \frac{NH}{2\Omega a} \right)^{1/2}, \quad (11.13)$$

as shown in Fig. 11.7. For values of parameters typical for Earth this gives a latitude of around  $20^\circ$ .

### 11.3.1 Discussion

A parameter that all these results depend on is the rotation rate,  $\Omega$ , and with the value this takes for Earth, along with reasonable values for the stratification and depth of the atmosphere, it is hard for the Hadley Cell to continue polewards of about  $30^\circ$ . On Earth, then, the Hadley Cell is a tropical phenomenon. On more slowly rotating planets the Hadley Cell may extend further polewards, as we discover in Chapter 13, but for the rest of this chapter we focus on the dynamics of Earth's tropics. Finally, we remark that even if the angular momentum conserving solution did not hold and the air were stationary in radiative equilibrium, the horizontal temperature gradient and associated vertical shear would, of themselves, be baroclinically unstable.

## 11.4 THE TROPICAL CIRCULATION

If the Hadley Cell determines how far the tropics extend, let us now look at the circulation that take place within the tropics. We first look at the quasi-horizontal circulation and then at the phenomenon we most associate with the tropics, moist convection.

In earlier chapters we derived the quasi-geostrophic equations, appropriate when the Rossby number is small and so for midlatitude motion. Is such an approach possible for the tropics? The answer is, 'well, in part'. It is possible to derive some reduced sets of equations for the tropics, and that will be the main topic of this section. However, these reduced sets have not proven nearly as useful for the tropics as quasi-geostrophy has been for the midlatitudes, in part because it is harder to make relevant equations that are simple, or simple equations that are relevant. In any case, let us begin with the stratified primitive equations.

### 11.4.1 Balanced, Adiabatic Flow

We will present a scaling for tropical flow side-by-side with the corresponding scaling for midlatitude flow. We begin with the hydrostatic primitive equations for adiabatic, frictionless flow, using the Boussinesq approximation since the essential dynamics, and no more, are contained there. (A largely equivalent derivation could use pressure coordinates.) As in (5.31b) the equations are

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} = -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad (11.14a,b)$$

$$\frac{Db}{Dt} + N^2 w = 0, \quad \nabla \cdot \mathbf{v} = 0. \quad (11.14c,d)$$

We interpret the buoyancy as a perturbation potential temperature,  $b = g \delta\theta/\theta_0$ , and  $N^2 = d\bar{b}/dz$  where  $\bar{b}(z)$  is a reference stratification. Let us suppose that the basic variables scale according to

$$(x, y) \sim L, \quad z \sim H, \quad (u, v) \sim U, \quad w \sim W, \quad t \sim \frac{L}{U}, \quad (11.15)$$

$$\phi \sim \Phi, \quad b \sim B, \quad f \sim f_0.$$

The quantity  $B$  is representative of horizontal variations in buoyancy. Vertical variations scale differently, hence their separate representation in (11.14c). By choosing the time  $t$  to scale advectively we are implicitly eliminating gravity waves. The nondimensional numbers that arise are the Rossby, Burger and Richardson numbers,

$$Ro = \frac{U}{f_0 L}, \quad Bu = \left(\frac{L_d}{L}\right)^2 = \left(\frac{NH}{f_0 L}\right)^2, \quad Ri = \left(\frac{NH}{U}\right)^2, \quad (11.16)$$

and evidently

$$Bu = Ri \times Ro^2. \quad (11.17)$$

The Rossby number  $Ro$  is generally small in midlatitudes for large-scale flow, but in the tropics it is  $\mathcal{O}(1)$  or larger. The Richardson number is usually large in both midlatitudes and tropics except in regions of active convection where  $N$  is very small. If we take  $N = 10^{-2} \text{ s}^{-1}$ ,  $H = 10^4 \text{ m}$  and  $U = 10 \text{ m s}^{-1}$  then  $Ri = 100$ . In fact, on large-scales the atmosphere is sufficiently stratified that the Richardson number is usually such that  $1/(Ro Ri)$  is small in both midlatitudes and tropics.

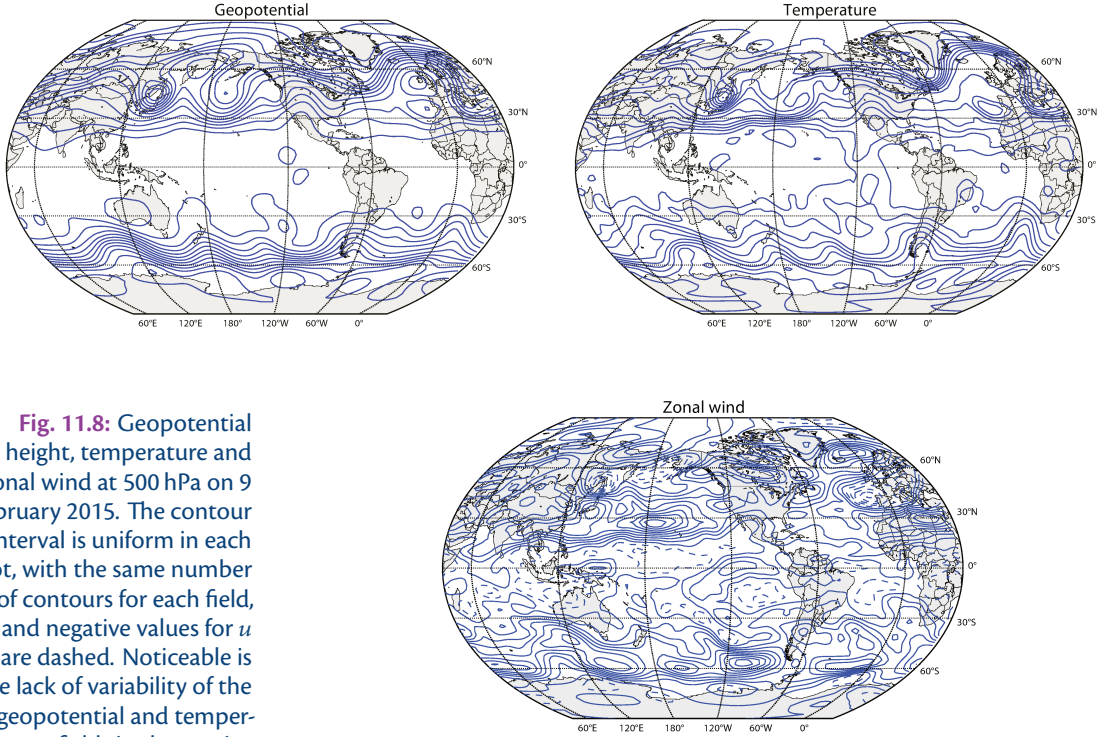
The difference between the tropics and midlatitudes is mainly apparent from the dominant balance in the momentum equation. At small Rossby number we have the familiar geostrophic balance, and with hydrostatic balance we obtain the scaling:

$$\mathbf{f} \times \mathbf{u} \approx -\nabla_z \phi, \quad \frac{\partial \phi}{\partial z} = b, \quad \implies \quad \Phi = f_0 UL, \quad B = \frac{f_0 UL}{H}. \quad (11.18)$$

In the tropics the advective term, or the advectively-scaled time derivative, balances the pressure gradient meaning that  $D\mathbf{u}/Dt \sim \nabla_z \phi$  and, since we still have  $\partial \phi / \partial z = b$  we find

$$\Phi = U^2, \quad B = \frac{U^2}{H}. \quad (11.19)$$

If  $U$  is of similar magnitude in the tropics and midlatitudes (and in the absence of a dynamical analysis this is an assumption), then *variations of pressure and temperature are smaller in the tropics than in midlatitudes*. This is an important and not-quite obvious result and it is the essence of the *weak temperature gradient approximation*, discussed more in Section 11.4.2.



**Fig. 11.8:** Geopotential height, temperature and zonal wind at 500 hPa on 9 February 2015. The contour interval is uniform in each plot, with the same number of contours for each field, and negative values for  $u$  are dashed. Noticeable is the lack of variability of the geopotential and temperature fields in the tropics.

If we were to carry through the derivation in pressure coordinates (see the shaded box on page 42) with geopotential and temperature as the variables we would find

$$\text{Midlatitudes: } \Phi = f_0 UL, \quad T_s = \frac{f_0 UL}{R}, \quad (11.20a)$$

$$\text{Tropics: } \Phi = U^2, \quad T_s = \frac{U^2}{R}, \quad (11.20b)$$

where  $\Phi$  is now the scaling for geopotential,  $T_s$  is the scaling for temperature and  $R$  is the ideal gas constant. For  $U = 10 \text{ m s}^{-1}$ ,  $L = 10^6 \text{ m}$  and, for midlatitudes only,  $f_0 = 10^{-4} \text{ s}^{-1}$ , we obtain

$$\text{Midlatitudes: } \Phi \sim 1000 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 3 \text{ K}, \quad (11.21a)$$

$$\text{Tropics: } \Phi \sim 100 \text{ m}^2 \text{ s}^{-2}, \quad T_s \sim 0.3 \text{ K}. \quad (11.21b)$$

The implications of these results are illustrated in Fig. 11.8, which shows a snapshot of observed contours of geopotential height, temperature and zonal wind; the large-scale variability of geopotential and temperature is evidently much smaller in the tropics than midlatitudes.

### Vertical velocity

The obvious scaling for the vertical velocity is that suggested by the mass continuity equation, namely  $W = UH/L$ . However, as we know from

Chapter 5, the vertical velocity may be much less than this estimate in a stratified, rotating fluid, and here we start with the thermodynamic equation for adiabatic flow,

$$\mathbf{u} \cdot \nabla b + wN^2 = 0, \quad \implies \quad W = \frac{UB}{LN^2}. \quad (11.22)$$

Using (11.18) and (11.19) gives

$$\text{midlatitudes: } W = \frac{f_0 U^2}{HN^2} = \frac{f_0 L}{U} \frac{U^2}{N^2 H^2} \frac{UH}{L} = (Ro Ri)^{-1} \left( \frac{UH}{L} \right), \quad (11.23a)$$

$$\text{Tropics: } W = \frac{U^3}{LHN^2} = \frac{U^2}{N^2 H^2} \frac{UH}{L} = Ri^{-1} \left( \frac{UH}{L} \right), \quad (11.23b)$$

where  $Ri \equiv N^2 H^2 / U^2$  is the Richardson number, with typical values of  $\mathcal{O}(10)$ – $\mathcal{O}(100)$  for large-scale flow. Thus, again perhaps non-intuitively, the vertical velocity is, for adiabatic flow, *smaller* in the tropics than in midlatitudes, by an order of a midlatitude Rossby number. In midlatitudes the scaling for  $W$  is more commonly written as

$$W = \frac{f_0 U^2}{HN^2} = Ro \frac{L^2}{L_d^2} \frac{UH}{L} = \frac{Ro}{Bu} \frac{UH}{L}, \quad (11.23c)$$

so that for scales comparable to the midlatitude deformation radius (i.e., with  $Bu \sim 1$ ) the vertical velocity is an order Rossby-number smaller than mass-continuity scaling suggests.

### Vorticity

Cross-differentiating the horizontal momentum equation gives, as in (5.58), the vertical component of the vorticity equation with associated scalings, using (11.23) for  $W$ ,

$$\frac{D}{Dt}(\zeta + f) = -(\zeta + f) \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \left( \frac{\partial u}{\partial z} \frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \frac{\partial w}{\partial x} \right),$$

$$\text{Tropics: } \left( \frac{U}{L} \right)^2 \sim \left( \frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left( \frac{1}{Ri} \frac{U}{L} \right) \quad \left( \frac{U}{H} \right) \left( \frac{1}{Ri} \frac{UH}{L^2} \right),$$

$$\text{midlatitudes: } \left( \frac{U}{L} \right)^2 \sim \left( \frac{U}{L} + \frac{1}{Ro} \frac{U}{L} \right) \left( \frac{Ro}{Bu} \frac{U}{L} \right) \quad \left( \frac{U}{H} \right) \left( \frac{Ro}{Bu} \frac{UH}{L^2} \right). \quad (11.24a,b,c)$$

In the tropical case with  $Ro = \mathcal{O}(1)$  or larger *all* the terms on the right-hand side are much smaller than the terms on the left-hand side, whereas in the midlatitude case vortex stretching by the Coriolis term,  $f(\partial u / \partial x + \partial v / \partial y)$ , is the same order (because  $f$  is big and the divergence is small). Thus, in the tropical case the vorticity equation simplifies severely and at lowest order becomes the two-dimensional vorticity equation,

$$\frac{D}{Dt}(\zeta + f) = 0. \quad (11.25)$$

The large-scale velocity is, at this order, purely rotational and is given by a streamfunction  $\psi$  such that

$$\nabla^2 \psi = \zeta, \quad (u, v) = \left( -\frac{\partial \psi}{\partial u}, \frac{\partial \psi}{\partial x} \right). \quad (11.26)$$

This equation nominally holds independently at each vertical level, although the underlying assumptions take the depth scale to be large.

#### *A few comments*

The relative weakness of large-scale horizontal gradients of pressure and temperature in the tropics, for a given velocity field, is a robust result of the scaling analysis and borne out in observations. However, (11.26) is not well satisfied by tropical motion for the following reasons:

- The scaling does not take into account heating, or diabatic sources, which may be expected to be particularly important in tropical regions.
- The smallness of the vertical velocity requires that the Richardson number,  $N^2 H^2 / U^2$  be large; that is, that stratification be strong. This is only true in regions that are not actively convecting; in convective regions  $N$  may be small.
- Equation (11.26) tells us nothing about the vertical structure and so, unlike quasi-geostrophy in midlatitudes, is not sufficiently complete to be a useful prognostic, or even diagnostic, equation for tropical motion.

Let us now look at how diabatic effects might affect large-scale motion.

#### **11.4.2 ♦ Large-Scale Flow with Diabatic Sources**

In the tropics, heat sources such as condensational heating are very important and should be explicitly included in the reduced equations. Let us see if and how this is possible, using the shallow water equations for illustration. The underlying physical assumption is that in an air column adiabatic cooling associated with vertical motion balances diabatic heating, and in the shallow water equations this becomes a balance between the heating and the divergence.

#### *Diabatic balanced shallow water equations*

On an  $f$ -plane the shallow-water equations may be written in vorticity-divergence form as

$$\frac{\partial h}{\partial t} + \nabla \cdot (\mathbf{u}h) = Q, \quad (11.27a)$$

$$\frac{\partial \zeta}{\partial t} + \nabla \cdot [\mathbf{u}(\zeta + f_0)] = -r\zeta, \quad (11.27b)$$

$$\frac{\partial \delta}{\partial t} + \nabla^2 \left( \frac{1}{2} \mathbf{u}^2 + gh \right) - \hat{\mathbf{k}} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] = -r\delta, \quad (11.27c)$$



where  $\delta = \partial u / \partial x + \partial v / \partial y$  is the divergence,  $Q$  is the mass or heating source,  $r$  is a frictional coefficient and other notation is standard. The height field is a proxy for both pressure and temperature and we assume that horizontal gradients are weak, by which we mean that the dominant balance in (11.27a) is characterized by the scaling

$$H\Delta = Q_0, \quad (11.28)$$

where  $H$  is the mean thickness of the layer,  $\Delta$  is a scaling for the divergence and  $Q_0$  is the magnitude of the heating. We then choose the velocity scale,  $U$ , and the vorticity scale,  $Z$ , to be

$$U = \frac{Q_0 L}{H}, \quad Z = \Delta = \frac{Q_0}{H}. \quad (11.29)$$

Finally, the magnitudes of horizontal deviations in the height field,  $\mathcal{H}$ , are determined from the divergence equation. Depending on whether rotation is or is not important we deduce

$$\mathcal{H} = \frac{f_0 U L}{g} = \frac{Q_0 f_0 L^2}{g H}, \quad \text{or} \quad \mathcal{H} = \frac{U^2}{g} = \frac{Q_0^2 L^2}{g H^2}. \quad (11.30a,b)$$

The height field may then be separated into a mean and deviation,  $h = H + \eta$ , where  $\eta = \mathcal{H} \hat{\eta}$ . These scalings involve the heating in an essential way and so are fundamentally different from the adiabatic scaling of the previous section.

Using the above scalings (11.27) may be written in nondimensional form as

$$Bu^{-1} \left[ \frac{1}{f_0 T} \frac{\partial \hat{\eta}}{\partial \hat{t}} + Ro \nabla \cdot (\hat{\mathbf{u}} \hat{\eta}) \right] + \hat{\delta} = \hat{Q}, \quad (11.31a)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\zeta}}{\partial \hat{t}} + \nabla \cdot [\hat{\mathbf{u}} (\hat{\zeta} + \hat{f}_0)] = -\frac{r}{f_0} \hat{\zeta}, \quad (11.31b)$$

$$\frac{1}{f_0 T} \frac{\partial \hat{\delta}}{\partial \hat{t}} + \nabla^2 \left( \frac{1}{2} \hat{\mathbf{u}}^2 + \hat{\eta} \right) - \hat{\mathbf{k}} \cdot \nabla \times [\hat{\mathbf{u}} (\hat{\zeta} + \hat{f}_0)] = -\frac{r}{f_0} \hat{\delta}, \quad (11.31c)$$

where  $T$  is the scaling for time,  $Bu = (L_d/L)^2$  with  $L_d = \sqrt{gH}/f_0$ , and  $\hat{f}_0 = 1$ . The Rossby number is given by  $Ro = Q_0/f_0 H$  and is not necessarily small.

### Reduced equations

Let us now suppose that the mass source determines the divergence in (11.31a). The condition for this is that

$$\max \left( \frac{1}{f_0 T}, Ro \right) Bu^{-1} \ll 1, \quad (11.32)$$

which means that the scale of motion cannot be too large and the time scale cannot be too short. If (11.32) is satisfied then the dimensional height equation, (11.27a), becomes

$$\nabla \cdot \mathbf{u} = \frac{Q}{H}. \quad (11.33a)$$

This value of the divergence is used in (11.27b) and (11.27c) which, retaining all terms since none are obviously small, become

$$\frac{\partial \zeta}{\partial t} + \mathbf{u} \cdot \nabla(\zeta + f_0) + (\zeta + f) \frac{Q}{H} = -r\zeta, \quad (11.33b)$$

$$g\nabla^2 h = \hat{\mathbf{k}} \cdot \nabla \times [\mathbf{u}(\zeta + f_0)] - \frac{1}{H} \frac{\partial Q}{\partial t} - r\delta - \nabla^2 \frac{\mathbf{u}^2}{2}. \quad (11.33c)$$

The equation set (11.33) has but one prognostic equation, namely (11.33b), and so is truly balanced and may be thought of as a generalization of (11.25) to the case with non-zero heating. The divergence equation is a nonlinear balance equation except now with a diabatic term on the right-hand side. The divergent flow itself is computed using the height equation, by an assumed balance between adiabatic cooling and diabatic heating. The relationship between velocity and geopotential (or pressure) is the same as in the adiabatic case, because this arises through the momentum equation. Thus, even in the presence of a heating, gradients of geopotential and temperature remain relatively weak, a result that ultimately arises from the smallness of the Coriolis parameter. The importance of the result lies in what it implies about the response of the atmosphere to a localized heating: the equations provide a scaling for the response of the velocity, and suggest that the response may become spread out over a sufficient area to keep the temperature gradients small. This, taken with the scaling arguments of Section 5.3.1, is called the ‘weak temperature gradient approximation’.

## 11.5 EFFECTS OF MOISTURE

Moisture affects nearly every facet of tropical dynamics, and without it the tropics would be a very different place — no towering cumulonimbus clouds for example. However, convection *would* still take place since the basic state set up by the radiative forcing would still be convectively unstable.

In the rest of this chapter we talk about the two things that one immediately notices about the deep tropics — moisture and convection.

### 11.5.1 Measures of Moisture

There are various measures of the amount of moisture in the atmosphere, so let us summarize them. The *absolute humidity* is the amount of water vapour per unit volume, with units of  $\text{kg m}^{-3}$ , or informally  $\text{g m}^{-3}$ . The *mixing ratio*,  $w$ , is the ratio of the mass of water vapour,  $m^v$ , to that of dry air,  $m^d$ , in some volume of air and is thus

$$w \equiv \frac{m^v}{m^d} = \frac{\rho^v}{\rho^d}. \quad (11.34)$$

It is a nondimensional measure but it is often expressed in terms of grams per kilogram. In the atmosphere values range from close to zero to about  $2 \times 10^{-2}$  ( $20 \text{ g kg}^{-1}$ ) in the tropics on a humid day.

The *specific humidity*,  $q$ , is the ratio of the mass of water vapour to the total mass of air — dry air plus water vapour — and so is

$$q \equiv \frac{m^v}{m^d + m^v} = \frac{w}{1 + w} \quad \text{and} \quad w = \frac{q}{1 - q}. \quad (11.35a,b)$$

The specific humidity is just the mass concentration of water vapour in air. In most circumstances in Earth's atmosphere  $m^v \ll m^d$  so that  $q \approx w$ , usually to an accuracy of about one percent. In most of this chapter we ignore the differences between  $w$  and  $q$ , but this is not appropriate for all planetary atmospheres.

The partial pressure of water vapour in air, or the vapour pressure,  $e$ , is the pressure exerted by water molecules and is proportional to the number of molecules, and so the number of moles, of water vapour in the volume. It is given by

$$e = \frac{n^v}{n^d + n^v} P = \frac{m^v/\mu^v}{m^d/\mu^d + m^v/\mu^v} P, \quad (11.36)$$

where  $n^v$  and  $n^d$  are the number of moles of water vapour and dry air in the mixture, and  $\mu^v$  and  $\mu^d$  are the respective molecular weights, and  $P$  is the total pressure. Using (11.34) we can write (11.36) as

$$e = \frac{wp}{w + \epsilon} \quad \text{or} \quad w = \frac{\epsilon e}{p - e}, \quad (11.37)$$

where  $\epsilon = \mu^v/\mu^d \approx 0.62$ . In terms of  $q$  instead of  $w$  these expressions are

$$e = \frac{qP}{q + \epsilon(1 - q)} \quad \text{and} \quad q = \frac{\epsilon e}{p - e(1 - \epsilon)}. \quad (11.38)$$

In Earth's atmosphere  $w \ll 1$  so that

$$e \approx w \frac{p}{\epsilon} = 1.61wp \quad \text{and} \quad q \approx w \approx \epsilon \frac{e}{p}. \quad (11.39)$$

If the mixing ratio of water vapour is  $10 \text{ g kg}^{-1}$  (a typical tropical value) and  $p = 1000 \text{ hPa}$  then  $e \approx 16 \text{ hPa}$ .

The *relative humidity*,  $\mathcal{H}$ , is the ratio of the actual vapour pressure to the saturation vapour pressure,  $e_s$ , which is the maximum vapour pressure that can occur at a given temperature before condensation occurs. Thus,  $\mathcal{H} = e/e_s \approx q/q_s$  where  $q_s$  is the specific humidity at saturation — which we now explain.

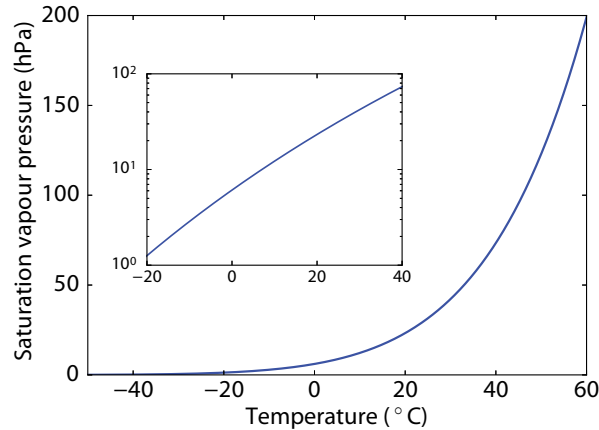
### Saturation and the Clausius–Clapeyron relation

At any given temperature, there is a maximum vapour pressure beyond which condensation normally occurs and this is known as the *saturation vapour pressure*, denoted  $e_s$ . This value is given by the *Clausius–Clapeyron* equation, which, to a very good approximation for Earth's atmosphere, is

$$\frac{de_s}{dT} = \frac{Le_s}{R^v T^2}. \quad (11.40)$$

Here,  $L$  is the latent heat of condensation, which is the amount of heat released per unit mass when water vapour condenses,  $R^v$  is the gas constant for water vapour and  $T$  is the temperature. If  $L$  is a constant (in

**Fig. 11.9:** The saturation vapour pressure of water vapour as a function of temperature. The inset plots the same quantities on a log scale over a slightly smaller range.



fact it varies by about 10% as temperature varies from 0°C to 100°C) then (11.40) can be integrated to give

$$e_s = e_0 \exp \left[ \frac{L}{R^v} \left( \frac{1}{T_0} - \frac{1}{T} \right) \right], \quad (11.41)$$

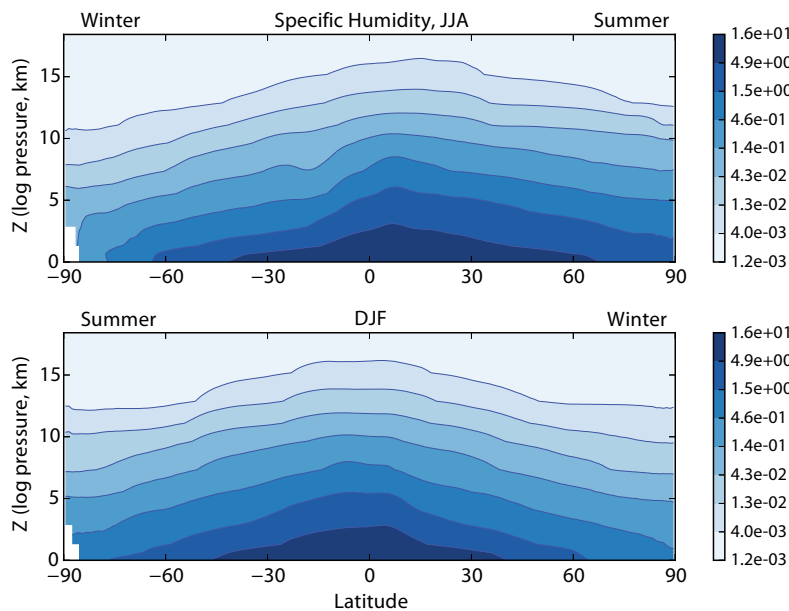
where, for water,  $T_0 = 273 \text{ K}$ ,  $e_0 = 6.12 \text{ hPa}$ ,  $L = 2.44 \times 10^6 \text{ J kg}^{-1}$  and  $R^v = 462 \text{ J kg}^{-1} \text{ K}^{-1}$ . As Fig. 11.9 indicates, the saturation vapour pressure goes up approximately exponentially with temperature. The saturation vapour pressure is a function *only* of temperature; in particular it is not a function of the pressure and its value is independent of the amount of dry air present. Suppose that a parcel of moist air is cooled. The saturation vapour pressure falls according to (11.41) and at some temperature (called the dew point) the actual vapour pressure will become equal to the saturation vapour pressure. The volume is then saturated and any further cooling leads to condensation. This is how rain forms — a parcel of moist air ascends, cools, saturates and condenses.

Why don't other gases cause rain? Why does water, and not nitrogen or oxygen, condense? It is because the other gases are much more volatile than water. For nitrogen, for example, the parameters in (11.41) are such that the saturation vapour pressure at common temperatures is very much higher than the actual pressure. However, on some planets other gases may condense. Thus, temperatures at the winter pole on Mars are sufficiently low that carbon dioxide will condense out, and in the frigid atmosphere of Titan (a moon of Saturn) methane is a condensate.

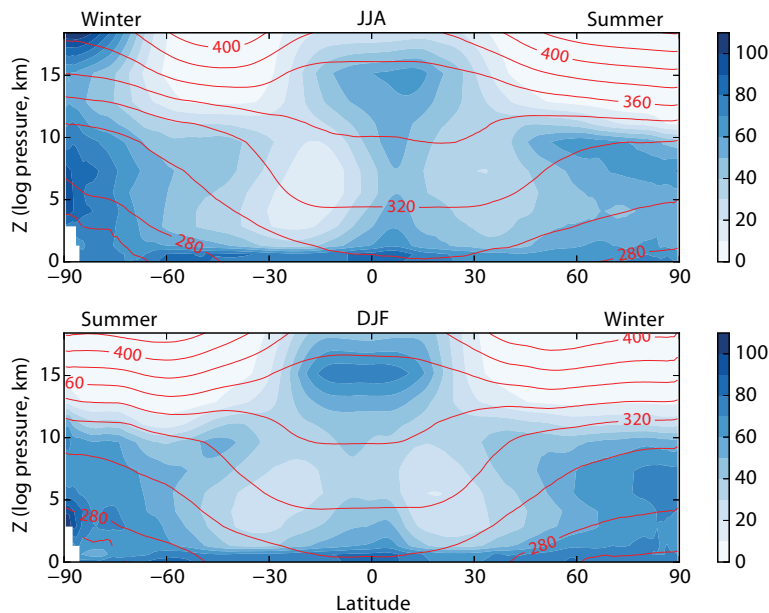
### 11.5.2 Distribution of Water Vapour in the Atmosphere

The strong dependence of  $e_s$  on temperature means that the absolute amount of water vapour in the atmosphere is strongly tied to temperature, as we see in Fig. 11.10. One could easily imagine that this were a plot of temperature, not moisture and the warm tropics hold a lot of moisture. However, the relative humidity field looks quite different, as we see in Fig. 11.11. What determines its distribution?

The amount of water vapour that a volume can hold is independent of the presence or otherwise of dry air in that volume. Rather, it is a function of temperature, and saturation vapour pressure increases approximately exponentially with temperature as in Fig. 11.9.



**Fig. 11.10:** Zonally-averaged specific humidity distribution (g/kg) in the atmosphere, as retrieved from a microwave satellite, for boreal summer in 2008 (top) and boreal winter 2008–2009 (bottom).



**Fig. 11.11:** Zonally-averaged relative humidity distribution in the atmosphere (in percent, shading), and isolines of equivalent potential temperature (contours). (Equivalent potential temperature is a modification (generally a small one, except sometimes in the tropics) of potential temperature to account the effects of water vapour.)

The relative humidity of a parcel at a particular location is, to a decent approximation, determined by the temperature at that location and the saturation specific humidity at the position of last saturation. To see this, imagine a moist, but unsaturated, parcel of air moving toward a colder region. At some location the parcel becomes saturated and excess moisture condenses and (we suppose) falls to earth as rain. If the parcel moves to still cooler locations more moisture falls out, but the parcel remains

saturated. At some location,  $\mathbf{x}$  say, we may suppose that the parcel starts to move to a warmer location and the condensation halts. The amount of moisture in the parcel,  $q$ , then remains fixed at the amount it had at  $\mathbf{x}$ , namely  $q_s(\mathbf{x})$ . At a new position,  $\mathbf{x}'$ , the relative humidity is thus given by

$$\mathcal{H} = \frac{e(\mathbf{x}')}{e_s(\mathbf{x}')} \approx \frac{q(\mathbf{x}')}{q_s(\mathbf{x}')} = \frac{q_s(\mathbf{x}')}{q_s(\mathbf{x})}. \quad (11.42)$$

This argument neglects the effects of re-evaporation of liquid water, which in some circumstances is quite important, but it does capture the main process.

In the tropics the air typically flows equatorward at low levels, picking up moisture from the ocean and the land surface, before rising in the ascending branch of the Hadley Cell. As it ascends it cools and the air becomes saturated. When the air descends in the subtropics it is moving to a warmer environment and so its relative humidity falls, as given by (11.42). The other features of Fig. 11.11 can be explained by similar reasoning, although the trajectories of parcels in the midlatitudes are more complex because they are affected by baroclinic instability. However, one robust feature can be readily explained and that is the very low relative humidity of the stratosphere. The dryness arises because the tropopause is a local minimum of temperature, so that air passing beyond the tropopause does not condense in the stratosphere and its relative (and absolute) humidity are very small. In fact most air entering the stratosphere does so in equatorial regions, where the tropopause is high and cold and serves as a *cold trap* for the moisture.

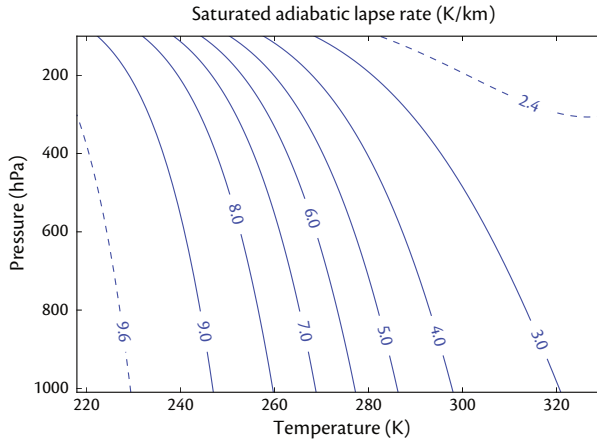
## 11.6 TROPICAL CONVECTION

Convection — often manifested in towering cumulonimbus clouds — is perhaps the most obvious feature of the tropics. Moisture has a large destabilizing effect on the tropics, as we can see by a simple qualitative argument. We saw in Section 3.4 that a column of dry air will be convectively unstable if its lapse rate exceeds (i.e., the temperature falls off more rapidly than) the dry adiabatic lapse rate,  $g/c_p$ , which is about  $10^\circ\text{C}/\text{km}$ . Now consider a column of air that is stable by this criterion but that contains water vapour (or any other condensable for that matter), and suppose we raise a parcel of air. If the water does not condense then the parcel is stable and it sinks back to its original position. But if the parcel cools sufficiently that it becomes saturated then some water will condense and the parcel will warm. It is then possible that the parcel warms to the extent that it is now more buoyant than its surroundings and it will then be unstable, and continue to rise. Let us use a parcel argument similar to that used in Section 3.4 to quantify this.

### 11.6.1 Saturated Adiabatic Lapse Rate

The first law of thermodynamics may be written as

$$c_p dT - \alpha dp = dQ, \quad (11.43)$$



**Fig. 11.12:** Contours of saturated adiabatic lapse rate (K/km) as a function of pressure and temperature, calculated using (11.49) with  $q_s = \epsilon e_s/p$ , and  $e_s$  given by the solution of the Clausius–Clapeyron equation, (11.41).

where  $dQ$  is the heat source. In an adiabatic process, such as may occur in a dry atmosphere,  $dQ = 0$  so that  $c_p dT = \alpha dp$ , and in a hydrostatic atmosphere  $\alpha dp = -g dz$ . Hence an adiabatic profile is characterized by  $c_p dT = -g dz$ , giving the dry adiabatic lapse rate  $dT/dz = -g/c_p$ .

In a moist process there exists the possibility of condensation so that (11.43) becomes

$$c_p dT - \alpha dp = -L dq. \quad (11.44)$$

Here  $-L dq$  is the amount of the heat released by the condensation of a small amount of water vapour  $dq$ , and the negative sign arises because the amount of water vapour in the fluid parcel is then decreasing. For a hydrostatic atmosphere the above equation becomes, treating  $c_p$  and, a little less accurately,  $L$ , as constants,

$$d(c_p T + gz + Lq) = 0. \quad (11.45)$$

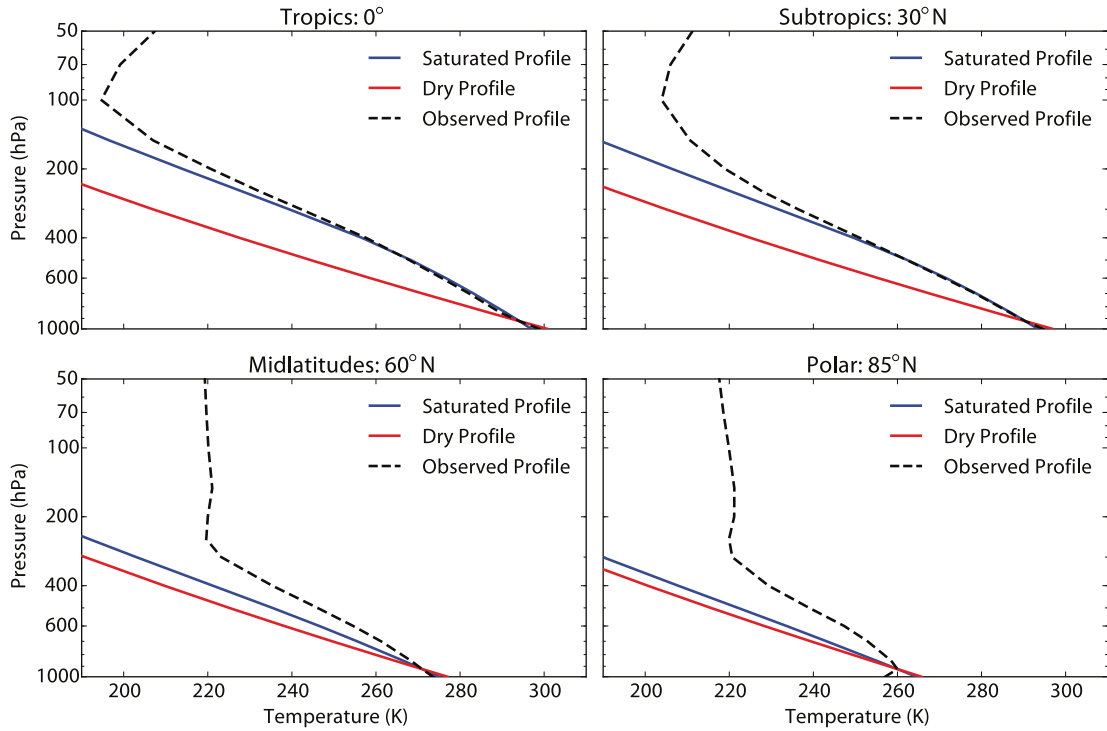
The quantity in brackets,  $\mathcal{M} = c_p T + gz + Lq$ , is known as the moist static energy (and  $c_p T + gz$  is the dry static energy). A parcel that is moving will conserve its moist static energy even if there is condensation. However, condensation will only occur if  $q = q_s$  and, given that fact, we can obtain an expression for the profile of temperature as follows.

As a parcel ascends  $d\mathcal{M}/dz = 0$  and using (11.45) an ascending parcel then has a lapse rate given by

$$c_p \frac{dT}{dz} = -L \frac{dq}{dz} - g. \quad (11.46)$$

If the air is moist but not saturated then an ascending parcel will follow the dry adiabatic lapse rate (because then  $dq/dz = 0$ ) but if it is saturated (with  $q = q_s$ ) then as the parcel ascends it will cool and some water vapour will condense. Since  $q_s \approx \epsilon e_s/p$ , and so is a function of temperature and pressure, we have

$$\frac{dq_s}{dz} = \left( \frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left( \frac{\partial q_s}{\partial p} \right)_T \frac{dp}{dz} = \left( \frac{\partial q_s}{\partial T} \right)_p \frac{dT}{dz} + \left( \frac{q_s}{p} \right) \rho g, \quad (11.47)$$



**Fig. 11.13:** The annually- and zonally-averaged observed temperature profiles at various latitudes, along with profiles constructed by integrating the saturated and dry adiabatic lapse rates, chosen to coincide with the observed temperature at 925 hPa, or about 750 m.

using hydrostasy. Using the Clausius–Clapeyron equation and the ideal gas equation of state gives

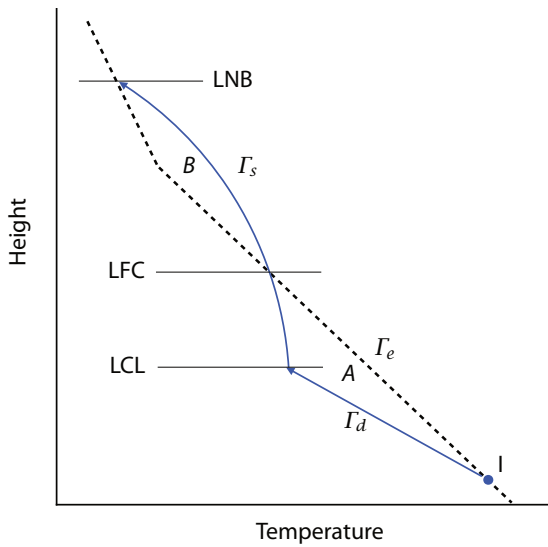
$$\frac{dq_s}{dz} = \frac{Lq_s}{R^v T^2} \frac{dT}{dz} + \frac{gq_s}{R^d T}. \quad (11.48)$$

Using (11.46) and (11.48) we obtain an expression for the lapse rate,  $\Gamma_s$ , of an ascending saturated parcel,

$$\Gamma_s = -\left. \frac{dT}{dz} \right|_{ad} = \frac{g}{c_p} \frac{1 + Lq_s/(R^d T)}{1 + L^2 q_s/(c_p R^v T^2)}. \quad (11.49)$$

This profile is called the *saturated adiabatic lapse rate*, plotted in Fig. 11.12. It is ‘adiabatic’ because we may regard the condensation process as one that is internal to the fluid parcel, with energy being transferred from one form to another during the condensation process without any external energy source. (However, we have not properly accounted for the presence of liquid water in our derivation, so the profile is sometimes called a ‘pseudo-adiabatic lapse rate’.) The lapse rate is a function of temperature and pressure because  $q_s = \epsilon e_s/p$  and  $e_s$  is given by the solution of the Clausius–Clapeyron equation, (11.41). Values of  $\Gamma_s$  are typically around  $6 \text{ K km}^{-1}$  in the lower atmosphere although since  $dq_s/dT$  is an increasing function of  $T$ ,  $\Gamma_s$  decreases with increasing temperature and can be as low as  $3 \text{ K km}^{-1}$ . The second term in the numerator of (11.49) is usually





**Fig. 11.14:** Schematic of a conditional instability in an atmosphere with environmental lapse rate  $\Gamma_e$  (dashed line). A parcel at I is forced to rise, and it does so along the dry adiabat,  $\Gamma_d$ , until it is saturated at the lifting condensation level, LCL. It will then rise along the saturated adiabat,  $\Gamma_s$ , and after reaching the level of free convection (LFC) it is convectively unstable. The parcel continues to rise along the saturated adiabat until it reaches the level of neutral buoyancy (LNB)

quite small (around 0.1, although it can become large at very high temperatures) but the second term in the denominator is positive and order unity. Since  $g/c_p$  is the dry adiabatic lapse rate, the saturated adiabatic lapse rate is smaller than the dry adiabatic lapse rate, as can be seen directly from (11.46) since  $dq^v/dz < 0$ .

Using parcel theory, just as in Section 3.4, a saturated column of air will be stable or unstable depending on whether the lapse rate is less than or greater than (11.49); that is

$$\text{Stability : } -\frac{\partial \tilde{T}}{\partial z} < \Gamma_s, \quad \text{Instability : } -\frac{\partial \tilde{T}}{\partial z} > \Gamma_s, \quad (11.50a,b)$$

where  $\tilde{T}$  is the environmental temperature.

### 11.6.2 Quasi-Equilibrium and Conditional Instability

The lapse rate of the atmosphere rarely exceeds the saturated adiabatic lapse rate, as we see in Fig. 11.13. In the tropics and subtropics the lapse rate is, on average, very close to the saturated adiabat up to about 300 hPa (about 9 km), whereas in midlatitudes it is considerably less, and so more stable, because of the upward transport of heat by baroclinic eddies, which we come to in the next chapter. In the tropics, however, moist convection is the primary determinant of the lapse rate. In the absence of convection temperature would fall off much more rapidly with height in the lower atmosphere, because most of the heating of the atmosphere occurs close to the surface — the ground is heated by the sun, and the ground in turn heats the lower atmosphere. This heating produces an unstable lapse rate and convection then acts, typically on a timescale of hours, to neutralize that lapse rate. *Convective quasi-equilibrium* is the posited state in which the forcing of a convectively unstable profile by large-scale dynamics and/or radiation is statistically balanced by convection. For quasi-

*The maintenance of the tropical lapse rate by convection remains an active topic of research. The degree to which the convection simply acts rather passively to maintain a moist adiabatic lapse rate in response to changes in the large-scale circulation, as well as the degree to which convection actually determines that large-scale circulation, are argued about to this day.*

equilibrium to hold the convection should act on a shorter timescale than changes in the large-scale circulation.

Convection only directly determines the lapse rate over the small fraction of the tropics where convection actually occurs, but nevertheless the average lapse rate over the entire tropics is close to moist neutral. This is because the gravity waves emanating from convective regions adjust the tropics to have a weak horizontal temperature gradient, in a process akin to geostrophic adjustment, hence maintaining approximately the same vertical profile even away from regions of active convection.

If the air is not saturated then the critical lapse rate is not the saturated adiabat, but it is the much larger dry adiabat, and it is common for a parcel to be stable if dry but unstable if saturated, and this leads to the notion of *conditional stability*, whereby a parcel is stable to an infinitesimal perturbation but unstable to a finite perturbation. Consider an environment in which the lapse rate lies between the moist and dry rates, as in Fig. 11.14, and consider a parcel near the surface at position I. Suppose the parcel is adiabatically lifted (perhaps mechanically) then its temperature profile follows the dry adiabat until it is saturated at, by definition, the *lifting condensation level* (LCL). However, the parcel is actually negatively buoyant and so would sink unless the parcel is forced to continue rising, but if it does continue to rise it will be along a saturated adiabat and eventually, at the *level of free convection* (LFC) it will become buoyant and convectively unstable. It will continue to rise until its buoyancy no longer exceeds that of the environment, at the *level of neutral buoyancy*, which may well be at or close to the tropopause where the temperature starts to increase again. (The height of the tropopause is not independent of the convection itself, a matter we discuss in Section 13.3.3.)

Evidently, to trigger such an instability a *finite* perturbation is needed (for example a flow over a hill) and this is known as *conditional instability*. Convection itself can in fact provide the trigger for more convection, leading us into the aforementioned quasi-equilibrium state, but the interested reader may pursue that topic elsewhere, perhaps starting with the references below.

## Notes and References

The angular momentum conserving model of the Hadley Cell stems from Schneider (1977) and Held & Hou (1980). The notion that the Hadley Cell and the radiative equilibrium temperature distribution itself are baroclinically unstable is implicit in the discussion of Lorenz (1967). The weak temperature and pressure gradients in the tropics are consequences of the scaling of Charney (1963) and the topic was further developed by Sobel *et al.* (2001) and others. Moist thermodynamics and convection are dealt with in more detail in a number of books, for example Emanuel (1994), Bohren & Albrecht (1998) and Ambaum (2010).

## Problems

Atmospheric problems are given at the end of Chapter 13.